The Collected Mathematical Papers of Arthur Cayley Vol. III

Arthur Cayley

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THE COLLECTED MATHEMATICAL PAPERS

OF

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The present volume contains 64 papers numbered 159 to 222 originally published in the years 1857 to 1862: the chronological order is slightly departed from for the sake of including a series of papers in the Memoirs and the Monthly Notices of the Royal Astronomical Society: there are thus some earlier papers which have been left over for the next volume.

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159. On some Integral Transformations.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 4–6.]

Suppose that x, a, b, c and x', a', b', c' have the same anharmonic ratios, or what is the same thing, let these quantities satisfy the equation

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ xx', & aa', & bb', & cc' \\ x, & a, & b, & c \\ x', & a', & b', & c' \end{vmatrix} = 0;$$

this equation may be represented under a variety of different forms, which are obtained without difficulty; thus, if for shortness

$$K = a(b'-c')(x-a') + b(c'-a')(x-b') + c(a'-b')(x-c'),$$

then

$$\begin{aligned} Kx' &= -\{bc(b-c)(x-a') + ca(c-a)(x-b') + ab(a-b)(x-c')\}, \\ K(x-a) &= (c-a)(a-b)(b'-c')(x-a'), \\ K(x-b) &= (a-b)(b-c)(c'-a')(x-b'), \\ K(x-c) &= (b-c)(c-a)(a'-b')(x-c'). \end{aligned}$$

Consider x, x' as variables; then

$$K^2\,dx' = -(b-c)(c-a)(a-b)(b'-c')(c'-a')(a'-b')\,dx;$$

let d, d' be any corresponding values of x, x'; then

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ a, & b, & c, & d \\ a', & b', & c', & d' \\ aa', & bb', & cc', & dd' \end{vmatrix} = 0,$$

and we have

$$K(x-d) = D(x'-d'),$$

where

$$D = (c-a)(a-b)(b-c)\cdot A,$$

and

$$A = \frac{(a'-d')(b'-c')}{(b-d)(c-a)} = \frac{(c'-d')(a'-b')}{(c-d)(a-b)} = \frac{(a-d')(b'-c')}{(b-d)(c'-a')} = \frac{(c-d')(a'-b')}{(c-d)(a'-b')}.$$

Suppose $\alpha+\beta+\gamma+\delta=-2$; then

$$\int (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta \, dx = J \int (x'-a')^\alpha (x'-b')^\beta (x'-c')^\gamma (x'-d')^\delta \, dx',$$

where

$$J = (b-c)^{\beta+\gamma+1}(c-a)^{\gamma+\alpha+1}(a-b)^{\alpha+\beta+1}(b'-c')^{\alpha+\delta+1}(c'-a')^{\delta+\alpha+1}(a'-b')^{\alpha+\beta+1}.$$

We may in particular take for a', b', c', d' the systems b, a, d, c; c, d, a, b and d, c, b, a respectively; this gives, writing successively y, z, w instead of x',

$$\begin{aligned}
&\int (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta \, dx \\
&\qquad = M \int (y-a)^\beta (y-b)^\gamma (y-c)^\delta (y-d)^\alpha \, dy \\
&\qquad = N \int (z-a)^\gamma (z-b)^\delta (z-c)^\alpha (z-d)^\beta \, dz \\
&\qquad\qquad = P \int (w-a)^\delta (w-b)^\alpha (w-c)^\beta (w-d)^\gamma \, dw,
\end{aligned}$$

where

$$M = (a-c)^{\alpha+\delta+1}(a-d)^\alpha \cdot (a-b)^{\alpha+\beta+1}(a-c)^{\alpha+\gamma+1}(b-d)^{\beta+\delta+1} \cdot (c-),$$

the relations between the variables x, y, z, w being

$$\begin{aligned}
x &= \frac{(c+d)ab-(a+b)cd-(ab-cd)y}{ab-cd-(a+b-c-d)y} \\
&= \frac{(b+d)ac-(a+c)bd-(ac-bd)z}{ac-bd-(a+c-b-d)z} \\
&= \frac{(a+d)bc-(ad-bc)w}{ad-bc-(a+d-b-c)w}.
\end{aligned}$$

The formula is given in the paper "On an Integral Transformation," *Camb. and Dub. Math. Jour.* t. III. (1848), p. 286 [62], which was suggested to me by Hermann's transformation for elliptic functions, (*Crelle*, t. XXIII. (1846), p. 330).

Suppose now that the values of a', b', c', d' are 0, 1, ∞, ξ; we have in this case

$$x' = \frac{a(b-c)+c(a-b)y}{a(b-c)+(c-a)y},$$

and representing the denominator by K, then

$$\begin{aligned} K(x-a) &= (a-b)(a-c)y, \\ K(x-c) &= (a-c)(b-c), \\ K(x-d) &= (a-b)(c-d)(y-\xi), \end{aligned}$$

where

$$\xi = \frac{(a-d)(b-c)}{(a-c)(b-d)},$$

and we have

$$K^2\,dx = (a-b)(a-c)(b-c)\,dy,$$

whence

$$\int (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta\,dx = \\ (-)^\gamma \frac{(a-b)^{\alpha+\beta+\delta+1}(a-c)^{\alpha+\gamma+1}}{(b-c)^{\alpha+\gamma+1}} \int y^\alpha (1-y)^\beta (y-\xi)^\delta\,dy.$$

It is easy, by means of this equation, to generalise a remarkable formula given by M. Serret in his memoir, "Sur la Representation geometrique des Fonctions elliptiques et ultra-elliptiques," *Liouville*, t. XI. and XII. [1846 and 1847], and *Recueil des Savans etrangers*, t. XL. [1851].[1]

In fact, suppose that the indices α, β, γ, δ are integers, and that two of these indices, e.g. γ, δ, are negative, the remaining two indices being positive, then writing γ', δ' instead of γ, δ the integral

$$\int \frac{(x-a)^\alpha (x-b)^\beta\,dx}{(x-c)^{\gamma'}(x-d)^{\delta'}},$$

where $\gamma' + \delta' = \alpha + \beta + 2$, depends on the integral

$$\int \frac{y^\alpha (1-y)^\beta\,dy}{(y-\xi)^{\delta'}}.$$

Suppose that the fraction under the integral sign is resolved into simple fractions, each of these fractions will be integrable algebraically, except the fraction having for its denominator the simple power

[1] M. Serret has reproduced the theorem in his very interesting and instructive treatise, "Cours d'Algebre superieure," deuxieme edition, Paris, 1854, [quatrieme edition, Paris, 1877].

$y-\xi$, the integral of which is a logarithm. The coefficient of this fraction is at once found by writing in the numerator $y=\xi$ for y; and expanding in ascending powers of $y-\xi$ and equating this coefficient to zero, we have

$$\alpha(\xi-1)+\beta\xi=0,$$

which [observing that $(\gamma'-1)+(\delta'-1)=\alpha+\beta$] is easily seen to be equivalent to

$$\frac{\alpha}{\xi}+\frac{\beta}{\xi-1}+\frac{\gamma'-1}{\xi}+\frac{\delta'-1}{\xi-1}=0.$$

Hence if the function

$$\frac{d}{d\xi}R=\frac{\alpha}{\xi}+\frac{\beta}{\xi-1}+\frac{\gamma'-1}{\xi}+\frac{\delta'-1}{\xi-1}$$

satisfy this condition, the indefinite integral

$$\frac{(a-b)(c-d)}{J}\int\frac{(x-a)^{\alpha}(x-b)^{\beta}\,dx}{(x-c)^{\gamma'}(x-d)^{\delta'}},$$

where $\gamma'+\delta'=\alpha+\beta+2$, will be expressible as a rational algebraical fraction.

It may be noticed, that in the general case, observing that $x=a$, $x=b$ give $y=0$, $y=1$, the integral

$$\int_a^b(x-a)^{\alpha}(x-b)^{\beta}(x-c)^{\gamma}(x-d)^{\delta}\,dx$$

depends on

$$\int_0^1 y^{\alpha}(1-y)^{\beta}\,dy,$$

or, putting $u=\frac{y}{y-\xi}$, upon

$$\int u^{\alpha}(1-u)^{\beta}\left(1-\frac{u}{\xi}\right)^{\gamma}du,$$

which is expressible by means of a hypergeometric series having u for its argument or fourth element.

2, Stone Buildings, Lincoln's Inn, Feb. 1854.

160. On a Theorem relating to Reciprocal Triangles.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 7–10.]

The following theorem is, I assume, known; but the analytical demonstration of it depends upon a formula in determinants which is not without interest.

The theorem referred to may be thus stated: "A triangle and its reciprocal are in perspective;" where by the reciprocal of a triangle is meant the triangle the sides of which are the polars of the angles of the first-mentioned triangle with respect to a conic; and triangles are in perspective when the three lines forming the corresponding angles meet in a point, or what is the same thing, when the three points of intersection of the corresponding sides lie in a line.

Let the equation of the conic be

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

and take (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$ for the coordinates of the angles of the triangle, then if K be the determinant, and (A, B, C), (A', B', C'), (A'', B'', C'') the inverse system, i.e. if

$$\begin{aligned}
KA &= \beta'\gamma'' - \beta''\gamma', & KA' &= \beta''\gamma - \beta\gamma'', & KA'' &= \beta\gamma' - \beta'\gamma,\\
KB &= \gamma'\alpha'' - \gamma''\alpha', & KB' &= \gamma''\alpha - \gamma\alpha'', & KB'' &= \gamma\alpha' - \gamma'\alpha,\\
KC &= \alpha'\beta'' - \alpha''\beta', & KC' &= \alpha''\beta - \alpha\beta'', & KC'' &= \alpha\beta' - \alpha'\beta,
\end{aligned}$$

equations which may be represented in the notation of matrices by the single equation

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}^{-1} = \begin{vmatrix} A, & A', & A'' \\ B, & B', & B'' \\ C, & C', & C'' \end{vmatrix};$$

then the equations of the sides of the triangle are

$$\begin{aligned}
Ax + By + Cz &= 0,\\
A'x + B'y + C'z &= 0,\\
A''x + B''y + C''z &= 0,
\end{aligned}$$

and the coordinates of the angles of the reciprocal triangle may be taken to be (A, B, C), (A', B', C'), (A'', B'', C''); the equations of the lines joining the corresponding angles of the two triangles are therefore

$$\begin{aligned}(B\gamma - C\beta)x + (C\alpha - A\gamma)y + (A\beta - B\alpha)z = 0,\\ (B'\gamma' - C'\beta')x + (C'\alpha' - A'\gamma')y + (A'\beta' - B'\alpha')z = 0,\\ (B''\gamma'' - C''\beta'')x + (C''\alpha'' - A''\gamma'')y + (A''\beta'' - B''\alpha'')z = 0;\end{aligned}$$

the condition that these lines may meet in a point is therefore

$$\begin{vmatrix} B\gamma - C\beta, & C\alpha - A\gamma, & A\beta - B\alpha \\ B'\gamma' - C'\beta', & C'\alpha' - A'\gamma', & A'\beta' - B'\alpha' \\ B''\gamma'' - C''\beta'', & C''\alpha'' - A''\gamma'', & A''\beta'' - B''\alpha'' \end{vmatrix} = 0,$$

an equation which is satisfied identically when A, B, C; A', B', C'; A'', B'', C'' are replaced by their values.

To prove this I transform the different quantities which enter into the determinant as follows: putting

$$\begin{aligned} F &= \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'',\\ G &= \alpha''\alpha + \beta''\beta + \gamma''\gamma,\\ H &= \alpha\alpha' + \beta\beta' + \gamma\gamma', \end{aligned}$$

we have

$$\begin{aligned} K(B\gamma - C\beta) &= \gamma(\gamma'\alpha'' - \gamma''\alpha') - \beta(\alpha'\beta'' - \alpha''\beta')\\ &= \alpha''(\beta\beta' + \gamma\gamma') - \alpha'(\beta\beta'' + \gamma\gamma'')\\ &= \alpha''(\alpha\alpha' + \beta\beta' + \gamma\gamma') - \alpha'(\alpha\alpha'' + \beta\beta'' + \gamma\gamma'')\\ &= \alpha''H - \alpha'G, \quad \text{etc.}; \end{aligned}$$

and the equation becomes

$$\begin{vmatrix} \alpha''H - \alpha'G, & \beta''H - \beta'G, & \gamma''H - \gamma'G \\ \alpha F - \alpha''H, & \beta F - \beta''H, & \gamma F - \gamma''H \\ \alpha'G - \alpha F, & \beta'G - \beta F, & \gamma'G - \gamma F \end{vmatrix} = 0.$$

Now the minor

$$(\beta F - \beta''H)(\gamma G - \gamma F) - (\gamma F - \gamma''H)(\beta G - \beta F)$$

is equal to

$$GH(\beta'\gamma'' - \beta''\gamma') + HF(\beta''\gamma - \beta\gamma'') + FG(\beta\gamma' - \beta'\gamma),$$

i.e. to

$$K(GHA + HFA' + FGA'').$$

And expressing the other minors in a similar form, the equation to be proved is

$$\begin{aligned}(GHA + HFA' + FGA'')(B\gamma - C\beta)& \\ + (GHB + HFB' + FGB'')(C\alpha - A\gamma)& \\ + (GHC + HFC' + FGC'')(A\beta - B\alpha) &= 0,\end{aligned}$$

i.e.

$$HF\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ \alpha, & \beta, & \gamma \end{vmatrix} + FG\begin{vmatrix} A'', & B'', & C'' \\ A, & B, & C \\ \alpha', & \beta', & \gamma' \end{vmatrix} = 0.$$

The first determinant is

$$-\{\alpha(BC' - B'C) + \beta(CA' - C'A) + \gamma(AB' - A'B)\} = -K,$$

and the second determinant is

$$\{\alpha'(B''C - BC'') + \beta'(C''A - CA'') + \gamma'(A''B - AB'')\} = K,$$

and we have therefore identically

$$-HF + FG = -HFG + FGH = 0. \quad \text{Q.E.D.}$$

The corresponding theorem in geometry of three dimensions is that a tetrahedron and its reciprocal have to each other a certain relation, viz. the four lines joining the corresponding angles are generating lines of a hyperboloid, or, what is the same thing, the four lines of intersection of corresponding faces are generating lines of a hyperboloid. The demonstration would show how the theorem in determinants is to be generalised.

2, Stone Buildings, Lincoln's Inn, February, 1855.

161. A Problem in Permutations.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 79.]

The game called Mousetrap gives rise to a singular problem in permutations.

A set of cards, ace, two, three, etc., say up to thirteen, are arranged in a circle with their faces upwards; you begin at any card, and count one, two, three, etc., and if upon counting suppose the number five, you arrive at the card five, that card is thrown out; and beginning again with the next card, you count one, two, three, etc., throwing out if the case happen a new card as before, and so on until you have counted up to thirteen, without coming to a card which ought to be thrown out.

It is easy to see that, whatever the number of the cards is, they may be so arranged as to be all thrown out in the order of their numbers; but that it is not possible in general to arrange the cards so that all the cards, or any specified cards, may be thrown out in a given order. Thus, if all the cards are to be thrown out in the order of their numbers, the arrangements in the case of a single card, two, three, etc. cards, are

$$1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad \text{etc.}$$

The determination in general of the arrangements by which one or more specified cards are thrown out in a given order, is the *Problem in Permutations* proposed for solution.

The particular case which it is proper to consider is that of a single card; say the card k is to be thrown out at the mth counting; the problem is, required the arrangement or arrangements by which this is effected.

Let the cards be k, a_2, a_3, …, a_n, and suppose that the counting begins at k; then, since the card k is not thrown out until the mth counting, the number k must not occur in the first $m-1$ places; and when the number k does occur, the card k is thrown out, and the counting begins again at the next card; hence the cards after k in the arrangement are a_{m+1}, …, a_n.

The problem may be stated as follows: to find the arrangements of k, a_2, a_3, …, a_n such that the number k does not occur in the first $m-1$ places, and that the number a_2 does not occur in the first m places, nor the number a_3 in the first $m+1$ places, …, nor the number a_{n-m+1} in the first $n-1$ places.

It is clear that if $m = n$, there is but one arrangement, viz. that in which the cards are in the order of their numbers; and if $m > n$, there is no arrangement.

The problem may also be stated as follows: required the number of permutations of $1, 2, 3, \ldots, n$ such that the number 1 is not in the first $k-1$ places, the number 2 is not in the first k places, $\ldots$, the number $n-k+1$ is not in the first $n-1$ places.

Feb. 1857.

162. Two Letters on Cubic Forms.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 85–87 and 90–91.]

Letter I

Cher Monsieur Hermite,

Il y a longtemps que j'ai voulu vous ecrire, mais j'en ai ete empeche je ne sais comment; j'ai assez a vous dire par rapport aux covariants, mais a present je vais vous parler des formes cubiques a deux indeterminees.

Il me semble que l'on peut simplifier la theorie de Eisenstein, et l'etendre au cas d'un determinant negatif quelconque, de la maniere que voici.

Soit $(a, b, c, d \,|\, x, y)^3$ une forme cubique, je represente par $(a, b, c, d \,|\, x, y)^3$ la forme quadratique derivee $(ac - b^2, \frac{1}{2}(ad - bc), bd - c^2 \,|\, x, y)^2$. Cela etant, soit (A, B, C) une forme representative (reduite et proprement primitive) au determinant $-D$; a moins que $(A, B, C)^2 = (A, -B, C)$, c'est-a-dire, a moins que (A, B, C) ne soit une forme laquelle par sa triplication produit la forme principale, il n'existe pas de forme cubique (a, b, c, d) telle que $-(a, b, c, d \,|\, x, y)^3 = (A, B, C \,|\, x, y)^2$, ou, si l'on veut, telle que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$; mais en supposant que l'on ait $(A, B, C)^2 = (A, B, C)$ on peut trouver une seule forme cubique qui satisfait a l'equation dont il s'agit. J'ecarte, cela va sans dire, l'une ou l'autre des deux formes (a, b, c, d) et $(-a, -b, -c, -d)$.

En effet on a identiquement

$$(b^2-ac, -\tfrac{1}{2}(bc-ad), c^2-bd \,|\, bxx+cxy+cxy+dyy,\ axx+bxy+bxy+cyy)^2$$

$$= (b^2-ac, \tfrac{1}{2}(bc-ad), c^2-bd \,|\, x, y)^2 \times (b^2-ac, \tfrac{1}{2}(bc-ad), c^2-bd \,|\, x, y)^2;$$

donc, en supposant que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$, il s'ensuit que $(A, B, C)^2 = (A, -B, C)$.

Letter II

Je suppose donc $(A, B, C)^2 = (A, B, C)$, et je dis qu'il ne peut pas y avoir deux formes cubiques, (a, b, c, d) et (a_1, b_1, c_1, d_1), qui aient la propriete dont il s'agit; car, en ecrivant

$$\xi = bxx + cxy + cxy + dyy, \quad \xi_1 = b_1xx + c_1xy + c_1xy + d_1yy,$$
$$\eta = axx + bxy + bxy + cyy, \quad \eta_1 = a_1xx + b_1xy + b_1xy + c_1yy,$$

$(A, -B, C \mid \xi, \eta)^2 = (A, -B, C \mid \xi_1, \eta_1)^2$, ce qui implique d'abord que ξ_1, η_1 soient des fonctions lineaires de ξ, η.

Mais (A, B, C) etant une forme reduite et proprement primitive au determinant D, il n'existe pas de transformation de la forme quadratique en elle-meme, hormis $\xi_1 = \xi$, $\eta_1 = \eta$. Le cas $D = 1$ doit se traiter a part; dans ce cas particulier il n'y a que la forme cubique $(0, 1, 0, 1)$. Donc etc.

Enfin, si $(A, B, C)^2 = (A, -B, C)$, il existe une forme cubique (a, b, c, d) telle que $b^2 - ac = A$, etc.; car en cherchant par la methode de Gauss les valeurs des coefficients p, p', p'', p''' et q, q', q'', q''' qui donnent cette transformation, on obtient d'abord $p' = p'''$, $q' = q'''$. On peut donc representer ces coefficients par b_1, c_1, c_1, d_1; a, b, b, c; savoir, on peut trouver a, b, c, b_1, c_1, d_1 de maniere que

$$(A, B, C \mid b_1xx{+}c_1xy{+}c_1y{+}d_1yy,\ axx{+}bxy{+}bxy{+}cyy)^2 = (A, B, C \mid x, y)^2{\cdot}(A, B, C \mid$$

Cela etant, les equations de Gauss donnent

$$\begin{aligned} A &= b_1^2 - ac_1, & A &= b^2 - ac, \\ 2B &= cb_1 - ad_1, & -2B &= cb + ad - 2bc_1, \\ C &= cc_1 - bd_1, & C &= c^2 - bd, \end{aligned}$$

et de la on obtient

$$b(c - c_1) - a(b - b_1) = 0, \quad c(c - c_1) - b(c - c_1) = 0,$$

c'est-a-dire, ou

$$\frac{b}{c} = \frac{a}{b} = \frac{c_1}{c} = \frac{b_1}{c_1} = \text{etc.},$$

ce qui n'est pas vrai (car cela donnerait $A = 0$, $B = 0$, $C = 0$), ou $b - b_1 = 0$, $c - c_1 = 0$. Donc $b_1 = b$, $c_1 = c$; et en ecrivant d au lieu de d_1, on voit que l'equation de transformation devient

$$(A, B, C \mid bxx{+}cxy{+}cxy{+}dyy,\ axx{+}bxy{+}bxy{+}cyy)^2 = (A, B, C \mid x, y)^2{\cdot}(A, B, C \mid x,$$

ou

$$A = b^2 - ac, \quad 2B = bc - ad, \quad C = c^2 - bd.$$

Je ne sais pas si je vous ai mentionne que j'ai calcule les formes quadratiques pour les treize numeros 307, etc. aux determinants irreguliers. Pour $D = 307$ ces formes sont:

Ordre P.P., 1 genre, 9 classes, c'est-a-dire:

Composition.

$$\begin{array}{c}(1,0,307), \quad (7,1,44), \quad (7,-1,44), \quad 8, \quad 8B, \quad *8, \quad 307 \\ (4,1,77), \quad (11,-1,28), \quad (17,4,19), \quad 8, \quad 88; \quad \text{etc.}\end{array}$$

ou $\mathbf{S}3 = 1$, $\mathbf{S}$, $s = 1$.

Ordre I.P., 1 genre, 3 classes, c'est-a-dire:

$$(2,1,154), \quad (14,1,22), \quad (14,-1,22).$$

A chaque forme de l'ordre P.P. il correspond donc une forme et une seule forme cubique au determinant 1228. Ces formes sont

$$\begin{array}{c}(0,1,0,-307), \quad (1,1,-6,8), \quad (1,-1,-6,-8), \\ (0,2,1,-38), \quad (1,-3,-2,8), \quad (4,1,-4,-3), \\ (0,2,-1,-38), \quad (4,-1,-4,3), \quad (1,3,-2,8).\end{array}$$

Je serai bien aise d'avoir de vos nouvelles, et je vous prie de me croire votre tres-devoue

A. CAYLEY.

Cher Monsieur Hermite,

On demontre sans peine la proposition avancee dans ma derniere lettre, savoir qu'en supposant

$$\begin{aligned}\xi &= bxx + cxy + cxy + dyy, \quad & \xi_1 &= b_1xx + c_1xy + c_1xy + d_1yy, \\ \eta &= axx + bxy + bxy + cyy, \quad & \eta_1 &= a_1xx + b_1xy + b_1xy + c_1yy,\end{aligned}$$

$(A, -B, C \mid \xi, \eta)^2 = (A, -B, C \mid \xi_1, \eta_1)^2$, on doit avoir

$$\xi_1 = \alpha\xi + \beta\eta, \quad \eta_1 = \gamma\xi + \delta\eta,$$

ou α, β, γ, δ sont des entiers.

En effet en trouve d'abord en eliminant par ex. xy et xy,

$$\xi_1 = \lambda + \mu\xi + \nu xx + \rho yy, \quad \eta_1 = \lambda' + \mu'\xi + \nu' xx + \rho' yy,$$

ou α, β, etc., λ, μ, etc., sont des quantites rationnelles. En substituant ces valeurs de ξ_1, η_1, on aura

$$(A, -B, C \mid \lambda, \mu)^2 = 0, \quad (A, -B, C \mid \lambda', \mu')^2 = 0.$$

Donc le determinant n'etant pas un carre, $\lambda = 0$, $\nu = 0$, $\rho = 0$, $\mu' = 0$, etc.; et

$$\xi_1 = \alpha\xi + \beta\eta, \quad \eta_1 = \gamma\xi + \delta\eta,$$

ou α, β, γ, δ sont des quantites rationnelles. Donc

$$b_1 = \alpha b + \beta a, \quad a_1 = \gamma a + \delta b,$$

et de meme pour les autres coefficients; donc α, β seront des quantites rationnelles ayant pour denominateur l'une quelconque a volonte des trois quantites $b^2 - ac$, $c^2 - bd$, $ad - bc$, c'est-a-dire, des quantites A, B, C; et, puisque (A, B, C) est une forme P.P., cela ne peut arriver a moins que α, β ne soient des entiers; de meme, γ, δ seront des entiers.

Je remarque de plus les equations

$$a\beta + b(\alpha - \delta) - c\gamma = 0, \quad b\beta + c(\alpha - \delta) - d\gamma = 0,$$

lesquelles donnent

$$\beta : \alpha - \delta : \gamma = bd - c^2 : bc - ad : ac - b^2 = C : -2B : A,$$

cela fait voir que la transformation en elle-meme de la forme $(A, B, C \mid \xi, \eta)^2$ a moyen de $\xi_1 = \alpha\xi + \beta\eta$, $\eta_1 = \gamma\xi + \delta\eta$ est une transformation propre.

J'aime cependant mieux la maniere dont vous vous etes servi pour deduire d'une forme cubique donnee toutes les autres formes cubiques qui correspondent a la meme forme quadratique. Il serait facile de la meme maniere, etant donnee une transformation quelconque d'une forme quadratique dans le produit de deux autres formes quadratiques, d'en deduire toutes les autres transformations; car il y a pour la fonction axx' etc. un covariant qui correspond au cubicovariant de la fonction ax^3 etc....

Je suis votre tres-devoue

A. CAYLEY,
2, Stone Buildings, Lincoln's Inn,
6 Mars, 1855.

163. On Hansen's Lunar Theory.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 112–125.]

The following paper was written in order to exhibit, in as clear a form as may be, the investigation of the remarkable equations for the motion of the moon established in Hansen's *Fundamenta Nova Investigationis Orbitae quam Luna perlustrat*, etc., Gotha, 1838.

I have availed myself for this purpose of the remarks in Jacobi's two letters in answer to a letter of Hansen's, *Crelle*, t. XLII. [1851], p. 12; it may be convenient to remark that the quantity there represented by A, and which does not occur in Hansen's own investigation, is in this paper represented by Ω.

The position of the moon referred to the earth as centre is determined by r, the radius vector, L, the longitude, Λ, the latitude. Suppose, moreover, that the attractive force at distance unity, $= K(M+E)$, is represented by n^2a^3, then the principal function will be $V = \frac{\mu}{r}$, and the disturbing function R may be represented by $n^2a^3\Omega$; the expression for the half of the *vis viva* is

$$\frac{1}{2}\left\{\left(\frac{dr}{dt}\right)^2 + r^2\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 + r^2\left(\frac{d\Lambda}{dt}\right)^2\right\} = \frac{\mu}{r} - n^2a^3\Omega,$$

and the equations of motion are therefore

$$\begin{aligned}\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 - r\left(\frac{d\Lambda}{dt}\right)^2 &= \frac{\mu}{r^2} - n^2a^3\frac{d\Omega}{dr},\\ \frac{d}{dt}\left(r^2\cos^2\Lambda\frac{dL}{dt}\right) &= -n^2a^3\frac{d\Omega}{dL},\\ \frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) + r^2\cos\Lambda\sin\Lambda\left(\frac{dL}{dt}\right)^2 &= -n^2a^3\frac{d\Omega}{d\Lambda},\end{aligned}$$

where Ω is considered as a function of r, L, Λ.

Consider the orbit as an ellipse, then putting

a, the mean distance,

n, the mean motion $= \sqrt{\dfrac{K(M+E)}{a^3}}$,

e, the excentricity,

ϵ, the mean anomaly at epoch,

ω, the distance of perigee from node,

θ, the longitude of node,

i, the inclination,

Φ, the distance from node,

Ψ, the reduced distance from node, $= L - \theta$,

U, the excentric anomaly,

f, the true anomaly, $= \Phi - \omega$,

the elements of the orbit are a, e, ϵ, ω, θ, i, and we have from the theory of elliptic motion,

$$nt + \epsilon = U - e \sin U, \quad r = a(1 - e\cos U) = \frac{a(1-e^2)}{1 + e\cos f}.$$

Moreover i is the angle at the base of a right-angled spherical triangle, the base, perpendicular, and hypothenuse of which are Ψ, Λ, Φ, hence

$$\tan\Psi = \cos i \tan\Phi, \quad \sin\Lambda = \sin i \sin\Phi, \quad \sin\Psi = \cot i \tan\Lambda, \quad \cos\Phi = \cos\Lambda\cos\Psi.$$

Considering the elements as constant, we have

$$\begin{aligned}
\frac{dr}{dt} &= \frac{nae\sin f}{\sqrt{1-e^2}}, \\
\frac{dL}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}, \\
\frac{d\Lambda}{dt} &= \frac{na^2\sqrt{1-e^2}\cos i}{r^2}, \\
\frac{d\Phi}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}.
\end{aligned}$$

Hence also

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{dr}{dt}\right) &= \frac{n^2a^3e\cos f}{r^2},\\
\frac{d}{dt}\left(r^2\cos^2\Lambda\frac{dL}{dt}\right) &= 0,\\
\frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) &= \frac{n^2a^4(1-e^2)\cos^2 i\sin\Lambda}{\cos^3\Lambda\cdot r^2},\\
\frac{d}{dt}\left(r^2\frac{d\Phi}{dt}\right) &= \frac{2n^2a^3e\sin f}{r}.
\end{aligned}
$$

The foregoing values show that the equations of motion, neglecting the terms which involve the disturbing functions, are satisfied by the elliptic values of r, L, Λ: and in order to satisfy the actual equations of motion, we have only to consider the elements as variable and to write

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 - r\left(\frac{d\Lambda}{dt}\right)^2 &= \frac{\mu}{r^2} - n^2a^3\frac{d\Omega}{dr},\\
\frac{d}{dt}\left(r^2\cos^2\Lambda\frac{dL}{dt}\right) &= -n^2a^3\frac{d\Omega}{dL},\\
\frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) + r^2\cos\Lambda\sin\Lambda\left(\frac{dL}{dt}\right)^2 &= -n^2a^3\frac{d\Omega}{d\Lambda},
\end{aligned}
$$

where the differentiations relate only to the elements, or, what is the same thing, to t in so far only as it enters through the variable elements: the system is at once transformed into

$$
\begin{aligned}
\frac{dr}{dt} &= \frac{nae\sin f}{\sqrt{1-e^2}}\frac{d\epsilon}{dt} + \dots,\\
\frac{dL}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}\frac{d\epsilon}{dt} + \dots,\\
\frac{d\Lambda}{dt} &= \frac{na^2\sqrt{1-e^2}\cos i}{r^2}\frac{d\epsilon}{dt} + \dots
\end{aligned}
$$

Now $\Psi = L - \theta$, or (supposing, as before, that the differentiations relate to t, only in so far as it enters through the variable elements) $d\Psi = -d\theta$, and thence $d\theta = -\frac{d\Psi}{\sin\Psi}$; we have also $d\Phi = -\cos i\,d\theta$.

The equations containing $\frac{dL}{dt}$ and $\frac{d\Lambda}{dt}$ give

$$\cos i\, d\Psi - \frac{na^2\sqrt{1-e^2}}{r^2}\, dt = \frac{n^2a^3}{na^2\sqrt{1-e^2}}\frac{d\Omega}{dL}\, dt,$$

$$\sin i\, d\Psi + \frac{na^2\sqrt{1-e^2}\cos i}{r^2}(\cos\Psi\cos i\, d\theta + \sin\Psi\sin i\, di) = \frac{n^2a^3}{na^2\sqrt{1-e^2}}\frac{d\Omega}{d\Lambda}\, dt;$$

or, expressing $d\theta$ by means of $d\Psi$ and reducing, the second of these equations becomes

$$\sin\Psi\, d\Psi + \frac{na^2\sqrt{1-e^2}}{r^2}\cos\Psi\, d\theta = \frac{n^2a^3}{na^2\sqrt{1-e^2}}\frac{d\Omega}{d\Lambda}\, dt,$$

and combining this with the first of the two equations, and observing that

$$\cos^2 i + \sin^2 i = 1, \quad \cos^2\Lambda\cos^2\Psi = \cos^2\Phi,$$

we find

$$na\sin i\cos^2\Lambda\frac{d\Psi}{dt} + \cos i\cos\Lambda\frac{d\theta}{dt} = \frac{n^2a^3}{na^2\sqrt{1-e^2}}\frac{d\Omega}{d\Lambda}.$$

Now, considering a as a function of r, θ, i, Φ, then Λ, L are given as functions of θ, i, Φ by the equations

$$\sin\Lambda = \sin i\sin\Phi, \quad \tan\Psi = \cos i\tan\Phi, \quad \Psi = L - \theta,$$

and after some simple reductions,

$$\begin{aligned}
\frac{da}{dr} &= \frac{2a}{r},\\
\frac{da}{d\theta} &= -\tan\Phi(\sin i\cos^2\Lambda\frac{dL}{d\theta} + \cos i\cos\Lambda\frac{d\Lambda}{d\theta}),\\
\frac{da}{d\Phi} &= \frac{r}{a}\frac{da}{dr} + \sin i\cos\Lambda\frac{d\Lambda}{d\Phi},
\end{aligned}$$

whence also

$$\frac{da}{d\theta} = \cos i\frac{da}{d\Phi} - \sin i\cot\Phi\frac{da}{di}.$$

We have therefore

$$
\begin{aligned}
\frac{da}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}\frac{d\epsilon}{dt} + \dots,\\
\frac{de}{dt} &= \frac{na\sqrt{1-e^2}\sin f}{r}\frac{d\omega}{dt} + \dots,\\
\frac{d\epsilon}{dt} &= -\frac{na\cot i}{\sqrt{1-e^2}}\frac{di}{dt} + \dots,\\
\frac{d\omega}{dt} &= \frac{na\sqrt{1-e^2}}{e}\frac{de}{dt} + \dots,\\
\frac{d\theta}{dt} &= -\frac{na\sqrt{1-e^2}\sin\Psi}{r^2\sin i}\frac{di}{dt} + \dots,\\
\frac{di}{dt} &= \frac{na\cot i}{r^2\sqrt{1-e^2}}\frac{d\Omega}{dt} + \dots
\end{aligned}
$$

Suppose now that we have ρ, a radius vector, ϕ, for Φ, ϑ, a true anomaly, etc., ν, an excentric anomaly, etc., i.e. let ρ, ϕ, ϑ, ν be what the radius vector, the true anomaly and the excentric anomaly become when the time t, in so far as it enters directly, and not through the variable elements, is replaced by a new variable τ.

We have

$$nt+\epsilon = \nu - e\sin\nu, \quad \tan\frac{\vartheta}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{\nu}{2}, \quad \rho = a(1-e\cos\nu) = \frac{a(1-e^2)}{1+e\cos\vartheta};$$

and of course the differential coefficients of ρ, ϕ with respect to τ may be at once deduced from the corresponding expressions for the differential coefficients of r, f with respect to t, the elements being considered as constant.

Now, using l to denote a logarithm, and supposing that the differentiations affect only the elements, we have

$$\frac{d\rho}{\rho} + \frac{d\vartheta}{\tan\vartheta} = \frac{da}{a} - \frac{e\,de}{1-e^2} + \frac{e\sin\vartheta\,d\omega}{1+e\cos\vartheta},$$

and

$$\rho\cos\phi\,d\theta = \rho\sin\phi\,d\omega = \dots$$

Consider now the point in which the orbit is intersected by any orthogonal trajectory to the successive positions of the orbit, or to fix the ideas, the orthogonal trajectory passing through T, the point in question may, for want of a recognised name, be called the "departure point;" and the angular distances in the orbit measured from this point may be termed "departures;" the expression, "the departure," is to be understood as meaning the departure of the moon.

Write now

χ, the departure of the perigee,

ψ, the departure, $= f + \chi$,

σ, the departure of the node, $= \Psi - \psi$,

Θ, the longitude in orbit of departure point, $= \theta - \sigma$.

It should be remarked that χ is not properly an element, i.e. it is not a function of a, e, ϵ, ω, θ, i without t, and in like manner σ and Θ (which depend upon χ) are not elements.

We have from the geometrical definition

$$d\chi = d\omega + \cos i \, d\theta;$$

and therefore

$$d\sigma = \cos i \, d\theta, \quad d\Theta = (1 - \cos i) \, d\theta.$$

Moreover $\psi = \Phi + \sigma$, which gives (assuming that the differentiations are performed with respect to t, only in so far as it enters through the variable elements) $d\psi = d\Phi + d\sigma = d\Phi + \cos i \, d\theta$, i.e. $d\psi = 0$, an equation which might have been assumed for the purpose of defining the departure point; the equation, in fact, expresses that the departure ψ is measured from a point not actually fixed, but such that the increment of ψ in the interval of time dt is the angular distance between two consecutive positions of the moon.

We have, as above noticed, $d\sigma = \cos i \, d\theta$, and thence and from what has preceded

$$\frac{d\sigma}{dt} = \frac{na \cot \Phi}{\sqrt{1-e^2}} \frac{d\epsilon}{dt} + \ldots, \qquad \frac{da}{dt} = \frac{na \cot i}{r^2\sqrt{1-e^2}} \frac{d\Omega}{dt} + \ldots$$

Now the position of the moon can be determined by means of the quantities r, ψ, Θ, σ, i; hence Ω (which has been considered as a function of r, Φ, θ, i) may, if we please, be considered as a function of r, ψ, Θ, σ, i and from the differential relations

$$dr = dr, \quad d\psi = d\Phi + \cos i \, d\theta, \quad d\Theta = (1 - \cos i) \, d\theta, \quad d\sigma = d\omega + \cos i \, d\theta, \quad di = di,$$

we find
$$\frac{d\Omega}{dr}dr + \frac{d\Omega}{d\psi}d\psi + \frac{d\Omega}{d\Theta}d\Theta + \frac{d\Omega}{d\sigma}d\sigma + \frac{d\Omega}{di}di$$
we have therefore
$$\frac{d\Omega}{d\theta} = \frac{1}{\cos i}\frac{d\Omega}{d\Theta} = \frac{1}{\cos i}\frac{d\Omega}{d\sigma} = -\tan i \cot\Phi\frac{d\Omega}{di}.$$

Or in virtue of a preceding equation
$$\frac{d\Omega}{d\theta} = \frac{1}{\cos i}\frac{d\Omega}{d\Theta} = \frac{1}{\cos i}\frac{d\Omega}{d\sigma} = -\tan i \cot\Phi\frac{d\Omega}{di}.$$

And effecting the substitutions, and collecting the results,
$$\begin{gathered}
\frac{na^2\sqrt{1-e^2}}{r^2}\,dt = n^2a^3\frac{d\Omega}{d\psi}\,dt, \\
\frac{dr}{dt} = 0, \\
\frac{nae\sin f}{\sqrt{1-e^2}}\frac{d\psi}{dt} = -\frac{n^2a^3}{r^2}\frac{d\Omega}{d\epsilon}, \\
\frac{na\cot i}{\sqrt{1-e^2}}\frac{d\Theta}{dt} = -\frac{n^2a^3}{r^2}\frac{d\Omega}{d\sigma}, \\
\frac{na\cot i}{r^2\sqrt{1-e^2}}\frac{di}{dt} = \frac{n^2a^3}{r^2}\frac{d\Omega}{d\sigma},
\end{gathered}$$
where Ω is considered as a function of r, ψ, Θ, σ, i.

Instead of a, i we may introduce the new quantities p, q defined by the equations
$$p = \sin i \sin\sigma, \quad q = \sin i \cos\sigma;$$
this gives
$$\sin^2 i = p^2 + q^2, \quad \sigma = \tan^{-1}\frac{p}{q}, \quad \text{and} \quad \cos i = \sqrt{1-p^2-q^2},$$
and retaining in the formulas the sine and cosine of i, to avoid the introduction of irrational functions of p^2+q^2, we have
$$\frac{1}{\cos i}\frac{d\Omega}{d\theta} = \frac{1}{\cos i \sin^2 i}(q\,dp - p\,dq),$$
i.e.
$$\frac{d\Omega}{d\theta} = \frac{1}{\cos i}(q\,dp - p\,dq),$$

which determines Ω by means of p and q.

We have moreover

$$dp = \sin i \cos\sigma\, d\sigma + \cos i \sin\sigma\, di,$$
$$dq = \sin i \sin\sigma\, d\sigma + \cos i \cos\sigma\, di,$$

from which equations and the foregoing values of di and $d\sigma$ we find the values of dp and dq; the other equations of the system remain unaltered, and we have therefore

$$\frac{dp}{dt} = \frac{na\sqrt{1-e^2}}{r^2}\frac{d\Omega}{d\psi} + \dots,$$
$$\frac{dq}{dt} = -\frac{na\sqrt{1-e^2}}{r^2}\frac{d\Omega}{d\psi} + \dots$$

These are the remarkable equations of Hansen's Lunar Theory. The quantities ρ, ϕ, ϑ, ν are the "partial" elements, and the actual elements are obtained by combining the partial elements with the variations due to the disturbing function.

Feb. 1857.

164. On Gauss' Method for the Attraction of Ellipsoids.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 162–166.]

The following is the method employed in Gauss' Memoir "Theoria attractionis corporum sphaeroidicorum ellipticorum homogeneorum methodus nova tractata," *Werke*, t. V. pp. 1–22.

The attraction of an ellipsoid upon an external point can be found by the following theorem, due to Maclaurin: — The attraction of an ellipsoid of revolution upon an external point in the direction of the axis is equal to the attraction of a confocal ellipsoid passing through the given point, upon the corresponding point of the latter ellipsoid, in the direction of the corresponding axis.

This theorem may be generalised as follows: — The attraction of any ellipsoid upon an external point in the direction parallel to a given diameter is equal to the attraction of a confocal ellipsoid passing through the given point, upon the corresponding point of the latter ellipsoid, in the direction of the corresponding diameter.

To prove this, consider the potential V of the ellipsoid; we have

$$V = \iiint \frac{dx'\, dy'\, dz'}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}},$$

the integral being extended over the volume of the ellipsoid

$$\frac{x'^2}{A^2} + \frac{y'^2}{B^2} + \frac{z'^2}{C^2} = 1.$$

The attraction in the direction of x is $-\frac{dV}{dx}$. Now if we put

$$\frac{x'}{A} = \xi', \quad \frac{y'}{B} = \eta', \quad \frac{z'}{C} = \zeta',$$

the integral becomes

$$V = ABC \iiint \frac{d\xi'\, d\eta'\, d\zeta'}{\sqrt{(x-A\xi')^2 + (y-B\eta')^2 + (z-C\zeta')^2}},$$

the integral being now extended over the sphere $\xi'^2 + \eta'^2 + \zeta'^2 = 1$.

Differentiating with respect to x, we have

$$-\frac{dV}{dx} = ABC \iiint \frac{x - A\xi'\, d\xi'\, d\eta'\, d\zeta'}{\{(x-A\xi')^2 + (y-B\eta')^2 + (z-C\zeta')^2\}^{\frac{3}{2}}}.$$

Now let A, B, C be considered as variables connected by the condition that the point (x, y, z) lies on the ellipsoid, i.e.

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1,$$

and consider the variation of $-\frac{dV}{dx}$ when A, B, C are changed to $A + \delta A$, $B + \delta B$, $C + \delta C$, the point (x, y, z) remaining the same.

We have

$$\frac{x^2}{A^3}\delta A + \frac{y^2}{B^3}\delta B + \frac{z^2}{C^3}\delta C = 0.$$

Also, after some reductions,

$$\delta\left(-\frac{dV}{dx}\right) = \frac{BC\delta A}{A}\iiint \frac{P\, d\xi'\, d\eta'\, d\zeta'}{\{(x - A\xi')^2 + \ldots\}^{\frac{5}{2}}},$$

where P is a certain quadratic function of ξ', η', ζ'.

But the integral $\iiint P\, d\xi'\, d\eta'\, d\zeta' / \{(x - A\xi')^2 + \ldots\}^{\frac{5}{2}}$ vanishes by symmetry when the ellipsoid is one of revolution; and hence the theorem of Maclaurin follows at once.

For the general case, the theorem follows by observing that the variation $\delta(-\frac{dV}{dx})$ can be made to vanish by a proper choice of the variations δA, δB, δC; in fact, the condition for this is precisely that the ellipsoids should be confocal.

Theorem. The attraction is also given by the formula

$$-\frac{dV}{dx} = 2\pi ABC\int_0^\infty \frac{x\, d\lambda}{(A^2 + \lambda)\sqrt{(A^2 + \lambda)(B^2 + \lambda)(C^2 + \lambda)}}.$$

Consider now an ellipsoid, the semi-axes of which are A, B, C, and putting a, b, c for the direction-cosines of the radius vector r, we have

$$x = ar, \quad y = br, \quad z = cr,$$

and

$$\frac{a^2}{A^2} + \frac{b^2}{B^2} + \frac{c^2}{C^2} = \frac{1}{r^2}.$$

The attraction in the direction of the radius vector is

$$Xa + Yb + Zc = -a\frac{dV}{dx} - b\frac{dV}{dy} - c\frac{dV}{dz},$$

which gives

$$Xa+Yb+Zc = 2\pi ABC \int_0^\infty \frac{r\,d\lambda}{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)} \cdot \left\{ \frac{a^2}{A^2+\lambda} + \frac{b^2}{B^2+\lambda} + \frac{c^2}{C^2+} \right.$$

Now from the equation

$$\frac{a^2}{A^2+\lambda} + \frac{b^2}{B^2+\lambda} + \frac{c^2}{C^2+\lambda} = \frac{1}{r^2+\lambda},$$

we find

$$\frac{\partial\lambda}{\partial A} = -\frac{2A\,a^2}{(A^2+\lambda)^2} \Big/ \left\{ \frac{a^2}{(A^2+\lambda)^2} + \frac{b^2}{(B^2+\lambda)^2} + \frac{c^2}{(C^2+\lambda)^2} \right\}.$$

But

$$X = -\frac{dV}{dx} = 2\pi ABC \int_0^\infty \frac{x\,d\lambda}{(A^2+\lambda)\sqrt{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)}};$$

which gives

$$X = \frac{2\pi ABC}{\sqrt{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)}} \cdot \frac{\partial}{\partial A} \int_0^\infty \frac{d\lambda}{\sqrt{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)}}.$$

Now from the equation

$$\frac{x^2}{A^2+\lambda} + \frac{y^2}{B^2+\lambda} + \frac{z^2}{C^2+\lambda} = 1,$$

we find

$$\frac{\partial\lambda}{\partial A} = -\frac{2A\,x^2}{(A^2+\lambda)^2} \Big/ \left\{ \frac{x^2}{(A^2+\lambda)^2} + \frac{y^2}{(B^2+\lambda)^2} + \frac{z^2}{(C^2+\lambda)^2} \right\}.$$

And consequently

$$X = 2\pi ABC \int_0^\infty \frac{x\,d\lambda}{(A^2+\lambda)\sqrt{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)}}.$$

And thence, effecting the integration,

$$X = \frac{2\pi ABC}{\sqrt{(A^2+\lambda)(B^2+\lambda)(C^2+\lambda)}} \cdot x,$$

where λ refers to the ellipsoid whose semi-axes are $\sqrt{A^2+\lambda}$, $\sqrt{B^2+\lambda}$, $\sqrt{C^2+\lambda}$.

2, Stone Buildings, Lincoln's Inn, March, 1857.

The Collected Mathematical Papers of Arthur Cayley

Vol. III (1856–1860)

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165. On some Geometrical Theorems relating to a Triangle circumscribed about a Conic.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 169–175.]

The following investigations were suggested to me by Sir F. Pollock's interesting paper "On a Geometrical Theorem relating to an Equilateral Triangle circumscribed about a Circle," [*Quart. Math. Jour.* t. I. (1857), pp. 167–169].

If on the sides of a triangle ABC, there be taken points α, β, γ, such that $A\alpha$, $B\beta$, $C\gamma$ meet in a point; and if on each side of the triangle there be taken two points forming with the two angles on the same side an involution having the first-mentioned point on the same side for a double point; then if three of the six points lie in a line, the two lines are said to be *harmonically related* with respect to the triangle ABC and point O. Call these the lines (r), (s). The triangle ABC and point O give rise to a determinate conic; viz., the conic touching the sides at the points α, β, γ.

The harmonic of the point O with respect to the triangle is a line (o), which is also the polar of O with respect to the conic. The conic meets this line in two points (as the figure is drawn, imaginary ones), I, J.

Suppose now that the harmonic lines (r), (s) are harmonically related to the points I, J, (i.e. let the lines (r), (s) and the lines through their point of intersection and I, J, be a harmonic pencil, (or, what is the same thing, let I, J, and the points in which the line of junction meets the lines (r), (s) be a harmonic range), then,

Theorem. *The point of intersection of the lines (r), (s) lies on the conic.*

In order to prove this, take $x = 0$, $y = 0$, $z = 0$, for the equations of BC, CA, AB, and $x = y = z$ for the point O; the equations of the harmonic lines will be

$$ax + by + cz = 0, \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0,$$

the equation of the conic is

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

the equation of the line (o) is $x + y + z = 0$.

The coordinates of the points I, J, are

$$x : y : z = 1 : \omega : \omega^2, \quad \text{and} \quad x : y : z = 1 : \omega^2 : \omega,$$

where ω is an imaginary cube root of unity.

The equations of the lines joining the point of intersection of (r), (s) with the points I, J, are

$$\frac{ax + by + cz}{a + b\omega + c\omega^2} = \frac{bcx + cay + abz}{bc + ca\omega + ab\omega^2}$$

and

$$\frac{ax + by + cz}{a + b\omega^2 + c\omega} = \frac{bcx + cay + abz}{bc + ca\omega^2 + ab\omega},$$

and these will form with the lines $ax+by+cz = 0$, $bcx+cay+abz = 0$, a harmonic pencil, if

$$(a+b\omega+c\omega^2)(bc+ca\omega^2+ab\omega)+(a+b\omega^2+c\omega)(bc+ca\omega+ab\omega^2) = 0;$$

i.e., if

$$bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b = 0,$$

which is, therefore, the condition in order that (r), (s), I, J may be harmonically related.

The coordinates of the point of intersection of (r), (s) are

$$x : y : z = a(b^2 - c^2) : b(c^2 - a^2) : c(a^2 - b^2);$$

and substituting these values in $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy$, we see, *a priori*, that the result contains the factor

$$2abc + bc^2 + b^2c + ca^2 + c^2a + ab^2 + a^2b = (b + c)(c + a)(a + b);$$

in fact $b + c = 0$ gives $x : y : z = b(b^2 - a^2) : b(b^2 - a^2)$, i.e. $x = 0$, $y + z = 0$, which makes $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy$ vanish; and so for the other factors.

Effecting the substitution, we find

$$x^2+y^2+z^2-2yz-2zx-2xy = (2abc+bc^2+b^2c+ca^2+c^2a+ab^2+a^2b)\times(2abc+bc^2+b^2c+$$

which equals 0, on account of the second factor.

Hence the point of intersection of the lines (r), (s) lies upon the conic. Q.E.D.

Consider, next, a side of the triangle: then,

Theorem. *The following four points on a side of the triangle, viz. the point of contact, the point of intersection with the line* (o), *and the points of intersection with the harmonic lines* (r), (s), *form a harmonic range.*

In fact, for the side $x = 0$ the points in question are given by means of the equations

$$y - z = 0, \quad y + z = 0, \quad by + cz = 0, \quad \frac{y}{b} + \frac{z}{c} = 0,$$

and forming the equations the theorem is at once seen to be true. Q.E.D.

It is interesting to give the analytical expressions for some other of the points and lines of the figure. The harmonic lines (r), (s), meet in a point of the conic given by the equations

$$x : y : z = a(b^2 - c^2) : b(c^2 - a^2) : c(a^2 - b^2).$$

This point may be spoken of as the *common point of intersection* of the harmonic lines with the conic. The foregoing values give

$$\frac{x}{a} = b^2 - c^2, \quad \frac{y}{b} = c^2 - a^2, \quad \frac{z}{c} = a^2 - b^2,$$

which may also be written

$$x = Pa, \quad y = Qb, \quad z = Rc,$$

if for shortness

$$P = b^2 - c^2, \quad Q = c^2 - a^2, \quad R = a^2 - b^2,$$

formulae which will be presently useful.

Each harmonic line meets the conic in another point, which may be termed the *simple point of intersection*. For the line $ax+by+cz=0$, the coordinates of the simple point of intersection are given by

$$x:y:z=-a(b+c)^2:-b(c+a)^2:-c(a+b)^2;$$

or, what is the same thing, by

$$\frac{x}{-Pa}:\frac{y}{-Qb}:\frac{z}{-Rc}=\frac{1}{P}:\frac{1}{Q}:\frac{1}{R};$$

and, in like manner, the coordinates of the simple point of intersection of the line $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=0$ are given by

$$\frac{x}{a}:\frac{y}{b}:\frac{z}{c}=(b-c)^2:(c-a)^2:(a-b)^2.$$

Hence the equation of the line joining the two simple points of intersection is

$$\begin{vmatrix} x & y & z \\ -Pa & -Qb & -Rc \\ \frac{a}{P} & \frac{b}{Q} & \frac{c}{R} \end{vmatrix}=0,$$

or expanding and reducing

$$\frac{x}{a}(Q-R)+\frac{y}{b}(R-P)+\frac{z}{c}(P-Q)=0.$$

The equation of the tangent to the conic at the common point of intersection is evidently

$$Px+Qy+Rz=0.$$

The last-mentioned lines, together with the harmonic lines (r), (s), viz. the lines $ax+by+cz=0$, $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=0$, may be considered as the sides of an inscribed quadrilateral; the equation of the conic must therefore be expressible in a form in which this is put in evidence; to do this, I first form the equation

$$A\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)^2+B(ax+by+cz)^2+C\left[\frac{x}{a}(Q-R)+\frac{y}{b}(R-P)+\frac{z}{c}(P-Q)\right]^2+D(P$$

which may also be written

$$\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)^2\left(A+Ba^2+Ca^2\frac{(Q-R)^2}{a^4}+Da^2P^2\right)+\&\text{c.}=0,$$

where

$$A=-2a^2b^2c^2,\quad B=\frac{1}{2}(b^2-c^2)(c^2-a^2)(a^2-b^2),\quad C=-\tfrac{1}{2}a^2b^2c^2,\quad D=\tfrac{1}{2},$$

and then putting

$$M=-2a^2b^2c^2,\quad N=\frac{1}{2}(b^2-c^2)(c^2-a^2)(a^2-b^2),$$

an equation which gives

$$\Lambda=\frac{1}{2}(b^2-c^2)(c^2-a^2)(a^2-b^2),$$

$$A=P^2-(Q-R)^2=-2(QR+RP+PQ),$$

and it is easy by means of these relations to verify the identical equation

$$\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)^2\Lambda+(ax+by+cz)(bcx+cay+abz)\cdot 2+\left[\frac{x}{a}(Q-R)+\frac{y}{b}(R-P)+\frac{z}{c}(P-\right.$$

or, writing for Λ its value $\Lambda=0$, the equation of the conic takes the form

$$M(ax+by+cz)\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)-(Pax+Qby+Rcz)(Px+Qy+Rz)=0,$$

which is as it should be.

It may be added, that the common point of intersection and the points in which the harmonic lines (r), (s) meet a side of the triangle lie in a conic passing through the points I, J, such that with respect to this conic the point of contact is the pole of the line (o). Thus for the side $x=0$ the equation of the conic in question is

$$bc(b-c)(c-a)(a-b)\,yz+\text{\&c.}=0,$$

where

$$bc(a-b)(c-a)-2bc\,yz+\text{\&c.}=0,$$

and similarly for the other two conics.

In the case where the triangle is an equilateral triangle and the line (o) is the line at ∞, the conic becomes a circle, the points I, J, are the circular points at infinity, and lines harmonic in respect to the points I, J, become lines at right angles to each other: the foregoing results agree, therefore, with Sir F. Pollock's Theorem.

166. Note on the Homology of Sets.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 178.]

Let L denote a set of any four elements a, b, c, d, and in like manner L_1, L_2, &c. sets of the four elements α, β, γ, δ; a_1, b_1, c_1, d_1, &c.; then we may establish a relation of homology between four sets L, L_1; L_2, L_3, and four other sets Λ, Λ_1, Λ_2, Λ_3; viz., considering the corresponding anharmonic ratios of the different sets, we may suppose a relation of homology between these ratios.

Thus considering the set L, write

$$x = (a-b)(c-d), \quad y = (a-c)(d-b), \quad z = (a-d)(b-c),$$

then $x+y+z=0$ and the anharmonic ratios of the set are $x:y:z$; we may, if we please, take $x:y$ as the anharmonic ratio of the set. And in like manner taking $\xi:\eta$ as the anharmonic ratio of the set α, β, γ, δ, &c., the assumed relation between the sets L, L_1, L_2, L_3 and the sets Λ, Λ_1, Λ_2, Λ_3 will be

$$\begin{vmatrix} x & x_1 & x_2 & x_3 \\ y & y_1 & y_2 & y_3 \\ \xi & \xi_1 & \xi_2 & \xi_3 \\ \eta & \eta_1 & \eta_2 & \eta_3 \end{vmatrix} = 0,$$

and it is to be observed, that this relation is independent of the particular ratio $x:y$ which has been chosen as the anharmonic ratio of the set; in fact, if we write $x = \eta\xi$, $y = \xi\zeta$, &c., then reducing the result by means of an elementary property of determinants, the equation will preserve its original form, but will contain the ratios $y:z$; $\eta:\zeta$, &c., instead of the ratios $x:y$; $\xi:\eta$, &c.

167. Apropos of Partitions.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 183–184.]

Let

$$\Pi(1-x^a)^{-1} = (1-x^a)^{-1}(1-x^b)^{-1}\ldots(k \text{ factors})$$

and assume that

$$\Pi(1-x^a)^{-1} = \frac{1}{\sqrt{(1-z)^k}}\, e^{-\frac{Az}{1-z}-\frac{Bz^2}{(1-z)^2}-\&\text{c.}},$$

the part of $\frac{1}{\sqrt{(1-z)^k}}$ which involves negative powers of $1-x$, then

$$\Pi(1-x^a)^{-1} = \text{Coefficient } z^n \text{ in } \frac{1}{\sqrt{(1-z)^k}}\, e^{-\frac{Az}{1-z}-\&\text{c.}},$$

$$= \text{Coefficient } z^{-1} \text{ in } \frac{1}{\sqrt{(1-z)^k}}\, \frac{e^{-\frac{Az}{1-z}-\&\text{c.}}}{z^{n+1}},$$

which suggests the question of the expansion in powers of z, of the function

$$\frac{1}{\sqrt{(1-z)^k}}\, e^{-\frac{Az}{1-z}-\frac{Bz^2}{(1-z)^2}-\&\text{c.}}.$$

Now by the definition of Bernoulli's numbers

$$\frac{t}{e^t-1} = 1 - \frac{t}{2} + \frac{B_1t^2}{2!} - \frac{B_2t^4}{4!} + \frac{B_3t^6}{6!} - \&\text{c.},$$

$$\frac{1}{1-e^{-t}} = \frac{1}{t} + \frac{1}{2} + \frac{B_1t}{2!} - \frac{B_2t^3}{4!} + \frac{B_3t^5}{6!} - \&\text{c.},$$

from which it is easy to deduce

$$\log\frac{1}{1-e^{-t}} = \log\frac{1}{t} + \frac{t}{2} - \frac{B_1t^2}{2\cdot 2!} + \frac{B_2t^4}{4\cdot 4!} - \&\text{c.},$$

and, writing $-z$ for t,

$$\log\frac{1}{1-e^{z}} = \log\frac{1}{-z} - \frac{z}{2} - \frac{B_1z^2}{2\cdot 2!} - \frac{B_2z^4}{4\cdot 4!} - \&\text{c.}$$

168. A Demonstration of the Fundamental Property of Geodesic Lines.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 185–186.]

This is the translation of a paragraph in M. Bertrand's Memoir "Démonstration géométrique de quelques théorèmes relatifs à la théorie des Surfaces," *Liouv.* t. XIII. (1848), p. 73, in which he shows that the theorem that the geodesic lines on a surface have at each point their osculating plane normal to the surface is an immediate consequence of Meusnier's theorem.

The geodesic lines on a surface have at each point their osculating plane normal to the surface.

In fact, let AB be a geodesic line on the surface S, A and B two infinitely near points of this line, AT the tangent at A, BM a line parallel to AT. The plane TAB is the osculating plane at A; it is required to show that this plane is normal to the surface. Suppose that it is not, and let AP be the normal section made in the surface by a plane through AT perpendicular to the osculating plane. According to Meusnier's theorem, the radius of curvature of the section AP is equal to the radius of curvature of any other section through the same tangent, multiplied by the cosine of the angle between the two sections. Now the section AB has for its radius of curvature the quantity $\dfrac{AB^2}{2BM}$; if AP were not normal to the osculating plane, its radius of curvature would be smaller than this, and would therefore be a finite quantity. This is impossible, since the two points A and P coincide (because, the line AT being tangent to the surface, the section AP touches the surface, and its radius of curvature at the point A is therefore infinite). The supposition that the osculating plane is not normal to the surface is therefore inadmissible.

169. Eisenstein's Geometrical Proof of the Fundamental Theorem for Quadratic Residues.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 186–191.]

(Translated from the German.)

I give, after the memoir of M. Gauss: "Theorematis fundamentalis in doctrina de residuis quadraticis demonstrationes et ampliationes novae," *Werke*, t. II. pp. 47–64, the geometrical proof indicated at the foot of p. 64 of the same work, and I add some remarks relating to the extension of the theorem in the case of two positive odd numbers.

Let p, q be two different positive odd primes; and let

$$AD = FB = p, \quad AF = DB = q, \quad AC = EG = \tfrac{1}{2}(p-1), \quad AE = CG = \tfrac{1}{2}(q-1).$$

The rectangle $ADBF$ is divided by lines parallel to AD and AF into pq equal rectangles, of which, neglecting those through the points C and E, there remain $(p-1)(q-1)$. The diagonal AB does not pass through any point of intersection of these lines, because p and q are prime to each other. Let l be any number in the series $1, 2, 3, \ldots, \frac{1}{2}(p-1)$; the lth line parallel to AD, counting from AD, meets the diagonal AB in a point K between B and A, such that $AK : AB = l : p$, and therefore $AK = \dfrac{l \cdot q}{p}$; hence the part of this line intercepted between AD and the diagonal AB contains $\left[\dfrac{lq}{p}\right]$ rectangles. If m is any number in the series $1, 2, 3, \ldots, \frac{1}{2}(q-1)$, the mth line parallel to AF (counting from AF) contains $\left[\dfrac{mp}{q}\right]$ rectangles between AF and the diagonal AB.

Now let μ be the number of remainders which the numbers $q, 2q, 3q, \ldots, \frac{1}{2}(p-1)q$ leave on division by p, and which are greater than $\frac{1}{2}p$, and let ν be the analogous number relative to p and q; then

$$\mu \equiv \sum_{l=1}^{\frac{1}{2}(p-1)} \left[\frac{lq}{p}\right] \pmod 2, \quad \nu \equiv \sum_{m=1}^{\frac{1}{2}(q-1)} \left[\frac{mp}{q}\right] \pmod 2,$$

formulae for the number of lattice points below and above the diagonal respectively.

For the number of lattice points below the diagonal is $\sum_{l=1}^{\frac{1}{2}(p-1)} \left[\frac{lq}{p}\right]$, and the number above is $\sum_{m=1}^{\frac{1}{2}(q-1)} \left[\frac{mp}{q}\right]$; and it is easy to see that

$$\sum_{l=1}^{\frac{1}{2}(p-1)} \left[\frac{lq}{p}\right] + \sum_{m=1}^{\frac{1}{2}(q-1)} \left[\frac{mp}{q}\right] = \frac{p-1}{2} \cdot \frac{q-1}{2}.$$

Now by the properties of quadratic residues, $\left(\frac{q}{p}\right) = (-1)^{\mu}$ and $\left(\frac{p}{q}\right) = (-1)^{\nu}$, so that

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\mu+\nu} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

which is the fundamental theorem.

170. On Schellbach's Solution of Malfatti's Problem.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 222–226.]

The problem of Malfatti is: "To describe in a given triangle three circles, each of which touches the other two and also two sides of the triangle." Schellbach's solution is given in Crelle, t. XLV. pp. 376–378, but a remark essential to the complete solution is there omitted.

Let the equations of the sides of the triangle be $x = 0$, $y = 0$, $z = 0$; and let the circles be

$$u_1^2 - 4yz = 0,$$
$$u_2^2 - 4zx = 0,$$
$$u_3^2 - 4xy = 0,$$

where ν_1, ν_2, ν_3 are linear functions of x, y, z. The equations of the common tangents to the circles taken in pairs are

$$\nu_1 + \nu_2 + 2\sqrt{xz} = 0, \quad \nu_2 + \nu_3 + 2\sqrt{xy} = 0, \quad \nu_3 + \nu_1 + 2\sqrt{yz} = 0,$$

and these must be respectively identical with

$$x = 0, \quad y = 0, \quad z = 0.$$

Hence we must have

$$\nu_1 + \nu_2 + 2\sqrt{xz} = \lambda x, \quad \nu_2 + \nu_3 + 2\sqrt{xy} = \mu y, \quad \nu_3 + \nu_1 + 2\sqrt{yz} = \nu z,$$

and consequently

$$\nu_1 = \tfrac{1}{2}(\lambda x - \mu y + \nu z) - \sqrt{xz} + \sqrt{xy} - \sqrt{yz},$$

where

$$\lambda = m + n, \quad \mu = n + l, \quad \nu = l + m,$$

i.e.

$$2\sqrt{mn} = l, \quad 2\sqrt{nl} = m, \quad 2\sqrt{lm} = n,$$

which are given by Schellbach. The condition that ν_1 shall be linear gives

$$\sqrt{xz} - \sqrt{xy} + \sqrt{yz} = \sqrt{x}(\sqrt{z} - \sqrt{y}) + \sqrt{yz},$$

and writing

$$\nu_1 = \tfrac{1}{2}(\lambda x - \mu y + \nu z) - (\sqrt{xz} - \sqrt{xy} + \sqrt{yz}),$$

the circle is $\nu_1^2 - 4yz = 0$.

171. Note on Mr Salmon's Equation of the Orthotomic Circle.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 267–269.]

Mr Salmon has given in his *Higher Plane Curves* the equation of the orthotomic circle of a given cubic. I propose to obtain the equation by a different method.

Let the equation of the cubic be

$$U = (x + y + z)^3 + 6kxyz = 0,$$

and let it be required to find the equation of the circle which cuts this cubic at right angles. The general equation of a circle is

$$x^2 + y^2 + z^2 - 2yz \cos A - 2zx \cos B - 2xy \cos C = 0,$$

or as it may be written,

$$V = x^2 + y^2 + z^2 - 2yz \cos A - 2zx \cos B - 2xy \cos C = 0.$$

The condition that two curves cut at right angles is that their tangents at each point of intersection are at right angles. Now the equation of the pair of tangents to the cubic at a point (x, y, z) is given by the Hessian condition; and for orthotomy, we must have the sum of the angles of intersection equal to π.

Now $V = 0$ may be written

$$V = (x+y+z)^2-2\sum yz(1+\cos A) = (x+y+z)^2-4\sum yz \cos^2 \tfrac{A}{2} = 0,$$

and if we write for shortness

$$f = (x + y + z)^3 + 6kxyz,$$

then the orthotomic circle is found from the condition that the polar conic of any point on $V = 0$ with respect to $f = 0$ shall be a rectangular hyperbola.

The polar conic of a point (ξ, η, ζ) with respect to the cubic is

$$(x\partial_\xi + y\partial_\eta + z\partial_\zeta)^2 f = 0,$$

which expands to

$$6(x + y + z)(\xi + \eta + \zeta) + 2k(\eta\zeta x + \zeta\xi y + \xi\eta z) = 0.$$

For this to be a rectangular hyperbola, we must have the sum of the coefficients of x^2, y^2, z^2 equal to zero when referred to its principal axes; or equivalently, the condition is that the curve and circle cut orthogonally.

The condition of orthotomy may be expressed directly. If $S = 0$ is the circle, $U = 0$ the cubic, then at any common point the radii of curvature R_1, R_2 of the two curves satisfy

$$\frac{1}{R_1^2} + \frac{1}{R_2^2} - \frac{2\cos\omega}{R_1 R_2} = \frac{1}{\rho^2},$$

where ω is the angle between the curves and ρ is the radius of curvature of one in terms of the other. For orthogonal intersection, $\cos\omega = 0$.

The actual calculation gives for the orthotomic circle

$$x^2+y^2+z^2-2yz\cos A-2zx\cos B-2xy\cos C = \lambda\left[(x+y+z)^2 + \mu xyz\right],$$

where λ and μ are determined by the condition that the circle passes through the foci of the cubic.

Mr Salmon's result is equivalent to the following: if we write the cubic in the form

$$ax^3+by^3+cz^3+3a_2x^2y+3a_3x^2z+3b_1y^2x+3b_3y^2z+3c_1z^2x+3c_2z^2y+6dxyz = 0,$$

then the orthotomic circle is

$$\begin{vmatrix} a & a_2 & a_3 & c_1 \\ b_1 & b & b_3 & c_2 \\ a & a_2 & a_3 & \alpha \\ \beta & \beta' & \beta'' & \gamma \end{vmatrix} = 0,$$

where α, β, β', β'', γ are linear functions of x, y, z.

The equation may be simplified by the following consideration. If A, B, C, F, G, H are the coefficients of x^2, y^2, z^2, yz, zx, xy in the equation of the orthotomic circle, then we have

$$(A,B,C,F,G,H) = (A',B',C',F',G',H') + \lambda(a,b,c,2f,2g,2h),$$

where (A',B',C',F',G',H') is a fixed conic and λ is a parameter.

The result is that the orthotomic circle of the cubic $U = 0$ is given by the equation

$$V = AU + B\,H(U) = 0,$$

where $H(U)$ is the Hessian of U, and A, B are properly determined constants.

172. Note on the Logic of Characteristics.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 282–283.]

The theory of characteristics, as developed by Monge and applied by him to the integration of partial differential equations of the first order, depends essentially on the following principles.

Let $u = 0$ be an integral of a partial differential equation; and let the equation be such that through any point of a given surface there passes a characteristic. Then if we have two integrals $u = 0$, $v = 0$, the system of equations $u = 0$, $v = 0$ represents the curve of intersection of the two surfaces; and if this curve lies on a surface which is also an integral surface, then the characteristics at each point of the curve lie on this surface.

In the theory of characteristics, we consider a partial differential equation of the form

$$F(x, y, z, p, q) = 0,$$

where $p = \partial z/\partial x$, $q = \partial z/\partial y$. The characteristics are the curves defined by the system of ordinary differential equations

$$\frac{dx}{\partial F/\partial p} = \frac{dy}{\partial F/\partial q} = \frac{dz}{p\,\partial F/\partial p + q\,\partial F/\partial q} = -\frac{dp}{\partial F/\partial x + p\,\partial F/\partial z} = -\frac{dq}{\partial F/\partial y + q\,\partial F/}$$

Now the logical principle involved is this: if two integral surfaces touch at a point, they touch along the whole characteristic through that point. For at the point of contact, the values of x, y, z, p, q are the same for both surfaces; hence the direction of the characteristic is the same; and therefore the two surfaces touch along the whole characteristic.

This leads to the construction of the general integral. Given a partial differential equation, if we can find a complete integral containing two arbitrary constants, we may obtain the general integral by taking an arbitrary relation between these constants and forming the envelope of the resulting one-parameter family of surfaces.

More precisely, let $V(x, y, z, a, b) = 0$ be a complete integral, containing two arbitrary constants a, b. Let $b = \phi(a)$ be an arbitrary relation between them. Then the envelope of the family $V(x, y, z, a, \phi(a)) = 0$ is obtained by eliminating a between this equation and

$$\frac{\partial V}{\partial a} + \frac{\partial V}{\partial b}\phi'(a) = 0.$$

The resulting equation represents a surface generated by characteristics; and since the relation ϕ is arbitrary, we obtain in this way the general integral of the partial differential equation.

The logic of characteristics depends therefore on the geometrical fact that the contact of two integral surfaces along a characteristic is propagated along the whole characteristic; and this is the fundamental principle which Monge established.

173. On Laplace's Method for the Attraction of Ellipsoids.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 285–341.]

The problem of the attraction of ellipsoids has been treated by many geometers, but the methods of Laplace are in many respects the most satisfactory. I propose in the present paper to develop these methods with some additional details, and to show how they lead to the complete solution of the problem.

I. General Theory.

Let the equation of the ellipsoid be

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1,$$

and let it be required to find the attraction at an external or internal point (x, y, z). Laplace's method consists in introducing a new variable θ defined by the equation

$$\frac{x^2}{a^2+\theta}+\frac{y^2}{b^2+\theta}+\frac{z^2}{c^2+\theta}=1,$$

which, for an external point, has a single positive root; and for an internal point, we take $\theta = 0$.

The potential V at the point (x, y, z) is given by

$$V=\pi abc\int_\theta^\infty \frac{ds}{\sqrt{(a^2+s)(b^2+s)(c^2+s)}}\left(1-\frac{x^2}{a^2+s}-\frac{y^2}{b^2+s}-\frac{z^2}{c^2+s}\right),$$

where $\theta = 0$ for an internal point, and θ is the positive root of the above equation for an external point.

The components of attraction are given by

$$X=-\frac{\partial V}{\partial x},\quad Y=-\frac{\partial V}{\partial y},\quad Z=-\frac{\partial V}{\partial z}.$$

II. Development of the Potential in Series.

For many purposes it is useful to develop the potential in a series of spherical harmonics. Let r, θ, ϕ be the spherical coordinates of the attracted point; then we may write

$$V = \sum_{i=0}^{\infty} \frac{U_i}{r^{i+1}},$$

where U_i is a spherical harmonic of degree i.

To find U_i, we use the formula

$$U_i = \iiint \rho\, r'^i\, P_i(\cos\gamma)\, dx'\, dy'\, dz',$$

where ρ is the density, r' is the distance of the element from the centre, γ is the angle between the radius vectors, and P_i is the Legendre function.

For a homogeneous ellipsoid, ρ is constant, and we have

$$U_i = \rho \int_0^a \int_0^b \int_0^c r'^i\, P_i(\cos\gamma)\, dx'\, dy'\, dz'.$$

By Laplace's method, this may be reduced to the evaluation of the integral

$$\int_0^\infty \frac{ds}{\sqrt{(a^2+s)(b^2+s)(c^2+s)}}\, \frac{1}{(a^2+s)^\alpha (b^2+s)^\beta (c^2+s)^\gamma},$$

where $\alpha + \beta + \gamma = i$.

III. The Elliptic Integrals.

The fundamental integral may be reduced to the standard forms of elliptic integrals. Let

$$F = \int_0^\infty \frac{ds}{\sqrt{(a^2+s)(b^2+s)(c^2+s)}},$$

then by the substitution $s = c^2 \tan^2\phi - a^2$ (supposing $a > b > c$), we obtain

$$F = \frac{2}{\sqrt{a^2-c^2}} \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1-k^2\sin^2\phi}},$$

where $k^2 = \dfrac{a^2-b^2}{a^2-c^2}$; so that F is expressed by the complete elliptic integral of the first kind.

In like manner, the integrals which occur in the expressions for the components of attraction may be reduced to the complete elliptic integrals of the first and second kinds. If we write

$$A = abc\int_0^\infty \frac{ds}{(a^2+s)\Delta}, \quad B = abc\int_0^\infty \frac{ds}{(b^2+s)\Delta}, \quad C = abc\int_0^\infty \frac{ds}{(c^2+s)\Delta},$$

where $\Delta = \sqrt{(a^2+s)(b^2+s)(c^2+s)}$, then the components of attraction at an internal point are

$$X = -\frac{3M}{a^2}x, \quad Y = -\frac{3M}{b^2}y, \quad Z = -\frac{3M}{c^2}z,$$

where M is the mass, and the factors $\dfrac{3M}{a^2}$, &c. are expressible in terms of the elliptic integrals.

IV. Extension to Heterogeneous Ellipsoids.

The method may be extended to ellipsoids whose density varies according to any law of the form $\rho = f(m)$, where m is a function of the coordinates such that the level surfaces are similar concentric ellipsoids. Let

$$m = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2},$$

and let $\rho = \phi(m)$. Then the potential may be found by the same method, the only difference being that the integrals with respect to s must be multiplied by the appropriate function of the density.

If the density is a function of the distance from the centre only, say $\rho = \psi(r)$, then the level surfaces are spheres, and the problem reduces to that of the attraction of a spherically symmetrical body. But if the density varies in such a way that the level surfaces are ellipsoids, then Laplace's method still applies, and the potential may be found by quadratures.

V. Particular Cases.

When the ellipsoid reduces to a spheroid, the elliptic integrals degenerate into elementary functions. For an oblate spheroid ($a = b > c$), we have

$$A = B = \frac{3M}{2a^2c}\left(\frac{\sqrt{1-e^2}}{e^3}\arcsin e - \frac{1-e^2}{e^2}\right),$$

$$C = \frac{3M}{a^2c}\left(\frac{1}{e^2} - \frac{\sqrt{1-e^2}}{e^3}\arcsin e\right),$$

where e is the eccentricity of the meridian section.

For a prolate spheroid ($a > b = c$), the formulae are similar, with the appropriate changes.

When the ellipsoid reduces to a sphere, $a = b = c$, and we have

$$A = B = C = \frac{M}{a^3},$$

and the attraction at an internal point is

$$X = -\frac{M}{a^3}x, \quad Y = -\frac{M}{a^3}y, \quad Z = -\frac{M}{a^3}z,$$

which agrees with the known result for a sphere.

VI. Conclusion.

Laplace's method has the advantage of being equally applicable to internal and external points, and of reducing the problem in all cases to the evaluation of definite integrals of a single variable. The resulting expressions involve in general elliptic integrals, but in particular cases these degenerate into elementary functions. The method is also applicable to heterogeneous ellipsoids, provided the strata of equal density are similar concentric ellipsoids.

174. On the Oval of Descartes.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 320–328.]

This is an extract of the passage in the Geometry (La Géométrie de M. Descartes, 12° Paris, 1705, pp. 79–92) in which he applies his method of coordinates to this now well-known curve: the marginal reference is "Explication de quatre nouveaux genres d'Ovales qui servent à l'Optique."

The paper was intended to be introductory to a discussion in some detail of the history and properties of the Cartesian ovals; but the matter which I had collected has since appeared in the memoir "Sur les courbes planes du quatrième ordre et du premier genre" by M. Chasles, (*Comptes Rendus*, t. XLIV. pp. 977–982, 1093–1100, 1133–1139), and it is only in the historical point of view that there is anything to add.

Descartes' ovals are defined by the property that, if s and s' denote the distances from two fixed foci, then

$$ms \pm ns' = c,$$

where m, n, c are constants. When $m = n$, the curve becomes an ellipse or hyperbola. The general equation in Cartesian coordinates is of the fourth degree; but when one of the foci is at infinity, the equation reduces to the third degree, and the curve is the "parabolic oval" of Newton.

The construction given by Descartes is as follows. Let F, F' be the foci; from F as centre describe a circle with radius r; and let P be a point such that $m \cdot FP \pm n \cdot F'P = c$. Then if we take r such that $mr = c$, the point P lies on the oval. The different species of ovals correspond to the different arrangements of the foci and the different signs in the equation.

Descartes enumerates four kinds of ovals, according to the relative positions of the foci and the constants. In the first kind, both foci are on the same side of the tangent; in the second, they are on opposite sides; in the third and fourth kinds, one or both of the foci are virtual.

The application to optics is immediate: if light be refracted at the surface of the oval, the rays from one focus converge to the other; and by properly choosing the constants, any desired relation between the object and image distances may be obtained. This was the original purpose of Descartes' investigation, and it led him to the construction of lenses bounded by Cartesian surfaces.

It may be remarked that the ovals of Descartes are particular cases of the curves now called bicircular quartics; they are in fact the only bicircular quartics which have two real foci in the ordinary sense of the word.

175. On the Porism of the In-and-Circumscribed Triangle.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 344–354.]

The porism of the in-and-circumscribed triangle in its most general form relates to a triangle the angles of which lie in fixed curves, and the sides of which touch fixed curves; but at present I consider only the case in which the angles lie in one and the same fixed curve, which for greater simplicity I assume to be a conic. We have therefore a triangle ABC, the angles of which lie in a conic S, and the sides of which touch a conic Σ.

I. First Lemma.

Lemma. *If two conics S, Σ be such that a triangle can be inscribed in S and circumscribed about Σ, then there is an infinity of such triangles.*

This is the fundamental lemma of the porism. To prove it, let ABC be one such triangle; let the sides BC, CA, AB touch Σ at a, b, c respectively. Then if through any point A' on S we draw a tangent to Σ meeting S again at B', and through B' draw another tangent to Σ meeting S at C', then the line $C'A'$ will also touch Σ.

The proof depends on the theory of the correspondence of points on a conic. The points A and B correspond to each other, since they are joined by a line touching Σ; and the correspondence is a $(1, 1)$ correspondence, since through each point there pass two tangents to Σ. The fixed points of this correspondence are the points where S meets Σ (if they meet); and the lemma is a consequence of the theory of such correspondences.

II. Second Lemma.

Lemma. *If A, B, C are three points on a conic S, and a, b, c the points where the opposite sides touch the inscribed conic Σ, then the lines Aa, Bb, Cc meet in a point.*

This point is called the *pole* of the triangle with respect to the conic Σ. Its existence is a consequence of Brianchon's theorem; for the hexagon $ABcCaB$ has its diagonals meeting in a point, which is precisely the point of concurrence of Aa, Bb, Cc.

III. Condition of the Porism.

The condition that two conics S and Σ admit of an in-and-circumscribed triangle may be expressed in various forms. The most elegant is that given by Cayley: if the equation of S be $S = 0$ and that of Σ be $\Sigma = 0$, then the condition is that the discriminant of $S + \lambda\Sigma = 0$ shall be a perfect square.

If we write the discriminant in the form

$$\Delta(\lambda) = \begin{vmatrix} a+\lambda A & h+\lambda H & g+\lambda G \\ h+\lambda H & b+\lambda B & f+\lambda F \\ g+\lambda G & f+\lambda F & c+\lambda C \end{vmatrix},$$

then the condition is that $\Delta(\lambda)$ shall be a perfect square in λ; i.e., if $\Delta(\lambda) = I + J\lambda + K\lambda^2 + L\lambda^3$, then we must have

$$JL^2 - 4IKL + K^2 = 0,$$

or equivalently, the discriminant of the cubic $\Delta(\lambda)$ must vanish.

IV. Degenerate Cases.

When the conic Σ reduces to a triangle, the porism becomes: to inscribe in S a triangle whose sides pass through three fixed points. This is always possible, and there is an infinity of solutions.

When the conic S reduces to two right lines, the porism becomes: to circumscribe about Σ a triangle whose vertices lie on two fixed lines. This again is always possible.

When both conics degenerate, we have the porism of two triangles: to find a triangle inscribed in one and circumscribed about the other.

V. Extension to Polygons.

The porism may be extended to polygons of any number of sides. The condition for a quadrilateral is that the discriminant of $S + \lambda\Sigma = 0$ shall have two double roots; for a pentagon, that it shall have three double roots; and so on. In general, for an n-gon, the condition is that the discriminant shall have $n - 2$ double roots.

The proof depends on the theory of the periodicity of the correspondence defined by the two conics. If the correspondence is of period n, then there exists an in-and-circumscribed n-gon; and the condition for periodicity is precisely the condition that the discriminant shall have the requisite number of double roots.

VI. Analytical Theory.

Let the equation of the conic S be

$$S = (a, b, c, f, g, h \parallel x, y, z)^2 = 0,$$

and that of Σ be

$$\Sigma = (A, B, C, F, G, H \parallel x, y, z)^2 = 0.$$

Then the condition for the porism of the triangle is that the equation

$$\begin{vmatrix} a + \lambda A & h + \lambda H & g + \lambda G \\ h + \lambda H & b + \lambda B & f + \lambda F \\ g + \lambda G & f + \lambda F & c + \lambda C \end{vmatrix} = 0$$

shall have equal roots; i.e., if I, J, K are the invariants of the two conics, then

$$I^2J^2 - 6IJK - 4J^3 + K^2 = 0,$$

where

$$I = abc + 2fgh - af^2 - bg^2 - ch^2,$$

$$J = aBC + bCA + cAB + 2fFG + 2gGH + 2hHF - AF^2 - BG^2 - CH^2,$$

and K is the corresponding invariant of the second degree.

VII. Conclusion.

The porism of the in-and-circumscribed triangle is one of the most beautiful results in the theory of conics. It was first stated by Poncelet, and proved by him by synthetic methods; but the analytical proof, as developed above, shows more clearly the nature of the conditions involved, and leads naturally to the extension to polygons of any number of sides.

176. Note on Jacobi's Canonical Formulae for Disturbed Motion in an Elliptic Orbit.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 355–356.]

Consider a body (afterwards called the disturbed body) revolving about a central body under the influence of their mutual attraction and of any disturbing forces. Then referring the disturbed body to axes through the central body and parallel to fixed lines in space, write x, y, z, for the coordinates of the disturbed body, r, r', for the radii vectores of the disturbed and disturbing bodies, R for the disturbing function; then the equations of motion are

$$\frac{d^2x}{dt^2}+\frac{\mu x}{r^3}=\frac{\partial R}{\partial x},\quad \frac{d^2y}{dt^2}+\frac{\mu y}{r^3}=\frac{\partial R}{\partial y},\quad \frac{d^2z}{dt^2}+\frac{\mu z}{r^3}=\frac{\partial R}{\partial z},$$

where μ is the sum of the masses of the two bodies.

We may, to fix the ideas, take the plane of xy to be the plane of the ecliptic and the axis of x to be the line through the first point of Aries, or origin of longitude.

Jacobi's canonical elements may be taken to be,

1. The constant of *vis viva* or invariable part of the half-square of the velocity, which is equal to the sum of the masses divided by twice the mean distance and with the sign minus.

2. The constant of areas, which is equal to $\sqrt{\mu a(1-e^2)}$.

3. The mean anomaly at the epoch.

4. The longitude of the perihelion.

5. The longitude of the node.

6. The inclination.

The peculiarity of Jacobi's method is that the elements are so chosen that the equations of motion assume the canonical form

$$\frac{dc_i}{dt}=\frac{\partial H}{\partial w_i},\quad \frac{dw_i}{dt}=-\frac{\partial H}{\partial c_i},$$

where c_i are the constant elements, w_i are angles, and H is the Hamiltonian function.

The advantage of this form is that it exhibits the duality between the coordinates and momenta, and leads immediately to the method of variation of arbitrary constants. If the disturbing function R is small, we may integrate by successive approximations, treating the elements as variable; and the canonical form ensures that the equations for the variations of the elements are themselves canonical.

Jacobi has shown that this form of the equations is intimately connected with the theory of the elliptic functions; and that the integrals of the undisturbed motion may be expressed in terms of the theta-functions, which lead naturally to the canonical elements.

177. Solution of a Mechanical Problem.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 405–406.]

A heavy plane is supported by parallel elastic strings of small extensibility; and the strings are of the same length and extensibility: required the position of equilibrium.

Imagine the plane horizontal, and let n be the number of strings, (a, b), (a', b'), &c. the coordinates of the points of attachment; ξ, η the coordinates of the centre of gravity of the plane; W the weight.

Then if l be the natural length of each string, e the extensibility, z the depression of the plane below the points of attachment, the tension of each string is

$$T = \frac{e}{l}(z - l) = \lambda(z - l),$$

and the equations of equilibrium are

$$\sum T = W, \quad \sum T\,a = W\xi, \quad \sum T\,b = W\eta.$$

Since the strings are of equal length and extensibility, λ is the same for all; and if z is the same for all (which it is, the plane being rigid), we have

$$n\lambda(z - l) = W, \quad \lambda(z - l)\sum a = W\xi, \quad \lambda(z - l)\sum b = W\eta.$$

Hence

$$z = l + \frac{W}{n\lambda},$$

and the centre of gravity of the points of attachment must coincide with the projection of the centre of gravity of the plane; i.e.

$$\frac{\sum a}{n} = \xi, \quad \frac{\sum b}{n} = \eta.$$

If these conditions are not satisfied, the plane will not remain horizontal, but will take an inclined position. To find the inclination, let z now denote the varying height of the plane above a fixed horizontal plane; then if the equation of the plane is

$$z = \alpha x + \beta y + \gamma,$$

the length of the string at (a, b) is $z(a, b) - l$, and the tension is $\lambda(z - l)$. The equations of equilibrium are

$$\sum \lambda(\alpha a + \beta b + \gamma - l) = W,$$

$$\sum \lambda a(\alpha a + \beta b + \gamma - l) = W\xi,$$

$$\sum \lambda b(\alpha a + \beta b + \gamma - l) = W\eta,$$

$$\sum \lambda(\alpha a + \beta b + \gamma - l)(b_r a - a_r b) = 0,$$

where the last equation expresses that the resultant couple vanishes.

These equations determine α, β, γ; and the position of equilibrium is thus completely determined. The problem is interesting as showing how the principle of virtual velocities leads directly to the equations of equilibrium in a case where the constraints are not all of the same nature.

177. Solution of a Mechanical Problem

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 405–406.]

[Continued from p. 78.]

Combining with these the equations

$$x\cos\alpha + y\sin\alpha - p = 0,$$

and eliminating linearly $\cos\alpha$, $\sin\alpha$, p and W, we have

$$\begin{vmatrix} x, & y, & 1 \\ \xi, & \eta, & 1 \\ A, & H, & L \\ H, & B, & M \\ L, & M, & N \end{vmatrix} = 0$$

for the equation of the required line $x\cos\alpha + y\sin\alpha - p = 0$.

Replacing L, M, N by their values, the equation is readily transformed into

$$\begin{vmatrix} x, & y, & 1 \\ \xi, & \eta, & 1 \\ a, & b, & \frac{1}{\Delta} \end{vmatrix} = 0,$$

where $\Delta = AB - H^2$.

178. On the A Posteriori Demonstration of the Porism of the In-and-Circumscribed Triangle

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 31–38.]

In my former paper "On the Porism of the In-and-circumscribed Triangle" (Journal, t. I. p. 344 [175]) the two porisms (the homographic and the allographic) were established *a priori*, i.e. by means of an investigation of the order of the curve enveloped by the third side of the triangle.

I propose in the present paper to give the *a posteriori* demonstration of these two porisms; first according to Poncelet, and then in a form not involving (as do his demonstrations) the principle of projections.

My objection to the employment of the principle may be stated as follows:—viz. that in a systematic development of the subject, the theorems relating to a particular case and which are by the principle in question extended to the general case, are not in anywise more simple or easier to demonstrate than are the theorems for the general case; and, consequently that the circuitity of the method can and ought to be avoided.

The porism (homographic) of the in-and-circumscribed triangle, viz.

If a triangle be inscribed in a conic, and two of the sides envelope conics having double contact with the circumscribed conic, then will the third side envelope a conic having double contact with the circumscribed conic.

The following is Poncelet's demonstration, the Nos. are those of the *Traité des Propriétés Projectives* [Paris, 1822]:

No. 431. If a triangle be inscribed in a circle and two of the sides are parallel to given lines, then the third side envelopes a concentric circle.

This is evident, for, the angle in the segment subtended by the third side being constant, the length of the third side is constant; hence, the length of the perpendicular from the centre upon the third side is also constant, and the third side envelopes a concentric circle.

Hence, by the principle of projections:

If a triangle be inscribed in a conic and two of the sides pass through given points, the remaining side envelopes a conic having double contact with the circumscribed conic, the line through the two *points being the chord of contact.*

No. 434*. Conversely, if there be a triangle inscribed in a conic and the first side envelope a conic having double contact with the circumscribed conic, and the second side pass through a fixed point in the chord of contact, then will the third side also pass through a fixed point in the chord of contact.

No. 437. In particular, if there be a triangle inscribed in a conic and two of the sides pass through fixed points, then will the third side pass through a fixed point, viz. the point forming with the other two points a conjugate system.

No. 439. It follows that: *If there be a triangle inscribed in a conic and the first side passes through a fixed point, and the second side envelopes a conic having double contact with the circumscribed conic, then will the third side envelope a conic having double contact with the circumscribed conic.*

For the chord of contact meets the polar of the fixed point with respect to the circumscribed conic in a point; the line joining this point with the third angle (i.e. the angle opposite the third side) of the triangle meets the conic in a variable point; and joining this variable point with the first and second angles of the triangle we have a new triangle; two of the sides of this new triangle (by Nos. 434 and 437) pass through fixed points; hence the remaining side, i.e. the third side of the original triangle, touches a conic having double contact with the circumscribed conic.

We have thus passed from the case of the two sides passing through fixed points to that of one of the two sides enveloping a conic having double contact with the given conic and the other of them passing through a fixed point; and, by a repetition of the reasoning, Poncelet passes to the general case, viz.

If there be a triangle inscribed in a conic, and two of the sides envelope conics having double contact with the circumscribed conic, then will the third side envelope a conic having double contact with the circumscribed conic.

But it is somewhat more simple to omit the intermediate case of a conic and point, and pass directly, by the reasoning of No. 439, from the case of two points to that of two conics.

The porism (allographic) of the in-and-circumscribed triangle, viz.

If a triangle be inscribed in a conic and two of the sides envelope conics meeting the circumscribed conic in the same four points, then the third side will touch a conic meeting the circumscribed conic in the four points.

The following is Poncelet's demonstration:

No. 433. In the particular case of the homographic porism, viz. that in which two of the sides of the triangle pass through fixed points and the remaining side envelopes a conic having double contact with the circumscribed conic it is easy to see that the lines joining the angles of the triangle with the two fixed points and with the point of contact on the third side, meet in a point; this follows at once by the principle of projection from the case in No. 431, viz. the case of a triangle inscribed in a circle when two of the sides are parallel to given lines and the third side touches a concentric circle.

Hence,

No. 531. *If there be a triangle inscribed in a conic, and two of the sides envelope fixed curves, and the third side envelopes a certain curve; the lines joining the angles of the triangle with the points of contact meet in a point.*

In fact, attending only to the infinitesimal variation of the position of the triangle, the curves enveloped by the first and second sides may be replaced by the points of contact on these sides, and the curve enveloped by the third side may be replaced by a conic having double contact with the circumscribed conic, and the general case thus follows at once from the particular one.

Nos. 162 and 163.

Lemma (I). *If, on the sides of a triangle ABC, there are taken any three points L, M, N in the same line, and the harmonics A', B', C' of these points (i.e. the harmonic of each point with respect to the two vertices on the same side of the triangle), then the lines AA', BB', CC' will meet in a point; and, moreover, if AL, BM, CN are bisected in F, G, H (or, what is the same thing, if $FA^2 = FB \cdot FC$, $GB^2 = GC \cdot GA$, $HC^2 = HA \cdot HB$), then will the three points F, G, H lie in a line.*

This is, in fact, the theorem No. 164, "In any complete quadrilateral the middle points of the three diagonals lie in a line."

It is now easy to prove a particular case of the allographic porism, viz.

No. 531. *If there be a triangle inscribed in a circle, such that two of the sides envelope circles having a common secant (real or ideal) with the circumscribed circle, then the third side will envelope a circle having the same common secant with the circumscribed circle.*

For let MN be the common secant, and let the sides BC, CA, AB of the triangle touch the two circles in the points M, N; M, N; M, N respectively. Let the lines AM, AN meet the circumscribed circle in the points A', A''; and let the chords $A'M$, $A'N$ meet BC in the points F, F' respectively; and similarly let the chords $A''M$, $A''N$ meet BC in the points F, F''.

Then considering the triangle formed by the lines AA', BB', CC' and the points M, N on the sides of this triangle; the lines AM, AN meet the circumscribed circle in A', A''; hence the points F, G, H and A', B', C' are points on the sides of the triangle ABC, such that F, G, H lie in a line, and AA', BB', CC' meet in a point.

And by a property of the circle

$$FA'^2 = FM \cdot FN = FB \cdot FC, \quad GB'^2 = GM \cdot GN = GC \cdot GA;$$

whence by the lemma (or rather its converse) $HC'^2 = HA' \cdot HB'$ and by a property of the circle $HA' \cdot HB' = HM \cdot HN$; and therefore, $HC'^2 = HM \cdot HN$, a property which can only belong to a circle having, with the other circles, the common secant MN: the particular case is thus demonstrated.

And the principle of projections leads at once to the general case of the allographic porism.

To exhibit the demonstrations in a form independent of the principle of projections, it will be convenient to enunciate the following three lemmas: the first of them being, in fact, the theorem contained in No. 434, as generalised by No. 531; the second of them a theorem connected with and including the properties of the circle assumed in Poncelet's demonstration of the allographic porism; and the third of them a theorem derivable by the principle of projections from the theorem in Nos. 162 and 163.

Lemma (I). *If there be a triangle inscribed in a conic, such that two of the sides envelope given curves and the third side envelopes a curve; then the lines joining the angles of the triangle with the points of contact of the opposite sides meet in a point.*

Lemma (II). *If there be three conics meeting in the same four points, then any line meets the conic in six points forming a system in involution.*

Corollary (1). *If the line be a tangent to one of the conies, then the point of contact is the double or sibi-conjugate point of the involution formed by the points of intersection with the other two conics.*

And conversely if the curve enveloped by the line is not given, but the preceding property holds for all positions of the tangent line; then the curve enveloped by such line is a conic passing through the points of intersection of the two given conics.

Corollary (2). *If one of the conics be a pair of coincident lines, then the other two conics are conics having double contact, with the line in question for their chord of contact; any line meets the chord of contact in a point which is a double or sibi-conjugate point of the involution formed by the points of intersection with the other two conics; and if the line be a tangent to one of the conics, then the point of contact and the point of intersection with the chord of contact are corresponding points of the involution.*

The third lemma is a theorem (first explicitly stated, so far as I am aware, by Steiner, *Lehrsätze* 24 and 25, Crelle, t. III. [1828], p. 212, and demonstrated by Bauer, t. XIX. [1839], p. 227) which, in a note in the *Phil. Mag.*, Augt. 1853 [118], I have called the Theorem of the harmonic relation of two lines with respect to a quadrilateral.

Lemma (III). *If on each of the diagonals of a quadrilateral there be taken two points harmonically related with respect to the angles upon this diagonal; then if three of the points lie in a line, the other three points will also lie in a line: the two lines are said to be harmonically related with respect to the quadrilateral.*

The relation may be exhibited under a different form. The three diagonals of the quadrilateral form a triangle, the sides of which contain the six angles of the quadrilateral; and considering only three of the six angles (one angle on each diagonal) these three angles are points which either lie in a line, or else are such that the lines joining them with the opposite angles of the triangle meet in a point. Each of the three points is, with respect to the involution formed by the two angles of the triangle and the two points harmonically related thereto, a double or sibi-conjugate point, and we have thus a theorem of the harmonic relation of two lines to a triangle and line, or else to a triangle and point, viz.

Theorem. *If on the sides of a triangle there be taken three points which either lie in a line or else are such that the lines joining them with the opposite angles of the triangle meet in a point; and if on each side of the triangle there be taken two points forming with the two angles on the same side an involution having the first-mentioned point on the same side for a double or sibi-conjugate point; then if three of the six points lie in a line, the other three of the six points will also lie in a line; the two lines are said to be harmonically related with respect to the triangle and line, or triangle and point, as the case may be.*

Demonstration of the homographic porism.

Let ABC be the triangle, A', B', C' the points of contact on the three sides, then by Lemma I. the lines AA', BB', CC' meet in a point.

Take a pair of lines passing through the points of intersection of the circumscribed conic with the two given conics enveloped by the sides CA, CB, and let one of these lines meet the sides of the triangle in the points F, G, H, and the other of them meet the sides of the triangle in the points F', G', H'.

Then considering the following three conics, viz. the last-mentioned pair of lines, the circumscribed conic, and the conic enveloped by the side CA; these are conics passing through the same four points, and the side CA is a tangent to one of them: hence by Lemma II. Coroll. 1, G, G', C, A' will be an involution having the point B' for a double or sibi-conjugate point, and similarly F, F', C, B' are an involution having the point A' for a double or sibi-conjugate point.

It follows from Lemma III. that H, H', A', B' will be an involution having C' for a double or sibi-conjugate point.

Hence by Lemma II. Coroll. 1 (the two given conics being the before-mentioned pair of lines and the circumscribed conic) the curve enveloped by the side AB will be a conic passing through the points of intersection of the pair of lines and the circumscribed conic, or, what is the same thing, the points of intersection of the circumscribed conic and the conics enveloped by the other two sides.

Demonstration of the allographic porism.

Let ABC be the triangle, A', B', C' the points of contact on the three sides, then by Lemma I. the lines AA', BB', CC' meet in a point.

Take a pair of lines passing through the points of intersection of the circumscribed conic with the two given conics enveloped by the sides CA, CB, and let one of these lines meet the sides of the triangle in the points F, G, H, and the other of them meet the sides of the triangle in the points F', G', H'.

Then considering the following three conics, viz. the last-mentioned pair of lines, the circumscribed conic, and the conic enveloped by the side CA; these are conics passing through the same four points, and the side CA is a tangent to one of them: hence by Lemma II. Coroll. 1, G, G', C, A' will be an involution having the point B' for a double or sibi-conjugate point, and similarly F, F', C, B' are an involution having the point A' for a double or sibi-conjugate point.

It follows from Lemma III. that H, H', A', B' will be an involution having C' for a double or sibi-conjugate point.

Hence by Lemma II. Coroll. 1 (the two given conics being the before-mentioned pair of lines and the circumscribed conic) the curve

enveloped by the side AB will be a conic passing through the points of intersection of the pair of lines and the circumscribed conic, or, what is the same thing, the points of intersection of the circumscribed conic and the conics enveloped by the other two sides.

2, Stone Buildings, Oct. 2, 1856.

179. On Certain Forms of the Equation of a Conic

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 44–48.]

To find the general equation of a conic which passes through two given points and touches a given line.

Let the coordinates of the given points be (α, β, γ), $(\alpha', \beta', \gamma')$, and the equation of the given line be $\lambda x + \mu y + \nu z = 0$.

Then writing

$$\begin{vmatrix} x, & y, & z \\ \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \end{vmatrix} = 0,$$

for the equation of the line through the two points, and

$$u = \begin{vmatrix} x, & y, & z \\ \alpha, & \beta, & \gamma \\ \lambda, & \mu, & \nu \end{vmatrix}, \quad w = \begin{vmatrix} x, & y, & z \\ \alpha', & \beta', & \gamma' \\ \lambda, & \mu, & \nu \end{vmatrix},$$

the equation of any conic through the two points is

$$sV + u\nu + vw = 0,$$

where $V = \lambda x + \mu y + \nu z$ and s is an arbitrary parameter.

Writing

$$A = (\lambda\alpha+\mu\beta+\nu\gamma)\cdot s, \quad B = (\lambda\alpha'+\mu\beta'+\nu\gamma')\cdot s, \quad C = -(\lambda\alpha+\mu\beta+\nu\gamma)\cdot s,$$

we have

$$(\lambda x + \mu y + \nu z) \cdot sV + Au + Bv + Cw = 0,$$

and consequently the equation $\lambda x + \mu y + \nu z = 0$ is equivalent to $Au + Bv + Cw = 0$, and we have only to express that the line represented by this equation touches the conic $uw - v^2 = 0$.

Combining the two equations, we find $Au + Cw + B\sqrt{uw} = 0$, that is $(Au + Cw)^2 - B^2uw = 0$, an equation which must have equal roots; and the condition for this is obviously

$$4AC - B^2 = 0.$$

Or putting the condition under the form $-B + 2\sqrt{AC} = 0$ and substituting for A, B, C their values, the condition becomes

$$\mu = \sqrt{\left(\frac{-V}{u}\right)} \text{ as usual.}$$

We have consequently

$$\sqrt{(\lambda\alpha + \mu\beta + \nu\gamma)(\lambda\alpha' + \mu\beta' + \nu\gamma')} = 0,$$

and the equation of the conic is

$$\begin{vmatrix} x, & y, & z \\ \alpha, & \beta, & \gamma \\ \lambda, & \mu, & \nu \end{vmatrix} \begin{vmatrix} x, & y, & z \\ \alpha', & \beta', & \gamma' \\ \lambda, & \mu, & \nu \end{vmatrix} = 0.$$

[179]

[The original text continues with further derivations of equations for conics under various conditions, including the equation of a circle touching three given lines. The derivations involve extensive algebraic manipulations with determinants and symmetric functions. The key results include:]

Hence also the equation of a circle touching the three lines

$$x\cos\alpha+y\sin\alpha-p=0,\quad x\cos\beta+y\sin\beta-q=0,\quad x\cos\gamma+y\sin\gamma-r=0,$$

is

$$\sin(\beta-\gamma)\sqrt{(x\cos\alpha+y\sin\alpha-p)}+\sin(\gamma-\alpha)\sqrt{(x\cos\beta+y\sin\beta-q)}+\sin(\alpha-\beta)\sqrt{(x$$

To rationalise the equation, I remark that an equation

$$\sqrt{A}+\sqrt{B}+\sqrt{C}=0$$

gives in general

$$(1,1,1,1,1,1)(A,B,C)^2=0,$$

and that

$$(1,1,1,1,1,1)[2p\sin^2(\beta-\gamma),2q\sin^2(\gamma-\alpha),2r\sin^2(\alpha-\beta)]^2$$

or as it may also be written

$$(1,1,1,1,1,1)$$

is identically equal to

$$\{p(\sin\beta-\sin\gamma)+q(\sin\gamma-\sin\alpha)+r(\sin\alpha-\sin\beta)\}^2$$
$$+\{p(\cos\beta-\cos\gamma)+q(\cos\gamma-\cos\alpha)+r(\cos\alpha-\cos\beta)\}^2$$
$$-\{p\sin(\beta-\gamma)+q\sin(\gamma-\alpha)+r\sin(\alpha-\beta)\}^2.$$

Hence if we replace p, q, r by

$$x\cos\alpha+y\sin\alpha-p,\quad x\cos\beta+y\sin\beta-q,\quad x\cos\gamma+y\sin\gamma-r,$$

the last-mentioned expression equated to zero will give the equation of the circle, and we obtain

$$\{\sqrt{V}+p(\sin\beta-\sin\gamma)+q(\sin\gamma-\sin\alpha)+r(\sin\alpha-\sin\beta)\}^2$$

$$+\{\sqrt{V}p(\cos\beta - \cos\gamma) - q(\cos\gamma - \cos\alpha) - r(\cos\alpha - \cos\beta)\}^2$$
$$-\{p\sin(\beta-\gamma) + q\sin(\gamma-\alpha) + r\sin(\alpha-\beta)\}^2 = 0,$$

where $V = \sin(\beta-\gamma)+\sin(\gamma-\alpha)+\sin(\alpha-\beta)$, and we have thus the equation of the circle in the usual form with the coordinates of the centre and the radius put in evidence.

The condition that there may be a circle touching the four lines

$$Ax+By+C=0, \quad A'x+B'y+C'=0, \quad A''x+B''y+C''=0, \quad A'''x+B'''y+C'''=0,$$

is by the general formula shown to be

$$\begin{vmatrix} A, & B, & C, & A^2+B^2 \\ A', & B', & C', & A'^2+B'^2 \\ A'', & B'', & C'', & A''^2+B''^2 \\ A''', & B''', & C''', & A'''^2+B'''^2 \end{vmatrix} = 0,$$

which is in fact obvious from other considerations.

C. M.

180. Note on the Reduction of an Elliptic Orbit to a Fixed Plane

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 49–54.]

The principal object of the present Note is to obtain an expression for the quantity ϵ which I call the *modified mean longitude at epoch*, viz. taking as the elements the longitude θ of the node, inclination ϕ and any four elements which determine the motion in the plane of the orbit, then the longitude measured in the fixed plane (or reduced longitude) will be a function of the form

$$nt + \epsilon + \text{periodic terms},$$

where ϵ is a determinate function of the elements, and it is proposed to find the expression of this function.

But as the corresponding formulæ relating to the eccentricity and longitude of the pericentre are not in general given as part of the theory of elliptic motion, but occur only, so far as I am aware, in works on the lunar theory, I have thought it desirable to include these formulæ and take as the subject of this Note the reduction of an elliptic orbit to a fixed plane.

Write a_1, the semiaxis-major, e_1, the eccentricity $(= \sin K_1)$, ϖ_1, the longitude of pericentre in orbit, ϵ_1, the mean longitude in orbit at epoch, θ_1, the longitude of node, ϕ_1, the inclination $(= \tan^{-1}\gamma)$, and moreover the mean motion $n = \sqrt{(\mu/a_1^3)}$.

[180]

[The note continues with detailed derivations of the formulae for reducing the elements of an elliptic orbit to a fixed plane. The key equations include:]

By the definition of a we have $a(1-e^2) = a_1(1-e_1^2)$.

Hence

$$\cos\chi = \frac{1+e\cos(v-\varpi)}{1+e_1\cos(v_1-\varpi_1)},$$

which, combined with the equation

$$\tan\chi = \tan\phi\sin(v-\theta),$$

determines the position of the body in terms of the modified elements and of the reduced longitude v.

Introducing into the two equations s ($=\tan\chi$) and γ ($=\tan\phi$) in the place of χ and ϕ, they become

$$\frac{s}{\sqrt{(1+s^2)}}\{1+e\cos(v-\varpi)\} - e\sin(v-\varpi) = 0,$$

$$s = \gamma\sin(v-\theta),$$

which is the form in which the equations occur in the lunar theory.

Proceeding now to the formulæ which involve the time, it is to be remarked that the true anomaly and the quotient of the radius vector by the semiaxis-major are given functions of the eccentricity and the mean anomaly, and calling for a moment the last-mentioned quantities e, ζ, I represent the functions in question by $\mathrm{elta}(e,\zeta)$, $\mathrm{elqr}(e,\zeta)$, or more simply when the mean anomaly only is attended to by elta, elqr. I have found this notation very convenient as a means of dispensing with the introduction of the eccentric anomaly.

The reduced longitude is found in terms of the time by means of the equations

$$\tan(v-\theta) = \sec\phi\tan(v_1-\varpi_1),$$

$$v_1-\varpi_1 = \mathrm{elta}(nt+\epsilon_1-\varpi_1),$$

the former equation gives, as is well known,

$$v-\theta = v_1-\theta-\tan^2\tfrac{1}{2}\phi\sin(2v_1-2\theta)+\tan^4\tfrac{1}{2}\phi\sin(4v_1-4\theta)-\&c.,$$

(where the successive coefficients are the reciprocals of the natural numbers) we have therefore

$$v = v_1-\tan^2\tfrac{1}{2}\phi\{\sin(2v_1-2\varpi_1)\cos(2\varpi_1-2\theta)+\cos(2v_1-2\varpi_1)\sin(2\varpi_1-2\theta)\}$$

$+\frac{1}{2}\tan^4\frac{1}{2}\phi\{\sin(4v_1-4\varpi_1)\cos(4\varpi_1-4\theta)+\cos(4v_1-4\varpi_1)\sin(4\varpi_1-4\theta)\}-\&c.,$

and for the present purpose it is only necessary to attend to the non-periodic part of the function on the right-hand side.

Now $v_1-\varpi_1=\text{elta}(nt+\epsilon_1-\varpi_1)$, the non-periodic part of which is $nt+\epsilon_1-\varpi_1$.

And the non-periodic part of $\sin p(v_1-\varpi_1)$ is given by the equation (62) of Hansen's Memoir "Entwickelung des Products u. s. w." *Abhand. der K. Sachs. Gesellschaft zu Leipzig*, t. II. (1853).

In fact, Hansen's

$$\cos p(v_1-\varpi_1)=\left(\frac{V}{\tan\frac{1}{2}K_1}\right)^{\tan^2\frac{1}{2}\phi}(1+\mu\cos K_1),$$

$$\sin p(v_1-\varpi_1)=0.$$

Hence, substituting these values and putting for the non-periodic part of v the assumed value $nt+\epsilon$, we find

$$\epsilon=\epsilon_1-\frac{\tan^2\frac{1}{2}\theta\tan^2\frac{1}{2}K_1\sin(2\varpi_1-2\theta)}{1+\tan^2\frac{1}{2}\phi\tan^2\frac{1}{2}K_1\cos(2\varpi_1-2\theta)}$$

$$-\frac{\frac{1}{2}\tan^4\frac{1}{2}\phi\tan^4\frac{1}{2}K_1\sin(4\varpi_1-4\theta)}{1+2\tan^2\frac{1}{2}\phi\tan^2\frac{1}{2}K_1\cos(2\varpi_1-2\theta)+\tan^4\frac{1}{2}\phi\tan^4\frac{1}{2}K_1}-\&c.$$

in which formula the values of $\tan\frac{1}{2}\phi$, $\tan\frac{1}{2}K_1$ (in terms of γ, e_1) are

$$\tan\tfrac{1}{2}\phi=\frac{\gamma}{1+\sqrt{(1+\gamma^2)}},\quad \tan\tfrac{1}{2}K_1=\frac{e_1}{1+\sqrt{(1-e_1^2)}},$$

and that of $\cos K_1$ is $\sqrt{(1-e_1^2)}$.

We have thus the required expression for the modified mean longitude at epoch, and all the modified elements are now expressed in terms of the original elements.

[The note concludes with a theorem relating to the expansion of certain functions in multiple cosines.] The following investigation leads to a theorem which it is, I think, worth while to notice.

We have

$$dt=\frac{a^{\frac{3}{2}}(1-e^2)^{\frac{3}{2}}\,dv}{\sqrt{(\mu)}\cos\phi\{\sec\chi+e\cos(v-\varpi_1)\}},$$

and thence

$$\frac{a_1^{\frac{3}{2}}(1-e_1^2)^{\frac{3}{2}}\,dv}{\sqrt{(\mu)}\cos\phi\{\sec\chi + e\cos(v-\varpi_1)\}} = \frac{a_1^{\frac{3}{2}}(1-e_1^2)^{\frac{3}{2}}\,dv}{\sqrt{(\mu)}\cos\phi\{\sqrt{(1+\gamma^2\sin^2(v-\theta_1))} + e_0\cos(v - \ldots}$$

or as it may be written

$$\frac{(1-e_1^2)^{\frac{3}{2}}\,dv}{\sqrt{(1+\gamma^2)}\{\sqrt{(1+\gamma^2\sin^2(v-\theta_1))} + e_0\cos(v-\varpi_1)\}}.$$

But it is easy to see that if the mean longitude $nt+\epsilon$ is expanded in terms of v, the relation between these quantities must be of the form $nt+\epsilon = v +$ periodic terms. It follows that in the preceding equation the non-periodic part of the function which multiplies dv (the expansion being in multiple cosines of v) must be equal to $(1-e_1^2)^{\frac{3}{2}}/(1+\gamma^2)^{\frac{1}{2}}$.

Hence, putting for e_1 its value, we find that the non-periodic part of

$$\frac{1}{\sqrt{(1+\gamma^2)}\{\sqrt{(1+\gamma^2\sin^2(v-\theta))} + e\cos(v-\varpi)\}}$$ —expanded in multiple cosines of

is

$$\frac{(1-e_1^2)^{\frac{3}{2}}}{(1+\gamma^2)^{\frac{1}{2}}},$$

a theorem which might, it is probable, be verified without much difficulty.

2, Stone Buildings, October, 1856.

181. On Sir W. R. Hamilton's Method for the Problem of Three or More Bodies

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 66–73.]

The problem of three or more bodies is considered by Sir W. R. Hamilton in his two well-known memoirs on a general method in Dynamics, *Phil. Trans.* 1834 and 1835, and the differential equations for the relative motion, with respect to the central body, of all the other bodies are obtained in a form containing a single disturbing function only.

Several methods of integration are given or indicated, among others, one which is in fact the method of the variation of the elements as applied to the particular form of the equations of motion. But the investigation shows (and Sir W. R. Hamilton notices this as a defect in his theory, as compared with the ordinary theory of the variation of the elements), that in the method in question, the elements are not osculating elements, i.e. that the positions only, and not the velocities of the bodies, can be calculated as if the elements remained constant during an element of time.

The peculiar advantage of the method is of course the having a single disturbing function only, and this seems so important, that if I may venture to express an opinion, I cannot but think that the method will ultimately be employed for the purposes of Physical Astronomy. But, however this may be, it has appeared to me that it may be useful to present the method in a separate and distinct form, disengaged from the general theory as an illustration of which it was given by the author; and this is what I propose now to do.

Consider a central body M, and two other bodies M_1, M_2, and let the coordinates of M referred to a fixed origin be x, y, z, and the coordinates of M_1, M_2 referred to the body M as origin be x_1, y_1, z_1 and x_2, y_2, z_2 respectively. Then the coordinates of M_1, M_2 referred to the fixed origin, are $x+x_1$, $y+y_1$, $z+z_1$ and $x+x_2$, $y+y_2$, $z+z_2$ respectively, and if as usual T denotes the Vis-viva or half sum of each mass into the square of its velocity, and U denote the force function, then we have

$$T = \tfrac{1}{2}M(\dot{x}^2+\dot{y}^2+\dot{z}^2)+\tfrac{1}{2}M_1\{(\dot{x}+\dot{x}_1)^2+(\dot{y}+\dot{y}_1)^2+(\dot{z}+\dot{z}_1)^2\}+\tfrac{1}{2}M_2\{(\dot{x}+\dot{x}_2)^2+(\dot{y}+\dot{y}_2)^2+$$

$$U = \frac{MM_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}}+\frac{MM_2}{\sqrt{(x_2^2+y_2^2+z_2^2)}}+\frac{M_1M_2}{\sqrt{\{(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2\}}},$$

and the equations of motion are as usual

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{x}} - \frac{\partial T}{\partial x} = \frac{\partial U}{\partial x}, \quad \&c.$$

If we assume that the centre of gravity of the bodies is at rest, then we have

$$Mx + M_1(x + x_1) + M_2(x + x_2) = 0, \quad \&c.,$$

and consequently

$$x = -\frac{M_1x_1 + M_2x_2}{M + M_1 + M_2}, \quad \&c.$$

Now the value of T is

$$T = \tfrac{1}{2}(M+M_1+M_2)(\dot{x}^2+\dot{y}^2+\dot{z}^2)+\dot{x}(M_1\dot{x}_1+M_2\dot{x}_2)+\dot{y}(M_1\dot{y}_1+M_2\dot{y}_2)+\dot{z}(M_1\dot{z}_1+M_2\dot{z}_2$$
$$+\tfrac{1}{2}M_1(\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2) + \tfrac{1}{2}M_2(\dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2),$$

or, putting for $\dot{x}$, $\dot{y}$, $\dot{z}$ their values,

$$T = \frac{M_1(M_2 + M)}{2(M + M_1 + M_2)}(\dot{x}_1^2+\dot{y}_1^2+\dot{z}_1^2)+\frac{M_2(M_1 + M)}{2(M + M_1 + M_2)}(\dot{x}_2^2+\dot{y}_2^2+\dot{z}_2^2)$$
$$+\frac{M_1M_2}{(M + M_1 + M_2)}(\dot{x}_1\dot{x}_2 + \dot{y}_1\dot{y}_2 + \dot{z}_1\dot{z}_2),$$

and with this new value of T the equations of motion still are

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{x}_1} - \frac{\partial T}{\partial x_1} = \frac{\partial U}{\partial x_1}, \quad \&c.$$

[181]

Suppose now that the differential coefficients of T, with respect to $\dot{x}_1$, $\dot{y}_1$, $\dot{z}_1$; $\dot{x}_2$, $\dot{y}_2$, $\dot{z}_2$, are respectively P_1, Q_1, R_1; P_2, Q_2, R_2, i.e. write

$$P_1 = \frac{\partial T}{\partial \dot{x}_1}, \quad Q_1 = \frac{\partial T}{\partial \dot{y}_1}, \quad R_1 = \frac{\partial T}{\partial \dot{z}_1}, \quad \&\text{c.},$$

and imagine T expressed as a function of P_1, Q_1, R_1; P_2, Q_2, R_2, and when this is done put $H = T - U$ (so that H stands for a function of P_1, Q_1, R_1; P_2, Q_2, R_2; x_1, y_1, z_1; x_2, y_2, z_2), then the equations of motion in Sir W. R. Hamilton's form are

$$\frac{dx_1}{dt} = \frac{\partial H}{\partial P_1}, \quad \frac{dy_1}{dt} = \frac{\partial H}{\partial Q_1}, \quad \frac{dz_1}{dt} = \frac{\partial H}{\partial R_1}, \quad \frac{dP_1}{dt} = -\frac{\partial H}{\partial x_1}, \quad \frac{dQ_1}{dt} = -\frac{\partial H}{\partial y_1}, \quad \frac{dR_1}{dt} =$$

Now from the last given value of T

$$P_1 = \frac{M_1(M_2 + M)}{M + M_1 + M_2}\dot{x}_1 + \frac{M_1 M_2}{M + M_1 + M_2}\dot{x}_2,$$

$$P_2 = \frac{M_2(M_1 + M)}{M + M_1 + M_2}\dot{x}_2 + \frac{M_1 M_2}{M + M_1 + M_2}\dot{x}_1,$$

and thence

$$\dot{x}_1 = \frac{P_1}{M_1} - \frac{P_2}{M_1 + M_2 + M}, \quad \dot{x}_2 = \frac{P_2}{M_2} - \frac{P_1}{M_1 + M_2 + M},$$

and consequently

$$T = \frac{1}{2(M + M_1 + M_2)}\{(P_1 + P_2)^2 + (Q_1 + Q_2)^2 + (R_1 + R_2)^2\},$$

or, reducing,

$$T = \frac{1}{2}\left(\frac{P_1^2 + Q_1^2 + R_1^2}{M_1} + \frac{P_2^2 + Q_2^2 + R_2^2}{M_2}\right) - \frac{1}{2(M + M_1 + M_2)}\{(P_1+P_2)^2+(Q_1+Q_2)$$

and consequently

$$U = \frac{MM_1}{\sqrt{(x_1^2 + y_1^2 + z_1^2)}} + \frac{MM_2}{\sqrt{(x_2^2 + y_2^2 + z_2^2)}} + \frac{M_1 M_2}{\sqrt{\{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\}}},$$

and H having this value, the equations of motion are as before mentioned.

[181]

Instead of H write $H+\Gamma$ where

$$\Gamma = \frac{MM_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}} + \frac{MM_2}{\sqrt{(x_2^2+y_2^2+z_2^2)}},$$

and

$$H = \frac{1}{2}\left(\frac{P_1^2+Q_1^2+R_1^2}{M_1} + \frac{P_2^2+Q_2^2+R_2^2}{M_2}\right) - \frac{1}{2(M+M_1+M_2)}\{(P_1+P_2)^2+(Q_1+Q_2)$$

and the function Γ is to be treated as a disturbing function.

The equations of motion for the body M_1 become

$$\frac{dx_1}{dt} = \frac{1}{M_1}P_1 - \frac{1}{M+M_1+M_2}(P_1+P_2), \qquad \frac{dP_1}{dt} = -\frac{MM_1x_1}{(x_1^2+y_1^2+z_1^2)^{\frac{3}{2}}} + \frac{\partial\Gamma}{\partial x_1},$$

$$\frac{dy_1}{dt} = \frac{1}{M_1}Q_1 - \frac{1}{M+M_1+M_2}(Q_1+Q_2), \qquad \frac{dQ_1}{dt} = -\frac{MM_1y_1}{(x_1^2+y_1^2+z_1^2)^{\frac{3}{2}}} + \frac{\partial\Gamma}{\partial y_1},$$

$$\frac{dz_1}{dt} = \frac{1}{M_1}R_1 - \frac{1}{M+M_1+M_2}(R_1+R_2), \qquad \frac{dR_1}{dt} = -\frac{MM_1z_1}{(x_1^2+y_1^2+z_1^2)^{\frac{3}{2}}} + \frac{\partial\Gamma}{\partial z_1},$$

and there is of course a precisely similar system of equations of motion for the body M_2.

If we neglect Γ, the left-hand equations show that P_1, Q_1, R_1 denote the velocities or differential coefficients $\dot{x}_1$, $\dot{y}_1$, $\dot{z}_1$ multiplied by the constant factor MM_1, and substituting these values in the right-hand equations, we obtain the ordinary equations for the elliptic motion of the body M_1; and similarly for the body M_2.

We may, if we please, complete the solution by the method of the variation of the arbitrary constants.

Suppose for this purpose that a_1, b_1, c_1, e_1, f_1, g_1 are the elements for the elliptic motion of the body M_1, then treating these elements as variable we must have

$$\frac{\partial x_1}{\partial a_1}\frac{da_1}{dt} + \frac{\partial x_1}{\partial b_1}\frac{db_1}{dt} + \cdots + \frac{\partial x_1}{\partial g_1}\frac{dg_1}{dt} = \frac{\partial\Gamma}{\partial P_1},$$

$$\frac{\partial P_1}{\partial a_1}\frac{da_1}{dt} + \frac{\partial P_1}{\partial b_1}\frac{db_1}{dt} + \cdots + \frac{\partial P_1}{\partial g_1}\frac{dg_1}{dt} = -\frac{\partial\Gamma}{\partial x_1},$$

and it appears from these equations that as already noticed the disturbed values of the velocities are not (as they are in the ordinary theory) identical with the undisturbed values.

The disturbing function Γ may be considered as a function of the elements of the two orbits and of the time, and it is easy to obtain, as in the ordinary theory, the values of the differential coefficients $\frac{da_1}{dt}$, &c. in the form

$$\frac{da_1}{dt} = \frac{\partial\Gamma}{\partial b_1} + (a_1, b_1, c_1, \ldots)\frac{\partial\Gamma}{\partial e_1} + \cdots ,$$

where

$$(a_1, b_1) = \frac{\partial(a_1, b_1)}{\partial(x_1, P_1)} = \frac{\partial a_1}{\partial x_1}\frac{\partial b_1}{\partial P_1} - \frac{\partial a_1}{\partial P_1}\frac{\partial b_1}{\partial x_1},$$

if for shortness

$$\frac{\partial(a_1, b_1)}{\partial(x_1, P_1)} = \frac{\partial a_1}{\partial x_1}\frac{\partial b_1}{\partial P_1} - \frac{\partial a_1}{\partial P_1}\frac{\partial b_1}{\partial x_1}.$$

It will be remembered that in the ordinary theory, if H denote Lagrange's disturbing function ($H = R$ if R is the disturbing function of the *Mécanique Céleste*) the corresponding formulæ are

$$\frac{da_1}{dt} = \frac{\partial H}{\partial[a_1, b_1]} + \cdots ,$$

where $[a_1, b_1]$ denotes the Poisson combination of the elements a_1, b_1.

C. III.

182. On Lagrange's Solution of the Problem of Two Fixed Centres

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 76–83.]

The following variation of Lagrange's Solution of the Problem of Two Fixed Centres(1), is, I think, interesting, as showing more distinctly the connection between the differential equations and the integrals.

The problem referred to is as follows: viz. to determine the motion of a particle acted upon by forces tending to two fixed centres, such that r, q being the distances of the particle from the two centres respectively, and α, β, γ being constants, the forces are $\dfrac{\alpha}{r^2}$ and $\dfrac{\beta}{q^2}$.

Take the first centre as origin and the line joining the two centres as axis of x; and let h be the distance between the two centres, then writing for symmetry

$$x_1 = x, \quad x_2 = h - x$$

(so that x_1 is the coordinate corresponding to the first centre as origin, and x_2 the coordinate corresponding to the second centre as origin) the distances are given by the equations

$$r^2 = x_1^2 + y^2 + z^2, \quad q^2 = x_2^2 + y^2 + z^2,$$

and the equations of motion are

$$\frac{d^2x}{dt^2} = -\frac{\alpha x_1}{r^3} + \frac{\beta x_2}{q^3}, \quad \frac{d^2y}{dt^2} = -\frac{\alpha y}{r^3} - \frac{\beta y}{q^3}, \quad \frac{d^2z}{dt^2} = -\frac{\alpha z}{r^3} - \frac{\beta z}{q^3}.$$

(1) Lagrange's Solution was first published in the *Anciens Mémoires de Turin*, t. IV., [1766–69], and is reproduced in the *Mécanique Analytique*.

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and we obtain at once the integral of Vis-viva, viz. multiplying the three equations by $\frac{dx}{dt}$, $\frac{dy}{dt}$, $\frac{dz}{dt}$ respectively, adding and integrating (observing that $\frac{dx_1}{dt} = -\frac{dx_2}{dt} = \frac{dx}{dt}$), we have

$$\frac{1}{2}\left\{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right\} = \frac{\alpha}{r} + \frac{\beta}{q} + H, \qquad (1,\ a)$$

and with equal facility, the equation of areas round the line joining the two centres, viz. multiplying the second and third equations by z, y, adding and integrating, we have

$$y\frac{dz}{dt} - z\frac{dy}{dt} = B. \qquad (2,\ a)$$

So far Lagrange: to obtain a third integral I form the equation

$$\frac{dx}{dt}\left(\frac{d^2x}{dt^2}\right) + \frac{dy}{dt}\left(\frac{d^2y}{dt^2}\right) + \frac{dz}{dt}\left(\frac{d^2z}{dt^2}\right) - \frac{dx}{dt}\left(-\frac{\alpha x_1}{r^3} + \frac{\beta x_2}{q^3}\right).$$

The terms independent of the forces are

$$\frac{dx}{dt}\frac{d^2x}{dt^2} + \frac{dy}{dt}\frac{d^2y}{dt^2} + \frac{dz}{dt}\frac{d^2z}{dt^2} - \frac{dx}{dt}\frac{d^2x}{dt^2},$$

which are equal to

$$\frac{1}{2}\frac{d}{dt}\left\{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right\} - \frac{dx}{dt}\frac{d^2x}{dt^2},$$

and the terms depending on the forces are readily reduced to the form

$$-\alpha\frac{dr}{dt}\cdot\frac{x_1}{r^2} + \beta\frac{dq}{dt}\cdot\frac{x_2}{q^2}.$$

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which is equal to

$$-\alpha\frac{d}{dt}\left(\frac{1}{r}\right)+\beta\frac{d}{dt}\left(\frac{1}{q}\right),$$

and similarly the term multiplied by β is

$$\beta\frac{d}{dt}\left(\frac{x_2}{q}\right),$$

lastly, the term multiplied by γ is $\gamma\dfrac{d}{dt}(y\frac{dz}{dt}-z\frac{dy}{dt})$.

The preceding combination of the differential equations gives therefore an equation integrable *per se*, and effecting the integration we have

$$\frac{dy}{dt}\frac{d^2y}{dt^2}+\frac{dz}{dt}\frac{d^2z}{dt^2}-x\frac{d^2x}{dt^2}=\frac{\alpha x_1}{r}+\frac{\beta x_2}{q}+K,$$

which is the third integral equation.

It may be convenient to mention here (what appears by the comparison of the formulas obtained in the sequel with the corresponding formulae of Lagrange) that the value of Lagrange's constant of integration C is

$$C=K-2Hh^2-B^2.$$

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Making use of the ordinary transformation

$$y = \rho\cos\phi, \quad z = \rho\sin\phi,$$

the integral equations may be written under the forms

$$\frac{1}{2}\left\{\left(\frac{dx}{dt}\right)^2 + \left(\frac{d\rho}{dt}\right)^2 + \rho^2\left(\frac{d\phi}{dt}\right)^2\right\} = \frac{\alpha}{r} + \frac{\beta}{q} + H, \qquad (1,\ b)$$

$$\rho^2\frac{d\phi}{dt} = B, \qquad (2,\ b)$$

and

$$\rho\frac{d\rho}{dt}\frac{dx}{dt} - x\left\{\left(\frac{d\rho}{dt}\right)^2 + \rho^2\left(\frac{d\phi}{dt}\right)^2\right\} = \frac{\alpha x_1}{r} + \frac{\beta x_2}{q} + K. \qquad (3,\ b)$$

and observing that $y^2 + z^2$, x_1, x_2 are in fact functions of r, q, it is clear that the determination of r, q in terms of t depends upon the first and third equations alone.

Moreover the form of the equations shows that we can at once eliminate dt and thus obtain a differential equation between r, q alone.

It would be difficult to discover *a priori*, before actually obtaining the differential equation in question, that it would be possible to effect the separation of the variables, but we know that this can be done by taking instead of r, q the new variables $u = r + q$, $s = r - q$.

In order to complete the solution the first step is to introduce the variables r, q into the first and third equations: for this purpose we have

$$\frac{dx}{dt} = \frac{1}{2h}\left\{(r^2 - q^2 + h^2)\frac{dr}{dt} - (r^2 - q^2 - h^2)\frac{dq}{dt}\right\},$$

if for shortness

$$V = 2r^2q^2 + 2h^2r^2 + 2h^2q^2 - h^4 - r^4 - q^4,$$

and consequently

$$\rho^2 = \frac{V}{4h^2}, \quad \frac{dx}{dt} = \frac{1}{2h}\left(\frac{dr}{dt} - \frac{dq}{dt}\right)(r^2 - q^2) + \cdots$$

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Substituting these values in the two equations, we find for the first equation

$$V(r\,dr-q\,dq)^2+\{(h^2-r^2+q^2)\,r\,dr+(h^2+r^2-q^2)\,q\,dq\}^2 = 2h^2\left\{\frac{\alpha}{r}+\frac{\beta}{q}-\gamma(r^2+q^2)+ \right. \tag{1, c}$$

and for the second equation

$$V^2(r\,dr-q\,dq)^2+2V(r\,dr-q\,dq)(r^2-q^2)\{(h^2-r^2+q^2)\,r\,dr+(h^2+r^2-q^2)\,q\,dq\}$$

$$+(r^2-q^2)^2\{(h^2-r^2+q^2)\,r\,dr+(h^2+r^2-q^2)\,q\,dq\}^2$$

$$= 2h^2\left\{\alpha r(r^2-q^2+h^2)+\beta q(r^2-q^2-h^2)-\gamma V(r^2+q^2)+2KV-2B^2\{h^4-(r^2 \right. \tag{2, c}$$

The first equation is easily reduced to

$$4h^2\{r^2q^2(dr^2+dq^2)+(h^2-r^2-q^2)rq\,dr\,dq\} = 2h^2\left\{\frac{\alpha}{r}+\frac{\beta}{q}-\gamma(r^2+q^2)+2HV-2h^2 \right.$$

the second equation gives

$$8h^4(h^2-r^2-q^2)r^2q^2(dr^2+dq^2)+4h^4\{(h^2-r^2-q^2)^2+4q^2r^2\}rq\,dr\,dq = \cdots$$

and the function on the left-hand side is

$$8h^4(h^2-r^2-q^2)r^2q^2(dr^2+dq^2)+4h^4\{(h^2-r^2-q^2)^2+4q^2r^2\}rq\,dr\,dq.$$

Hence putting for a moment

$$M = \frac{\alpha}{r}+\frac{\beta}{q}-\gamma(r^2+q^2)+2HV-2h^2H,$$

$$N = \frac{\alpha r}{r^2-q^2+h^2}+\frac{\beta q}{r^2-q^2-h^2}-\gamma V(r^2+q^2)+2KV-2B^2\{h^4-(r^2-q^2)^2\},$$

we have

$$2rq(dr^2+dq^2)+(h^2-r^2-q^2)\,2rq\,dr\,dq = M\,dt^2,$$

$$2(h^2-r^2-q^2)\,2r^2q^2(dr^2+dq^2)+\{(h^2-r^2-q^2)^2+4q^2r^2\}\,2rq\,dr\,dq = N\,dt^2,$$

and thence recollecting that $-(h^2-r^2-q^2)^2+4q^2r^2 = V$, we find

$$V\cdot 2rq\,dr\,dq = \{(h^2-r^2-q^2)\,2M-N\}\,dt^2,$$

$$V\cdot 2r^2q^2(dr^2+dq^2) = [-\{(h^2-r^2-q^2)^2+4q^2r^2\}\,M+(h^2-r^2-q^2)\,N]\,dt^2,$$

and substituting for M, N their values, the functions on the right-hand side contain V as a factor, and dividing by V, we obtain

$$2rq\,dr\,dq = \left[-\frac{\alpha}{r}(3r^2+q^2-h^2) - \frac{\beta}{q}(r^2+3q^2-h^2) - \gamma\{r^6+q^6+15r^2q^2(r^2+q^2) -$$

(1, d)

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$$2r^2q^2(dr^2{+}dq^2) = \big[2\alpha r(r^2+3q^2-h^2)+2\beta q(3r^2+q^2-h^2)-\gamma\{r^6+q^6+15r^2q^2(r \qquad (2,\ d)$$

and by comparing the first of these formulae with the corresponding formulae of Lagrange, we find, as already observed, that the relation between the constant K and Lagrange's constant C is $K = C + 2Hh^2 + B^2$.

And substituting this value of K, the two equations become identical with those of Lagrange[1].

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[1] The formula referred to are the formulae (b), (c), *Mec. Anal.* t. II. page 112 of the second edition and page 97 of the third edition, but there is an inaccuracy in the formulae (c), B'' ought to be changed into $B - \frac{1}{2}P$; the error is continued in the subsequent formulae and moreover the constant term Ch^2 is omitted on the right-hand side of the formulas (e) and in the subsequent formulae, i.e. in the functions of s, u, the term $-B^2$ should be $-B^2 - Ch^2$.

The equation $y\dfrac{dz}{dt} - z\dfrac{dy}{dt} = B$, (putting $y = \sqrt{(y^2+z^2)}\cos\phi$, $z = \sqrt{(y^2+z^2)}\sin\phi$) gives at once $(y^2+z^2)\,d\phi = B\,dt$, and substituting for y^2+z^2 its value $= \dfrac{V}{4h^2}$, we find

$$d\phi = \frac{4h^2B\,dt}{V},$$

which is the third of Lagrange's equations.

To complete the solution, the combination of the first and second equations gives

$$r^2q^2(dr^2{-}dq^2)^2 = \big[\alpha\{(r-q)^3-h^2(r-q)\}-\beta\{(r-q)^3-h^2(r+q)\}-\gamma\{(r-q)^6$$

and thence putting $r+q=s$, $r-q=u$ and writing for shortness

$$\Phi(s) = \alpha(s^3{-}sh^2){+}\beta(s^3{-}sh^2){-}\gamma(s^6{-}h^2s^4{-}h^4s^2{+}h^6){+}H(s^4{-}2h^2s^2{+}h^4){+}2K(s^2{-}$$

and

$$\Psi(u) = -\gamma(u^8-h^2u^4-h^4u^2+h^6)+2K(u^2-h^2)-\cdots$$

we have

$$\frac{1}{4}(s^2-u^2)^2\left(\frac{ds^2}{\Phi(s)}-\frac{du^2}{\Psi(u)}\right)=0, \qquad (1,\ e)$$

$$\frac{s^2\,ds^2}{\Phi(s)} - \frac{u^2\,du^2}{\Psi(u)} = \frac{4h^2 B^2\,dt^2}{(s^2-u^2)^2}, \tag{3, e}$$

and thence finally

$$\frac{s^2\,ds^2}{\Phi(s)} - \frac{u^2\,du^2}{\Psi(u)} = \frac{4h^2 B^2\,dt^2}{(s^2-u^2)^2},$$

so that the problem is reduced to quadratures, the functions to be integrated involving the square roots of two rational and integral functions of the sixth degree.

2, Stone Buildings, 10th Nov., 1856.

183. Note on Certain Systems of Circles

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 83–88.]

It will be convenient to remark at the outset that two concentric circles, the radii of which are in the ratio of $1 : i$ (i being as usual the imaginary unit), are orthotomic[1], and that the most convenient quasi representation of a circle, the centre of which is real and the radius a pure imaginary quantity, is by means of the concentric orthotomic circle.

This being premised consider a circle and a point C. The points of contact of the tangents through C to the circle may be termed the *taction points*; the points where the chord through C perpendicular to the line joining C with the centre meets the circle, may be termed the *section points*. It is clear that, for an exterior point, the taction points are real and the section points imaginary, while, for an interior point, the section points are real and the taction points imaginary.

A circle having C for its centre and passing through the taction points (in fact the orthotomic circle having C for its centre) is said to be the *taction circle*. A circle having C for its centre and passing through the section points is said to be the *section circle*. Of course for an exterior point the taction circle is real and the section circle imaginary; while for an interior point the taction circle is imaginary and the section circle is real. It is proper also to remark that the taction circle and the section circle are concentric orthotomic circles.

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[1] Two concentric circles are, it is well known, conics having a double contact at infinity, and it appears at first sight difficult to reconcile with this, the idea of two particular concentric circles being orthotomic. The explanation is that any two lines through a circular point at infinity may be considered as being at right angles to each other, and therefore any line through a circular point at infinity may be considered as being at right angles to itself. The two concentric circles in question have, in fact, at each circular point at infinity a common tangent, but this common tangent must be considered as being at right angles to itself. The paradox disappears entirely upon a homographic deformation of the figure; two lines KL, KM are then defined to be at right angles when joining K with the fixed points I, J, the four lines KL, KM, KI, KJ are a harmonic pencil; but when K coincides with I, then KI is indeterminate and may be taken to be the fourth harmonic of the pencil, i.e. any two lines IL, IM through

the point I may be considered as being at right angles.

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Passing now to the case of two systems of orthotomic circles, let MM', NN' be lines at right angles to each other intersecting in R, and let M, M' be real or pure imaginary points on the line MM', equidistant from R. Imagine a system of circles, each of them having its centre on the line NN' and passing through the points M, M' (so that MM' is the radical axis of these circles). There are always on the line NN' two pure imaginary or real points N, N' equidistant from R, such that the circles, each of them having its centre on MM' and passing through the points N, N' (NN' being therefore the radical axis of these circles), are orthotomic to the first-mentioned system of circles, [the four points M, M', N, N' being thus as I have since termed them, two pairs of antipoints]. Moreover if R be made the centre of a circle passing through M, M', then the concentric orthotomic circle passes through N, N'; this is in fact only a particular case of the general property.

Suppose now that M, M' being given as the points of intersection of two circles having their centres on NN', it is required to find a circle having for its centre a given point C on NN' and passing through the points M, M'.

In the case of M, M' being real, the required circle is obviously given and is always real. But if M, M' are imaginary; then if about any point of MM' as centre a circle be described orthotomic to one of the circles, it will be orthotomic to the other circle, and will meet NN' in the real points N, N'.

Now if ρ be the radius of the required circle (i.e. of the circle having C for its centre and passing through the points M, M'), then

$$\rho^2 = (RC)^2 + (RM)^2 = (RC)^2 - (RN)^2.$$

Hence if $RC > RN$ or if C lies outside the space NN', ρ^2 is positive or the required circle is real, and the radius is at once constructed from the preceding expression $\rho^2 = RC^2 - RN^2$.

But if $RC < RN$ or C lies within the space NN', then the required circle is imaginary, but the concentric orthotomic circle is at once constructed from the formula $\rho^2 = RN^2 - RC^2$.

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Suppose now the point C is a centre of similitude of the two circles.

The circle having C for its centre and passing through the points M, M' is a taction circle of all the taction circles of the two circles, it may be termed the *tactaction circle*. The concentric orthotomic of the circle having C for its centre and passing through the points M, M' is a section circle of all the taction circles of the two circles, it may be termed the *sectaction circle*.

Consider first the case where the circles intersect in a pair of real points; here the two centres of similitude are on opposite sides of R; the tactaction circles are both real, the sectaction circles both imaginary.

Secondly, the case where the two circles are wholly exterior each to the other, the two centres of similitude lie on the same side of R, viz. the centre of inverse similitude between R and N, the centre of direct similitude beyond N. Hence the tactaction circle corresponding to the centre of direct similitude and the sectaction circle corresponding to the centre of inverse similitude are real, the other tactaction circle and sectaction circle are imaginary.

Thirdly, the case where one of the circles is wholly interior to the other; here the two centres of similitude are still on the same side of R, but the centre of direct similitude lies between R and N, and the centre of inverse similitude lies beyond N. Hence the sectaction circle corresponding to the centre of direct similitude and the tactaction circle corresponding to the centre of inverse similitude are real, the other sectaction circle and tactaction circle are imaginary.

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To obtain a distinct idea of the methods made use of in Gaultier's "Mémoire sur les moyens généraux de construire graphiquement un cercle déterminé par trois conditions," (*Jour. Ecole Polyt.* t. IX. [1813], p. 124), and in Steiner's "Geometrische Betrachtungen," Crelle, t. I. [1826], p. 161; it should be remarked that both of these geometers, confining as they do their attention to real circles, do not consider the section circle of an exterior point, or the taction circle of an interior point. The taction circle of an exterior point, or the section circle of an interior point, is Gaultier's "Cercle radical," and Steiner's "Potenzkreis," and Steiner also speaks of the radius of this circle as the "Potenz" of its centre in relation to the given circle.

The nature of the Cercle radical, or Potenzkreis, (i.e. whether it is a taction circle or a section circle) is of course determined as soon as it is known whether the centre is an exterior or an interior point, and Gaultier distinguishes the two cases as the "radical réciproque" and the "radical simple," and in like manner Steiner speaks of the Potenz as being "reell" or "imaginár."

[The note continues with further geometric constructions and algebraic derivations relating to systems of circles, their equations, and properties of the taction and section circles in various configurations.]

[The algebraic equations of the tactaction circles are derived as follows. For two circles with centres (α,β), (α',β') and radii c, c', the tactaction circle corresponding to the centre of direct similitude has equation:]

$$(c'-c)(x^2+y^2)-2(\alpha c'-\alpha' c)x-2(\beta c'-\beta' c)y+c'(\alpha^2+\beta^2)-c(\alpha'^2+\beta'^2)+cc'(c'-c)=0,$$

[and similar equations for other configurations.]

Consider any three circles and combining them in pairs, by what has preceded the equations of the tactaction circles corresponding to the centres of direct similitude will be

$$(c''-c')(x^2+y^2)-2(\alpha' c''-\alpha'' c')x-2(\beta' c''-\beta'' c')y+c''(\alpha'^2+\beta'^2)-c'(\alpha''^2+\beta''^2)+c'c''(c''-$$

$$(c-c'')(x^2+y^2)-2(\alpha'' c-\alpha c'')x-2(\beta'' c-\beta c'')y+c(\alpha''^2+\beta''^2)-c''(\alpha^2+\beta^2)+c''c(c-c'')=$$

$$(c'-c)(x^2+y^2)-2(\alpha c'-\alpha' c)x-2(\beta c'-\beta' c)y+c'(\alpha^2+\beta^2)-c(\alpha'^2+\beta'^2)+cc'(c'-c)=0,$$

and representing these equations by $U=0$, $U'=0$, $U''=0$, we have identically

$$c(c''-c')U+c'(c-c'')U'+c''(c'-c)U''=0;$$

hence the three tactaction circles pass through the same two points, or what is the same thing, have a common radical axis.

C. III.

184. A Theorem Relating to Surfaces of the Second Order

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 140–142.]

Given a surface of the second order $(a, b, c, d, f, g, h, l, m, n)(x, y, z, w)^2 = 0$; and a fixed plane $\alpha x + \beta y + \gamma z + \delta w = 0$, imagine a variable plane $\xi x + \eta y + \zeta z + \omega w = 0$, subjected to the condition that it always touches a surface of the second order, or what is the same thing such that the parameters ξ, η, ζ, ω satisfy a condition $(a, b, c, d, f, g, h, l, m, n)(\xi, \eta, \zeta, \omega)^2 = 0$.

The given surface of the second order, and the variable plane meet in a conic, and the fixed plane and the variable plane meet in a line, it is required to find the locus of the pole of the line with respect to the conic.

The pole in question is the point in which the variable plane is intersected by the polar of the line with respect to the surface of the second order: this polar is the line joining the pole of the fixed plane with respect to the surface of the second order, and the pole of the variable plane with respect to the surface of the second order.

Let α_1, β_1, γ_1, δ_1 be given linear functions of α, β, γ, δ, and ξ_1, η_1, ζ_1, ω_1 be given linear functions of ξ, η, ζ, ω, viz., if $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N})$ are the inverse system to $(a, b, c, d, f, g, h, l, m, n)$, then let

$$\alpha_1 = \mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma + \mathfrak{L}\delta, \quad \text{\&c.},$$

and in like manner,

$$\xi_1 = \mathfrak{A}\xi + \mathfrak{H}\eta + \mathfrak{G}\zeta + \mathfrak{L}\omega, \quad \text{\&c.};$$

then the coordinates of the pole of the fixed plane are as $\alpha_1 : \beta_1 : \gamma_1 : \delta_1$, and the coordinates of the pole of the variable plane are as $\xi_1 : \eta_1 : \zeta_1 : \omega_1$; whence the equations of the polar are

$$\begin{vmatrix} x, & y, & z, & w \\ \alpha_1, & \beta_1, & \gamma_1, & \delta_1 \\ \xi_1, & \eta_1, & \zeta_1, & \omega_1 \end{vmatrix} = 0,$$

a system of equations which may be thus represented

$$U_1 = 0, \quad U_2 = 0,$$

where K is the discriminant of the system $(a, b, c, d, f, g, h, l, m, n)$.

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Write

$$\bar{x} = ax + hy + gz + lw, \quad \bar{y} = hx + by + fz + mw,$$

$$\bar{z} = gx + fy + cz + nw, \quad \bar{w} = lx + my + nz + dw,$$

the last preceding system of equations may be written

$$\xi = \lambda\bar{x} + \mu\alpha, \quad \eta = \lambda\bar{y} + \mu\beta,$$

$$\zeta = \lambda\bar{z} + \mu\gamma, \quad \omega = \lambda\bar{w} + \mu\delta,$$

equations in which λ, μ, are indeterminate, and where x, y, z, w may be considered as current coordinates, and this system represents the polar above referred to.

Combining the equations in question with the equation $\xi x + \eta y + \zeta z + \omega w = 0$, of the variable plane, we have

$$\lambda(\bar{x}x + \bar{y}y + \bar{z}z + \bar{w}w) + \mu(\alpha x + \beta y + \gamma z + \delta w) = 0,$$

i.e.

$$\lambda(a, \ldots)(x, y, z, w)^2 + \mu(\alpha x + \beta y + \gamma z + \delta w) = 0,$$

or what is the same thing

$$\lambda : \mu = -(\alpha x + \beta y + \gamma z + \delta w) : (a, \ldots)(x, y, z, w)^2,$$

and substituting these values in the expressions for ξ, η, ζ, ω we have ξ, η, ζ, ω in terms of the coordinates x, y, z, w of the pole above referred to, i.e., if for shortness,

$$U = (a, b, c, d, f, g, h, l, m, n)(x, y, z, w)^2, \quad P = \alpha x + \beta y + \gamma z + \delta w,$$

and combining with these equations the equation $(a, \ldots)(\xi, \eta, \zeta, \omega)^2 = 0$, we have

$$(a, \ldots)(UP_x - P\bar{x},\ UP_y - P\bar{y},\ UP_z - P\bar{z},\ UP_w - P\bar{w})^2 = 0$$

for the required locus of the pole of the line of intersection of the variable plane the fixed plane, with the conic of intersection of the given surface of the second order and the variable plane.

The locus in question is a surface of the fourth order and it may be remarked that this surface touches the given surface of the second order along the conic of intersection with the fixed plane.

1th April, 1857.

185. Note on the Circular Relation of Prof. Möbius

[From the *Quarterly Mathematical Journal*, vol. II. (1858), p. 162.]

Theorem. *Given the points A, B, C; P, and the points A', B', C' describe the circles α, β, γ, ω as follows: viz.*

$$\alpha \textit{ through } (B, C, P), \quad \beta \textit{ through } (C, A, P),$$
$$\gamma \textit{ through } (A, B, P), \quad \omega \textit{ through } (A, B, C),$$

and the circles α', β', γ', ω' as follows: viz.,

α' through (B', C') cutting ω at the angle at which α cuts ω;
β' through (C', A') cutting ω at the angle at which β cuts ω;
γ' through (A', B') cutting α at the angle at which γ cuts ω; and
ω' through (A', B', C').

then will α', β', γ' meet in a point P', i.e. we shall have the points A', B', C'; P' such that the circles α', β', γ', ω' pass through (B', C'', P'), (C', A', P'), (A', B', P'), (A', B', C').

We may construct in this manner two figures, such that to three points of the first figure there correspond in the second figure three points which may be taken at pleasure, but these once selected to every other point of the first figure there will correspond in the second figure a perfectly determinate point.

And the two figures will be such that whenever in the first figure four or more points lie in a circle, then in the second figure the corresponding points will also lie in a circle.

The relation in question is due to Prof. Möbius, who has termed it the *Kreis-verwandschaft* (circular relation) of two plane figures.

See his paper Crelle, t. LII. [1856], pp. 218–228, extracted from the *Berichte* of the Royal Saxon Society of Sciences of the 5th Feb. 1853, [and *Werke*, t. II. pp. 207–217].

186. On the Determination of the Value of a Certain Determinant

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 163–166.]

Considering the determinant

$$\begin{vmatrix} \theta, & 1, & n, \\ \theta-2, & 2, & n-1, \\ \theta-4, & 3, & n-2, \\ \vdots & \vdots & \vdots \\ \theta-2x+2, & x-1, & n-x+2, \\ \theta-2x, & x, & n-x+1 \end{vmatrix}$$

let the successive diagonal minors be U_0, U_1, U_2, ..., U_x, ..., it is easy to find

$$U_2 = \theta(\theta-2) - 3(n-2)\theta,$$

$$U_3 = (\theta-1)(\theta^2-9) - 6(n-3)(\theta-1) + 3(n-3)(n-1),$$

which in fact suggests the law, viz.

$$U_x = (n-x+1)(\theta+x-3)(\theta+x-5)\dots(\theta-x+5)(\theta-x+3)$$

$$-\frac{x(x-1)}{2}(n-x+1)(n-x+3)(\theta+x-5)\dots(\theta-x+5)\cdot\frac{\theta(\theta-1)(\theta-2)(\theta-3)}{2\cdot 4}$$

$$+\&\text{c.}$$

to $s = \frac{1}{2}x$ or $\frac{1}{2}(x-1)$, as x is even or odd, where the general term is

$$(-)^s\frac{x(x-1)\dots(x-2s+1)}{2\cdot 4\dots 2s}(n-x+1)(n-x+3)\dots(n-x+2s-1)$$

$$\times(\theta+x-2s-1)(\theta+x-2s-3)\dots(\theta-x+2s+1).$$

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And of course if x denote the number of lines or columns of the determinant, then U_x is the value of the determinant.

This theorem, or a particular case of it, is due to Prof. Sylvester: I have not been able to find an easier demonstration than the following one, which, it must be admitted, is somewhat complicated.

I observe that U_x satisfies the equation

$$U_x - \theta U_{x-1} + (x-1)(n-x+2)U_{x-2} = 0.$$

Hence writing $x-1$ and $x-2$ for x, we have the system

$$U_x - \theta U_{x-1} + (x-1)(n-x+2)U_{x-2} = 0,$$

$$\theta U_{x-1} - U_{x-2} + (x-2)(n-x+3)U_{x-3} = 0,$$

$$U_{x-2} - \theta U_{x-3} + (x-3)(n-x+4)U_{x-4} = 0,$$

or, eliminating U_{x-1} and U_{x-3},

$$\{(x-1)(n-x+2)+(x-2)(n-x+3)-\theta^2\}U_{x-2}-(x-2)(x-3)(n-x+3)(n-x+4)U_{x-4} =$$

Suppose, for shortness,

$$(\theta+x-1)(\theta+x-3)(\theta+x-5)\ldots(\theta-x+5)(\theta-x+3)(\theta-x+1) = H_x,$$

and assume

$$U_x = A_{x,0}H_x - A_{x,1}H_{x-2} + \ldots + (-)^s A_{x,s}H_{x-2s}\ldots,$$

where $A_{x,s}$ is independent of θ, then U_x contains the term $(-)^s A_{x,s}H_{x-2s}$, U_{x-2} contains the term $(-)^s A_{x-2,s}H_{x-2s-2}$, which is to be multiplied by $(x-1)(n-x+2)+(x-2)(n-x+3)-\theta^2$.

This multiplier may be written under the form

$$(x-1)(n-x+2)+(x-2)(n-x+3)-(x-2s-1)^2-\{\theta^2-(x-2s-1)^2\} = M_{x,s}-\{\theta^2-(x-2$$

if, for shortness,

$$M_{x,s} = (x-1)(n-x+2)+(x-2)(n-x+3)-(x-2s-1)^2.$$

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Now $\{M_{x,s} - [\theta^2 - (x - 2s - 1)^2]\}$ multiplied into $(-)^s A_{x-2,s} H_{x-2s-2}$ gives rise to the terms

$$(-)^s M_{x,s} A_{x-2,s} H_{x-2s-2} - (-)^s A_{x-2,s} H_{x-2s}$$

(since $[\theta^2 - (x - 2s - 1)^2] H_{x-2s-2} = H_{x-2s}$), or, what is the same thing,

$$(-)^s M_{x,s} A_{x-2,s} H_{x-2s-2} + (-)^{s+1} A_{x-2,s} H_{x-2s},$$

and moreover U_{x-2} contains the term $(-)^{s-1} A_{x-2,s-1} H_{x-2s}$, or, what is the same thing, $(-)^{s+1} A_{x-2,s-1} H_{x-2s}$.

Hence we must have

$$A_{x,s} - (A_{x-2,s} + M_{x,s} A_{x-2,s}) + (x-2)(x-3)(n-x+3)(n-x+4) A_{x-4,s} = 0,$$

where

$$M_{x-1} = (x-1)(n-x+2) + (x-2)(n-x+3) - (x-2s+1)^2.$$

This may be satisfied by assuming

$$A_{x,s} = B_{x,s}(n-x+1)(n-x+3)\ldots(n-x+2s-1);$$

for then

$$A_{x-2,s} = B_{x-2,s}(n-x+3)\ldots(n-x+2s-1)(n-x+2s+1),$$

$$A_{x-2,s-1} = B_{x-2,s-1}(n-x+3)\ldots(n-x+2s-1),$$

$$A_{x-4,s-2} = B_{x-4,s-2}(n-x+4)(n-x+5)\ldots(n-x+2s-1),$$

and consequently

$$B_{x,s}(n-x+1) - B_{x-2,s} - M_{x,s} B_{x-2,s-1} + B_{x-4,s-2}(x-2)(x-3)(n-x+4) = 0;$$

and if this equation be satisfied independently of n, we must have

$$B_{x,s} - B_{x-2,s} - (2x-3) B_{x-2,s-1} + (x-2)(x-3) B_{x-4,s-2} = 0,$$

$$B_{x,s} - (2s+1) B_{x-2,s} - \{5x - 8 - (x-2s+1)^2\} B_{x-2,s-1} + 4(x-2)(x-3) B_{x-4,s-2} = 0,$$

and these are both satisfied by

$$B_{x,s} = \frac{x(x-1)\ldots(x-2s+1)}{2 \cdot 1 \cdot 2 \cdot 3 \ldots s};$$

in fact, substituting this value and omitting the factor $\dfrac{1}{2 \cdot 1 \cdot 2 \cdot 3 \ldots s}$, the first equation becomes

$$x(x-1) - (x-2s)(x-2s-1) - (2x-3)2s + 4s(s-1) = 0,$$

and the second equation becomes

$$x(x-1)-(2s+1)(x-2s)(x-2s-1)-\{5x-8-(x-2s+1)^2\}2s+16s(s-1) = 0,$$

which are each of them an identical equation, the first being

$$x^2 - x - x^2 + (4s+1)x - 2s(2s+1) - 4sx + 6s + 4s(s-1) = 0,$$

and the second being

$$x^2-x-(2s+1)\{x^2-(4s+1)x+2s(2s+1)\}+2s\{x^2-(4s+3)x+(2s-1)^2+8\}+16s(s-1) =$$

as may be easily verified.

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Hence writing for $B_{x,s}$ its value and recapitulating, the equation

$$U_x+\{(x-1)(n-x+2)+(x-2)(n-x+3)-\theta^2\}U_{x-2}+(x-2)(x-3)(n-x+3)(n-x+4)U_{x-}$$

is satisfied by

$$U_x = A_{x,0}H_x-A_{x,1}H_{x-2}\ldots+(-)^s A_{x,s}H_{x-2s}\ldots \text{to } s=\tfrac{1}{2}x \text{ or } \tfrac{1}{2}(x-1),$$

as x is even or odd, where

$$H_x = (\theta+x-1)(\theta+x-3)(\theta+x-5)\ldots(\theta-x+5)(\theta-x+3)(\theta-x+1),$$

$$A_{x,s} = \frac{x(x-1)\ldots(x-2s+1)}{2\cdot 1\cdot 2\cdot 3\ldots s}(n-x+1)(n-x+3)\ldots(n-x+2s-1);$$

and since for $x = 0, 1, 2, 3$ the values of the expression U_x coincide with those of the first four diagonal minors, the expression gives in general the value of the diagonal minor, or when x denotes the number of lines or columns of the determinant, then the value of the determinant.

2, Stone Buildings, 1st April, 1857.

187. On the Sums of Certain Series Arising from the Equation $x = u + tf(x)$

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 167–171.]

Lagrange has given the following formula for the sum of the inverse nth powers of the roots of the equation $x = u + tf(x)$,

$$\Sigma(x^{-n}) = u^{-n} + (-nu^{-n-1}f_u)'\frac{t}{1} + (-nu^{-n-1}f_u)''\frac{t^2}{1 \cdot 2} + \text{\&c.} \quad (1)$$

where n is a positive integer and the series on the second side of the equation is to be continued as long as the exponent of u remains negative (*Théorie des Équations Numériques*, p. 225).

Applying this to the equation $x = 1 + te^x$, we have

$$\Sigma(x^{-n}) = 1 + \frac{-n \cdot 1}{1}t + \frac{(-n)(-n-1) \cdot 1}{1 \cdot 2}t^2 + \text{\&c.},$$

to be continued while the exponent of 1 remains negative.

Let $n = \mu s + \rho$, ρ being not greater than $s - 1$, the series may always be continued up to $q = \mu$, and no further.

In fact writing the above value for n and putting $q = \rho + \theta$, the general term is

$$\frac{(n-q)(n-q-1)\dots(n-q-s+1)}{1 \cdot 2 \dots s} \cdot \frac{q(q-1)\dots 1}{1 \cdot 2 \dots q}t^q.$$

Now if $\rho+\theta-\theta(s-1)$ is negative or zero, the term is to be rejected on account of the index of 1 not being negative, and if this quantity be positive, then since $\rho - \theta s + 1$ is necessarily negative for any value of θ greater than zero, the factorial $(\rho - \theta s + \mu + \theta - 1)\dots(\rho - \theta s + 1)$ begins with a positive and ends with a negative factor, and since the successive factors diminish by unity, one of them is necessarily equal to zero, or the term vanishes; hence the series is always to be continued up to $q = \rho$.

Hence

$$\Sigma(x^{-(\mu s+\rho)}) = 1 + \frac{(\mu s + \rho)\{(-\mu-1)s - \rho\}\dots\{(-\mu-1)s - \rho - s + 1\}}{1 \cdot 2 \dots s}t + \text{\&c.} \quad (3)$$

continued to $q = \mu$.

By taking the terms in a reverse order, it is easy to derive

$$\Sigma(x^{-n}) = \frac{(-)^q n(n-1)\dots(n-q+1)}{1\cdot 2\dots q}(q+1)\dots(q+s)t^q + \&\text{c.} \quad (4)$$

continued to $q=\rho$.

Suppose in particular $s=2$, and $t=-\dfrac{1}{4\alpha}$, so that the equation in x becomes

$$x = 1 - \frac{1}{4\alpha}e^x,$$

or

$$\frac{x-1}{x}e^{-x} = \frac{1}{4\alpha},$$

whence $x=\dfrac{\alpha}{\alpha+1}$ or $x=\dfrac{-\alpha}{\alpha+1}$; or substituting in (2), we find

$$\left(\frac{\alpha+1}{\alpha}\right)^n + \left(\frac{-\alpha-1}{\alpha}\right)^n = 1 + \frac{n(n-3)}{1\cdot 2}\frac{1}{\alpha} + \&\text{c.},$$

continued to the term involving $\left(\dfrac{1}{\alpha}\right)^{\frac{1}{2}n-1}$ or $\left(\dfrac{1}{\alpha}\right)^{\frac{1}{2}(n-1)}$.

Put $a=\dfrac{\alpha}{\alpha+1}$; and therefore $\dfrac{a}{a+b}=\dfrac{\alpha}{\alpha+1}$, $\dfrac{b}{a+b}=\dfrac{1}{\alpha+1}$, $\dfrac{ab}{(a+b)^2}=\dfrac{\alpha}{(\alpha+1)^2}$; we obtain

$$\frac{a^n+b^n}{(a+b)^n-a^n-b^n} = \frac{n-3}{1}\frac{ab}{(a+b)^2} + \frac{n(n-4)(n-5)}{1\cdot 2}\frac{a^2b^2}{(a+b)^4} + \&\text{c.} \quad (7)$$

to be continued as long as the exponent of $(a+b)$ on the second side is negative.

[187]

This formula, which is easily deducible from that for the expansion of $\cos n\theta$ in powers of $\cos\theta$, is employed by M. Stern, Crelle, t. XX. [1840], in proving the following theorem:

If

$$S = 1 - \frac{n(n-3)}{1\cdot 2\cdot 3} + \frac{n(n-4)(n-5)}{1\cdot 2\cdot 3\cdot 4\cdot 5} - \&\text{c.} \tag{8}$$

continued to the first term that vanishes, then according as n is of the form $6k+3$, $6k\pm 1$, $6k$ or $6k\pm 2$,

$$S = \frac{2^n}{n}, \quad S = 0, \quad S = -\frac{2^n}{n}, \quad S = \frac{2^{n+1}}{n}, \tag{9}$$

which is in fact immediately deduced from it by writing $b = \omega a$, ω being one of the impossible cube roots of unity.

Substituting the above values of x in the equation (4),

$$\frac{a^n + (-a)^n}{(a+1)^{n+1}} = \frac{n(n-\rho-1)\dots 1}{1\cdot 2\dots\rho}(\rho+1)(\rho+2)t^\rho + \&\text{c.} \tag{10}$$

$$\frac{a^{n-1} + (-a)^{n-1}}{(a+1)^{n}} = \frac{(n-1)(n-\rho-1)\dots 1}{1\cdot 2\dots\rho}(\rho+1)(\rho+2)t^\rho + \&\text{c.} \tag{11}$$

whence

$$\frac{A}{a^n + (-a)^n} + \frac{B}{a^{n-1} + (-a)^{n-1}} = V \text{ suppose,} \tag{12}$$

where A and B refer to the variable ρ. The summation is readily effected by means of the formulae

$$\sum_{\rho=0}^{p} \frac{(n-\rho-1)\dots 1}{1\cdot 2\dots\rho}(\rho+1)(\rho+2)t^\rho = \frac{1}{(1-t)^{n+1}},$$

and we thence find

$$\frac{1}{a^n + (-a)^n} + \frac{1}{a^{n-1} + (-a)^{n-1}} = \frac{1}{(1+a)^{n+1}} + \frac{1}{(1+a)^n}, \tag{13}$$

a formula of Euler's (*Pet. Trans.* 1811) demonstrated likewise by M. Catalan (Liouville, t. IX. [1844], pp. 161–174) by induction. It may be expressed also in the slightly different form

$$\frac{1}{a^n + (-a)^n} = \frac{1}{(1+a)^{n+1}}\left\{1 + \frac{n}{1}a + \frac{n(n-1)}{1\cdot 2}a^2 + \&\text{c.}\right\}. \tag{14}$$

The two series (13), (14) are each of them supposed to contain $\rho + 1$ terms, ρ being an integer; but since the terms after these all of them vanish, the series may be continued indefinitely. Suppose the two sides expanded in powers of ρ, the coefficients will be separately equal, and thus the identity of the two sides will be independent of the particular values of ρ, or the equations (13), (14), and similarly, (10), (11), (12) are true for any values of ρ whatever.

It is to be observed that the series for negative values of ρ do not differ essentially from those for the corresponding positive values; as may be seen immediately by writing $-\rho$ for ρ, and $-\alpha$ for α.

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Suppose next $s = 3$, or that the equation in x is $x^3 = 1 + te^x$; to rationalise the roots of this, assume $t = \dfrac{(\beta^2-1)^2}{4\beta^3}$, then values of x are

$$x = \frac{2(\beta-1)}{\beta^2-1}, \quad x = \frac{2(\omega\beta-1)}{\omega^2\beta^2-1}, \quad x = \frac{2(\omega^2\beta-1)}{\omega\beta^2-1},$$

and hence

$$x^n = \frac{1}{4(\beta^2-1)}\{(-)^n(\beta+1)^n + (-)^n(\beta-1)^n\} + \&\text{c.} \tag{15}$$

where $t = \dfrac{(\beta^2-1)^2}{4\beta^3}$, and the series is to be continued up to the term involving β^{n-1} or β^{n-2}.

Again, from the formula (4) we deduce the three following forms,

$$\Sigma(x^{-n}) = \frac{(-)^q n(n-1)\dots(n-q+1)}{1\cdot 2\dots q}(q+1)\dots(q+3)\frac{1}{(4\alpha)^q} + \&\text{c.}, \tag{16}$$

all of them continued up to $q = \rho$.

2, Stone Buildings, 1st April, 1857.

The Collected Mathematical Papers of Arthur Cayley, Vol. III

Pages 151–200

Transcribed from the Cambridge University Press edition

193. On Rodrigues' Method for the Attraction of Ellipsoids

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 202–205.]

(Conclusion from previous pages.)

The points P, P' will be in like manner corresponding points on the surface (rA, rB, rC) and on the confocal surface $\{r(A+\delta A),\, r(B+\delta B),\, r(C+\delta C)\}$; and if the normal distance at the point P of the first two surfaces is δN, then the normal distance at the point P' of the second two surfaces will be $r\,\delta N$. The decrement of MP will be equal to the normal distance $r\,\delta N$ of the two surfaces at the point P multiplied into the cosine of the angle MQ, and we have, by a property of confocal surfaces,

$$A\,\delta A = B\,\delta B = C\,\delta C = \rho\,\delta N = \theta\,\delta\theta \quad \text{(suppose)},$$

we have therefore

$$d\,MP = -\frac{2\rho}{r}\cos MQ.$$

Hence from the equation

$$\frac{V}{ABC} = \iint \frac{r^2\,dr\,dS}{MP},$$

we deduce

$$\delta\frac{V}{ABC} = \iint r^2\,dr\,dS \cdot \delta\frac{1}{MP} = \iint \frac{2\rho\, r^2\,dr\,dS}{MP^2}\cos MQ = 0.$$

But we have

$$\frac{r\,dr}{ABC} = \frac{d\theta}{\rho\theta},$$

and the equation thus becomes

$$\delta\frac{V}{ABC} = \frac{2}{\rho\theta}\int d\theta \int \frac{r^2\,dS}{MP^2}\cos MQ = 0.$$

It may be proper to remark here by way of recapitulation that the course of the investigation has been as follows: viz. that, with a view to obtaining the potential V of an attracting ellipsoid, we have found the increment of $\dfrac{V}{ABC}$ in passing from the ellipsoidal surface (A, B, C) to the ellipsoidal surface $(A+\delta A, B+\delta B, C+\delta C)$, each of them confocal with the surface of the attracting ellipsoid; and that for finding such increment we have had to consider the two surfaces (rA, rB, rC) and $\{r(A+\delta A),\ r(B+\delta B),\ r(C+\delta C)\}$ confocal to each other and respectively similar to the first-mentioned two surfaces.

Resuming the formula just obtained, the integral with respect to dS is taken over the entire surface of the internal similar ellipsoid (rA, rB, rC), and if the attracted point is external to the ellipsoid (A, B, C) it will be external to the interior similar ellipsoid (rA, rB, rC): hence in this case the double integral vanishes for all values of r, or we have

$$\delta \frac{V}{ABC} = 0,$$

that is the function $\dfrac{V}{ABC}$, which represents the ratio of the potential to the mass, is not altered in passing from the ellipsoid (A, B, C) to the confocal ellipsoid $(A + \delta A, B + \delta B, C + \delta C)$, or, what is the same thing, the potentials (and therefore the attractions) of confocal ellipsoids upon the same external point are proportional to their masses; this is in fact Maclaurin's theorem for the attraction of ellipsoids upon an external point.

But if the attracted point is interior to the ellipsoid (A, B, C), then writing

$$M = \rho r \cdot ABC, \quad \text{where } r \text{ is less than unity,}$$

the double integral is $= 0$ from $r = 0$ to $r = r$, and is $= 4\pi$ from $r = r$ to $r = 1$, and we have

$$\delta \frac{V}{ABC} = \frac{8\pi}{\rho\theta}\, \delta\theta, \quad \text{or} \quad \delta \frac{V}{ABC} = -\delta \frac{8\pi}{\rho\theta},$$

that is, the right-hand side of the equation is the increment (or taken with its sign reversed so as to be positive, it is the decrement) of the function $\dfrac{V}{ABC}$ in passing from the ellipsoid (A, B, C) to the confocal ellipsoid $(A + \delta A, B + \delta B, C + \delta C)$, where

$$\delta \frac{8\pi}{\rho\theta} = \frac{8\pi}{\rho\theta}\, \delta\theta.$$

The preceding formula gives at once the potential for an interior point; in fact taking α, β, γ for the semiaxes of the ellipsoid and writing

$$A^2 = \alpha^2 + \theta, \quad B^2 = \beta^2 + \theta, \quad C^2 = \gamma^2 + \theta,$$

and using the ordinary symbol d instead of δ, we have

$$d\frac{V}{ABC} = -\frac{8\pi\, d\theta}{\sqrt{\{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)\}}},$$

and integrating from $\theta = 0$ to $\theta = \infty$, we have

$$\frac{V}{ABC} = \int_\theta^\infty \frac{8\pi\, d\theta}{\sqrt{\{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)\}}}.$$

where V is now the potential for the ellipsoid whose semiaxes are α, β, γ; and we have therefore

$$V = \alpha\beta\gamma \int_0^\infty \frac{8\pi\, d\theta}{\sqrt{\{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)\}}}.$$

To find the potential for an external point it is only necessary to remark that by the theorem above demonstrated $V \div \alpha\beta\gamma$ is equal to the corresponding function for the confocal ellipsoid through the attracted point, that is for the ellipsoid whose semiaxes are $\sqrt{(\alpha^2+\theta_1)}$, $\sqrt{(\beta^2+\theta_1)}$, $\sqrt{(\gamma^2+\theta_1)}$, where θ_1 is a positive quantity such that

$$\frac{x^2}{\alpha^2+\theta_1} + \frac{y^2}{\beta^2+\theta_1} + \frac{z^2}{\gamma^2+\theta_1} = 1;$$

hence in the value of $V \div \alpha\beta\gamma$ we have only to write the above values in the place of α, β, γ; and if we then write $\theta - \theta_1$ in the place of θ, the limits will be ∞, θ_1, and the expression for the potential is

$$V = \pi\alpha\beta\gamma \int_{\theta_1}^\infty \frac{d\theta}{\sqrt{\{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)\}}};$$

this completes the investigation.

194. Note on the Theory of Attraction

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 338–339.]

Imagine a closed surface, the equation of which contains the two parameters m, h. Call this the surface (m, h), and suppose also that for shortness the shell of uniform density included between the surfaces (m, h), (n, h) is called the shell (m, n, h). Suppose now that the surface is such:

1. That the infinitesimal shell $(m, m + dm, h)$ exerts no attraction upon an internal point.
2. That the equipotential surfaces of the shell in question for external points are the surfaces (m, k), where k is arbitrary.

Then, first, the attraction of the shell on a point of the equipotential surface (m, k) is proportional to the normal thickness at that point of the shell $(m, m + \delta m, k)$; or (more precisely) taking the density of the attracting shell as unity, the attraction is $= 4\pi\times$ mass of shell $(m, m + dm, h)$ into normal thickness of shell $(m, m + \delta m, k)$ divided by mass of the last-mentioned shell.

In fact the shell $(m, m+\delta m, k)$ exerts no attraction on an internal point, consequently if over the surface (m, k) we distribute the mass of the original shell $(m, m+dm, h)$ in such manner that the density at any point is proportional to the normal thickness of the shell $(m, m+\delta m, k)$ the distribution will be such that the attraction on an internal point may vanish; but in order that this may be the case, the density must be equal to $\frac{1}{4\pi}$ into the attraction upon that point of the shell $(m, m + \delta m, k)$. Hence the attraction is proportional to the normal thickness, and if the whole mass distributed over the surface (m, k) is precisely equal to the mass of the shell $(m, m + dm, h)$, then the density at any point must be equal to the mass into normal thickness divided by mass of $(m, m + \delta m, k)$, and attraction $= 4\pi$ into density, $= 4\pi\times$ mass of shell $(m, m + dm, h)$ into normal thickness of shell $(m, m + \delta m, k)$ divided by mass of the last-mentioned shell.

And, secondly, the attractions of the solids bounded by the two surfaces (n, h), (n, h') respectively upon the same exterior point are proportional to their masses. For the solid (n, h) may be divided into shells $(m, m + dm, h)$ and for this shell the equipotential surface is (m, k) and the attraction of the shell varies as mass of $(m, m+dm, h)$ into normal thickness of the shell $(m, m+\delta m, k)$. But in like manner the solid (n, h') may be divided into shells $(m, m + dm, h')$ and the attraction of the shell varies as mass of $(m, m + dm, h')$ into normal thickness of the shell $(m, m + \delta m, k)$ and the attractions are in each case in the direction normal to the shell (m, k), and therefore in the same direction; that is, the attraction of the shell $(m, m + dm, h)$ is in the same direction as that of the shell $(m, m+dm, h')$ and the two attractions are proportional to the masses. Hence integrating from $m = 0$ (if for this value the included space is zero) to n, the attractions of the solids (n, h), (n, h') are composed of elements proportional and parallel, the elements of the attraction of (n, h) to the elements of the attraction of (n, h'); and consequently the total attractions are in the same direction and proportional to the masses.

Thirdly, the attractions of the two surfaces (m, h), (n, h) upon the same interior point are equal.

A surface having the properties in question is of course the ellipsoidal surface

$$\frac{x^2}{a^2+h} + \frac{y^2}{b^2+h} + \frac{z^2}{c^2+h} = m,$$

where if m varies (h being constant) the several surfaces are similar to each other, but if h varies (m being constant) the several surfaces are confocal to each other: for it is in fact well known that the infinitesimal shell bounded by similar ellipsoidal surfaces has the properties assumed with respect to the shell $(m, m+dm, h)$. The first theorem in effect reduces the problem of the determination of an ellipsoid upon an exterior point to a single integration, and constitutes the foundation of Poisson's method for the attraction of ellipsoids. The second theorem (Maclaurin's theorem for the attraction of ellipsoids on the same external point) shows that the attraction of an ellipsoid upon an external point can be found by means of the attraction of the confocal ellipsoid through the attracted point; and by the third theorem the attraction of an ellipsoid upon an interior point is equal to that of the similar ellipsoid through the attracted point; hence the second and third theorems reduce the determination of the attraction of an ellipsoid upon an external or internal point to that of an ellipsoid upon a point on the surface.

2, Stone Buildings, W.C., 7th April, 1858.

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195. Report on the Recent Progress of Theoretical Dynamics

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mentioned special problems, I shall have frequent occasion to allude to: I mean the problem of the variation of the elements of a planets orbit, which has a close historical connexion with the general theories which form the subject of this report. The so-called ideal coordinates of Hansen, and the principles of his method of integration in the planetary and lunar theories, have a bearing on the general subject, and might have been considered in the present report; but on the whole I have considered it better not to do so. 1. Lagrange, Me canique Analytique, 1788. The equations of motion are obtained, as before mentioned, by means of the principle of virtual velocities and dAlemberts principle. In their original forms they involve the coordinates x, y, z of the different particles m or dm of the system, quantities which in general are not independent. But Lagrange introduces, in place of the coordinates x, y, z of the different particles, any variables or (using the term in a general sense) coordinates f, i/r, 0,... whatever, determining the position of the system at the time t: these may be taken to be independent, and then if , -//, </> ,... denote as usual the differential coefficients of f, ty, </>,... with respect to the time, the equations of motion assume the form d L dT$_d T +$
$Hdtddoorfownheenand H, th^e, saampm; lte; , ...fuanrcetiotnheV, partthieanltdhif$
$/ >, cdT_d T dV_+$ $Inthese equations, T, orthe visviva function, is the visviva of the syste$
$/ >, ...; and (when such function exists) V, or the force function^), is a function dependi$
$, ...; the two functions T and V are given functions, by means of which the equations of m$
$, ..., of their differential coefficients f \quad > \quad ty, <$
$, and of the time t; and it is of these cond order in regard to the dV if fies renatia glivceone$
$/ >, c.o.o.r; dianantdes (whenty, such ...fuanncdtioonfetxhiestst) imtehet. for T chei$
$e; , yoccurindynamicalproblems, butinanextensiveclassofsuchproblemstheformsa$
$, ...; the equations of motion V = -1UT, heandsigtnocaatltlriUbut(eidnstteoadVofiV$

158 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 have in this case an integral T+ V= h, which is the equation of vis viva, and the problems are distinguished as those in which the principle of vis viva holds good. It is to be noticed also that in this case since t does not enter into the

differential equations, the integral equations will contain t in the form t + c, that is, in connexion with an arbitrary constant c attached to it by addition. 2. The above-mentioned form is par excellence the Lagrangian form of the equations of motion, and the one which has given rise to almost all the ulterior developments of the theory but it is proper just to refer to the form in which the equations are ; in the first instance obtained, and which may be called the unreduced form, viz. the equations for the motion of a particle whose rectangular coordinates are x, y, z, are d-x dL dM where L = 0, M = 0,... are the equations of condition connecting the coordinates of the different points of the system, and X, f'...*areindeterminatemultipliers.3.Theideaofaforcefunctionseemstohaveoriginat*
thisseemstobethefirstattemptatthemethodofthevariationofthearbitraryconstants

observed that the variation of one of the elements, viz. the mean distance, was expressible in a remarkable form by means of the differential coefficients of the disturbing function taken with respect to the time t, in so far as it entered into the function through the coordinates of the disturbed planet. I am not able to say at what time, or whether by Euler, Lagrange, or Laplace, it was observed that such differential coefficient with respect to the time was equivalent to the differential coefficient of the disturbing function with respect to one of the elements. But however this may be, the notion of the representation of the variations of the elements by means of the differential coefficients of the disturbing function with respect to the elements had presented itself a posteriori, and was made use of in an irregular manner prior to the year 1800, and therefore some eight years at any rate before the establishment by Lagrange of the general theory to which these forms belong. 4. Poissons memoir of the 20th of June, 1808, " On the Secular Inequalities of the Mean Motion of the Planets," was presented by him to the Academy at the age of twenty-seven years. It contains, as already remarked, an expression in finite terms for the variation of the longitude of the epoch. But the memoir is to be considered rather as an application of known methods to an important problem of physical astronomy, than as a completion or extension of the theory of the variation of the planetary elements. The formulae made use of are those involving the differential coefficients of the disturbing function with respect to the coordinates; and there is nothing which can be considered an anticipation of Lagranges idea of the investigation, a, priori, of expressions involving the differential coefficients with respect to the elements. But, as well for its own sake as historically, the memoir is a very important one. Lagrange, in his memoir of the 17th of August, 1808, speaks of it as having recalled his attention to a subject with which he had previously occupied himself, but which he had quite lost sight of; and Arago records that, on the death of Lagrange, a copy in his own handwriting of Poissons memoir was found among his papers and ; the memoir is referred to in, and was probably the occasion of, Laplaces memoir also of the 17th of August, 1808. 5. With respect to Laplaces memoir of the 17th of August, 1808, it will be sufficient to quote a sentence from the introduction to Lagranges memoir : "Ayant montre a M. Laplace mes formules et mon analyse, il me montra de son cote en meme temps des formules analogues qui donnent les variations des elemens elliptiques par les differences partielles dune meme fonction, relatives a ces elemens. J ignore comment il y est parvenu ; mais je presume quil les a trouvees par une combinaison adroite des formules quil avait donnees dans la Me canique Celeste." This is, in fact, the character of Laplaces analysis for the demonstration of the formulas. 6. In Lagranges memoir of the 17th of August, 1808, " On the Theory of the Variations of the Elements of the Planets, and in particular on the Variations of the Major Axes of their Orbits," the question treated of appears from the title. The author obtains formulae for the variations of the elements of the orbit of a planet in terms of the differential coefficients of the disturbing function with respect to the elements; but the method is a general one, quite independent of the particular form

160 REPORT OX THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 of the integrals, and the memoir may be considered as the foundation of the general theory. The equations of motion are considered under the form, d?x l+$^{m}_{d}lX\ dtz\quad dxd?y_l + m_dQy\ di?r * \quad dyl + md?z_dl\ dtfT*\quad dzanditisassumedthatthetermsinflbeingneglected, theproblemisco$ $0, Sy = 0, bz =; andthentheequationsofmotiongive, , dx_dl\ dy_dfl\ dz_dldtdxdtdydtdzeq$ $C{-}dl^aI{-}TSa{+}C{-}dlToITTb{+}, c.Thedifferentialcoefficients{-}, c., canofcoursebe$ $j{-}, c.; and, byasimplecombinationoftheseveralequations, Lagrangededucesexpress$ $dja{-}, c., intermsofdafsee.viz., ; db + (, a, o)Nd - c + (, a, /_*).df + (., g), dq + (, a, h,).d - h, where(*)f > y, 9O > z8(a, b)8(a, 6)8(a, 6)inwhich, forshortness, 8387((xa, , x6 = -))stand, st - ordd - xa = .dduxo - -djdxadd- = xo.1Thesearesubstantiallytheformulaeof Lagrange; butIhaveintroducedhereande$

The form of the expressions shows at once that (a, 6) =- (b, a), so that the number of the symbols (a, b) is in fact fifteen.

Lagrange proceeds to show, that the differential coefficient with respect to t of the expression represented by the symbol (a, b) vanishes identically; and it follows, that the coefficients (a, 6) are functions of the elements only, without the time t. The general formulae are applied to the problem in hand; and, in consequence of the vanishing of several of the coefficients (a, b), it is easy in the particular problem $^{t}opassfromtheexpressionsfor -$ $a^{f} - t, c.intermsofat, c.tothosefordt, c.intermsof- =$ $-, c.Theauthorthusobtainsanelegantsystemofformulaeforthevariationsoftheele$ $dtdrdr\ dr\ drwhereTandVareofthedegreeofgeneralityconsideredintheMecanique$ $V = R, viz., intheformdRd_{dR}$;lt;mdtdrdrdrand,asintheprecedingmemoir,expressionsareinvestigatedfor

162 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 ment the investigation is simplified, and the true form of the functions (a, 6) obtained ddTT viz., writing - d = r - = p, ... t„hen V*"> "J o o (/ a, b L) O 3/ (a, 6 L) if, for shortness, 9 (r, p)$_{r}trdpdpdr, 9(a,6)dac?65adbTherepresentationof- = -, , - =$ $-, , c.bysinglelettersismadebyLagrangeintheAddition, dTdTNo.26(Lagrangewrite$ $a^{v}t = T, -c7t-s7 = T", c.), butquiteincidentallyinthatnumberonly, forthesakeofth$ $Ihavenoticedthis, asthestepisanimportantone.8.Itispropertoremarkthat, inorderto$ $drtf\ ArS - d = r-, -S2r/AA - d = -amrp; gt; +... J =$ $A, where, ifAa, A6, ...denoteanyarbitraryincrementswhateveroftheconstantsofin$ $7, butwhichitispropertomention, viz. : d - djla -$ $d, t = djda - r6 - dd7r - R7 + ...o\ rd - dja - d -$ $drrR - i...ITnffact, int, , hecformul, a8, . - dd3rR -$ $stand, sfor\ f - djdadd^{r}R-, d - djta - + - ddnbd - d^{R}r -$ $ddjtTb + ..\ Jd, t., and, 8_{r}for(fi - tr * (-ijc.i + (-I7T7 *$ $(-i7h7+)dt; and, onsubstitutingthesevalues, theidentityofthetwoexpressionsissee$ $V = a), andctheconstantattachedbyadditiontothetime, then -$ $dyt- = -d7c-, which, heobserves, isanequationremarkableas$

195] well from its simplicity and generality as because it can be obtained d priori, inde pendently of the variations of the other arbitrary constants : this is obviously the generalisation of the expression for the variation of the mean distance of a planet. 10. The consideration of Lagranges function (a, b) originated, as appears from what has preceded, in the theory of the variation of the elements but it is to be ; noticed, that the function (a, b) is altogether independent of the disturbing function, and the fundamental theorem that (a, b) is a function of the elements only, without the time, is a property of the undisturbed equations of motion. The like remark applies to Poissons function (a, b), in the memoir next spoken of. 11. Poisson s memoir of the 16th of October, 1809. The formulae of this memoir are. so to speak, the reciprocals of those of Lagrange. The relations between the differential coefficients -= c. of the disturbing function and the variations - c. da , dt , of the elements, depend with Lagrange, upon expressions for the coordinates and their differential coefficients in terms of the time and the elements with Poisson, on ; expressions for the elements in terms of the time, and of the coordinates and their differential coefficients. The distinction is far more important than would at first sight appear, and the theory of Poisson gives rise to developments which seem to have nothing corresponding to them in the theory of Lagrange. The reason is as follows : when the system of differential equations is completely integrated, it is of course the same thing whether we have the integral equations in the form made use of by Lagrange, or in that by Poisson, the two systems are precisely equivalent the one to the other; but when the equations are not completely integrated, suppose, for instance, we have an expression for one of the coordinates in terms of the time and the elements, it is impossible to judge whether this is or is not one of the integral equations ; the differential equations are not satisfied by means of this equation alone, but only by this equation with the assistance of the other integral equations. On the other hand, when we have an expression for one of the constants of integration in terms of the time and of the coordinates and their differential coefficients, it is possible, by mere substitution in the differential equations, and without the knowledge of any other integral equations, to see that the differential equations are satisfied, and that the assumed expression is, in fact, one of the system of integral equations. An expression of the form just referred to, viz., c = <f)(t, x, y,...x, 2/ ...), where the right-hand side does not contain any of the arbitrary constants, may, with great propriety, be termed an "integral," as distinguished from an integral equation, in which the constants and variables may enter in any conceivable manner; it is convenient also to speak of such equation simply as the integral c. [These locutions were introduced by Jacobi.] 12. Returning now to the consideration of Poissons memoir, the equations of motion are considered under the same form as by Lagrange, viz., putting T-V=R, under the form d dR dR$_d$$Cldtdfi\ d$ d

164 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 but Poisson writes dR$_S$$d$ "- thus, in effect, introducing a new set of variables, s, ... equal in number to the coordi n b a y tes the <j>, . i . n . t , ro b d u u t cti h o e n d t o h

e e s rei n n ot of co t m h p e let n e ew th v e ari t a r b a l n e s s fo s r , ma .. t . io i n n o t f he th p e lac di e ff o e f ren t t h i e al di e f q f u er a e t n i t o i n a s l t co i e m f e fic a ie f n t t e s rwa </am r p;gt; d ,. s .. b ; y th S i i s r v W e . ry R i . mp H o a r m t i a l n t t on t . ran P s o f i o s r s m o a n tio t n he w n as ass o u nl m y es ef t f h e a c t ted the a c u o n n d s i i s d t e u r r a b b e le d equations are integrated in the form above adverted to, viz., that the several elements a, b, ... are given as functions of the time t, and of the coordinates c. and their differential coefficients < , c. or, what is the form ultimately assume l d t;/> , ;, as functions of the time t, of the coordinates <f>, ... , and of the new variables s, c. ; and he then forms the functions , 6 7N 9 (a, b) + < ffl >=4r?) " where a (g > b) =da db$^{dbda9dsds(s,<f>)d<f>d<}$> (the notation is the abbreviated one before referred to), and he proves by differentiation that the differential coefficient of (a, b) with respect to the time vanishes : that is, that (a, b) which, by its definition is given as a function of t and of the coordinates <, ..., and of the new variables s, ..., is really a constant. Upon which Poisson remarks "On concoit que la constante...sera en general une fonction de a et b et des constantes arbitraires contenues dans les autres intdgrales des equations du mouve- ment; quelquefois il pourra arriver que sa valeur ne renferme ni la constante a ni la constante b ; dans dautres cas elle ne contiendra aucune constante arbitraire, et se reduira a une constante determinee ; mais afin c." 13. The importance of the remark seems to have been overlooked until the attention of geometers was called to it by Jacobi it has since been developed by ; Bertrand and Bour. It is clear from the definition that (a, 6) = (b, a). It may be as well to remark that the denominator of the functional symbol is (s, and not $(,s),whichwould < / >$ $)reversethesign.[ItmaybenoticedthatthroughouttheReport,IspeakoftheLagranges$ $y-,c.intermsof(Jjltd-f,lpc.,insteadoftheseexpressionshavingtobedeterminedfr$ $j-,WoC.intermsoff-d=t-,,c.$

15. Poisson applies his formulae to the case of a body acted upon by a central force varying as any function of the distance, and also to the case of a solid body revolving round a fixed point. There is, as Poisson remarks, a complete similarity between the formulas for these apparently very different problems, but this arises from the analogy which exists between the arbitrary constants chosen in the memoir for the two problems. The formulae obtained form a very simple and elegant system, and one which, although not

actually of the canonical form (the meaning of the term will be presently explained), might by a slight change be reduced to that form. 16. I may notice here a problem suggested by Poisson in a report to the Institute in the year 1830, on a manuscript work by Ostrogradsky on Celestial Mechanics, viz., in the case of a body acted upon by a central force, the effect of a disturbing function, which is a function only of the distance from the centre, is merely to alter the amount of the central force and the expressions for the variations of ; the elements should therefore, in the case in question, admit of exact integration ; the report is to be found in Crelle, t. vu. [1831], pp. 97 101. 17. The two memoirs of Lagrange and Poisson, which have been considered, establish the general theory of the variation of the arbitrary constants, and there is not, I think, very much added to them by Lagranges memoir of 1810, the second edition of the Me canique Analytique, 1811, or Poissons memoir of 1816. The memoir by Maurice, in 1844, belongs to this part of the subject, and as its title imports, it is in fact a development of the theories of Lagrange and Poisson. 18. There is, however, one important point which requires to be adverted to. Lagrange, in the memoir of 1810, and the second edition of the Mecanique Analytique, remarks, that for a particular system of arbitrary constants, viz., if a, ... denote the dT initial values of the coordinates , ... and X, ... denote the initial values of d r g p, , ... then the equations for the variations of the elements take the very simple form da.= *cZOdX =*
dldt d" di da"thisis, infact, theoriginalideaandsimplestexampleofasystemofcano
*damp;ylt;MddayddQzddzaandsimilarlyfordbCan(*tllisbeinso >*
a11tlieformulaewillsubsistasinthecaseofanactuallyexistingdisturbingfunction.

166 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 20. Cauchy, in a note in the Bulletin de la Societe Philomatique for 1819 (repro duced in the " Memoire sur Ilntdgration des Equations aux Derivees Partielles du Premier Ordre," Exer. dAnal. et de Physique Math., t. II. pp. 238272 (1841)), showed that the integration of a partial differential equation of the first order could be reduced A to that of a single system of ordinary differential equations. particular case of this general theorem was afterwards obtained by Jacobi in the course of his investi gations (founded on those of Sir W. R. Hamilton) on the equations of dynamics, and he was thence led to a slightly different form of the general theorem previously established by Cauchy, viz., Cauchys method gives the general, Jacobis the complete integral, of the

partial differential equation. The investigations of the geometers who have written on the theory of dynamics are based upon those of Sir W. R. Hamilton and Jacobi, and it is therefore unnecessary, in the present report, to advert more particularly to Cauchys very important discovery. 21. I come now to Sir W. R. Hamilton s memoirs of 1834 and 1835, which are the commencement of a second period in the history of the subject. The title of the first memoir shows the object which the author proposed to himself, viz., the discovery of a function by means of which the integral equations can be all of them actually represented. The method given for the determination of this function, or rather of each of the several functions which answer the purpose, presupposes the knowledge of the integral equations ; it is therefore not a method of integration, but a theory of the representation of the integral equations assumed to be known. I venture to dissent from what appears to have been Jacobis opinion, that the author missed the true application of his discovery ; it seems to me, that Jacobis investigations were rather a theory collateral to, and historically arising out of the Hamiltonian theory, than the course of development which was of necessity to be given to such theory. But the neAv form obtained in Sir W. R. Hamiltons memoirs for the equations of motion, is a result of not less importance than that which was the professed object of the memoirs. 22. Hamiltons principal function V. The formulae are given for the case of any number of free particles, but, for simplicity, I take the case of a single particle. The equations of motion are taken to be - dU d^{*y} = $indPdy_{d?y-dUrrn2dt*dzsothatthevisvivafunctionisandtheforcefunction,takenwithLagrangessign,wc}$

195] b T y = of, UH b + r , H c . of The the eq s ua a t m i e on fo o r f m vi f s or vi t v h a e is in T iti = al U v + al H ue , s an o d f t t h h i e s g c i o v o e r s din r a i t s e e s. to a T n he eq a u u a t t h io o n r) then writes an equation, the form of which implies that T is expressed as a function of the time and of the constants of integration a, b, c, a , b , c . The method of the calculus of variations leads to the equation 8V= m (x Sx+ ySy +z8z) -m (aSa+b8b+ cSc) + t8H, to understand which, it should be remarked that the coordinates x, y, z, and the velocities x, y , z, being functions of t and of a, b, c, a , b , c , then V is, in the first instance, given as a function of these quantities. But x, y, z being functions of a a, nd 6, t c, hu a s , V b , b c e , co t, me w s e a ma f y unc c t o i n o v n er o s f el x y , c y o , ns z i , de a, r b a , , c, b , t. c I a n s l f i u k n e cti m o a n n s ne of r x,Hy, is z, a a, fu b n , ct c, io t n , of x, y, z, a, b, c, t, and, eliminating t, we have V a function of x, y, z, a, b, c, H, which is the form in which in the last equation V is considered to be expressed. The equation then gives dV dV dV dx - = mx , - d r y my, , -r d = z = mz . dV dV dV - d = a - = ma , -dy,o = mo , - d 7 c = me , d-V=f dH H and, considering V as a known function of x, y, z, a, b, c, H, the elimination of gives a set of equations which are in fact the integral equations of the problem, viz., the first three equations and the last equation give equations containing x, y, z, x , y , z , t and a, b, c, that is, the intermediate integrals; the second three equations and the last equation, give equations containing x, y, z, t, a, b, c, a , b , c , that is, the final integrals. The function V satisfies the two partial differential equations which, if they could be integrated, would give V as a function of x, y, z, a, b, c, H, and thus determine the motion of the system. 23. Hamiltons principal function S. This is connected with the function V by the equation V=tH+ S;

or, what is the same thing, the new principal function 8 is defined by the equation t S=f (T+U)dt; J but S is considered (not like V as a function of x, y, z, a, b, c, H, but) as a function of x, y, z, a, b, c, t. The expression for the variation of S is 8S=-HSt +m (xSx+ ySy +zSz - m (aSa+ bSb + cSc) which is equivalent to the system dS dS dS = mx -; = my -r- = mz dx , dy , dz , dS dS dS dya- = ma„ -djbr = moi, , -djc- me, , dS H dt- the first three and the second three of which give, respectively, the intermediate and the final Hintegrals; the last equation leads only to the expression of the supernumerary constant in terms of the initial coordinates a, b, c, and it may be omitted from the system. The function S satisfies the partial differential equations dt 2m dy dS + d$^{)}(db) which, if they could be integrated, would give S as a function of x, y, z, a, b, c, t, an

then T, in its original form, is a function of 77,... 77 , ... , homogeneous of the second order as regards the differential coefficients 77, ... ; and, consequently, these being linear functions (without constant terms) of the new variables OT, the vis viva function T can be expressed as a function of 77, ... tzr, ... , homogeneous of the second order as regards the variables tr , ... And when T has been thus expressed, the equations of motion take the form dU d/n$_{d}H(fa_{dTdtdmdtdrjdrj}$ which is the required transformation. The force function U is independent of the

a of lre N a o d . y 1 kn 3 o , wn. a qu s ot; quo T t; h gi e ving f f un o c r t m i u o l n a H for is t c h o e nsi v d a e r r ia e t d io a n s o c f ons e i l s e t m i e n n g ts of mo t r wo e p a a n r a t l s, ogo o u n s e t o o f t t h h o e s m e being treated as a di-

sturbing function ; the equations of motion assume therefore the form
+ di dvrd−ndtdrjdrjHHThavewrittenH,Tinsteadoftheauthorsltt Thetermsinvolvingarein(Ithefirstinstanceneglec

195] constant. The investigation applies to the case where the attracting force is any function whatever of the distance, and the six elements ultimately adopted form a canonical system. 28. The precise relation of Sir W. R. Hamiltons form of the equations of motion to that of Lagrange, is best seen by considering Lagranges equations, not as a system of differential equations of the second order between the coordinates and the time t, but as a system of twice as many differential equations of the first order between the coordinates, their differential coefficients treated as a new system of variables, and the time. It will be convenient to write U, instead of Lagranges force-function V, and (to conform to the usage of later writers who have treated the subject in the most general manner) to represent the coordinates by q, ... , their differential coefficients by q , ... , and the new variables which enter into the Harniltonian form by p, ... ; then the Lagrangian system will be dq = , d$_d T_d$ Tdtdtdqdqdqorputting$T +$ $U = Z$(thisisthesameasLagrangessubstitution,$TV =$ R),thesystembecomesdq$_d Z_d Z$, $d^d i$ qdidq dqwhiletheHarniltoniansystemisdq $=$ dTdp $=_d$ TdU$_d$tdpdtdqdqorputtingasbeforeTU $=$ H,thesystemisdHdHdq$_d p_d$tdpdtdqwhere,intheLagrangiansystems,TandU,andco
transformation itself, as regards the actual value of the new function T (= T expressed in terms of the new variables), which enters into the transformed equations, depends essentially upon the special form just referred to of the function T, although, as will be seen in the sequel, there is a like transformation applying to the most general form of the function T. 29. In the greater part of what has preceded, and especially in the above- mentioned substitutions T+U=Z and T U= H, it is of course assumed that the force function U exists when there is no force function these substitutions cannot be made, but the forms c ; orresponding to the untransformed forms in T and U are as follows, viz. the Lagrangian form is dq,ddT$_d T_n$dt qdtdqdq ∗
andtheHamiltonianformisdq$_d$Tdpdtdpdtthatis,theonlydifferenceis,thatthefunc
x$_d$Ud∗y$_d$U wh(exr2e+yUn−) = isU+ah.funcAtsiosnumeso,tyhatwitahnooutther
Fis(x,thye,xe,quyat)i,otnheonfxv,isyvwiivlalingeneralbefunctionsofx,y,a,h,a
ydywillbeanexactdifferential,butitsdifferentialcoefficientswithrespecttoa,hwil
ffdrxd7x + dy, J(da dfa − dy∗)/, +Tm = (fdxd, x + dy, tj
−djjh−−djfridyyJ,atheorem,therelationofwhichtothegeneralsubjectwillpresent
The second result does not relate to the general subject, but I give it in a note for its own sake(J). 31. Poissons memoir of

1837. This contains investigations suggested by Sir W. R. Hamiltons memoir, and relating to the aid to be derived from a system of given integral equations (equal in number to the coordinates) in the determination of the dV principal function V. The equations -T- =mx , c. give d V= m (x dx+ ydy +z dz), or in the case of a system of points, dV= 2m (x dx + ydy-f z dz). If the points, instead of being free, are connected together by any equations of condition, then, by means of these equations, the coordinates x, y, z of the different points and their differential coefficients, x , y , z, can be expressed as functions of a certain number of independent t va a r k i e a s bl t e h s e < f />, or t m y, d 6, V = c. X , d<f a amp; n gt; d + Y o d f ty th + ei Z r dO di + ffe . r .. e , nti w a h l er c e oef X f , icie Y n , ts Z < ar , e -v/ f r u , nc 9 ti , on s c. o ; f d V lt;, i/ t r, he .. n . e amp;lt;/ q gt;, ua T l //, i . n .. . nu I m m b a e g r in t e o n t o h w e a ind s e y p s e t n e d m en o t f v i a n r t i e a g b r l a e l s s (amp;l o t;, ne ^{r}, of0..t.h; emthetnh, eebqyuatthieonaiodfovifstvhievsae)equations, < j >, i/r, ..., and, consequently, X, Y, Z, ...canbeexpressedasfunctions(oftheconstan
'0,
Hence,attendingonlytothevariables,dV=Xdcf>+Yd^r+ZdO+...isadifferentialexpressioninvolvingonlythevariabl

where T, I, m, ... are new arbitrary constants. But, as before remarked, the expression for dV is not always a complete differential. Poisson accordingly inquires into and determines (but not in a precise form) the conditions which must be satisfied, in order that the expression in question may be a complete differential. He gives, as an example, the case of the motion of a body in space under the action of a central force and, secondly, the case considered in Jacobis letter of 1836, which he refers to, vi ; z., here dV= xdcc + ydy, and when the two integral equations are one of them, t e h q e uat e i q o u n ati a o = ns F(x of , y v , is x v , iv y a) wh (a oc t /2 e + ve y r quot; 2) (= su U bje + ct h, o a n n ly d t t o he the oth r e es r tri o c f tio t n he t m hat an a y is in n t o eg t ral a function of x, y, a/2 + y2 , the necessity of which is obvious) the condition is satisfied per se, and, consequently, xdx+ ydy is a complete differential, and its integral gives V (in value, although as before remarked not in form) the principal function and such value of V gives the two integral equations obtained in Jacobis letter. 32. Jacobis note of the 29th of November, 1836, " On the Calculus of Variations, and the Theory of Differential Equations." The greater part of this note relates to the differential equations which occur in the calculus of variations, including, indeed, the differential equations of dynamics, but which belong to a different field of investigation. The latter part of the note relates more immediately to the differential equations of dynamics. The author remarks, that, in any dynamical problem of the motion of a single particle for which the principle of vis viva holds good, if, besides the integral of vis viva, there is given any other integral, the problem is reducible to the integration of an ordinary differential equation of two variables, and that it is always possible to integrate this equation, or at least discover by a precise and general rule the factor which renders it integrable. This would seem to refer to Jacobis researches on the theory of the ultimate multiplier, but the author goes on to refer to a preceding communication to the Academy of Paris (the before-mentioned letter of 1836), which does not belong (or, at least, does not obviously belong) to this theory. He speaks also of a class of dynamical problems, viz. that of the motion of a system of bodies which mutually attract each other, and which may besides be acted upon by forces in parallel lines, or directed to fixed centres, or even to centres the motion of which is given and, he remarks, in the solution of such a problem, the system ; of differential equations being in the first instance of the order 2?i (that is, being a system admitting of 2n arbitrary constants), then if one integral is known, it is possible by a proper choice of the quantities selected for variables to reduce the system to the order 2n 2. If another integral is known, the equation may in like manner be reduced to a system of the order 2n 4, and so on until there are no more equations to be integrated ; and thus the operations to be effected depend only upon quadratures. All this seems to refer to researches of Jacobi, which, so far as I am aware, have not hitherto been published. The results correspond with those recently obtained by Bour, post, Nos. 66 and 67. 33. Jacobis memoir of 1837. Jacobi refers to the memoirs of Sir W. R. Hamilton, and he reproduces, in a slightly different form, the investigation of the fundamental property of the principal function S. The case considered is that of a system of n

195] particles, the coordinates of which are connected together by any number of equations ; but it will be sufficient here to attend to the case of a single free particle. The equations of motion are assumed to be $_dUd?yd^{dU}$ $_{butUisconsideredasbeingafunctionofx,y,zandofthetimet,thatis,itisassumedthattheconditionofvisviva}$

i m n ay the be pr t e a s k e e n n t t c o ase b , e t a her

c e o f n o s re t , ant fou a r tta a c r h b e i d trar to y c t o h n e sta f n u t n s c . tio B n ut S o b n y e m of ere the a s d e dit c i o o n n, sta a n n t d s which disappears from the differential coefficients, and it is only necessary to attend to the other three arbitrary constants. 8 is consequently a function of t, x, y, z, and And of the arbitrary constants a, /3, being so, it is shown that the integrals of the problem are d d -= x S - = nui„ d - d r y S =my„ d - d r z S - = viz,1 , dS $dS_d S_T^* Ta$ 1dy *whereX/**
vareanyotherarbitraryconstants, viz., thefirstthreeequationsgivetheintermediate
35.*JacobisprincipalfunctionF.First, whentheconditionofvisvivaissatisfied.Here*
(*Yi−dx*)*wherehistheconstantofvisviva.Thepartialdifferentialequationcontainso*
wixd7y− = *my*, , −*djz−* =?*nz*, , $dV_d Z$ = *dV* =
whereT, A,[a] *renewarbitraryconstants.*36.*JacobisprincipalfunctionFSecondly, wh*
aUis+*notsatisfied.HereUcontainsthetimet, andwehavenosuchequationash, butal*

195] dV where, in the expression for U, it is assumed that t is replaced by -·*Thereare, consequently, fourindependentvariables, andacompletesolution*
djx− = *mx*, , −*dTy−* = *my*, , −*djz* − *dVdVdV* =
dHHwhereA, , //., *varearbitraryconstants, viz., eliminatingfromthefirstthreeeq*
U+*H, thatis, H*(= *TU*)*isthatfunctionwhich, whentheconditionofvisvivaissatisfi*

equations, a system which is also treated of in Jacobis memoir of 1844, "Theoria Novi Multiplicatoris c." I take the opportunity of referring here to a short note by Brioschi, "Intorno ad una Proprieta delle Equazioni alle Derivate Parziali del Primo Ordine," Tortolini, t. vi. pp. 426 429 (1855), where the theory of the integration of a partial differential equation of the first order is presented under a singularly elegant form. 38. Jacobis note of 1837, " On the Integration of the Differential Equations of Dynamics." Jacobi remarks that it is possible to derive from Lagranges form of the equations of motion an important profit for the integration of these equations, and he refers to his communication of the 29th of November 1839 to the Academy of Berlin, and to his former note to the Academy of Paris. He proceeds to say, that whenever the condition of vis viva holds good, he had found that it was possible in the integration of the equations of motion to follow a course such that each of the given integrals successively lowers by two unities the order of the system and that the like theorem ; holds good when the condition of vis viva is not satisfied, that is, when the force function involves the time (this seems to be a restatement, in a more general form, of the theorems referred to in the note of the 29th of November 1836 to the Academy of Berlin) and he mentions that he had been, by

his researches on the ; theory of numbers, led away from composing an extended memoir on the subject. The note then passes on to other subjects, and it concludes with two theorems, which are given without demonstration as extracts from the intended work he had before spoken of. These theorems are in effect as follows : I. Let d2x
$_dUd2y_dlld2z_dUdt2dxdt2dydt2dzbethe3ndifferentialequationsofthemotionofafree$
$y2+z2)dt = U+h, theequationofvisviva.LetVbeacompletesolutionofthepartialdif$
$(*(*.hi^V dx)\ dzJjthatis, asolutioncontaining, besidestheconstantsattachedtoVbyme$
$i1constantsa(a1a2, ...a3n-^), thenfirsttheintegralequationsaredVdV =$
$Q = da.dhwhere/3(/3i, /32, .../331l_i)andTarenewarbitraryconstants :$
$thisisinfactthetheoremalreadyquotedfromJacobismemoirof1837, anditisinthepre$

to as an easy generalisation of Sir W. R. Hamiltons formulae. But Jacobi proceeds (and this is given as entirely new) that the disturbed equations being rtl d dt f 2 a

$ddxU+d\ fQa,H\ ddfty?\ _{d\ dly7+d\ fOy,mdd-?z}$

$(dteU+dlftlethentheequationsforthevariationsoftheabovesyste$

of elliptic motion of the coordinates x, y, z and the velocities x, y , z with respect to the elements and he remarks, that if Enckes elements are replaced by a system ; of elements a, ft, 7, a. , ft, 7 which he mentions, connected with those of Encke by equations of a simple form, then considering first x, y, z, x , y , z as given functions of t and the elements, and afterwards the elements a, (3, 7, a , ft, y as given functions of t and x, y, z, x, y , z , there exists the remarkable theorem that the thirty-six fJn fJfi partial differential coefficients - Ci j v - C , - CL r O - C , c., and the thirty-six partial differential co- dec dec coefficients j- , -
$'c., areequaltoeachother, ordifferonlyintheirsign, viz.dxdo.dx_dadx_da!dx_dot.da.dxd$
$sixequationsinall, viz.thepaira, a.ofcorrespondingelementsmaybereplacedbythepa$
$andthenineachofthetwelveequationsy, yorz, zmaybewritteninsteadofx, x.Thelikec$
$!_, , c.oftheforcefunctionTwithrespecttop, ...; andforeverysystemofelementswhichp$
$1or1, asisknowntobethecasewiththelast-mentionedsystemofelements; inotherwor$
$suivantlememeprocedeonpourratrouverunequatrieme, unecinquiemeintegrate, ete$

195] les integrates, ou ce qui revient aii meme 1integration complete du probleme. Dans des cas particuliers on retombera sur une combinaison des integrales dejd trouvees avant qnon soit parvenu a toutes les integrales da probleme, mais alors les deux integrales donnees jouissent des proprietes particulieres dont on pent tirer un autre profit pour 1integration des equations dynamiques proposees. Cest ce qu on verra dans un ouvrage auquel je travaille depuis plusieurs annees et dont peut-etre je pourrai bientot faire commencer 1impression." 41. Liouvilles addition to Jacobis letter of 1840. This contains the demonstration of a theorem similar to that given in Jacobis letter of 1836, and Poisson s memoir of 1837, but somewhat more general the system considered is a system of four ; differential equations of the first order : W -x dx -

$^{d}Udy_{-}).dUdU_{d}t\ dx"dt\ dxdi\ dy"dtdyiwnhteergeralUisisUa=fau,ncatnidonifofthxe,rey,bxe,yan,otahned$

44. Jacobis memoir of 1842, "De Motu Puncti singularis": the author remarks, that the greater the difficulties in the general integration of the equations of dynamics, the greater the care which should be bestowed on the examination of the dynamical problems in which the integration can be reduced to quadratures; and the object of the memoir is stated to be the examination of the simplest case of all, viz. the problems relating to the motion of a single point. The first section, entitled, " De Extensione quadam Principii Virium vivarum," contains a remark which, though obvious enough, is of considerable importance: the forces X, Y, Z which act upon a particle, may be such that Xdx+ Ydy +Zdz is not an exact differential, so that if the particle were free, there would be no force function, and the equations of motion would not be expressible in the standard form. But if the point move on a surface or a curve, then in the former case Xdx+ Ydy + Zdz will be reducible to the form Pdp + Qdq, which will be an exact differential if a single condition (instead of the three conditions which are required in the case of a free particle) be satisfied, and in the latter case it will be reducible to the form Pdp, which is, per se, an exact differential. In the case of a surface, the requisite transformation is given by the Hamiltonian form of the equations of motion, which Jacobi demonstrates for the case in hand ; and then in the third section, with a view to its application to the particular case, he enumerates the general proposition "quse pro novo principio mechanico haberi potest," which is as follows : " Consider the motion of a system of material points subjected to any conditions, and let the forces acting on the several points in the direction of the axes be functions of the coordinates alone : if the determination of the orbits of the several points is reduced to the integration of a single differential equation of the first order between two variables, for this equation there may be found, by a general rule, a multiplier which will render it integrable by quadratures only." And for the particular case the theorem is thus stated : " Given three differential equations of the first order between the four quantities P ?1, ffs, Pi, 2, d gi : d , q2 : d, Pl : d , p2 = $^{d}T :^{d} T : -^{d}T^{+}r.$:

$-^{d}T^{+}nQ,, inwhichQ1Q2arefunctionsofqlq.2only; supposethattherearekknowntwo$
$integratedanequationofthefirstorder, .dq.2-rdqlbetweenthequantitiesqi >$
$q-2, bywhichisdeterminedtheorbitofthepointonthegivensurface :$
$Isaythattheleft-handsideoftheequationmultipliedbythefactordpidp2 _{dj)2dp^{d}zdtfdac}$

will be a complete differential, or will be integrable by quadratures alone," and the demonstration of the theorem is given. The remainder of the memoir, sections 4 to 7, is occupied by a very interesting discussion of various important special problems. 45. There is an important memoir by Jacobi, which, as it relates to a special problem, I will merely refer to, viz. the memoir " Sur FElimination des Noeuds dans le probleme des trois Corps," Crelle, t. xxvu. pp. 115 131 (1843). The solution is made to depend upon six differential equations, all of them of the first order except one, which is of the second order, and upon a quadrature. 46. Jacobis memoir of 1844, "Theoria Novi Multiplicatoris c." This is an elaborate memoir establishing the definition and developing the properties of the " multiplier" of a system of ordinary differential equations, or of a linear partial differential equation of the first order, with applications to various systems of differential equations, and in particular to the differential equations of dynamics. The definition of the multiplier is as follows, viz. the multiplier of the system of differential equations dx : dy : dz : dw...=X : Y : Z : W... or of the linear partial differential equation of the first order y df vdf df TIT df dx dy dz dw is a function J/, such that dMX dMY dMZ dMW dx dy dz dw One of the properties of the multiplier is that contained in the theorem of the ultimate multiplier, viz. that when all the integrals (except one) of the system of partial differential equations are known, and the system is thereby reduced to a single differential equation between two variables, then th M e multiplier (in the ordinary sense of the word) of this last equation is MV, where is the multiplier of the system, and V is a given derivative of the known integrals, so that the multiplier of the system being known, the integration of the last differential equation is reduced to a mere quadrature. To explain the theorem more particularly, suppose that the system of given integrals, that is, all the integrals (except one) of the system are represented = = by p a, q @ , ... , and let u, v be any two functions whatever of the variables, so that p, q,... u, v are in number equal to the system x, y, z, w, ... then if dx dy dz dw d d v x
$1_{y}ddyv_{ig}ddvzi, TyddwvI.ITthelastdifferentialequationtakestheformUdv-$
$Vdu = UNIVCEfRTHESITY0, OF$

184 REPORT OX THE RECENT PROGRESS OF THEORETICAL DYNAMICS. where it is assumed that T and T are, by the assistance of the given integrals, e l x a p s r t e - s m s e e n d tio a n s ed fu e n q c u t a i t o i n o s n o i f s Jf , V v , a wh n e d re the M co i n s st t a h n e ts mu o l f ti i p n l t i e e g r rat o i f on. the Th sy e st m e u m l , tip a l n i d er V of m t a h y e be expressed in either of the two forms a , $...u, r) and - *V_f 3(p.9, where the symbols on the right -$
$hand sides represent functional determinants; in the first form it" is assumed that r, y$
$the first of" j; atmp; hgte; , two forms, from its not involving this transformation bac$
$given the system of differential equations = XYZ dx : dy :$
$dz : die ::: W, and suppose that two of the integrals are p = a, q =$
$, the last equation to be integrated will be where, by the assistance of the given integrals,$
$a(*..v)?(*, /*) which supposes that x, y are expressed as functions of a, ft, z, v. 4a Jacob$
$ltahset equation is therefore equal to V. The two cases are to be distinguished in which t do$

case of a system the multiplier of which is known, viz., the theorem is, given all the integrals except one, the remaining integral can be found by quadratures only. Bat in the former case, which is the ordinary one, including all the problems in which the condition of vis viva, is satisfied, there is a further consequence deduced. In feet, the time t may be separated from the other variables, and the system of differential equations reduced to a system not involving the time, and containing a number of equations less by unity than the original system, and the theorem of the ultimate multiplier applies to this new system. But when the integrals of the new system have been obtained, the system may be completed by the addition of a single differential equation involving the time, and which is integrable by quadratures; the theorem consequently is, given all the integrals except two, these given integrals being independent of the time, the remaining integrals can be found by quadratures only. This is, in fact, the " Principium generale mechanicum" of the memoir of 1842. The last of the published writings of Jacobi on the subject of dynamics is the " Auszug zweier Schreiben des Professors Jacobi an Herrn Director Hansen," Crelle, t. XLII. pp. 12 31 (1851) : these relate chiefly to Hansens theory of ideal coordinatea. 49. The very interesting investigations contained in several memoirs by Liouville (Liouville, t. xi. xn. and xiv., and the additions to the "Connaissance des Temps" for 1849 and 1850) in relation to the cases in which the equations of motion of a particle or system of particles admit of integration, are based upon Jacobis theorv of the S function, that is, of the function which is the complete solution of a certain partial differential equation of the first order ; the equation is given, in the first instance, in rectangular coordinates, and the author transforms it by means of elliptic coordinates or otherwise, and he then inquires in what cases, that is, for what forms of the force function, the equation is one which admits of solution. A more particular account of these memoirs does not come within the plan of the present report. 50. Desboves memoir of 1848 contains a demonstration of the two theorems given in Jacobis note of 1837, in the Comptes Rendus ; and, as the title imports, there is an application of the theory to the problem of the planetary perturbations; the author refers to the above-mentioned memoirs of Liouville as containing a solution of the partial differential equation on which the problem depends, and also to a memoir of his own relating to the problem of two centres, where the solution is also given and from this he deduces the solution just referred to, and which is employed in ; the present memoir. Jacobis theorem gives at once the formulae for the variation of the arbitrary constants contained in the solution. The material thing is to determine the signification of these constants, which can of course be done by a comparison of the formulae with the known formulae of elliptic motion; the author is thus led to a system of canonical elements similar to, but not identical with, those obtained by Jacobi. 51. Serrets two notes of 1848 in the Comptes Rendus. These relate to the theory of Jacobis S function, that is, of the function considered as the complete solution of a given partial differential equation of the first order. In the first of the two notes, C. III. 24

186 REPORT OX THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 which relates to a single particle, the author gives a demonstration founded on a particular choice of variables, viz., those which determine orthotomic surfaces to the curve described by the moving point. The process seems somewhat artificial. 52. Sturms note of 1848, in the Comptes Rendus, relates to the theory of Jacobis 8 function, that is, of this function considered as the complete solution of a given partial differential equation of the first order. The force function is considered as involving the time t, which, however, is no more than had been previously done by Jacobi. 53. Ostrogradskys note of 1848. This contains an important step in the theory of the forms of the equations of motion, viz. it is shown how, in the case where the force function contains the time, the equations of motion may be transformed from the form of Lagrange to that of Sir W. R. Hamilton. If, as before, the force function (taken with T+

th U e = c Z ontrary sign to that o V f Lagrange) is represented by U, then putting, as before, (the author writes instead of Z), in the case under consideration Z will contain not only terms of the second order and terms of the order zero in the differential coefficients of the coordinates q, ..., but also terms of the first order, that iHs, Z will be of the form Z=Z, + Z 1 +Z , and putting H=Z,-Z li, this new function being expressed as a function of the coordinates q, ... and of $_{t}henewvariablesp, ..., thentheequationsofmotiontaketheHamiltonianform, viz.dH$

a problem in the calculus of variations, had previously been treated of by Jacobi, but the theorem is probably new. In conclusion, the author shows in a very elegant manner the interdependence of the theorem relating to Lagranges coefficients (a, b), and of the corresponding theorem for the coefficients of Poisson. 55. Bertrands memoir of 1851, " On the Integrals common to several Mechanical Problems," is one of great importance, but it is not very easy to explain its relation to other investigations. The author remarks that, given the integral of a mechanical problem, it is in general a question admitting of determinate solution to find the expression for the forces; in other words, to determine the problem which has given rise to the integral; at least, this is the case when it is assumed that the forces are functions of the coordinates, without the time or the velocities; and he points out how the solution of the question is to be obtained. But, in certain cases, the method fails, that is, it leads to expressions which are not sufficient for the determination of the forces ; these are the only cases in which the given integral can belong to several different problems ; and the method shows the conditions necessary in order that these cases may present themselves. It is to be remarked that the given integral must be understood to be one of an absolutely definite form, such for instance as the equations of the conservation of the motion of the centre of gravity or of areas, but not such as the equation of vis viva, which is a property common indeed to a variety of mechanical problems, but which involves the forces, and is therefore not the same equation for different problems. The author studies in particular the case where the system consists of a single particle ; he shows, that when the motion is in a plane, the integrals capable of belonging to two or more different problems are two in number, each of them involving as a particular case the equation of areas. When the point moves on a surface, he arrives at the remarkable theorem "In order that the equations of motion of a point moving on a surface may have an integral independent of the time, and common to two or more problems, it is necessary that the surface should be a surface of revolution, or one which is developable upon a surface of revolution." When the condition is satisfied, he gives the form of the integral, and the general expression of the forces in the problems for which such He integral exists. examines, lastly, the general case of a point moving freely in space. The number of integrals common to several problems is here infinite. After giving a general form which comprehends them all, the author shows how to obtain as many particular forms as may be desired: it is, in fact, only necessary to resolve any problem relative to motion in a plane, and to effect a certain simple transformation on the integrals; one thus obtains a new equation which is the integral of an infinite number of different problems relating to the motion in space. As an instance of the analytical forms on which these remarkable results depend, I quote the following, which is one of the most simple: "If an integral of the equations of motion of a point in a plane belongs to two different problems, it is of the form = F(<l> f a, y, t), where is the derivative with respect to t, of a function of x, y, which equated to zero gives the equation of a system of right lines."

56. Bertrands memoir of 1851, " On the Integration of the Differential Equations of Dynamics." The author refers to Jacobis note of 1840, in relation to Poissons theorem ; and after remarking that there are very few problems of which two integrals are not known, and which therefore might not be solved by the method if it never failed; he observes that unfortunately there are (as was known) cases of exception, and that, as his memoir shows, these cases are far more numerous than those to which the method applies; thus for example the equation of vis viva, combined with any other integral whatever, leads to an illusory result. The theorem of Poisson may lead to an illusory result in two ways ; either the resulting integral may be an identity = 0, or it may be an integral contained in the integrals already known, and which con sequently does not help the solution. It appears by the memoir that the two cases are substantially the same, and that it is sufficient to study the case in which the = two integrals lead to the identity 0. Suppose that one integral is given, the author shows that there always exist integrals which, combined with the given integral, lead to an illusory result, and he shows how the integrals which, combined with the given integral, leads to such illusory result, are to be obtained. For instance, in the case of a body moving round a fixed centre [and attracted to it by a force which is a function of

the distance], there are here two known integrals ; first, the equation of vis viva (but this, as already remarked, combined with any other integral whatever, leads to an illusory result) ; secondly, the equation of areas. The question arises, what are the integrals which, combined with the equation of areas, lead to an illusory result ? The integrals in question are, in fact, the other two integrals of the problem; so that the inquiry into the integrals which give an illusory result, leads here to the completion of the solution. The like happens in two other cases which are considered, viz. first, the problem of motion in piano when the force function is a homogeneous function of the co ordinates of the degree 2 secondly, the problem of two fixed centres. Indeed the ; case is the same for all problems whatever, where the coordinates of the points of the system can be expressed by means of two independent variables. The next problem considered is that where two bodies attract each other, and are attracted to a fixed centre. Suppose, first, the motion is in piano, then as in the former case all the integrals will be found by seeking for the integrals which, combined with the equation of areas, give an illusory result. When the motion is in space, the principle of areas furnishes three integrals (the equation of vis viva is contained in these three equations) ; the integrals which, combined with the integrals in question, give illusory results, are eight in number, and, to complete the solution, there must be added to these one other integral, which alone does not put the method in default. The problem of three bodies is then shown to be reducible to the last-mentioned problem ; and the same consequences therefore hold good with respect to the problem of three bodies, viz., there are eight integrals which, combined with the integrals furnished by the principle of areas, give illusory results. To complete the solution it would be necessary to add to these a ninth integral, which alone would not put the method into default. 57. The author remarks that it appears by the preceding enumeration that the method of integration, based on the theorem of Poisson, is far from having all the

195] importance attributed to it by Jacobi. The cases of exception are numerous ; they constitute, in certain cases, the complete solutions of the problems, and embrace in other cases eleven integrals out of twelve. But it would be a misapprehension of his meaning to suppose that, according to him, the cases in which Poissons theorem is usefully applicable ought to be considered as exceptions. The expression would not be correct even for the problems which are completely resolved in seeking for the integrals which put the method into default ; there exists for these problems, it is true, a system of integrals which give illusory results ; but these integrals, combined in a suitable manner, might furnish others to which the theorem could be usefully applied. The author remarks, that, in seeking the cases of exception to Poissons theorem, there is obtained a new method of integration, which may lead to useful results ; and, after referring to Jacobis memoir on the elimination of the nodes in the problem of three bodies, he remarks that, by his own new method, the problem is reduced to the integration of six equations, all of them of the first order ; so that he effectuates one more integration than had been done by Jacobi [this is incorrect] ; and he refers to a future memoir, (not, I believe, yet published) [? the Memoir of 1857] for the further development of his solution. 58. To give an idea of the analytical investigations, the equations of motion are considered under the Hamiltonian form dH dH dq

dp_{dt} dpdtdqHwhereisanyfunctionwhateverofq,...p,...withoutt,andthenagivenintegralbeing=thequestionisshowntore

190 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 60. Bertrands notes, VI. and VII., to the third edition of the Mecanique Analytique, 1853, contain a concise and elegant exposition of various theorems which have been considered in the present report. The latter of the two notes relates to the above- mentioned theorem of Poisson, and places the theorem in a very clear light, in fact, establishing its connexion with the theory of canonical integrals. Bertrand in fact shows, that, given any integral a of the differential equations (in the last-mentioned form, the whole number of equations being 2k), then the solution may be completed by joining to the integral a a system of integrals , ... / 9 afc

$_1$, which, combinedwiththeintegrala, givetoPoissonsequationanidenticalform, viz.

$(a, ft) = l, (, ft)0, ...(a, ft^{)}0.$ This, heremarks, shows, that, givenanyintegrala, thes

$-nr(a, ft, ...*_,)$, anditisatonceseenthat$(a, 17) = (a, ft)$ −= −con −

GtyCJjCLfiisequentlytheexpression(a, 77)willnotbeidenticallyconstantunlessfe

butofttheintegrals, innumberinfinite, whichresultfromthecombinationofa, withf

icombinedwiththeintegrala, giveidenticalresults.Onlytheintegralswhichcontainft

itisassumedthat1ThedateisthatofthecommunicationofthenotetotheBureauofLon

half of the integrals are known, and that the given integrals are such that for any two of them a, ft, Poissons coefficient (or, ft) is equal to zero; this being so, the expression pdq+ ... Hdt, where, by means of the known integrals, the variables p, ... are expressed in terms of q,...t, is a complete differential in respect to q, ... t, viz. it will be the differential of Sir W. R. Hamiltons principal function

V, which is thus determined by means of the known integrals, and the remaining integrals are then given at once by the general theory. 63. Professor Donkins memoir of 1854 arid 1855, Part I. (sections 1, 2, 3, articles 1 to 48). The author refers to the researches of Lagrange, Poisson, Sir W. R. Hamilton, and Jacobi, and he remarks that his own investigations do not pretend to make any important step in advance. The investigations contained in section 1, articles 1 to 14, establish by an inverse process (that is, one setting out from the integral equations) the chief conclusions of the theories of Sir W. R. Hamilton and Jacobi, and in particular those relating to the canonical system of elements as given by Jacobis theory. The theorem (3), article 1, which is a very general property of functional determinants, is referred to as probably new. The most important results of this portion of the memoir are recapitulated in section 4, in the form of seven theorems there given without demonstration ; some of these will be presently again referred to. Articles 17 and 18 contain, I believe, the only demonstration which has been given of the equivalence of the generalised Lagrangian and Hamiltonian systems. The transformation is as follows: the generalised Lagrangian system is $^{d}Z_{dZ_dt\ dq\ dqHwhe=re-ZZ+qispa+ny...f,unwchteiHroen,oofntthaendrigohft-ahmp;lt;y,a.n.d.gr,si.d.}$

192 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 64. Part II. (sections 4, 5, 6 and 7, articles 49 93, appendices). Section 4 con tains the seven theorems before referred to. Although not given as new theorems, yet, to a considerable extent, and in form and point of view, they are new theorems. Theorem 1 is a theorem standing apart from the others, and which is used in the demonstration of the transformation from the Lagrangian to the Hamiltonian system. X It is as follows: viz., if be a function of the n variables , ... , and if y, ... be n other variables connected with these by the n equations dX then will the values of se, ... , expressed by means of these equations in terms of be of the form y, ... , dT = -T- ... dy , and if p be any other quantity explicitly contained in X, then also dX dY +- =0, the differentiation with respect to p being in each case performed only so far as p appears explicitly in the function. The value of Y is given by the equation Y= - X + + xy ... where, on the right-hand side, x, ... are expressed in terms of y, ... Theorems 2, 3 and 4, and a supplemental theorem in article 50, relate to the deduction of the generalised Hamiltonian system of differential equations from the i i n n t s e t g e r a a d l of eq t u h a e tio a n u s tho a r s s su X m , ed aslt to ... b a; e n, k y n ly ow . n .. . yn, In a lt f

. a .. ct an (, w b r l i t ti .. n . g bn V , it q, i . s .. as p s , u . m .. ed b, t . h .. at a, V ... is , a given function of t, of the n variables q, ... , and of the n constants b,..., and that the n variables p, ... , and the n constants a,..., are determined by the conditions dV X < > -dV w ar-*-i so that in fact by virtue of these 2>i equations the 2n variables X, q,... p, ... may be considered as functions of t, and the 2n constants b,... a,... (hypothesis 1), or con versely, the 2n constants b,... a,..., may be considered as functions of t and of the 2n variables q, ... p,... (hypothesis 2).

Theorem 2 is as follows : viz., if from the 2n equations (1, 2) and their total differential coefficients with respect to t, the 2n constants be eliminated, there will result the following 2w simultaneous differential equations of the first order, viz.: dH dq = dt dp "
$_{d}t$ "" *dqHishereafunctionofq,...p,...(whichwillingeneralalsocontaintexplicitly),givenbytheequationdVH=*

In theorem 4 the authors symbol (p, q) has a signification such as Poissons (a, b), and if we write as before . 3 (a, 6) IIL rt i, Ut
I d *- (/ p, q) r -r j where d (a, 6) $_{d}adbdbdad(pq)dpdqdpdq(thisrefersofcoursetohypo$
$(b,a)(a,b)=1, butthatforanyotherpairsb,a,orforanypairswhateverb,bora,a,thec$
infact,thattheconstantsb,...anda,...formacanonicalsystemofelements.Theorem5
viz.,ifq,...p,...areany2/ivariablesconcerningwhichnosuppositionismade,exceptth
bp,...),< f > (q,...whichequationsareonlysubjecttotheconditionofbeingsufficient
(bitbj)maybesatisfiedidentically.Theorem7is,infact,thetheorempreviouslyestabl
0subsistidentically,theremainingnintegralscanbefoundasfollows :
Bymeansofthenintegrals,letthenvariables

H p, ... be expressed in terms of se, ... b, ... and t, and let stand for what H, as oriHginally given, becomes when q, ... are thus expressed. Then the values of p, ... and are the differential coefficients of one and the same function of p, ... and t , call this function V, then, since its differential coefficients are all given (by the dV dV = - V equations -y =p, ... , H, may be found by integration and it is therefore ; to be considered as a given function of p, ... and t and of the constants b, ... The remaining n integrals are given by the n equations dV = db where the n quantities a, ... are new arbitrary constants. 65. Section 5 of the memoir relates to the theory of the variation of the elements considered in relation to the following very general problem: viz., Q, ... P, ... being any functions whatever of the 2n variables q, ... p,... and t; it is required to express the integrals of the system 2?? differential equations dq

pdp dtdtW>inthesameformastheintegrals(supposedgiven)ofthestandardsystemdHdHdq$_{d}$p$_{d}$tdpdtdqbysu where K is any function of 77, ... p, ... which may also contain t explicitly, then will the transformed equations be drj $_{d<}>$ dvr$_d$ $< dtd - srdtdvjinwhichisdefinedbytheequation < f >$ $dtandistobeexpressedintermsofthenewvariables, thesubstitutionofthenewdKKva$ $H, sothat, inthiscase, thetransformationHiseffectedmerelybyexpressinginterms$

1.95] three bodies and the problem of rotation the former problem is specially discussed ; in section 7 but the results obtained (and which, as the author remarks, affords an ; example of the so-called "elimination of the nodes") do not come within the plan of the present report. 66. Bours memoir of 1855, " On the Integration of the Differential Equations of Analytical Mechanics." It has been already seen that the knowledge of half the entire system of the integrals of the differential equations (these known integrals satisfying certain conditions) leads by quadratures only to the knowledge of the remaining integrals; the researches contained in this most interesting and valuable memoir show that this theorem is, in fact, only the last of a series of theorems, here first established, relating to the successive reduction which results from the knowledge of each new integral. Speaking in general terms, it may be stated that the author operates on the linear partial differential equation of the first order, which is satisfied by the integrals of the differential equations and that he effectuates upon this equation a reduction of ; two unities in the number of variables for every suitable new integral which is obtained The author shows also that an equal or greater reduction may sometimes ( ). be obtained by means of integrals which appear at first foreign to his method. Before going further, it ma H y be convenient to remark that the author restricts himself to the case in which is independent of the time, and where, consequently, the con dition of vis viva is satisfied it was, however, remarked by Liouville that the analysis, ; H slightly modified, applies to the most general case where is any function of t and the variables, and it is possible that when the entire memoir is published (it is given in Liouvilles Journal as an extract), the theory will be exhibited under this more general form. 67. To give an idea of the analytical results, the equations are considered under the form d $_d - Hd_d HdP, tid^q i* \, d - qt; i - - -$ $d^q i(^{(}/_L UU(//IIH(where, asalreadyremarked, isindependentoft).Theintegralsadn$ $ff)istheequationofvisviva; ft(=Gt)istheintegralconjugatetothis, andtheonlyinte$ $u_3 anda2n_2 areconjugatewehavepairs :==== (,,a)(=$ $(!, fO)0, (ft)(=(i,))=0, (a,,a,)1, (a,,a,)0, ...(alf._)0.Theintegralsa1; a.,,...a.JH$ $*-!-dpi1Ihaveborrowedthisandthenextsentencefromliouvillesreport.Itwould, It$

198 REPORT ON THE RECENT PROGRESS OF THEORETICAL DYNAMICS. [195 Avhich is also satisfied by while, on the contrary, the first member of the equation (1), becomes unity for in other words (H, 6r) = 1. The equation (1) replaces the original differential equations; it is to the equation (1) that the theorems of Poisson and Bertrand may be supposed to be applied, and it is this equation (1) which is studied in the memoir, where it it shown how the order may be diminished when one or more integrals are known. =H In the first place, the integral a. which is known, may be made use of to eliminate one of the variables, suppose pn the result is found to be ; dqt dpi dpi dqj dqn which has the same integrals as the equation (1), except the integral of vis viva =H; it is this equation (2) which would have to be integrated if only the integral a were known. Suppose now there is known a new integral o;thisgivesrisetothepartialdifferentialequation*!drfotjd=V01i = iI7qid7pd7pid7q=whichissatisfiedbyTheequation(4)issatisfiedbymannerastheequ H)conductstotwodifferentequations, accordingasuofnktnhoewneqiunatteigroanl);Gsiosatlhsaoteliinmintahteed.secTohnidsfsoercmondoffotrhmeistrfaonusnfa ioi.dpidpidqiJwhichispreciselysimilartotheequation(1)(onlythenumberofvariable) = ().AndthetheoremsofPoissonandBertrandapplyequallytothisequation; theonly 2arethenobtainedbyquadraturesonly, inthemethodexplainedinthememoir, andwhic

195] 68. Liouvilles note of July, 1855, on the occasion of Bours memoir, mentions that the author of the memoir had recognized that, according t H o the remark made to him, his formulae subsist with even increased elegance when is considered as a function of t and the other variables. But (it is remarked) the general case can be always reduced to the particular one considered in the memoir, provided that the number of equations is augmented by two unities by the introduction of the new variables r and u, the former of them, r, equal to t + constant, so that dt $_d$r thelatterofthem, u, definedbytheequationdHdu$_d$rdtSupposeinfactthatV = H+u, then, sincerandudonotenterintoH, whichisafunctiononlyoftandthevariables dudrand, moreover, thedifferentialcoefficientswithrespecttot, q, ...p, ...ofthefunc dVdu =$_d$ Vdt$_d$rdudrdt = =$_d$ rdqdr dpVwhichisasystemcontainingtwomorevariable suchinferencedoesnotseemtobeintended, andwould, Ithink, beawrongone.69.Brios

but if the variables in the denominator are a non-corresponding pair out of the two 2 s ? e s r) i , es th q e , n ... th a e nd ex p p , re . s .. si , on or o e n lse the a l p e a f i t r -h o a u n t d s o i f de on i e s s e e q r u i a e l s t o o nly zer (o t . hat Th i i s, s b i o s th in q f s act or a b s o o t r h t of reciprocal theorem to the theorem which defines the canonical system of integrals. There are two or three memoirs of Brioschi in Crelles Journal connected with this note and the note of 1853; but as they relate professedly to skew determinants and not to the equations of dynamics, it is not necessary here to refer to them more particularly. 70. Bertrands memoir of 1857 forms a sequel to the memoir of 1851, on the common to several problems of mechanics. The author calls to mind that he integrals has shown in the first memoir, that, given an integral of a mechanical problem, and assuming only that the forces are functions of the coordinates, it is possible to determine the problem and find the forces which act upon each point; and (he proceeds) it is important to remark, that the solution leads often to contradictory results, that, in fact, an equation assumed at hazard is not in general an integral of any problem whatever of the class under consideration: and he thereupon proposes to himself in the present memoir to develope some of the consequences of this remark, and to seek among the most simple forms, the equations which can present themselves as integrals, and the problems to which such integrals belong. The various special results obtained in the memoir are interesting and valuable. 71. In what precedes I have traced as well as I have been able the series of investigations of geometers in relation to the subject of analytical dynamics. The various theorems obtained have been in general stated with sufficient fulness to render them to mathematicians; the attempt to state them in a uniform notation intelligible and systematic order would be out of the province of the present report. The leading steps are, first, the establishment of the Lagrangian form of the equations of motion; secondly, Lagranges theory of the variation of the arbitrary constants, a theory perfectly complete in itself; and it would not have been easy to see a priori that it would be less fruitful in results than the theory of Poisson ; thirdly, Poissons theory of the variation of the arbitrary constants, and the method of integration thereby afforded; fourthly, Sir W. R Hamiltons representation of the integral equations by means of a single characteristic function determinable a posteriori by means of the integral equations assumed to be known, or by the conditi R on of its simultaneous satisfaction of two partial differential equations; fifthly, Sir W. Hamiltons form of the equations of motion; sixthly, Jacobis reduction of the integration of the differential equations to the problem of finding a complete integral of a single partial differential equation, and the general theory of the connexion of the integration of a system of ordinary differential equations, and of a partial differential equation of the first order, a theory, however, of which Jacobi can only be considered as the second founder; seventhly, the notion (arising from the researches of Lagrange and Poisson) and ulterior development of the theory of a system of canonical integrals. I remark in conclusion, that the differential equations of dynamics (including in the expression, as I have done throughout the report, the generalized Lagrangian and

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Transcribed from the Cambridge University Press edition

195. Report on the Recent Progress of Theoretical Dynamics (concluded)

(Conclusion from previous pages.)

I remark in conclusion, that the differential equations of dynamics (including in the expression, as I have done throughout the report, the generalized Lagrangian and Hamiltonian forms) constitute really one of the chief classes of differential equations occupying the attention of geometers. The greater part of what has been done with respect to the general theory of a system of differential equations is due to Jacobi, and he has also considered in particular, besides the differential equations of dynamics, the Pfaffian system of differential equations (including therein the system of differential equations which arise from any partial differential equation of the first order), and the so-called isoperimetric system of differential equations, that is, the system arising from any problem in the calculus of variations. In a systematic treatise it would be proper to commence with the general theory of a system of differential equations, and as a branch of this general theory, to consider the generalized Hamiltonian system; and in relation thereto to develope the various theorems which have a dynamical application. It would be shewn that the generalized Lagrangian form would be transformed into the Hamiltonian form, but the first-mentioned form would, I think, properly be treated as a particular case of the isoperimetric system of differential equations.

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[*See* [298] *at the end of papers (Vol. IV.).*]

2, Stone Buildings, March 28, 1856.

196. Note sur un Problème d'Analyse Indéterminée

[From the *Nouvelles Annales de Mathématiques* (Terquem and Gerono), t. XVI. (1857), pp. 163–165.]

EULERA a donné dans le Mémoire: *Regula facilis problemata Diophantea per numeros integros expedite solvendi* (*Commentt. Arith. Coll.*, t. II., p. 263) la solution générale d'un cas de l'équation indéterminée

$$ax^2 + 2bx + cy + 2y = ay^2 + 2gy + d,$$

en supposant que l'on ait la solution $x = a$, $y = b$ du nombre que

$$aa^2 + 2ba + cy + 2 = ab^2 + 2gb + d,$$

et en posant

$$a^2 = ac + 1,$$

où c est une quantité quelconque, l'équation sera satisfaite par les valeurs

$$x = aa + \frac{ab}{2a}(a-1)\frac{f}{a} + (a-1)g,$$
$$y = -bn + a + \frac{(a-1)b}{2a} + \frac{a-1}{a}(fd).$$

En effet, en voit sans peine que ces valeurs donnent

$$ax^2 + 2bx + cy + 2 = a(a^2 + acb + ed)(ay^2 + 2gy + d) = 0.$$

En remplaçant à la place des coefficients a, β, γ, ξ, θ autant des nombres entiers tels que a_1^2 soit un entier positif non carré, on obtient une méthode pour trouver ente r des nombres qui s'il existe et le coefficient sont positifs ou négatifs; et si l'équation $a^2 = aN + 1$ est satisfaite par (a, a, c, q) et (a', a, c, q'), l'équation

$$ax^2 + \beta x + \gamma + \xi a' + xy + \rho yx + sxy + d, \quad (1')$$

est réduite à la forme connue.

L'équation indéterminée (2) existe dans celle-ci

$$(a,b,c,f,g,h)x,y_1^2 = 1, \quad \frac{(a,b,c,f,g,h)x,y,1) = 0}{0}, \quad (2)$$

en supposant que la forme quadratique

$$(a,b,c,f,g,h)x,y,z)^2$$

est l'équation indéterminée (2). En effet en supposant que la forme quadratique

$$(a,b,c,f,g,h)x,y,z)^2 = 0$$

se transforme en elle-même au moyen d'une substitution linéaire quelconque, on peut supposer que cette substitution soit telle que l'on ait $x' = x = 1$ et au moment du plus $b = 0$. L'équation (3) se réduit alors à montrer que $x' = x$ aurait *de* plus $b = 0$. L'équation (3) ne peut trouver par le méthode précédente. M. Hermite a donné cette méthode dans l'ordinaire,

$$\mathfrak{A} = ba - f\gamma, \quad \mathfrak{B} = pb - h\gamma,$$

$$\mathcal{B} = aa - hg^2, \quad \mathfrak{H} = ga - h\beta + h\beta.$$

Il faut pour cela écrire

$$ax = \mathfrak{A}^2, \quad ay = \mathfrak{B}a - p,$$

où

$$\mathfrak{B} = \mathfrak{A}^2 + \mathfrak{B}^2 + \mathfrak{H}^2.$$

ces équations sont identiquement satisfaites par

$$(a,b,c,f,g,h)x,y,z)^2, \quad (\mathfrak{B},b,c,f,g,h)x,y,z)^2,$$

et un moyen de cette équation et de ces valeurs

$$x' = \mathfrak{B} - \mathfrak{A}, \quad y' = \mathfrak{B}a + p,$$

on forme tout de suite équation et les valeurs

$$x = \mathfrak{B}\mathfrak{A} - p\mathfrak{B}, \quad y = \mathfrak{B}\mathfrak{A} - p\mathfrak{B} - p,$$

en forme telles que sur les trois équations

$pa = 0, \quad \mathfrak{B} = 0$ en ajoutant, on obtient $\mathfrak{A} = h\mathfrak{B},$ ce qui donne $x' = a$ et de là x'

Cela étant, les deux équations donnent, en remplaçant $\mathfrak{B}$ par sa valeur,

$$\alpha\mathfrak{A}^2 + (h - g\mathfrak{Y}) + a + g\mathfrak{B}^2 + a\mathfrak{B} = -\beta,$$

$$(h + g\mathfrak{Y})^2 - \alpha\mathfrak{B}^2 + a(g - g\mathfrak{X})^2 = -\beta,$$

et en remettant pour $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{Y}$, $\mathfrak{X}$ leurs valeurs

(page continues with formulas 4, 5, 6, 7 and conclusion)

London, 10 Mars, 1857.

197. Note on the Theory of Logarithms

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 275–280.]

An imaginary quantity $x + yi$ may always be expressed in the form

$$x + yi = r(\cos\theta + i\sin\theta) = re^{i\theta},$$

where r is positive, and θ is included between the limits $-\pi$ and $+\pi$. We have in fact

$$r = \sqrt{x^2 + y^2},$$

and when x is positive,

$$\theta = \tan^{-1}\frac{y}{x},$$

but when x is negative,

$$\theta = \tan^{-1}\frac{y}{x} \pm \pi,$$

where $\tan^{-1}$ denotes an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$, and where the upper or under sign is to be employed according as y is positive or negative. I use for convenience the mark $\pm$ to denote identity of sign; we may then write

$$\theta = \tan^{-1}\frac{y}{x} \pm \epsilon\pi,$$

where

$$x = +, \quad \epsilon = 0; \qquad x = -, \quad \epsilon = \pm 1 \pm y.$$

It should be remarked that θ has a unique value except in the single case $x = -$, $y = 0$, where θ is indeterminately $\pm\pi$. We have, in fact, $\theta = +\pi$ or $\theta = -\pi$ according as it is considered to the limit of $x + yi$, or $x + yi$, y being positive or negative, in such an arc is the case so $r + yi$ is to be regarded.

The preceding definition is, in fact, in the case of a positive, that given by M. Cauchy in the *Exercices de Mathématique*, vol. i. [1826]; and he has there shewn that x, x', $xx' - yy'$ being all of them positive, the above-mentioned expression reduces itself to zero. The general definition is that given to my *Mémoire sur quelques Formules du Calcul Intégral*, Liouville, vol. XII. [1847], p. 231 [40], but I was wrong in assuming that the expression always reduced itself to zero. We have, in fact, in general

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a+\beta}{1-a\beta},$$

when $1-a\beta$ is positive; but when $1-a\beta$ is negative (which implies that a, β have the same sign), then

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a+\beta}{1-a\beta} \pm \pi,$$

where the upper or under sign is to be employed according as a and β are positive or negative; or what is the same thing,

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a+\beta}{1-a\beta} \pm \epsilon\pi,$$

where

$$1-a\beta = +, \quad \epsilon = 0; \qquad 1-a\beta = -, \quad \epsilon = \pm 1 \pm a = \pm 1 \pm \beta.$$

This being premised, then writing

$$\begin{aligned}
\log(x+yi) &= \log\sqrt{x^2+y^2} + i\left(\tan^{-1}\frac{y}{x} \pm \epsilon\pi\right), \\
\log(x'+y'i) &= \log\sqrt{x'^2+y'^2} + i\left(\tan^{-1}\frac{y'}{x'} \pm \epsilon'\pi\right), \\
\log(x+yi)(x'+y'i) &= \log[(xx'-yy') + i(xy'+x'y)] \\
&= \log\sqrt{(xx'-yy')^2+(xy'+x'y)^2} + i\left(\tan^{-1}\frac{xy'+x'y}{xx'-yy'} \pm \epsilon''\pi\right)
\end{aligned}$$

we find

$$\log(x+yi) + \log(x'+y'i) - \log(x+yi)(x'+y'i) = (\epsilon+\epsilon'-\epsilon'')\pi i.$$

Hence, considering the different cases:

I. $x = +,\ x' = +,\ xx'-yy' = +,$

$$\begin{aligned}
\epsilon &= 0, \\
\epsilon' &= 0, \\
\epsilon'' &= 0,
\end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

II. $x = +,\ x' = +,\ xx' - yy' = -,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= 0, \\ \epsilon'' &= \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = \pm 1 \qquad \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

III. $x = +,\ x' = -,\ xx' - yy' = +,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= \pm 1 \pm y' = -\frac{y'}{x'} = -\frac{y}{x'}, \\ \epsilon'' &= 0, \\ \epsilon'' &= \pm 1 \qquad \equiv \left(-\frac{y'}{x'}\right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

IV. $x = +,\ x' = -,\ xx' - yy' = -,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= \pm 1 \pm y' = -\frac{y'}{x'}, \\ \epsilon'' &= \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \qquad \text{if } \frac{y'}{x'} = -\frac{y}{x'},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \qquad \text{if } \frac{y'}{x'} = -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

V. $x = -,\ x' = +,\ xx' - yy' = +,$

$$\begin{aligned} \epsilon &= \pm 1 \pm y = -\frac{y}{x} = -\left(\frac{y}{x}\right), \\ \epsilon' &= 0, \\ \epsilon'' &= 0, \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

VI. $x = -,\ x' = +,\ xx' - yy' = -,$

$$\begin{aligned} \epsilon &= \pm 1 \pm y = -\frac{y}{x}, \\ \epsilon' &= 0, \\ \epsilon'' &= \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \qquad \text{if } \frac{y}{x} = -\frac{y}{x},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \qquad \text{if } \frac{y}{x} = -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

VII. $x = -,\ x' = -,\ xx' - yy' = +,$

$$\begin{aligned} \epsilon &= \pm 1 \pm y = -\frac{y}{x}, \\ \epsilon' &= \pm 1 \pm y' = -\frac{y'}{x'}, \\ \epsilon'' &= 0, \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \qquad \text{if } \frac{y}{x} = -\frac{y'}{x'},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \qquad \text{if } \frac{y}{x} = \frac{y'}{x'}.$$

VIII. $x = -,\ x' = -,\ xx' - yy' = -,$

$$\begin{aligned} \epsilon &= \pm 1 \pm y = -\frac{y}{x}, \\ \epsilon' &= \pm 1 \pm y' = -\frac{y'}{x'}, \\ \epsilon'' &= \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = \pm 1 \qquad \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

Hence writing

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = E\pi i,$$

we have $E = 0$, except in the following cases, viz.

1. (See IV.) $x = +,\ x' = -,\ xx' - yy' = -,\quad \frac{y'}{x'} = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$ where

$$E = \pm 2 \pm \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

2. (See VI.) $x = -,\ x' = +,\ xx' - yy' = -,\quad \frac{y}{x} = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$ where

$$E = \pm 2 \pm \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

3. (See VII.) $x = -,\ x' = -,\ xx' - yy' = +,\quad \frac{y}{x} = \frac{y'}{x'},$ where

$$E = \pm 2 \pm \frac{y}{x}.$$

4. (See VIII.) $x = -,\ x' = -,\ xx' - yy' = -,$ where

$$E = \pm 2 \mp \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

It thus appears that when any one of the arguments $x+yi$, $x'+y'i$ are real and negative, or in the imaginary cases when the product of the proper expression for the arguments must be positive but their sums must be negative, then the equation

$$\log(x + yi) + \log(x' + y'i) = \log(x + yi)(x' + y'i)$$

ceases to hold in any case.

The preceding results do not apply to the case where any one of the arguments $x + yi$, $x' + y'i$, $(x + yi)(x' + y'i)$ is real and negative, in which case the formula $\log z = +\pi i$ being still negative, has in addition to its arbitrary positive or negative value $\pm\pi i$, what is the same thing the logarithm is absolutely indeterminate. The general methods are illustrated by applications to the problem of three points of rotation; the former problem is specially discussed in section 7., but the results obtained (and which, to the author remarks, afford an example of the so-called "elimination of the nodes") do not come within the plan of the present report.

2, Stone Buildings, March 18, 1856.

198. Note upon a Result of Elimination

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 378–379.]

If the quadratic function

$$(a,\ b,\ c,\ f,\ g,\ h)(x,\ y,\ z)^2$$

break up into factors, then representing one of these factors by $\xi x + \eta y + \zeta z$, and taking any arbitrary quantities a, β, γ, the factor in question, and therefore the quadratic function is reduced to zero by substituting $\beta\zeta - a\eta$, $\gamma\xi - a\zeta$, $a\eta - \beta\xi$ in the place of the ideas, a more general; the function then results of the operation of performed on U, and QPU denotes the result of the operation Q performed on PU, and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the operation QPU it is easy to see that we may write

$$QPU = (Q + P)U + (QP)U,$$

where on the right-hand side $(Q+P)$ and (QP) signify as follows: viz. $Q + P$ denotes the new algebraical product of Q and P, while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q + P)U$ and $(QP)U$ denote respectively the results of the operations $(Q + P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q + P)$ and $(P + Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expression for $RQPU$, &c. This, however, has at one effected by means of the analytical forms called *trees* (see figs. 1, 2, 3), which contain all the trees which can be formed with one branch, two branches, and three branches respectively.

[Figures of trees with single branches and combinations are described in the text.]

The coefficients a, b, c, d are linear functions of coordinates, the equation

$$aa - b^2 = 0, \quad bd - cd = 0, \quad bd - c^2 = 0$$

(equivalent, of course, to two equations) belong to a curve of the third order in space; the edge of regression of the developable surface obtained by putting the discriminant equal to zero, or which has for its apolar

$$-x^3d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = 0,$$

and, moreover, the above forms are the general representations of any curve of the third order, and developable surface of the fourth order. The question arises, to find the equation of the cone of the third order having an arbitrary point for its vertex, and passing through the curve of the third order. This may be done by Joachimsthal's method (vide. p. 21, q., f. be the values of a, b, c, d at the vertex of the cone, and is the equations of the cone, say in

$$aa - b^2 = 0, \quad bd - c^2 = 0,$$

for a, b, c, d write $aa + ka$, $ab + kb$, $ac + kc$, $ad + kd$; if then the equations so obtained in k are eliminated, the resulting equation will be that of the cone of the third order.

2, Stone Buildings, May 1, 1856.

199. Note on the Theory of Elliptic Motion

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 425–428.]

If, as usual, r, $\dot{r}$ denote the radius vector and longitude, and μ the central mass, then the Vis Viva and Force function are respectively

$$T = \tfrac{1}{2}(\dot{r}^2 + r^2\dot{\theta}^2),$$

$$U = \frac{\mu}{r},$$

and writing

$$\frac{dT}{d\dot{r}} = p, \qquad r^2\dot{\theta} = p,$$

$$\frac{dT}{d\dot{\theta}} = r^2\dot{\theta} = q,$$

we have $\dot{r}^2 = p$, $\dot{\theta}^2 = q$, and $T = \tfrac{1}{2}\left(p^2 + \dfrac{q^2}{r^2}\right)$, whence, putting $H = T - U$, the value of H is

$$H = \tfrac{1}{2}\left(p^2 + \frac{q^2}{r^2}\right) - \frac{\mu}{r};$$

and by Sir W. R. Hamilton's theory, the equations of motion are

$$\frac{dr}{dt} = \frac{dH}{dp}, \quad \frac{d\theta}{dt} = \frac{dH}{dq},$$

$$\frac{dp}{dt} = -\frac{dH}{dr}, \quad \frac{dq}{dt} = -\frac{dH}{d\theta},$$

or substituting for H its value, the equations of motion are

$$\frac{dr}{dt} = p,$$

$$\frac{d\theta}{dt} = \frac{q}{r^2},$$

$$\frac{dp}{dt} = \frac{q^2}{r^3} - \frac{\mu}{r^2},$$

$$\frac{dq}{dt} = 0.$$

Putting, as usual, $p = a^2(1 - e^2)$, and introducing the eccentric anomaly u, which is given as a function of t by means of the equation

$$nt + \xi = u - e\sin u,$$

$\left(\text{so that } \dfrac{du}{dt} = \dfrac{n}{1 - e\cos u}\right)$, the integral equations are

$$q = na^2\sqrt{1 - e^2},$$

$$p = \frac{ena\sin u}{1 - e\cos u},$$

$$r = a(1 - e\cos u),$$

$$\theta = \varpi + \tan^{-1}\left(\frac{\sqrt{1 - e^2}\sin u}{\cos u - e}\right),$$

where the constants of integration a, e, ϖ denote as usual the mean distance, the eccentricity, the mean anomaly at epoch, and the longitude of perisaster.

Suppose that q_0, p_0, r_0, θ_0, u_0 correspond to the time t_0 (q is constant, so that $q_0 = q$), and write

$$V = nt'(u - u_0 + e\sin u - e\sin u_0),$$

joining to this the equations

$$r = a(1 - e\cos u), \quad r_0 = a(1 - e\cos u_0),$$

$$\theta - \varpi = \tan^{-1}\left(\frac{\sqrt{1 - e^2}\sin u}{\cos u - e}\right) - \tan^{-1}\left(\frac{\sqrt{1 - e^2}\sin u_0}{\cos u_0 - e}\right),$$

u, u_0, θ will be functions of a, e, r, r_0, θ, and consequently ξ being throughout considered as a function of a, e, ξ, H, in this view, the characteristic function V of Sir W. R. Hamilton, and according to his theory we ought to have

$$dV = \tfrac{1}{2}n(u - u_0)da + p\,dr - p_0\,dr_0 + q\,d\theta - q_0\,d\theta_0.$$

To verify this, I form the equation

$$\begin{aligned} dV &= [nu(u - u_0 + e\sin u - e\sin u_0)\,da \\ &\quad + nt'\{(1 - e\cos u)\,du - (1 - e\cos u_0)\,du_0 - \sin u\,de + \sin u_0\,de\}] \\ &= \frac{n\sin u}{1 - e\cos u}\,[de - (1 - e\cos u)\,da - r\sin u\,du + a\sin u_0\,de] \\ &\qquad + na\sqrt{1-e^2}\left[d\theta - \frac{1}{\sqrt{1-e^2}(1 - e\cos u)}((1-e^2)\,du + ae\sin u\,de)\right. \\ &\qquad \left. - dH + \frac{1}{\sqrt{1-e^2}(1 - e\cos u_0)}((1-e^2)\,du_0 + ae\sin u_0\,de)\right], \end{aligned}$$

the coefficient of da on the right-hand side is

$$nu'(1 + e\cos u) - \frac{nut'\sin^2 u}{1 - e\cos u} - \frac{ane^2(1-e^2)}{1 - e\cos u},$$

which vanishes, and similarly the coefficient of du, also vanishes; the coefficient of de is the difference of two parts, the first of which is

$$\frac{ne\sin u}{1 - e\cos u}\,de - na\sin u\,e^2 du,$$

and the second is the form of the form u, by virtue of which the coefficient therefore is

$$\frac{1}{2}n(u - u_0 + e\sin u - e\sin u_0).$$

We have therefore

$$\begin{gathered} dV = \tfrac{1}{2}n(u - u_0 + e\sin u - e\sin u_0)\,da \\ + \frac{ne\sin u}{1 - e\cos u}\,dr - na\sqrt{1-e^2}\,d\theta, \\ - \frac{ne\sin u_0}{1 - e\cos u_0}\,dr_0 - na\sqrt{1-e^2}\,d\theta_0, \end{gathered}$$

or what is the same thing,

$$dV = \tfrac{1}{2}n(u - u_0)\,da + p\,dr - p\,d\theta\,dr_0 - q\,d\theta_0 - q\,d\theta_0,$$

the equation which was to be verified.

2, Stone Buildings, March 26, 1856.

200. On the Cones which pass through a given curve of the third order in space

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 20–22.]

THE following investigation is connected with the theory of the cubic

$$(a, b, c, d)(x, y)^3,$$

and in particular with a theorem that the determinant formed with the second differential coefficients of the discriminant gives the square of the discriminant.

Consider the coefficients a, b, c, d as linear functions of coordinates, the equations

$$ac - b^2 = 0, \quad bd - cd = 0, \quad bd - c^2 = 0$$

(equivalent, of course, to two equations) belong to a curve of the third order in space; the edge of regression of the developable surface obtained by putting the discriminant equal to zero, or which has for its apolar

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = 0,$$

and moreover, the above forms are the general representations of any curve of the third order, and developable surface of the fourth order. The question arises, to find the equation of the cone of the third order having an arbitrary point for its vertex, and passing through the curve of the third order. This may be done by Joachimsthal's method: let a, b, c, d be the values of a, b, c, d at the vertex of the cone, and in the equations of the cone, viz.

$$ac - b^2 = 0, \quad ad - bc = 0, \quad bd - c^2 = 0,$$

for a, b, c, d write $a + ka$, $b + kb$, $c + kc$, $d + kd$; if then the equations so obtained in k are eliminated, the resulting equation will be that of the cone of the third order.

The substitution in question gives

$$La + Ma = R + S'a + 0,$$

$$La' + 2Ma = M'a + 0,$$

where

$$\begin{aligned} L &= \tfrac{1}{2}(ac-b^2), & M &= -ac+bd-2hd, & N &= \tfrac{1}{2}(by-8'),\\ L &= \tfrac{1}{2}(ad-c), & M &= bd-bd-cd, & N &= \tfrac{1}{2}(ed-dc),\\ L'' &= \tfrac{1}{2}3M''M+2g', & M'' &= bd-bd+cd-3M'', & X'' &= \tfrac{1}{2}(cd-bb); \end{aligned}$$

the result of the elimination is

$$4(LX'-M)(LX''-2M'')-(LX''+LX'-2MM'')^2=0,$$

and we have

$$LX-M^2=4(a-b)(c-bd)-(4-bc)^2=ac+3b'(a'-b'),$$

$$LX''-M''=4(a-d)(bd-ce)-(ad-bc)^2=(ad-bc)^2,$$

$$LX''+LX'-2MM''=8(ad-bc)(bd-cd)+(4-bd)(ad-bc)+8M.M,$$

Write for shortness

$$by-cf=i,\quad cn-dy=m,\quad cd-bn=n,\quad ad-cf=w,$$

$$ad-by=w,\quad bd-cf=u,\quad sf-by=w,\quad dd-2f=y,$$

values which (if a, b, c, d are linear functions of coordinates x, y, z, w) might, u, v, w this is equal to

$$-16(P-cf-syh+day-4)bw+uw^2,$$

or, substituting and reducing,

$$P-cyh+lay=1bw-1ux+b'yw-3a'(P_x-dyx)+3y(d'a-4'8)bd,$$

$$+k(P-4yk)+3by^2+by^2+bd-(a8)k\Omega_b'^2+8y'd+4cd+8cd,$$

I do not however refer the result obtained.

Now putting

$$bd-c^2=p,\qquad bc-cd=q,\qquad ac-b^2=r,$$

and in like manner

$$8d-q'=D,\qquad By-cb=Q,\qquad ay-bg=R,$$

and reducing, the final result may be expressed in the form

$$\begin{aligned} P[\quad &= 2Dpq + (pn-2)q + p(2a-py)r] \\ +Q[\quad &-Wn + ((ph-2)r - py) \quad He] \\ +R[\quad &+ (Dy - 2(b)p - (Db - Ny) + 0] \\ &= 0, \end{aligned}$$

where a, b, c, d are current coordinates, and p, q, r are quadratic functions of a, b, c, d, the equation $P = 0$ of the second order in which the vertex (current point) of the cone (p, q, r) cone is the quadratic functions of a, b, c, d, the equation of the cone is in p, q, r, R, r are quadratic; and note VII this is equal to (suppose) is

$$F = a + ay + aav = R,$$

of the cone in question (so an arbitrary quantity), is

$$p + uq + sP'r = 0,$$

or so all length quantity

$$(bd - c^2) + (a + xc)(c - cd) + 8c^2(ac - b^2) = 0,$$

in fact this equation is evidently that of a surface of the second order passing through the curve; and there is no difficulty in shewing that it is a cone.

2, Stone Buildings, May 1, 1856.

201. Second Note on the Theory of Logarithms

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 354–363.]

THE theory of logarithms, as developed in my first note (*Phil. Mag.* April 1856), [197] may be exhibited in a clearer light by considering, instead of $\log a + \log b - \log ab = E\pi i$, the equation $\log \dfrac{x}{a} = \log b + \log a - E\pi i$, in which more readily enables the answering *à priori* for the discontinuity in the values of E. Writing then for b the complex values $x + yi$, $c + yi$, we have

$$\log \frac{x' + y'i}{x + yi} = \log(x' + y'i) - \log(x + yi) + E\pi i,$$

where, according to the assumed definition of a logarithm,

$$\log(x + yi) = \log \sqrt{x^2 + y^2} + i\left(\tan^{-1} \frac{y}{x} \pm \epsilon\pi\right),$$

in which $\tan^{-1}$ denote an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$, and $\tan^{-1} \dfrac{y}{x}$ is an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$. The coefficient ϵ is equal to zero when x is positive; but when x is negative, then $\epsilon = +1$ or -1, according as y is positive or negative, i.e. we have

$$\begin{array}{ll} x = +, & \epsilon = 0, \\ x = -, & \epsilon = \pm 1 \pm y; \end{array}$$

and of course the other logarithm in the equation have an analogous signification.

Hence, attending to the equation

$$\tan^{-1} a + \tan^{-1} \alpha = \tan^{-1} \frac{\beta + a}{1 + a\beta} \pm \epsilon''\pi,$$

where, when $1 + a\beta$ is positive, ϵ'' is equal to zero; but when $1 + a\beta$ is negative, ϵ'' is equal to $+1$ or -1, according as $\beta + a$ is positive or negative; or what is the same thing, (β being of opposite signs when $1 + a\beta$ is negative), or to

$$\begin{array}{ll} 1 + a\beta = +, & \epsilon'' = 0; \\ 1 + a\beta = -, & \epsilon'' = \pm 1 \pm a = \mp 1 \mp \beta. \end{array}$$

we find

$$E = \epsilon + \epsilon' - \epsilon'',$$

where ϵ, ϵ', ϵ'' are defined by the conditions

$$\begin{aligned}
x &= +, & \epsilon &= 0,\\
x &= -, & \epsilon &= \pm 1 \pm y,\\
x' &= +, & \epsilon' &= 0,\\
x' &= -, & \epsilon' &= \pm 1 \pm y',\\
xx' + yy' &= +, & \epsilon'' &= 0,\\
xx' + yy' &= -, & \epsilon'' &= \pm 1 \pm xy' = \mp 1 \mp x'y.
\end{aligned}$$

Suppose, to fix the ideas, that x, y are each of them positive; we have $x = 0$, and considering the several cases:

1. $x' = +,\ y' = +,$

Here $xx' + yy' = +$, $1 + \frac{y}{x}\frac{y'}{x'} = +$; and consequently not only $\epsilon' = 0$, but also $\epsilon'' = 0$, $\epsilon'' = 0$, and thence $E = 0$.

2. $x' = +,\ y' = -,$

Here $\epsilon = 0$. Moreover, xx' being positive, and $\epsilon yy'$ and $1 + \frac{y}{x}\frac{y'}{x'}$ will have the same sign. If they are both positive, $xx' + yy'$ is positive, and $1 + \frac{y}{x}\frac{y'}{x'}$ is negative. then $x' = 1 \pm xy' = +(\pm\frac{y}{x} - \frac{y'}{x'})$ (since xy' is negative) $= -\frac{y}{x}$, i.e. $\epsilon'' = +1$, and $1 + \frac{y'}{x'}\frac{y'}{x'}$ being positive, we have $\epsilon'' = 0$. Hence in each case $E = 0$.

3. $x' = -,\ y' = +,$

Here $\epsilon' = \pm 1 \pm y'$, i.e. $\epsilon = -1$. Also xx' and yy' being each negative, $xx' + yy'$ will be negative, and therefore

$$\epsilon'' = \pm 1 \pm xy' = -\left(\frac{y'}{x'}\right),$$

i.e. if $\frac{y'}{x'} > \frac{y}{x}$, then $\epsilon'' = -1$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $\epsilon'' = +1$. $1 + \frac{y}{x}\frac{y'}{x'}$ being positive, $\epsilon'' = 0$. Hence if $\frac{y'}{x'} > \frac{y}{x}$, $E - 1 - 1 = 0$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $E = 1 + 1 = 2$.

4. $x' = -,\ y' = -,$

Here $x' = \pm 1 \pm y'$, i.e. $\epsilon' = -1$. Also xx' and yy' being each negative, $xx' + yy'$ will be negative, and therefore

$$\epsilon'' = \pm 1 \pm xy' = -\left(\frac{y'}{x'}\right),$$

i.e. $\epsilon'' = -1$. Hence $1 + \frac{y}{x}\frac{y'}{x'} = -$. If $\frac{y'}{x'} > \frac{y}{x}$, $\epsilon'' = 0$. Hence in such case $E = 0$.

Consider (x, y) (x', y') as the rectangular coordinates of two points A, A'. In the case which has been considered, the point A has been taken in the positive quadrant; and the preceding discussion shews that we have always $E = 0$, except in the case in which the negative portion of the axis x. But when this happens, then if the line AA', considered as drawn from A to A', passes from above to below the axis of x, we have $E = +2$; but if the line AA', considered as drawn from A to A', passes from below to above the axis of x, then $E = -2$. So that treating the points A, A' as the geometrical representations of two values $x + yi$, $x' + y'i$, we have for the discontinuity in the value of the difference of the logarithms in passing in a continuous quadratic; and the preceding discussion shews that we have always $E = 0$, except in the case where the finite line AA' meets the negative portion of the axis of x, in which case we have $E = +2$ or $E = -2$. The same thing is true generally in whichever quadrant

A is situated, i.e. we have always $E = 0$, except in the case in which the finite line AA' meets the negative portion of the axis of x. But when this happens, then if the line AA', considered as drawn from A to A', passes from above to below the axis of x, we have $E = +2$; but if the line AA', considered as drawn from A to A', passes from below to above the axis of x, then $E = -2$.

It thus appears that when the values $x + yi$, $x' + y'i$ are exhibited by means of two points A, A' in a plane, the discontinuity in the value of E depends on the path of the line which joins the two points. So that treating the points A, A' as the geometrical representations of the values $x + yi$, $x' + y'i$, the difference of the logarithms is the result of integration $\int_A^{A'} \dfrac{dz}{z}$, the path of integration being the line from A to A'. Consider in general the definite integral

$$\int_A^{A'} du\, \phi u,$$

where ϕ, ϕ' are complex numbers of the form $x + yi$, $x' + y'i$; and take A, A' the geometrical representations of these limits, and the variable point P as the geometrical representation of the complex variable z. The value of the definite integral will depend in general not only upon the values which ϕ, ϕ' assume as constants, but also upon the path which the variable z supposes to be successively assumed in passing from A to A'.

To order therefore to give a precise signification to the notation, we must for the path of the point P, and it is natural to assume that the path is a right line (of course there exist also an infinity of paths which give the same value to the definite integral, or as we may call them, paths equivalent to the right line); but the consideration of these would be a tedious complication of the definition, and it is better to attend to the single path—the right line). The definition in an exact sense executed into an analytical mode of computing the negative part of the axis of x in which case we have $E = +2$ or $E = -2$ (according as the line goes from above to below or below to above the axis), then there may be considered as a more general spherical mass having a double natural connection of integration without limits. (For which is in fact identical with that between ϕ and ϕ', and in fact the equation between θ, θ' is

$$\tan\theta \tan\phi = \frac{1}{R},$$

which, if $\phi = \cos a$, gives $\theta = \sin(R - r)$.

Richelot has shewn, by precisely similar reasoning, that for circles of the sphere we have

$$\frac{d\phi}{\sqrt{\cos^2 r - (\cos R \sin u + \sin R \cos u \cos 2\phi)^2}} = \frac{d\phi'}{\sqrt{\cos^2 r - (\cos R \sin u + \sin R \cos u \cos 2\phi')^2}}$$

which is of the form

$$\frac{d\phi}{\sqrt{1 - (k + \mu\cos 2\phi)^2}} = \frac{d\phi'}{\sqrt{1 - (k + \mu\cos 2\phi')^2}},$$

where

$$k = \frac{\cos R \sin u}{\cos r}, \qquad \mu = \frac{\sin R \cos u}{\cos r}.$$

2, Stone Buildings, September 13, 1856.

202. Supplementary Remarks on the Porism of the In-and-Circumscribed Triangle

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 19–30.]

In my former papers (see *Phil. Mag.* August and November, 1853, [115, 116]) I established (as part of a more general one) the following theorem, viz. the condition that there may be inscribed in the conic $U = 0$ an infinity of triangles circumscribed about the conic $V = 0$, is that if we develope in ascending powers of k the square root of the discriminant of $kU + V$, the coefficient of k in this development must vanish. Thus writing

$$\text{disc.}(kV + U) = (K, \mathfrak{R}, \mathfrak{R}', K')(k, 1)^3,$$

the condition in question is found to be

$$36\mathfrak{R}^2 - 4KW = 0.$$

The following investigations, although relating only to particular cases of the theorem, are, I think, not without interest. If the equation of the conic containing the angles is

$$U = \xi ayx + \mathfrak{B}cx + \mathfrak{C}ay = 0,$$

and the equation of the conic touched by the sides is

$$V = a^2x^2 + b^2y^2 + c^2z^2 - 2bcxz - 2cayx - 2abxy = 0,$$

that is,

$$K = -A, \qquad \mathfrak{R} = -\tfrac{1}{3}(a+b+c), \qquad \mathfrak{R}' = -\tfrac{1}{3}(a+b+c)^2, \qquad K' = \text{disc.}.$$

[*The detailed investigation extends through the analysis of porism conditions for the in-and-circumscribed triangle, involving determinants, conics, and the formulae of Richelot for the sphere.*]

By comparing with the corresponding formula *in plano*, we arrive at Richelot's conclusion, that the formulæ for the sphere may be deduced from those *in plano* by writing in the place of $\frac{R}{r}$, $\frac{a}{r}$, the functions $\frac{\tan R}{\tan r}$, $\frac{\sin a}{\cos R \sin r}$, respectively.

2, Stone Buildings, October 1, 1856.

203. On the Theory of the Analytical Forms called Trees

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 172–176.]

A SYMBOL such as $AU_0 + BU_1 + \ldots$, where A, B, &c. contain the variables x, y, &c. in respect to which the differentiations are to be performed, partakes of the natures of an operand and operator, and may be therefore called an Operandator. Let P, Q, R, &c. be any operandators, and let U be a symbol of the same kind, or to the ideas a mere operand; PU denotes the result of the operation of P performed on U, and QPU denotes the result of the operation Q performed on PU; and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the operation QPU, it is easy to see that we may write

$$QPU = (Q + P)U + (QP)U,$$

where on the right-hand side $(Q+P)$ and (QP) signify as follows: viz. $Q + P$ denotes the mere algebraical product of Q and P, while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q + P)U$ and $(QP)U$ denote respectively the results of the operations $(Q + P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q + P)$ and $(P + Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expression for $RQPU$, &c. This, however, has been effected by means of the analytical forms called *trees*.

The inspection of these figures will at once shew what is meant by the term in question, and by the terms root, branches (which may be either main branches, intermediate, or free branches), and knots (which may be either the root itself, or proper knots, or the extremities of the free branches). To apply this to the question in hand, PU consists of a single term, represented by fig. 1 (*bis*); QPU consists, as we have seen, of two terms represented by the two parts of fig. 2 (*bis*); the first part represents the term $(Q + P)U$, and the second part represents the term $(QP)U$. And it is obvious then fig. 2 (*bis*) is at once formed from the figure 1 (*bis*) by adding on a branch terminated by Q at each of the knots: the upper part of fig. 1 (*bis*). In like manner $RQPU$ consists of six terms represented by the six parts of fig. 3 (*bis*), and this figure is at once formed from fig. 2 (*bis*) by adding on a branch terminated by R at each of the knots of each part of fig. 2 (*bis*). It is hardly necessary to remark that the first part of fig. 3 (*bis*) denotes what, in the notation first explained, would be denoted by $(R+Q+P)U$, the second term what would be like manner be denoted by $(RQ+P)U$, and so on; the last part being the term which would be denoted by $((RQ)P)U$, viz. the operation upon Q, giving the operandator RQ, which operates upon P, giving the operandator $(RQ)P$, which finally operates upon U.

The figures 1 (*bis*), 2 (*bis*), 3 (*bis*), contain the same trees as are contained in the corresponding figures 1, 2, 3, &c.; only, on account of the different modes of filling up, trees are considered as so many distinct trees in a figure of the second set which correspond to one and the same tree in the corresponding figure of the first set. A difference in the number of trees first occurs in the figures 3 and 3 (*bis*), the number of trees in figure 3 being 4, while the number of trees in figure 3 (*bis*) is 9. Again, on account of similarity, only the figures of the first set are considered as truly distinct from each other; thus, in fig. 3 the first tree is fill up in two way, viz. the third tree of fig. 3 (*bis*); in like manner U at the root and R, Q, P at the other knots in any one many ways; but the trees so obtained are considered as one and the same tree, for, then we must only fill up the first tree of fig. 3 (*bis*) in the order of fig. 3 (which figure for shortness in considered as one of fig. 3 (*bis*)). Again, in fig. 3, the second tree contains the latter contains six trees, viz. the first tree, the second, third and fourth tree, the fifth tree and the sixth tree of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3, &c. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U, R, Q, P in all the ways possible: thus the four orders of fig. 3 of with U, R, Q, P, may be filled up in one way; the second of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3, &c. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U, R, Q, P in all the ways possible: the only restriction is that the root to be filled in every way, but the trees so obtained are considered as one and the same tree, and there may only be no restriction in either form indifferently. The number of trees in the resulting fig. 3 (*bis*) which may then be filled in any direct manner are those in the annexed figures, and the number is therefore two. It is not difficult to see that we have in this case (R_r being the number of such trees with r free branches),

$$(1-x)^{-1}(1-x^2)^{-R_1}(1-x^3)^{-R_2}\cdots = 1+x+2R_2x^2+2R_3x^3+2R_4x^4+\&\text{c.};$$

To obtain the law for the number A_n of the trees with n branches, we consider how the trees with n branches can be formed by means of those of a smaller number of branches. A tree with n branches has either a single main branch, or else two main branches, three main branches, etc. to n main branches. If the tree has one main branch, it can only be formed by adding on to this main branch a tree with $(n-1)$ branches, i.e. A_n contains a part A_{n-1}. If the tree has two main branches, then $p+q$ being a partition of $n-2$, the tree can be formed by adding on to one main branch a tree of p branches, and to the other main branch a tree of q branches; the number of trees so obtained is A_pA_q; this, however, assumes that the parts p and q are unequal; if they are equal, it is easy to see that the number of trees is only $\frac{1}{2}A_p(A_p+1)$. Hence $p+q$ being any partition of $n-2$, A_n contains the part A_pA_q, if p and q are unequal, and the part $\frac{1}{2}A_p(A_p+1)$ if p and q are equal. In like manner, considering the trees with three main branches, then if $p+q+r$ is any partition of $n-3$, A_n contains the part $A_pA_qA_r$ if p, q, r are unequal; the part $\frac{1}{2}A_p(A_p+1)A_r$ if p and q are equal, and so on, until lastly we have a single tree with n main branches, or A_n contains the part A_1. A little consideration will shew that the preceding rule for the formation of the number A_n is completely expressed by the equation

$$(1-x)^{-A_1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}\cdots = 1+A_1x+A_2x^2+A_3x^3+A_4x^4+\&\text{c.},$$

and consequently that we may, by means of this equation, calculate successively for the different values of the number A_n of the trees with n branches. The calculation may be effected very easily as follows: [the table as originally printed contained at the end of it some errors of calculation which were corrected, B.A. Report for 1875, p. 258].

$A_1 =$	(1)	1	1	1	1	1	1	1	1	1	1	$(1 -$
$A_2 =$	1	(1)	2	2	3	3	4	4	5	5	6	$(1 -$
$A_3 =$	1	1	(2)	4	5	8	10	14	18	24	30	$(1 -$
$A_4 =$	1	1	2	(4)	6	10	14	22	30	44	60	$(1 -$
$A_5 =$	1	1	2	3	(9)	14	25	41	64	96	144	$(1 -$
$A_6 =$	1	1	2	3	5	(20)	29	56	95	153	247	$(1 - x$
$A_7 =$	1	1	2	3	5	9	(48)	75	142	247	410	$(1 - x$
$A_8 =$	1	1	2	3	5	9	20	(115)	181	348	624	$(1 - x$
$A_9 =$	1	1	2	3	5	9	20	48	(286)	460	884	$(1 - x$
$A_{10} =$	1	1	2	3	5	9	20	48	115	(719)	1842	$(1 - x$
$A_n =$	1	1	2	3	5	9	20	48	115	286	719	
for $n =$	1	2	3	4	5	6	7	8	9	10	11	

I have had occasion, for another purpose, to consider the question of finding the number of trees with a given number of free branches, bifurcations at least. Thus, when the number of free branches is three, the trees of the form in question are those in the annexed figure, and the number is therefore two. It is not difficult to see that we have in this case (R_r being the number of such trees with r free branches),

$$(1-x)^{-1}(1-x^2)^{-R_2}(1-x^3)^{-R_3}\cdots = 1+x+2R_2x^2+2R_3x^3+2R_4x^4+\&c.;$$

and a like process of development gives:

$R_n =$	1	2	5	12	33	90
for $n =$	2	3	4	5	6	7

I may mention, in conclusion, that I was led to the consideration of the foregoing theory of trees by Professor Sylvester's researches on the change of the independent variables in the differential calculus.

2, Stone Buildings, January 2, 1856.

204. On a Problem in the Partition of Numbers

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 245–248.]

It is required to find the number of partitions into a given number of parts, such that the first part is unity, and that no part is greater than twice the preceding part.

Commencing to form the partitions in question, those are

1 1 1 1 1 1 1 1 1 &c.;
1 2; 1 1 2; 1 1 1 2; 1 1 1 1 2; &c.;
1 1 2 3; 1 1 2 4; &c.;

and if we were to proceed to the 4-partitions, each 3-partition ending in 1 would give rise to two such partitions; each 3-partition ending in 2 to four such partitions; each 3-partition ending in 3 to six partitions; each 3-partition ending in 4 to eight such partitions. We form in this manner the Table:

Number of partitions	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Totals
1-partitions	1														1
2-partitions		1	1												2
3-partitions			1	2	1	1									5
4-partitions				1	2	3	3	3	2	2	1	1			18
5-partitions					1	2	3	5	6	8	10	11	10	11	100

and we are thus led to the series

1, 2, 5, 18, &c.,

where, considering it as the first term of each series, the first differences of any series are the terms twice repeated of the next preceding series: thus the differences of the fourth series are 1, 1, 2, 3, 4, 6, 6. It is moreover clear that the first half of each series is precisely the series which immediately precedes it; we need, in fact, only consider a single infinite series 1, 2, 5, 6, &c. It is to be remarked, moreover, that in the column of totals, the total of any term is precisely the first number in the next succeeding line.

Consider in general a series A, B, C, D, E, &c., and a series A', B', C', D', E', &c. derived from it as follows:

$$\begin{aligned}
A' &= 2A,\\
B' &= 2B,\\
C' &= 2A + 2B,\\
D' &= 2A + 2B + 2C,\\
E' &= 2A + 2B + 2C + 2D,\\
F' &= 2A + 2B + 2C + 2D + 2E,\\
&\text{\&c.};
\end{aligned}$$

viz. the first differences of the series 0, A', B', C', D', E', &c. are A, A, B, B, C, C, D, D, &c. Multiplying by 1, a, a^2, &c. and adding, we have

$$A' + B'a + C'a^2 + D'a^3 + E'a^4 + \text{\&c.} = 2(A + Ba + Ca^2 + \text{\&c.})$$

and if we form in a similar manner A'', B'', C'', D'', &c. from A', B', C', D', and so on, we have

$$A'' + B''a + C''a^2 + \text{\&c.} = \frac{1+a}{1-a}(A + Ba + Ca^2 + \text{\&c.}),$$

$$A''' + B'''a + C'''a^2 + \text{\&c.} = \frac{1+a}{1-a}\cdot\frac{1+a^2}{1-a^2}(A + Ba + Ca^2 + \text{\&c.});$$

and so on. While $A = 1$, and suppose that the process is repeated an indefinite number of times, we have

$$1 + 2a + 5a^2 + 2a^3 + \text{\&c.} = \frac{1+a}{1-a}\cdot\frac{1+a^2}{1-a^2}\cdot\frac{1+a^3}{1-a^3}\cdot\frac{1+a^4}{1-a^4}\ \text{\&c.}$$

and the coefficients 1, 2, 5, 18, &c. are precisely simple algorithms for the calculation of the coefficients of a, a^2, &c. which is the third notation of the following theorem:

$$1{+}2a{+}5a^2{+}18a^3{+}100a^4{+}\text{\&c.} = \frac{1+a}{(1+a)^2(1+a^2)} - x^{42}(1{+}a^2)^{-43} - x^{44}(1{+}a^3)^{-44}\text{\&c..}$$

which gives also to the following very simple algorithm for finding the number of coefficients of a, a^2, &c. (in question are positive numbers as is the case);

$$1 + 2a + 6a^2 + 2a^3 + \text{\&c.}\ ;$$

this, on multiplying by 1, a, a^2, &c. (in question are positive numbers and the doubling $3a^2$, $4a^3$, &c. and consequently, the operation $-3a^2(1+a^2)^{-3} - 4a^3(1+a^4)^{-3}$&c. (we may write the operation as a series, $1+2a+5a^2+6a^3+6a^4+$&c.), then this product is the same as the expression which represents the number of partitions of n, viz. as in the part question.

The proceeding expression for $1+2a+5a^2+6a^3+$&c. as part greater than twice the preceding part is equal to the number of partitions of $n+1$ into parts not greater than the previous one, may by itself become to the following:

$$1 + 2a + 5a^2 + 2a^3 + 4a^4 + \text{\&c.}$$

For example, the partitions of 5, 1, 2; are with the parts 1, 1, 7; 2, 3 = 1; the number of 3-partitions is 4; and by the second part of the theorem, the number of 4-partitions is

$$3(1 + 2 + 4 + 6) = 36.$$

the number of which are 1, 2, 4. Hence, by the first part of the theorem, the number of 5-partitions is 8, and by the second part of the theorem, the number of 3-partitions is 5.

2, Stone Buildings, March 17, 1857.

205. Note on the Summation of a Certain Factorial Expression

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 419–423.]

Mr Kirkman some months ago communicated to me a formula for the double summation of a factorial expression, to which formula he had been led by his researches on the partition of polygons. The formula is a slightly altered form is as follows: viz.

$$\sum_{m=0}^{m=n} \frac{(\rho+s+2)^m (s)^m}{(\rho)^m} \cdot \frac{(\rho+s-1-2)^{m-1}}{(s-1)^{m-1}} = \frac{2k}{(k+1)^{(k+1)}}(s)^{(k+1)}$$

where the summation extending from $z=0$ to $z=n-1$. In the particular case where $k=0$, this is a formula well known. In the same case the series except those in which $\rho = s$ vanish; and putting therefore $k=0$, and making a slight change in the form of the right-hand side, the formula becomes

$$\sum_{m=0}^{m=n} \frac{(2\rho+2k)^m}{\rho^m} \cdot \frac{(2\rho+1)^{(\rho-1)}}{(2\rho+1)^{(\rho-1)}} = \frac{(2\rho+1)^{(\rho-1)}}{(\rho-1)^{(\rho-1)}}$$

the summation extending from $z=0$ to $z=n-1$.

We have, in the notation of Gauss, $[m]^z = m(m-1)\ldots 2.1 = \Pi m$, and a factorial $[m]^z$ is expressed in terms of the function Π by the formula $[m]^z = \Pi m \div \Pi(m-n)$. Write also

$$\Pi(m-1) = \Pi(m) \div [m].$$

we have

$$\Pi m = m\Pi(m-1) = m\Pi(m-1),$$

$$\Pi(2m+1) = 2m\Pi(2m-1).$$

2, Stone Buildings, January 5, 1857. *[continues on the next page set]*

[End of pp. 201–250 chunk.]

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0.1 Supplementary Remarks on the Porism of the In-and-Circumscribed Triangle

[From the *Philosophical Magazine*, vol. xiii. (1857), pp. 19–30.]

In my former papers (see Phil. Mag. August and November, 1853, [115, 116]) I established (as part of a more general one) the following theorem, viz. the condition that there may be inscribed in the conic $U = 0$ an infinity of triangles circumscribed about the conic $V = 0$, is, that if we develope in ascending powers of k the square root of the discriminant of $kU + V$, the coefficient of k^2 in this development must vanish.

Thus writing

$$\text{disct.}(kV + U) = (K, \Theta, K' \mid k, 1)^3,$$

the condition in question is found to be

$$\Theta^2 - 4KK' = 0.$$

The following investigations, although relating only to particular cases of the theorem, are, I think, not without interest.

If the equation of the conic containing the angles is

$$U = 2ayz + 2bzx + 2cxy = 0,$$

and the equation of the conic touched by the sides is

$$V = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

we have

$$\text{disct.}(k, k, k, a - k, b - k, c - k \mid x, y, z)^2 = (K, \Theta, K' \mid k, 1)^3,$$

that is,

$$K = -4, \quad \Theta = \tfrac{1}{2}(a + b + c), \quad K' = -(a + b + c)^2, \quad K'' = 2abc;$$

and the equation $\Theta^2 - 4KK' = 0$ becomes

$$\tfrac{1}{4}(a + b + c)^4 - 128(a + b + c)^2 = 0,$$

which is satisfied identically. This is as it should be; for it is plain that there exists a triangle, viz. the triangle $(x = 0, y = 0, z = 0)$, inscribed in the conic $U = 0$, and circumscribed about the conic $V = 0$.

Suppose that the equation of the conic containing the angles is

$$U = 2ayz + 2bzx + 2cxy = 0,$$

and the equation of the conic touched by the sides is

$$V = ax^2 + by^2 + cz^2 = 0,$$

then the tangential equation of the last-mentioned conic is

$$bc\xi^2 + ca\eta^2 + ab\zeta^2 = 0;$$

and if we take for the angles of the triangle $x : y : z = 1 : 2\lambda : \lambda^2$, or $1 : 2\mu : \mu^2$, or $1 : 2\nu : \nu^2$, then the equation of the line joining the angles (μ), (ν) is

$$bc \cdot 4\mu\nu \cdot x + ca(\mu + \nu)^2 y + ab \cdot 4z = 0;$$

which will touch the conic $ax^2 + by^2 + cz^2 = 0$ if

$$bc \cdot 4\mu\nu + ca(\mu + \nu)^2 + ab \cdot 4 = 0;$$

and it is required to find under what circumstances the equations

$$\begin{aligned} bc \cdot 4\mu\nu + ca(\mu + \nu)^2 + ab \cdot 4 &= 0, \\ bc \cdot 4\nu^2\lambda^2 + ca(\nu + \lambda)^2 + ab \cdot 4 &= 0, \\ bc \cdot 4\lambda^2\mu^2 + ca(\lambda + \mu)^2 + ab \cdot 4 &= 0, \end{aligned}$$

become equivalent to two equations only.

The condition is of course included in the general formula; and putting

$$\text{disct.}(ka, kb + 1, kc, 0, -2, 0 \mid x, y, z)^2 = (K, \Theta, K' \mid k, 1)^3,$$

we must have $\Theta^2 - 4KK' = 0$.

The discriminant in question is

$$k^3abc + k^2ac - k \cdot 4b - 4 = 0,$$

where $K = -4$, $\Theta = \sqrt{ac}$, $K' = 16$, $K'' = 4$; the required condition is therefore $ac + 16b^2 = 0$, or say $b = \frac{i}{4}\sqrt{ac}$.

Substituting this value, the equations become

$$\begin{aligned} c\mu^2\nu^2 + i\sqrt{ca}(\mu+\nu)^2 + a &= 0,\\ c\nu^2\lambda^2 + i\sqrt{ca}(\nu+\lambda)^2 + a &= 0,\\ c\lambda^2\mu^2 + i\sqrt{ca}(\lambda+\mu)^2 + a &= 0; \end{aligned}$$

the first and second of these are

$$AB' + A'B - 2HH' = 0,$$

where

$$\begin{aligned} A &= (-i\sqrt{a} + \mu^2\sqrt{c})i\sqrt{a}, \quad H = i\sqrt{ca}\mu, \quad B = (i\sqrt{a} + \mu^2\sqrt{c})\sqrt{c},\\ A' &= (-i\sqrt{a} + \lambda^2\sqrt{c})i\sqrt{a}, \quad H' = i\sqrt{ca}\lambda, \quad B' = (i\sqrt{a} + \lambda^2\sqrt{c})\sqrt{c}; \end{aligned}$$

$$\begin{aligned} AB' + A'B - 2HH' &= 2i\sqrt{ac}(a - i\sqrt{ac}\lambda\mu + c\lambda^2\mu^2),\\ AB' - H'^2 &= i\sqrt{ac}(a - i\sqrt{ac}\mu^2 + c\mu^4),\\ A'B - H^2 &= i\sqrt{ac}(a - i\sqrt{ac}\lambda^2 + c\lambda^4); \end{aligned}$$

and the result of the elimination therefore is

$$(a - i\sqrt{ac}\lambda^2 + c\lambda^4)(a - i\sqrt{ac}\mu^2 + c\mu^4) - (a - i\sqrt{ac}\lambda\mu + c\lambda^2\mu^2)^2 = 0,$$

viz.

$$2\sqrt{ca}(\lambda-\mu)^2(c\lambda^2\mu^2 + i\sqrt{ca}(\lambda+\mu)^2 + a) = 0;$$

which agrees, as it should do, with the third equation.

To find the condition that it may be possible in the conic

$$x^2 + y^2 + z^2 = 0$$

to inscribe an infinity of triangles, each of them circumscribed about the conic

$$ax^2 + by^2 + cz^2 = 0 :$$

let the equations of the sides be

$$\begin{aligned} l\sqrt{ax} + m\sqrt{by} + n\sqrt{cz} &= 0,\\ l'\sqrt{ax} + m'\sqrt{by} + n'\sqrt{cz} &= 0,\\ l''\sqrt{ax} + m''\sqrt{by} + n''\sqrt{cz} &= 0; \end{aligned}$$

then the conditions of circumscription are

$$l^2+m^2+n^2=0,\quad l'^2+m'^2+n'^2=0,\quad l''^2+m''^2+n''^2=0;$$

and the conditions of inscription are

$$\begin{aligned}
bc(m'n''-m''n')^2+ca(n'l''-n''l')^2+ab(l'm''-l''m')^2&=0,\\
bc(m''n-mn'')^2+ca(n''l-nl'')^2+ab(l''m-lm'')^2&=0,\\
bc(mn'-m'n)^2+ca(nl'-n'l)^2+ab(lm'-l'm)^2&=0.
\end{aligned}$$

Now

$$(mn'-m'n)^2=(m^2+n^2)(m'^2+n'^2)-(mm'+nn')^2=l^2l'^2-(mm'+nn')^2=-(ll'+mm$$

and making the like change in the analogous expressions, and putting for shortness

$$bc+ca+ab=\alpha,\quad bc-ca+ab=\beta,\quad bc+ca-ab=\gamma,$$

the conditions in question become

$$\begin{aligned}
(ll''+mm''+nn'')(\alpha ll''+\beta mm''+\gamma nn'')&=0,\\
(l''l+m''m+n''n)(\alpha l''l+\beta m''m+\gamma n''n)&=0,\\
(ll'+mm'+nn')(\alpha ll'+\beta mm'+\gamma nn')&=0.
\end{aligned}$$

The proper solution is that given by the system of equations

$$\begin{aligned}
l^2+m^2+n^2&=0,\\
l'^2+m'^2+n'^2&=0,\\
l''^2+m''^2+n''^2&=0,\\
\alpha ll''+\beta mm''+\gamma nn''&=0,\\
\alpha l''l+\beta m''m+\gamma n''n&=0,\\
\alpha ll'+\beta mm'+\gamma nn'&=0;
\end{aligned}$$

and by writing

$$\frac{l}{\sqrt{a}}=f,\quad \frac{m}{\sqrt{b}}=g,\quad \frac{n}{\sqrt{c}}=h,\quad \&c.,$$

these equations become

$$\begin{aligned} Af^2 + Bg^2 + Ch^2 &= 0, \\ Af'^2 + Bg'^2 + Ch'^2 &= 0, \\ Af''^2 + Bg''^2 + Ch''^2 &= 0, \\ ff' + gg' + hh' &= 0, \\ f'f'' + g'g'' + h'h'' &= 0, \\ f''f + g''g + h''h &= 0. \end{aligned}$$

The first of which systems expresses that the points (f, g, h), (f', g', h'), (f'', g'', h'') are points in the conic $Ax^2 + By^2 + Cz^2 = 0$; and the second condition expresses that each of the points in question is the pole with respect to the conic $x^2 + y^2 + z^2 = 0$ of the line joining the other two points, i.e. that the three points are a system of conjugate points with respect to the last-mentioned conic.

The problem is thus reduced to the following one:

To find the condition in order that it may be possible in the conic $Ax^2 + By^2 + Cz^2 = 0$ *to inscribe an infinity of triangles such that the angles are a system of conjugate points with respect to the conic* $x^2 + y^2 + z^2 = 0$.

Before going further it is proper to remark that if, instead of assuming $\alpha ll'' + \beta mm'' + \gamma nn'' = 0$, we had assumed this, combined with the equations $l'^2 + m'^2 + n'^2 = 0$, $l''^2 + m''^2 + n''^2 = 0$, would have given $l : m : n = l' : m' : n'$, i.e. two of the angles of the triangle would have been coincident: this obviously does not give rise to any proper solution.

Returning now to the system of equations in f, g, h, &c., since the equations give only the ratios $f : g : h$, $f' : g' : h'$, $f'' : g'' : h''$, we may if we please assume

$$f^2 + g^2 + h^2 = f'^2 + g'^2 + h'^2 = f''^2 + g''^2 + h''^2 = 1,$$

which, combined with the second system of equations, gives

$$ff' + gg' + hh' + f''^2 + g''^2 + h''^2 = 1.$$

We have, consequently,

$$A + B + C = A(f^2 + f'^2 + f''^2) + B(g^2 + g'^2 + g''^2) + C(h^2 + h'^2 + h''^2)$$

$$= (Af^2 + Bg^2 + Ch^2) + (Af'^2 + Bg'^2 + Ch'^2) + (Af''^2 + Bg''^2 + Ch''^2),$$

i.e.

$$A + B + C = 0,$$

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for the condition that it may be possible in the conic $Ax^2+By^2+Cz^2=0$ to describe an infinity of triangles the angles of which are conjugate points with respect to the conic $x^2+y^2+z^2=0$.

The equation of the conic $Ax^2+By^2+Cz^2=0$ may be written in the form

$$(b^2c^2-c^2a^2-a^2b^2+2a^2bc)x^2+(c^2a^2-a^2b^2-b^2c^2+2ab^2c)y^2+(a^2b^2-b^2c^2-c^2a^2+2abc^2)z$$

which gives the values of A, B, C; or again in the form

$$2(bc+ca+ab)(bcx^2+cay^2+abz^2)-(bc+ca+ab)^2(x^2+y^2+z^2)+4abc(ax^2+by^2+cz^2)=0$$

where it should be observed that $bcx^2+cay^2+abz^2=0$ is the equation of the conic which is the polar of $ax^2+by^2+cz^2=0$ with respect to $x^2+y^2+z^2=0$.

It is very easy from the last form to deduce the equation of the auxiliary conic, when the conics $ax^2+by^2+cz^2=0$, $x^2+y^2+z^2=0$ are replaced by conics represented by perfectly general equations.

The condition $A+B+C=0$ gives, substituting the values of A, B, C,

$$b^2c^2+c^2a^2+a^2b^2-2abc(a+b+c)=0;$$

or in a more convenient form,

$$(bc+ca+ab)^2=4abc(a+b+c),$$

as the condition in order that it may be possible to inscribe in the conic $x^2+y^2+z^2=0$ an infinity of triangles, the sides of which touch the conic $ax^2+by^2+cz^2=0$: this agrees perfectly with the general theorem.

It is convenient to add (as a somewhat more general form of the equation $A+B+C=0$), that the condition in order that it may be possible in the conic $Ax^2+By^2+Cz^2=0$ to inscribe an infinity of triangles the angles of which are conjugate points with respect to the conic $A'x^2+B'y^2+C'z^2=0$, is

$$\frac{A}{A'}+\frac{B}{B'}+\frac{C}{C'}=0.$$

But the problem to find the condition in order that it may be possible in the conic $x^2+y^2+z^2=0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2+by^2+cz^2=0$, may, by the assistance of the geometrical theorem to be presently mentioned, be at once reduced to the problem:

To find the condition in order that it may be possible in the conic $Ax^2 + By^2 + Cz^2 = 0$ *to inscribe an infinity of triangles the sides of which are conjugate points with respect to a conic* $A'x^2+B'y^2+C'z^2 = 0$.

Theorem. *If the chord* PP' *of a conic* S *envelope a conic* σ*, the points* P, P' *are harmonics with respect to a conic* T *which has with* S, σ*, a system of common conjugate points.*

Take for the equation of S,

$$x^2 + y^2 + z^2 = 0,$$

and for the equation of σ,

$$ax^2 + by^2 + cz^2 = 0;$$

then if (x_1, y_1, z_1), (x_2, y_2, z_2) are the coordinates of the points P, P' respectively, we have

$$x_1^2 + y_1^2 + z_1^2 = 0, \quad x_2^2 + y_2^2 + z_2^2 = 0;$$

and the condition in order that the chord may touch the conic σ is

$$bc(y_1z_2 - y_2z_1)^2 + ca(z_1x_2 - z_2x_1)^2 + ab(x_1y_2 - x_2y_1)^2 = 0.$$

But we have

$$(y_1z_2-y_2z_1)^2 = (y_1^2+z_1^2)(y_2^2+z_2^2)-(y_1y_2+z_1z_2)^2 = x_1^2x_2^2-(y_1y_2+z_1z_2)^2$$

$$= -(x_1x_2 + y_1y_2 + z_1z_2)(x_1x_2 - y_1y_2 - z_1z_2);$$

and making the like change in the analogous quantities, and putting for shortness

$$\alpha = bc + ca + ab, \quad \beta = bc - ca + ab, \quad \gamma = bc + ca - ab,$$

the condition in question becomes

$$(x_1x_2 + y_1y_2 + z_1z_2)(\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2) = 0.$$

But the equation $x_1x_2 + y_1y_2 + z_1z_2 = 0$ must be rejected, as giving with the equations $x_1^2 + y_1^2 + z_1^2 = 0$, $x_2^2 + y_2^2 + z_2^2 = 0$ the relation $x_1 : y_1 : z_1 = x_2 : y_2 : z_2$; we have therefore

$$\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2 = 0,$$

which implies that the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are harmonics with respect to the conic

$$\alpha x^2 + \beta y^2 + \gamma z^2 = 0,$$

which is a conic having with S, σ, a system of common conjugate points.

The equation may also be written

$$(-bc + ca + ab)x^2 + (bc - ca + ab)y^2 + (bc + ca - ab)z^2 = 0;$$

or, as it may also be written,

$$(bc + ca + ab)(x^2 + y^2 + z^2) - 2(bcx^2 + cay^2 + abz^2) = 0;$$

and, as before remarked, $bcx^2 + cay^2 + abz^2 = 0$ is the equation of the conic which is the polar of $ax^2 + by^2 + cz^2 = 0$ with respect to $x^2 + y^2 + z^2 = 0$.

The condition in order that there may be inscribed in the conic $x^2 + y^2 + z^2 = 0$ an infinity of triangles the angles of which are conjugate points with respect to the conic $\alpha x^2 + \beta y^2 + \gamma z^2 = 0$, is

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 0;$$

or writing this equation under the form $\beta\gamma + \gamma\alpha + \alpha\beta = 0$, and substituting for α, β, γ their values, we have the equation already found, as the condition in order that it may be possible in the conic $x^2 + y^2 + z^2 = 0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2 + by^2 + cz^2 = 0$.

Theorem. *Let $ax^2 + by^2 + cz^2 = 0$ be the equation of a spherical conic, and let $(\xi : \eta : \zeta)$, a point on the conic, be the pole of a great circle cutting the conic in two points; the conic intersects upon the great circle an arc given by the equation*

$$\cos\delta = \frac{(b+c)\xi^2 + (c+a)\eta^2 + (a+b)\zeta^2}{\sqrt{(b+c)^2\xi^4 + (c+a)^2\eta^4 + (a+b)^2\zeta^4 - 4(bc\eta^2\zeta^2 + ca\zeta^2\xi^2 + ab\xi^2\eta^2)}};$$

hence if $a+b+c=0$, $\delta = 90$; or there may be inscribed in the conic an infinity of triangles having each of their sides equal to 90.

It is worth while, in connexion with the subject, and for the sake of a remark to which they give rise, to reproduce in a short compass some results long ago obtained by Jacobi and Richelot.

The following are Jacobi's formulae for the chords of a circle subject to the condition of touching another circle. Let

$$CP = R, \qquad cp = r, \qquad Cc = a,$$
$$\angle AcP = 2\phi, \qquad \angle acp = 2\psi,$$

then it is clear from geometrical considerations that

$$\frac{d\phi}{d\psi} = \frac{MA}{Ma}.$$

We have

$$MA^2 = cA^2 - cM^2 = a^2 + R^2 + 2aR\cos 2\phi - r^2 = (a+R)^2 - r^2 - 4aR\sin^2\phi = \{(a+R)^2 -$$

or

$$\frac{d\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{d\psi}{\sqrt{1-k'^2\sin^2\psi}},$$

where

$$k^2 = \frac{4aR}{(a+R)^2 - r^2} \quad \text{and therefore also} \quad k'^2 = \frac{4ar}{(a+r)^2 - R^2} = \frac{4ar}{(R+a)^2 - r^2}.$$

It will be convenient for comparison with the formulae of Richelot to write $\angle ACQ = 2\psi'$; this gives $2\psi' = \pi - 2\phi$, and the differential equation thus becomes

$$\frac{-d\psi'}{\sqrt{a^2 + R^2 - r^2 - 2aR\cos 2\psi'}} = \frac{d\psi}{\sqrt{a^2 + R^2 - r^2 - 2aR\cos 2\psi}}.$$

Or if $\tan\psi' = g\tan\theta$, and therefore

$$\cos 2\psi' = \frac{1 - g^2\tan^2\theta}{1 + g^2\tan^2\theta} = \frac{m - n\cos 2\theta}{m+n},$$

where $g^2 = \dfrac{m-n}{m+n}$, then this is

$$\frac{m-n}{m+n} = \frac{\cos^2\theta}{\cos^2\psi'} = \frac{\sin^2\theta}{\sin^2\psi'},$$

and we have also

$$\frac{m-n}{m+n}\frac{d\theta}{1 - \dfrac{2n}{m+n}\sin^2\theta};$$

and thence

$$\frac{\sqrt{m-n}\cos 2\psi'}{\sqrt{m+n}} = \frac{\sqrt{m-n}}{\sqrt{m+n}}\frac{1 - \dfrac{2n}{m+n}\sin^2\theta}{\sqrt{1 - \dfrac{2n}{m+n}\sin^2\theta}},$$

that is,

$$\frac{\sqrt{m-n}\cos 2\psi'}{\sqrt{m+n}\sqrt{1-k^2\sin^2\theta}} = \frac{\sqrt{m-n}}{\sqrt{m+n}}\sqrt{1-k^2\sin^2\theta},$$

where $k^2 = \dfrac{2n}{m+n}$ has the same value as before.

Hence the relation between θ, θ' is

$$\frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} = \frac{d\theta'}{\sqrt{1-k^2\sin^2\theta'}},$$

which is identical with that between ϕ and ψ; and in fact the equation between ϵ, ϕ is

$$\tan\epsilon\tan\phi = \frac{r}{R},$$

which, if $\phi = \operatorname{am} u$, gives $\epsilon = \operatorname{am}(K-u)$.

The differential equation contains only a single arbitrary parameter; hence the same differential equation might have been obtained from different values of a, R, r, the parameters which determine the circle enveloped by the moveable chord. The condition for this of course is $k^2 = k'^2$, that is

$$\frac{4aR}{(a+R)^2 - r^2} = \frac{4ar}{(R+a)^2 - r^2},$$

or as the equation may be written

$$(aa' - r^2 - R^2)(a - a') - R(a' - a)r^2 = 0;$$

this implies that the enveloped circles intersect the other circle in the same two points, or that all the circles have a common chord.

Suppose for $\psi = 0$, we have $\phi = \beta$, then it is easy to see geometrically that

$$\tan^2 \beta = \frac{(R-a)^2 - r^2}{(R+a)^2 - r^2};$$

let the corresponding value of θ be $\theta = \alpha$, i.e. suppose that for $\theta = 0$, we have $\epsilon = \alpha$, then

$$\tan^2 \alpha = \frac{(R+a)^2 - r^2}{(R-a)^2 - r^2} \cdot \frac{r^2}{R^2} = \frac{(R+a)^2 - r^2}{r^2} \cdot \frac{r^2}{(R-a)^2 - r^2},$$

i.e.

$$\tan^2 \alpha = \frac{(R+a)^2 - r^2}{(R-a)^2 - r^2} \cdot \frac{r^2}{R^2}, \quad \sec^2 \alpha = \frac{(R+a)^2 - r^2}{r^2} \cdot \frac{r^2}{R^2};$$

and α having this value, we have for the finite relation between θ, ϵ,

$$F\epsilon = F\theta + F\alpha.$$

Richelot has shown, by precisely similar reasoning, that for circles of the sphere we have

$$\frac{d\theta}{\sqrt{\cos^2 r - (\cos R \cos \alpha + \sin R \sin \alpha \cos 2\psi)^2}} = \frac{d\psi}{\sqrt{\cos^2 r - (\cos R \cos \alpha + \sin R \sin \alpha \, c}}$$

which is of the form

$$\frac{d\theta}{\sqrt{1 - (\lambda + \mu \cos 2\psi)^2}} = \frac{d\psi}{\sqrt{1 - (\lambda + \mu \cos 2\psi')^2}},$$

where

$$\lambda = \frac{\cos R \cos \alpha}{\cos r}, \quad \mu = \frac{\sin R \sin \alpha}{\cos r}.$$

It is very important to remark, that this equation contains the two parameters λ, μ, so that the same equation cannot be obtained with any new values of the parameters a, r; or the formulae in piano for three or more circles do not apply to circles of the sphere: the geometrical reason for this is as follows, viz. in the plane a circle is a conic passing through two fixed points (the circular points at ∞), and consequently any number of circles having a common chord are in fact to be considered as conics each of which passes through the same four points. But circles of the sphere are not spherical conics passing through two fixed points, but are merely spherical conics having a double contact with an absolute conic (the sphero-circle at infinity); and consequently circles of the sphere having a common spherical chord are not spherical conics passing through the same four points.

I am not sure whether this remark as to the ground of the distinction between the theory of circles in piano and that of circles on the sphere has been explicitly made in any of the treatises on spherical geometry.

To reduce the equation, write

$$\sin\theta = \frac{\sin R \sin\alpha}{\cos r}\sin\phi;$$

then, after a simple reduction,

$$\frac{d\theta}{\sqrt{\cos^2 r - (\cos R\cos\alpha + \sin R\sin\alpha\cos 2\psi)^2}} = \frac{d\phi}{\sqrt{1-k^2\sin^2\phi}},$$

or the relation between the two values of θ is

$$\frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} = \frac{d\theta'}{\sqrt{1-k^2\sin^2\theta'}},$$

where

$$k^2 = \frac{\tan^2 R\sin^2\alpha}{\tan^2 r\cos^2 R\sin^2 r}, \quad \text{that is} \quad k^2 = \left(\frac{\tan R\sin\alpha}{\tan r\cos R\sin r}\right)^2.$$

Suppose that for $\psi = 0$, we have $\psi' = \beta$, it is easy to see that

$$\cos^2\beta = \frac{\cos^2 r - \cos^2(R+\alpha)}{\cos^2 r - \cos^2(R-\alpha)};$$

let the corresponding value of θ be $\theta = \alpha$, i.e. suppose that for $\psi = 0$, we have $\theta = \alpha$, then

$$\begin{aligned}\cos^2\beta &= \frac{\cos^2 r - \cos^2(R+\alpha)}{\cos^2 r - \cos^2(R-\alpha)} = \frac{(\cos r - \cos(R+\alpha))(\cos r + \cos(R-\alpha))}{\cos^2 R\sin^2 r}\\ &= \frac{(\cos r + \sin R\sin\alpha)^2 - \cos^2 R\cos^2\alpha}{\cos^2 R\sin^2 r} = \frac{(\cos r\sin R + \sin\alpha)^2}{\cos^2 R\sin^2 r};\end{aligned}$$

i.e.

$$\tan^2\alpha = \frac{\tan^2 R\sin^2\alpha}{\tan^2 r\cos^2 R\sin^2 r} - 1, \quad \sec^2\alpha = \frac{\tan^2 R\sin^2\alpha}{\tan^2 r\cos^2 R\sin^2 r};$$

and α having this value, the finite relation between θ, θ' is

$$F\theta' = F\theta + F\alpha.$$

By comparing with the corresponding formula in piano, we arrive at Richelot's conclusion, that the formulae for the sphere may be deduced from those in piano by writing in the place of $\frac{a}{r}$, $\frac{R}{r}$, the functions $\frac{\tan R \sin \alpha}{\tan r}$, $\frac{\tan R \sin \alpha}{\cos R \sin r}$ respectively.

2, Stone Buildings,
October 1, 1856.

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0.2 On the Theory of the Analytical Forms Called Trees

[From the *Philosophical Magazine*, vol. xiii. (1857), pp. 172–176.]

A symbol such as $A\,dx + B\,dy + \ldots$, where A, B, &c. contain the variables x, y, &c. in respect to which the differentiations are to be performed, partakes of the natures of an operand and operator, and may be therefore called an *Operandator*.

Let P, Q, R, $\ldots$ be any operandators, and let U be a symbol of the same kind, or to fix the ideas, a mere operand; PU denotes the result of the operation P performed on U, and QPU denotes the result of the operation Q performed on PU; and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question.

Now considering the expression QPU, it is easy to see that we may write

$$QPU = (Q \times P)U + (QP)U,$$

where on the right-hand side $(Q \times P)$ and (QP) signify as follows: viz. $Q \times P$ denotes the mere algebraical product of Q and P, while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q \times P)U$ and $(QP)U$ denote respectively the results of the operations $(Q \times P)$ and (QP) performed each of them upon U as operand.

It is proper to remark that $(Q \times P)$ and $(P \times Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently.

But without a more convenient notation, it would be difficult to find the terms of an expression such as $SRQPU$. Imagine, however, a set of figures, termed *trees*, constituted with a set of knots connected by lines termed *branches*, in the following manner. The tree corresponding to the operandator P is to have at the extremity of a single branch the letter P; this single branch is to have at its other extremity a knot termed the *root*, which for greater distinctness is to be marked by the letter U representing the operand; that is, the tree is to be

P—U

(only in the actual figures, instead of the letter U the root is denoted by a thick dot); and in general the figure corresponding to any combination of operandators is to be obtained by taking the figure corresponding to the combination which stands on the right of the last prefixed operandator, and adding on at each of the knots of this figure a branch terminated by the last prefixed operandator; the figures thus obtained are of course to be all of them different, but the different figures may have branches in common; the different parts, or if any figure consists of different unconnected parts, then each of these parts is termed a *tree*.

The inspection of these figures will at once show what is meant by the term in question, and by the terms *root*, *branches* (which may be either main branches, intermediate branches, or free branches), and *knots* (which may be either the root itself, or proper knots, or the extremities of the free branches).

To apply this to the question in hand, PU consists of a single term represented by fig. 1 (bis); QPU consists, as above, of two terms represented by the two parts of fig. 2 (bis), viz. the first part represents the term $(Q \times P)U$, and the second part represents the term $(QP)U$. And it is obvious that fig. 2 (bis) is at once formed from the figure 1 (bis) by adding on a branch terminated by Q at each of the knots of the single part of fig. 1 (bis).

In like manner $RQPU$ consists of six terms represented by the six parts of fig. 3 (bis), and this figure is at once formed from fig. 2 (bis) by adding on a branch terminated by R at each knot of each part of fig. 2 (bis).

It is easy to see that the trees of the second set are connected in a definite manner with those of the first set. Thus the first set consists of the trees represented in the annexed figure, and the number is therefore two.

It is not difficult to see that we have in this case (B_r being the number of such trees with r free branches),

$$(1-x)^{-B_1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}(1-x^4)^{-B_4}\ldots = 1+x$$

and a like process of development gives:

$$B_r = \frac{1}{r}\binom{2r-2}{r-1} \quad \text{for } r = 1, 2, 3, \ldots$$

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0.3 On a Problem in the Partition of Numbers

[From the *Philosophical Magazine*, vol. xiii. (1857), pp. 245–248.]

It is required to find the number of partitions into a given number of parts, such that the first part is unity, and that no part is greater than twice the preceding part.

Commencing to form the partitions in question, these are

$$\begin{array}{lll} 1 & & \\ 1 & 1 & \\ 1 & 2 & \\ 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 4 \\ \&c. & & \end{array}$$

and if we were to proceed to the 4-partitions, each 3-partition ending in 1 would give rise to two such partitions; each 3-partition ending in 2 to four such partitions; each 3-partition ending in 3 to six partitions; and each 3-partition ending in 4 to eight such partitions.

We form in this manner the Table:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Σ
1	1																1
2	1	1															2
3	1	2	2														5
4	1	2	4	2													9
5	1	2	4	6	4												17
6	1	2	4	6	10	4											27
7	1	2	4	6	10	14	8										45
8	1	2	4	6	10	14	20	8									65
9	1	2	4	6	10	14	20	26	14								97
10	1	2	4	6	10	14	20	26	35	16							134
&c.																	

and we are thus led to the series

$$1,\ 1,\ 2,\ 1,\ 2,\ 4,\ 6,\ 1,\ 2,\ 4,\ 6,\ 10,\ 14,\ 20,\ 26,\ \&c.;$$

where, considering 1 as the first term of each series, the first differences of any series are the terms twice repeated of the next preceding series: thus the differences of the fourth series are $1, 1, 2, 2, 4, 4, 6, 6$.

It is moreover clear that the first half of each series is precisely the series which immediately precedes it; we need, in fact, only consider a single infinite series,

$$1,\ 2,\ 4,\ 6,\ 10,\ 14,\ 20,\ 26,\ 36,\ 46,\ 60,\ 74,\ 94,\ 114,\ 140,\ 166,\ \&\text{c.}$$

It is to be remarked, moreover, that in the column of totals, the total of any line is precisely the first number in the next succeeding line.

Consider in general a series A, B, C, D, E, &c., and a series A', B', C', D', E', &c. derived from it as follows:

$$A' = 2A, \quad B' = 2A, \quad C' = 2A+B, \quad D' = 2A+2B, \quad E' = 2A+2B+C, \quad F' = 2A$$

viz. the first differences of the series 0, A', B', C', D', E', &c. are A, A, B, B, C, C, &c.

Then multiplying by 1, x, x^2, &c. and adding, we have

$$A'+B'x+C'x^2+\&\text{c.} = (1+2x+2x^2+\ldots)(A+Bx+Cx^2+\&\text{c.}) = \frac{1+x}{1-x}(A+Bx+Cx^2+$$

and if we form in a similar manner A'', B'', C'', D'', &c., we find

$$1 + \Sigma_1 x + \Sigma_2 x^2 + \Sigma_3 x^3 + \&\text{c.} = \frac{(1+x)(1+x^2)(1+x^3)\ldots}{(1-x)(1-x^2)(1-x^3)\ldots},$$

and the coefficients 1, Σ_1, Σ_2, &c. are precisely those of the infinite series.

We have more simply

$$1 + \Sigma_1 x + \Sigma_2 x^2 + \&\text{c.} = \frac{1}{(1-x)(1-x^3)(1-x^7)(1-x^{15})\ldots}$$

which gives rise to the following very simple algorithm for the calculation of the coefficients:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	1
	1	2	4	6	10	14	20	26	36	46	60	74	94	114	14
	0	0	1	2	4	6	10	14	20	26	35	44	56	68	8
	1	2	4	6	10	14	20	26	35	44	56	68	84	100	12
	0	0	0	0	1	2	4	6	10	14	20	26	35	44	5
	1	2	4	6	10	14	20	26	35	44	56	68	84	100	12
	0	0	0	0	0	0	0	0	1	2	4	6	10	14	2
	| 1	| 2	| 4,	6	| 10,	14,	20,	26	| 36,	46,	60,	74,	94,	114,	14

&c.

The last line is marked off into periods of (reckoning from the beginning) 1, 2, 4, 8, &c.; and by what has preceded, the series which gives the number of 1-partitions, 2-partitions, 3-partitions, &c. is found by summing to the end of each period and doubling the results; we thus, in fact, obtain

$$(1),\ 2,\ 6,\ 26,\ 166,\ 1626,\ \&\text{c.} :$$

and the same series is also given by means of the last terms of the several periods.

The preceding expression for $1 + \Sigma_1 x + \Sigma_2 x^2 + \&\text{c.}$ shows that

$$\Sigma_m = \Sigma_{m-1} + \Sigma_{m-2} - \Sigma_{m-5} - \Sigma_{m-7} + \Sigma_{m-12} + \Sigma_{m-15} - \&\text{c.},$$

where the series on the right-hand side is to be continued so long as it is possible, and 1 is to be written in place of Σ_0.

C. III. 33

0.4 Note on the Summation of a Certain Factorial Expression

[From the *Philosophical Magazine*, vol. xiii. (1857), pp. 419–423.]

Mr Kirkman some months ago communicated to me a formula for the double summation of a factorial expression, to which formula he had been led by his researches on the partition of polygons. The formula in a slightly altered form is as follows: viz.

$$\Sigma\Sigma\frac{[r+k-x-y]^{k-1}\,[r-1-x]^{k-1-y}}{[r+k+2]^{k}\,[r]^{x}\,[y+1]^{y+1}\,[k-y]^{k-y}\,[k-1-y]^{k-1-y}}=\frac{2^{k}}{[k+1]^{k+1}\,[r]^{r}};$$

the summation extending from $x=0$ to $x=r-1$, and $y=0$ to $y=k-1$.

In the particular case when $k=r$, then all the terms of the series except those in which $y=x$ vanish; and putting therefore $k=r$ and $y=x$, and making a slight change in the form of the right-hand side, the formula becomes

$$\Sigma\frac{[2r-2x]^{r-1}}{[2r+1]^{r}}=\frac{2^{r}}{[r+1]^{r+1}\,[r]^{r}},$$

the summation extending from $x=0$ to $x=r-1$.

We have, in the notation of Gauss,

$$[m]^{m}=m(m-1)\dots 2\cdot 1=\Pi m,$$

and a factorial $[m]^{n}$ is expressed in terms of the function Π by the formula

$$[m]^{n}=\Pi m\div\Pi(m-n).$$

Write also

$$\Pi_1(m-\tfrac{1}{2})=(m-\tfrac{1}{2})(m-\tfrac{3}{2})\dots\tfrac{1}{2},$$

we have

$$\Pi(2m)=2^{2m}\Pi m\cdot\Pi_1(m-\tfrac{1}{2}),\quad \Pi(2m+1)=2^{2m+1}\Pi m\cdot\Pi_1(m+\tfrac{1}{2});$$

and transforming the factorials by these formulae, the series becomes

$$\Sigma\frac{\Pi_1(x-\frac{1}{2})\,\Pi_1(r-x-\frac{1}{2})}{\Pi(r+1)}=\frac{\Pi(r+\frac{1}{2})}{8r(r+\frac{1}{2})\Pi_1(r-\frac{1}{2})},$$

the summation, as before, from $x = 0$ to $x = r - 1$.

This may be written

$$\Sigma \frac{\Pi_1(x - \frac{1}{2})\,\Pi_1(r - x - \frac{1}{2})}{\Pi(r+1)} = \frac{\Pi_1(r - \frac{1}{2})}{8r(r + \frac{1}{2})},$$

or

$$\Sigma \frac{\Pi_1(x + \frac{1}{2})\,\Pi_1(r - x + \frac{1}{2})}{(r+2)(r+3)} = \frac{\Pi(r+1)}{8r(r + \frac{1}{2})} \cdot \frac{\Pi_1(r - \frac{1}{2})}{\Pi_1(r + \frac{1}{2})},$$

the summation from $x = 0$ to $x = r - 1$.

The general term does not vanish for $x = r$ or $x = r + 1$, but it vanishes for all greater values of x; hence if we add to the right-hand side the two terms corresponding to $x = r$ and $x = r + 1$, the summation may be extended from $x = 0$ to $x = r + 1$, or what is the same thing, from $x = 0$ indefinitely. The two terms in question are

$$\frac{-4r(r + \frac{1}{2})}{(r+2)(r+3)} \quad \text{and} \quad \frac{2}{(r+2)(r+3)}$$

and the resulting equation is

$$\Sigma \frac{\Pi_1(x + \frac{1}{2})\,\Pi_1(r - x + \frac{1}{2})}{(r+2)(r+3)} = \frac{-4r(r + \frac{1}{2}) + 2}{(r+2)(r+3)},$$

the summation from $x = 0$ indefinitely; or substituting for the functions Π and Π_1 their values, the formula is

$$\frac{\frac{3}{2} \cdot \frac{5}{2} \cdot (r - \frac{1}{2})}{(r+2)(r+3)} + \frac{\frac{3}{2} \cdot \frac{4}{2} \cdot \frac{5}{2} \cdot (r - \frac{1}{2})(r - \frac{3}{2})}{(r+2)(r+3)} + \frac{\frac{3}{2} \cdot \frac{4}{2} \cdot \frac{5}{2} \cdot \frac{6}{2} \cdot (r - \frac{1}{2})(r - \frac{3}{2})(r - \frac{5}{2})}{(r+2)(r+3)} + \&\text{c.} =$$

which is a formula obviously belonging to the theory of the hypergeometric series

$$F(\alpha, \beta, \gamma, x) = 1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma} x + \frac{\alpha(\alpha+1)\,\beta(\beta+1)}{1 \cdot 2 \cdot \gamma(\gamma+1)} x^2 + \&\text{c.};$$

but the form in which it is required to verify the formula is not one which can be very easily obtained from the ordinary formulae relating to the hypergeometric series.

I write for shortness

$$P = \frac{\Pi_1(x + \frac{1}{2})\,\Pi_1(r - x + \frac{1}{2})}{(r+2)(r+3)},$$

and introduce the new variable ξ connected with x by the equation $x+\frac{1}{2} = (r+1)\xi$, so that as x varies from 0 to r, ξ varies from $\dfrac{1}{2(r+1)}$ to $\dfrac{r+\frac{1}{2}}{r+1}$, the limits for $r = \infty$ being 0 and 1 respectively.

We have

$$P = \frac{\Pi(r+1)}{\Pi(x+\frac{1}{2})\,\Pi(r-x+\frac{1}{2})} \cdot \frac{\Pi_1(x+\frac{1}{2})\,\Pi_1(r-x+\frac{1}{2})}{(r+2)(r+3)},$$

and therefore

$$P = \frac{2^{2r+1}}{\pi} \cdot \frac{\Pi_1(x+\frac{1}{2})\,\Pi_1(r-x+\frac{1}{2})}{(r+2)(r+3)}.$$

But

$$\Pi_1(x+\tfrac{1}{2}) = \frac{\Pi(x+1)}{2^x\Pi_1(\frac{1}{2})}, \quad \Pi_1(r-x+\tfrac{1}{2}) = \frac{\Pi(r-x+1)}{2^{r-x}\Pi_1(\frac{1}{2})},$$

and

$$\Pi_1(\tfrac{1}{2}) = \frac{\sqrt{\pi}}{2},$$

whence

$$P = \frac{2^{2r+1}}{\pi} \cdot \frac{1}{(r+2)(r+3)} \cdot \frac{\Pi(x+1)\,\Pi(r-x+1)}{2^r \cdot \frac{\pi}{4}} = \frac{2^{r+3}}{\pi^2} \cdot \frac{\Pi(x+1)\,\Pi(r-x+1)}{(r+2)(r+3)}.$$

Hence multiplying by $d\xi$ and integrating from $\xi = 0$, and again multiplying by $d\xi$ and integrating from $\xi = 0$ to $\xi = \frac{1}{2}$, we find

$$\int_0^{\frac{1}{2}} d\xi \int_0^{\xi} P\,d\xi = \frac{2^{r+3}}{\pi^2(r+2)(r+3)} \int_0^{\frac{1}{2}} d\xi \int_0^{\xi} \Pi(x+1)\,\Pi(r-x+1)\,d\xi,$$

if for shortness

$$\frac{3\cdot 5}{0\cdot 4} \cdot \frac{7}{7} \cdot \frac{7}{7} = \frac{3\cdot 4\cdot 5\cdot 7}{0\cdot 4\cdot 7\cdot 7}.$$

Substituting for the definite integrals their values,

$$\frac{\Pi(-\gamma+\beta-1)}{\Pi(\alpha-1)\,\Pi(-\gamma-3)} = \frac{\Pi(-\gamma-3)}{\Pi(\alpha-1)\,\Pi(-\gamma-2)},$$

whence

$$\frac{\Pi(-\gamma+\beta-1)}{\Pi(\alpha-1)\,\Pi(-\gamma-3)} = \frac{\Pi(-\gamma-3)}{\Pi(\alpha-1)\,\Pi(-\gamma-2)}, \quad \text{c.}$$

The second and third terms are

$$\frac{1}{1} \quad \text{and} \quad \frac{1}{1},$$

which are

$$\frac{1}{1} \quad \text{and} \quad \frac{1}{1}.$$

For the reduction of the first term we have

$$\Pi(\alpha-\gamma-1) = [\beta-\gamma-1]^{\beta}\Pi(-\gamma-1);$$

and we thus find

$$S = \frac{\Pi_1(r+\tfrac{1}{2})}{8r(r+\tfrac{1}{2})\Pi_1(r-\tfrac{1}{2})},$$

where, as before,

$$\frac{3\cdot\gamma}{0\cdot 4}\cdot\frac{5\cdot 7\cdot\gamma}{7\cdot 7} = \frac{3\cdot 4\cdot 5\cdot 7\cdot\gamma}{0\cdot 4\cdot 7\cdot 7\cdot 7}.$$

This is the formula in hypergeometric series required for the present purpose, and it is certainly true when the series is finite.

Write now

$$\alpha = -r+\tfrac{1}{2}, \quad \beta = r+\tfrac{3}{2}, \quad \gamma = -r-2;$$

then the first term is

$$\frac{[1]^{r+2}}{[-4]^{r+2}} = \frac{[1]^{r+2}}{[-4]^{r+2}},$$

which vanishes on account of the numerator, and the second term is $\frac{1}{4}r(r+\frac{1}{2})$, and we have consequently

$$S - \tfrac{1}{4}r(r+\tfrac{1}{2}) = -\tfrac{1}{4}r(r+\tfrac{1}{2}),$$

which gives

$$\frac{4r(r+\frac{1}{2})}{(r+2)(r+3)}S = \tfrac{1}{4}r(r+\tfrac{1}{2}),$$

S being here the series in r, the sum of which was required, and the particular case of Mr Kirkman's formula is thus verified.

It is probable that the general case might be treated in an analogous manner by first grouping together the terms which correspond to a given difference $x-y$, and ultimately summing the sums of these partial series; but I have not examined this question.

2, Stone Buildings,
W.C., April 18, 1857.

C. III. 34

0.5 Note on a Theorem Relating to the Rectangular Hyperbola

[From the *Philosophical Magazine*, vol. xiii. (1857), p. 423.]

The following theorem is given in a slightly different form by Brianchon and Poncelet, Gergonne, vol. XI. [1820], p. 205, viz. Any conic whatever which passes through the three angles of a triangle and the point of intersection of the perpendiculars let fall from the angles of the triangle upon the opposite sides is a rectangular hyperbola; and there is an elegant demonstration depending on the properties of the inscribed hexagon.

The theorem is, however, a particular case of the following: viz.

"Any conic whatever which passes through the four points of intersection of two rectangular hyperbolas is a rectangular hyperbola."

And this, again, is a particular case of the following: viz.

If there be a conic Ω and a line P, then considering any two conics U, V such that the points of intersection of P, U are harmonics in respect to the points of intersection of P, Ω, and the points of P, V are also harmonics in respect to the points of intersection of P, Ω, then any conic whatever W which passes through the four points of intersection of U, V will have the like property, viz. the points of intersection of P, W will be harmonics in respect of the points of intersection of P, Ω; a theorem which is an immediate consequence of the theorem that three conics which intersect in the same four points are intersected by any line whatever in six points which are in involution.

2, Stone Buildings,
W.C., April 23, 1857.

C. III. 34

0.6 Analytical Solution of the Problem of Tactions

[From the *Philosophical Magazine*, vol. xiii. (1857), pp. 507–509.]

It is well known that the eight circles, each of which touches three given circles, are determined as follows: viz. considering any one in particular of the four axes of similitude of the given circles, and the perpendicular let fall from the radical centre (or centre of the orthotomic circle) of the given circles, there are two of the required tangent circles which have their centres upon the perpendicular, and pass through the points of intersection of the orthotomic circle and the axis of similitude, or in other words, the axis of similitude is a common chord (or radical axis) of the orthotomic circle and the two tangent circles.

This suggests the choice of the radical centre for the origin of coordinates; and the resulting formulae then take very simple forms, and the theorem is verified without difficulty.

Take then the centre of the orthotomic circle as the origin of coordinates, and let the radius of this circle be put equal to unity; then if (α, β), (α', β'), (α'', β'') are the coordinates of the centres of the given circles, the equations of these will be

$$\begin{gathered}
x^2 + y^2 + 1 - 2\alpha x - 2\beta y = 0,\\
x^2 + y^2 + 1 - 2\alpha' x - 2\beta' y = 0,\\
x^2 + y^2 + 1 - 2\alpha'' x - 2\beta'' y = 0;
\end{gathered}$$

and the radii of the circles will be $\sqrt{\alpha^2 + \beta^2 - 1}$, $\sqrt{\alpha'^2 + \beta'^2 - 1}$, $\sqrt{\alpha''^2 + \beta''^2 - 1}$.

It will be convenient to write

$$\gamma = \sqrt{\alpha^2 + \beta^2 - 1}, \quad \gamma' = \sqrt{\alpha'^2 + \beta'^2 - 1}, \quad \gamma'' = \sqrt{\alpha''^2 + \beta''^2 - 1},$$

where the three several signs + are fixed once for all in a determinate manner. If, however, all the signs are reversed, the result is the same, so that the system is one of four (not of eight) different systems.

The coordinates of a centre of similitude of the second and third circles are

$$\frac{\alpha'\gamma'' - \alpha''\gamma'}{\gamma' - \gamma''}, \quad \frac{\beta'\gamma'' - \beta''\gamma'}{\gamma' - \gamma''};$$

and forming the corresponding expressions for the coordinates of the centres of similitude of the third and first circles, and of the first and second circles, the three centres of similitude lie on a line which will be an axis of similitude: to find the equation, write

$$\begin{aligned}
A &= \beta'\gamma'' - \beta''\gamma' + \beta''\gamma - \beta\gamma'' + \beta\gamma' - \beta'\gamma, \\
B &= \gamma'\alpha'' - \gamma''\alpha' + \gamma''\alpha - \gamma\alpha'' + \gamma\alpha' - \gamma'\alpha, \\
C &= \alpha'\beta'' - \alpha''\beta' + \alpha''\beta - \alpha\beta'' + \alpha\beta' - \alpha'\beta, \\
V &= \alpha\beta'\gamma'' - \alpha\beta''\gamma' + \alpha'\beta''\gamma - \alpha'\beta\gamma'' + \alpha''\beta\gamma' - \alpha''\beta'\gamma,
\end{aligned}$$

values which, it will be observed, give

$$A\alpha+B\beta+C\gamma = V, \quad A\alpha'+B\beta'+C\gamma' = V, \quad A\alpha''+B\beta''+C\gamma'' = V;$$

then the equation of the axis of similitude is found to be

$$Ax + By - V = 0;$$

and hence the equation of the perpendicular let fall from the radical centre upon the axis of similitude is

$$Bx - Ay = 0.$$

It should therefore be possible to find two circles having their centres on the last-mentioned line and touching the three given circles.

Take $A\theta$, $B\theta$ as the coordinates of the centre of one of the two circles, and let r be its radius; the conditions of tangency are

$$r = \pm\sqrt{(A\theta-\alpha)^2 + (B\theta-\beta)^2} \mp \gamma,$$

where the sign $\pm$ has the same value in each expression. We have consequently

$$(r+\gamma)^2 = (A\theta-\alpha)^2 + (B\theta-\beta)^2;$$

or, observing that $A\alpha + B\beta = V - C\gamma$, and reducing,

$$r^2 - (A^2+B^2)\theta^2 + 2V\theta - 1 = 2\gamma(r + C\theta).$$

Forming the two analogous equations, the three equations will be satisfied if only

$$r^2 - (A^2+B^2-C^2)\theta^2 + 2V\theta - 1 = 0, \quad r + C\theta = 0.$$

Eliminating r, we have

$$(A^2+B^2-C^2)\theta^2-2V\theta+1=0,$$

which gives for the two values

$$(A^2+B^2-C^2)\theta=V\pm\sqrt{V^2-(A^2+B^2-C^2)},$$

and then r is determined linearly by the equation $r=\mp C\theta$.

The equation of the tangent circle is therefore

$$x^2+y^2-1-2\theta(Ax+By-V)=0;$$

or reducing,

$$x^2+y^2-1-2\theta(Ax+By-V)=0;$$

and recollecting that $Ax+By-V=0$ is the equation of the axis of similitude, the equation shows that the axis of similitude is a common chord or radical axis of the orthotomic circle and the two tangent circles.

2, Stone Buildings,
W.C., May 15, 1857.

C. III. 33

0.7 Note on the Equipotential Curve $\frac{m}{r}+\frac{m'}{r'}=C$

[From the *Philosophical Magazine*, vol. xiv. (1857), pp. 142–146.]

The equation $\frac{m}{r}+\frac{m'}{r'}=C$, where m, m', C are constants, and r, r' are the distances of a point P of the locus from two given points M, M' respectively, expresses that the potential of the attracting or repelling masses m, m' has a constant value at all points of the locus.

The locus is obviously a surface of revolution, having the line through the points M, M' for its axis; and instead of the surface, we may consider the section by a plane through the axis, or what is the same thing, we may consider r, r' as the distances in piano of a point P of the curve from the given points M, M': such curve may be termed the equipotential curve.

I propose in the present Note to investigate in a general manner, and without entering into any analytical detail, the general form of the curve corresponding to different values of the quantity C.

It is proper to remark, that the curve is not altered by changing the signs of each or any of the quantities m, m', C (in fact, analytically the distances r, r' are essentially ambiguous in sign), so that we may without loss of generality consider m, m', C as all of them positive.

The different branches of the complete analytical or geometrical curve have distinct mechanical significancies; thus, r, r' being positive, $\frac{m}{r}+\frac{m'}{r'}=C$ is the curve for which the potential of the attracting masses m, m' is equal to C; but $\frac{m}{r}-\frac{m'}{r'}=C$ is the curve for which the attracting mass m, and the repulsive mass m', have the potential C; but this is a distinction to which I do not attend.

I write $\frac{m}{a}=k$ instead of C, where a is the distance between the points M, M'; the equation thus becomes

$$\frac{m}{r}+\frac{m'}{r'}=\frac{k}{a},$$

where a is a positive distance, m, m', k may be considered as positive abstract numbers.

The curve is obviously a curve of the eighth order.

When k is large in comparison with m, m', then since r, r' cannot be both of them small in comparison of a (for if one be small, the other will be nearly equal to a), it is clear that one of these distances, for instance r, will be small, and the other, r', nearly equal to a.

We in fact have (neglecting in the first instance r in comparison with r') $\dfrac{m}{r} = \dfrac{k}{a}$, or more accurately, $\dfrac{m}{r} = \dfrac{k}{a} - \dfrac{m'}{r'} = \dfrac{k}{a} - \dfrac{m'}{a} = \dfrac{k-m'}{a}$, i.e. $r = \dfrac{ma}{k-m'}$, which shows that a part of the curve consists of two ovals, which are approximately concentric circles, radii $\sqrt{\dfrac{ma}{k-m'}}$, about the point M as centre.

In like manner a part of the curve consists of two ovals, which are approximately concentric circles, radii $\sqrt{\dfrac{m'a}{k-m}}$, about the point M' as centre.

I denote by A, B, the two ovals about M, viz. A is the exterior, and B the interior oval; and in like manner by A', B' the two ovals about M', viz. A' is the exterior, and B' the interior oval.

The distances inter se of the ovals A and B, or of the ovals A' and B', are small in comparison with the radii of these ovals respectively; and if, to fix the ideas, m be greater than m', then the ovals A', B' are greater than the ovals A, B.

It is easy to see that the curve will have a node or double point on the axis if $k = (\sqrt{m} \mp \sqrt{m'})^2$; and we must first consider the case $k = (\sqrt{m} + \sqrt{m'})^2$.

The node lies between the points M, M', and its distances from these points are respectively as $\sqrt{m} : \sqrt{m'}$, that is, it is nearest to M.

The transition from the original form is very obvious; the exterior ovals A, A' have gradually expanded until they come in contact, and at the instant of doing so the two ovals change themselves into a figure of eight, AA'.

The ovals B, B' also expand and change their form, but they preserve the general character of ovals enclosing the points M, M' respectively.

As k continues to diminish, the interior oval B' comes in contact with the exterior oval A, forming a node between the points M, M'; and at the instant of contact each of these ovals changes its form in the neighbourhood of the point of contact, the form of the portions which come into contact, so that the node is an ordinary double point.

The oval B' has, in fact, become what may be termed a re-entrant figure of eight, rO_1, the small part of which encloses the oval B which encloses the point M, while the large part encloses the point M'.

The curve consists of the exterior oval AA' (which has probably lost wholly or partially its hourglass form, and is more nearly an ordinary oval), of the re-entrant figure of eight, B', and of the enclosed oval B.

As k continues to diminish, the re-entrant figure of eight, B', breaks up into two detached ovals lB', mB', the larger of which, lB', encloses the other one and also the point M'; while the smaller one, mB', does not enclose M', but encloses the oval B which encloses M; the curve consists of the exterior oval AA', the ovals lB' and mB' which have arisen out of the oval B', and the oval B.

As k further decreases, the ovals AA' and lB' continually increase in magnitude, and the ovals mB' and B approximate more and more nearly together; and at length, when k becomes $= 0$, the ovals AA' and lB' disappear at infinity, while the ovals mB' and B unite themselves into a circle enclosing M', but not enclosing M: the equation of this circle is, in fact, $\frac{m'}{r'} = 0$; or what is the same thing, $r'^2 = \frac{m'}{m}a^2 - r^2$, and the points M, M' have, in relation to this circle, the well-known relation that each is the image of the other.

The preceding description is, I think, intelligible without the assistance of a series of figures illustrating the different forms of the curve, but there is no difficulty in actually tracing the curve for any particular values of the constant parameters.

Thus (taking the distance MM' for unity) suppose that the equation of the curve is $\frac{1}{r} + \frac{4}{r'} = 12$; (the value 12 was selected as a value not far from that for which the oval B' becomes a re-entrant figure of eight, though the change of form is so rapid that only the incipient tendency of the oval B' to take the form in question).

The form of the portion of the curve consisting of the two ovals B, B' will be that shown by the figure, which was constructed by points on a double scale with some accuracy.

The case $m = m'$ is an exception, and must be considered separately: the curve is here in all its changes symmetrical about a perpendicular to the axis midway between the two centres M, M'.

The curve in the first instance, i.e. when k is greater than $(\sqrt{m} + \sqrt{m'})^2 = 4m$, consists of the two ovals B, A about M, and the two ovals B', A' about M'.

As k decreases to $4m$, the two ovals A, A' gradually increase in magnitude, and at length come together, as before, into a figure of eight, AA'; and as k continues to diminish, the figure of eight opens out into an hourglass form AA', which continues increasing in magnitude, and degenerating into the form of an oval.

The interior ovals B, B' approach more and more nearly together, lengthen out in the direction perpendicular to the axis, and present to each other a more and more flattened portion.

The second value, $k = (\sqrt{m} - \sqrt{m'})^2$, which in the general case gives a node, in the present case only arises when $k = 0$; and there is not then any node, but the curve degenerates in a similar manner to what happens for $k = 0$ in the general case; viz. the oval AA' disappears at infinity, while the ovals B, B' coalesce together (their outer parts disappearing at infinity) into a pair of lines coincident with the perpendicular to the axis midway between the two centres.

2, Stone Buildings,
May 31, 1857.

C. III. 35

0.8 A Demonstration of Sir W. R. Hamilton's Theorem of the Isochronism of the Circular Hodograph

[From the *Philosophical Magazine*, vol. xiv. (1857), pp. 427–430.]

Imagine a body moving in piano under the action of a central force, and let h denote, as usual, the double of the area described in a unit of time; let P be any point of the orbit, then measuring off, on the perpendicular let fall from the centre of force on the tangent at P to the orbit, a distance OQ equal or proportional to h into the reciprocal of the perpendicular on the tangent, the locus of Q is the hodograph, and the points P, Q are corresponding points of the orbit and hodograph.

It is easy to see that the hodograph is the polar reciprocal of the orbit with respect to a circle having for its centre, and having its radius equal or proportional to $\sqrt{h}$.

And it follows at once that Q is the pole, with respect to this circle, of the tangent at P to the orbit.

In the particular case where the force varies inversely as the square of the distance, the hodograph is a circle.

And if we consider two elliptic orbits described about the same centre, under the action of the same central force, and such that the major axes are equal, then (as will be presently seen) the common chord or radical axis of the two hodographs passes through the centre of force.

Imagine an orthotomic circle of the two hodographs (the centre of this circle is of course on the common chord or radical axis of the two hodographs), and consider the arcs intercepted on the two hodographs respectively by the orthotomic circle; then the theorem of the isochronism of the circular hodograph is as follows, viz. the times of hodographic description of the intercepted arcs are equal; in other words, the times of description in the orbits, of the arcs which correspond to the intercepted arcs of the hodographs, are equal.

It was remarked by Sir W. R. Hamilton, that the theorem is in fact equivalent to Lambert's theorem, that the time depends only on the chord of the described arc and the sum of the two radius vectors.

And this remark suggests a mode of investigation of the theorem.

Consider the intercepted arc of one of the hodographs: the tangents to the hodograph at the extremities of this arc are radii of the orthotomic circle; i.e. the corresponding arc of the orbit is the arc cut off by the polar (in respect to the directrix circle by which the hodograph is determined) of the centre of the orthotomic circle; the portion of this polar intercepted by the orbit is the elliptic chord, and this elliptic chord and the sum of the radius vectors at the two extremities of the elliptic chord determine the time of description of the arc; and the values of these quantities, viz. the elliptic chord and the sum of the radius vectors, must be the same in each orbit.

The analytical investigation is not difficult.

I take as the equation of the first orbit,

$$r = \frac{a(1-e^2)}{1+e\cos(\theta-\varpi)},$$

then the polar of the orbit with respect to a directrix circle $r = c$ is

$$\frac{r^2}{c^2} - \frac{2r\cos(\theta-\varpi)}{a(1-e^2)} - \frac{1}{a(1-e^2)} = 0,$$

and putting $c^2 = k \cdot k\sqrt{a(1-e^2)}$ (where k is a constant quantity, i.e. it is the same in each orbit), the equation becomes

$$\frac{ek\sqrt{1-e^2}}{r} - \frac{2r\cos(\theta-\varpi)}{a(1-e^2)} - \frac{1}{a(1-e^2)} = 0.$$

But since a is supposed to be the same in each orbit, we may for greater simplicity write $k^2 = ma$; it will be convenient also to put $e = \sin\kappa$; we have then

$$r = \frac{a\cos^2\kappa}{1+\sin\kappa\cos(\theta-\varpi)}$$

for the equation of the orbit,

$$r^2 = ma\cos\kappa$$

for the equation of the directrix circle, and

$$\frac{r^2}{m} = \frac{ma\tan\kappa\cos(\theta-\varpi)}{1}$$

for the equation of the hodograph.

We have in like manner

$$r = \frac{a\cos^2\kappa'}{1+\sin\kappa'\cos(\theta-\varpi')}$$

for the equation of the second orbit,

$$r^2 = ma\cos\kappa'$$

for the equation of the corresponding directrix circle, and

$$\frac{r^2}{m} = \frac{ma\tan\kappa'\cos(\theta-\varpi')}{1}$$

for that of the hodograph.

The equations of the two hodographs give at once

$$\tan\kappa\cos(\theta-\varpi) = \tan\kappa'\cos(\theta-\varpi')$$

for the equation of the common chord or radical axis of the two hodographs, an equation which shows that, as already noticed, the common chord passes through the origin or centre of force.

This equation gives $\theta = \alpha$ if

$$\tan\kappa\cos(\alpha-\varpi) = \tan\kappa'\cos(\alpha-\varpi'),$$

i.e. α is a quantity such that the expressions $\tan\kappa\cos(\alpha-\varpi)$ and $\tan\kappa'\cos(\alpha-\varpi')$, which correspond to each other in the two orbits, are equal.

We may take R, α as the polar coordinates of the centre of the orthotomic circle (where R is arbitrary); the equation of the polar of this point with respect to the directrix circle $r^2 = ma\cos\kappa$, is then at once seen to be

$$r\cos(\theta-\alpha) = \frac{ma\cos\kappa}{R},$$

which is the equation of the line cutting off the arc of the elliptic orbit

$$r = \frac{a\cos^2\kappa}{1+\sin\kappa\sin(\theta-\varpi)}.$$

Writing $\theta-\varpi = \theta-\alpha+(\alpha-\varpi)$, the two equations give

$$\cos(\theta-\alpha) = \frac{A}{r},$$

if for shortness

$$A = \frac{ma\cos\kappa}{R}, \quad B = \frac{ma\cos\kappa\cos(\alpha-\varpi)}{a\cos^2\kappa} = \frac{m\tan\kappa\cos(\alpha-\varpi)}{\sin\kappa\sin(\alpha-\varpi)}, \quad C = \sin\kappa\sin(\alpha$$

we have therefore

$$\frac{A}{r} = \cos(\theta-\alpha),$$

or, what is the same thing,

$$(1-C^2)r^2 - 2BCr - (A^2+B^2) = 0;$$

and thence, if r', r'' are the two values of r,

$$r'+r'' = \frac{2BC}{1-C^2}, \quad r'r'' = -\frac{A^2+B^2}{1-C^2}.$$

Let θ', θ'' be the corresponding values of θ, we have

$$\theta'-\theta'' = \theta'-\alpha-(\theta''-\alpha),$$

and thence

$$\begin{aligned}\cos(\theta'-\theta'') &= \frac{A^2}{r'r''} + \left(\frac{B}{r'}+C\right)\left(\frac{B}{r''}+C\right)\\ &= \frac{A^2}{r'r''} + BC\left(\frac{1}{r'}+\frac{1}{r''}\right) + C^2,\end{aligned}$$

or adding unity to each side, multiplying by $r'r''$, and on the right-hand side substituting for $r'+r''$, $r'r''$ their values

$$(r'+r'')^2 - 2r'r''(1+\cos(\theta'-\theta''));$$

hence to prove the theorem, it is only necessary to show that $r'+r''$ and $r'r''(1+\cos(\theta'-\theta''))$ have the same values in each orbit, that is, that $\dfrac{2BC}{1-C^2}$ and $\dfrac{2C^2A^2}{1-C^2}$ have the same values in each orbit.

But observing that

$$1-\sin^2\kappa\sin^2(\alpha-\varpi) = \cos^2\kappa+\sin^2\kappa\cos^2(\alpha-\varpi) = \cos^2\kappa\{1+\tan^2\kappa\cos^2(\alpha-\varpi)\},$$

the values of these expressions are respectively

$$\frac{2a(m\tan\kappa\cos(\alpha-\varpi)-R)}{R^2}, \quad \frac{2a^2\{1+\tan^2\kappa\cos^2(\alpha-\varpi)\}}{R^2},$$

which contain only the quantities m, α, R, $\tan\kappa\cos(\alpha-\varpi)$, which are the same for each orbit, and the theorem is therefore proved, viz. it is made to depend on Lambert's theorem.

I may remark, that a geometrical demonstration which does not assume Lambert's theorem is given by Mr Droop in his paper "On the Isochronism of the Circular Hodograph," Quarterly Mathematical Journal, vol. I. [1857] pp. 374–378, where the dependence of the theorem on Lambert's theorem is also shown.

By what precedes, the theorem may be stated in a geometrical form as follows:

"Imagine two ellipses having a common focus, and their major axes equal; describe about the focus two directrix circles having their radii proportional to the square roots of the minor axes of the ellipses respectively; the polar reciprocal of each ellipse in respect to its own directrix circle will be a circle (the hodograph), and the common chord or radical axis of the two hodographs will pass through the focus.

Consider any point on the common chord, and take the polar with respect to each directrix circle; such polar will cut off an arc of the corresponding ellipse; and then, theorem, the elliptic chord, and the sum of the radius vectors through the two extremities of the chord, will be respectively the same for each ellipse."

2, Stone Buildings,
W.C., June 24, 1857.

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0.9 On the Cubic Transformation of an Elliptic Function

[From the *Philosophical Magazine*, vol. xv. (1858), pp. 363–364.]

Let

$$z = \frac{(a', b', c', d' \mid x, 1)^3}{(a, b, c, d \mid x, 1)^3}$$

be any cubic fraction whatever of x, then it is always possible to find quartic functions of z, x respectively, such that

$$\frac{dz}{dx} = \frac{\sqrt{(a', b', c', d', e' \mid z, 1)^4}}{\sqrt{(a, b, c, d, e \mid x, 1)^4}}.$$

This depends upon the following theorem, viz. putting for shortness,

$$U = (a, b, c, d \mid x, y)^3, \quad U' = (a', b', c', d' \mid x, y)^3,$$

and representing by the notation

$$\text{disct.}(\alpha U - \alpha' U', \beta U - \beta' U', \gamma U - \gamma' U', \delta U - \delta' U');$$

or more shortly by disct. $(\alpha U - \alpha' U', \ldots)$, the discriminant in regard to the facients (λ, μ) of the cubic function

$$(\alpha U - \alpha' U', \beta U - \beta' U', \gamma U - \gamma' U', \delta U - \delta' U' \mid \lambda, \mu)^3;$$

or what is the same thing, the cubic function

$$(\alpha, \beta, \gamma, \delta \mid \lambda, \mu)^3 \cdot (a', b', c', d' \mid x, y)^3 - (\alpha', \beta', \gamma', \delta' \mid \lambda, \mu)^3 \cdot (a, b, c, d \mid x, y)^3;$$

and by $J(U, U')$ the functional determinant, or Jacobian, of the two cubics U, U'; the theorem is that the discriminant contains as a factor the square of the Jacobian, or that we have

$$\text{disct.}\,(\alpha U - \alpha' U', \ldots) = \{J(U, U')\}^2 \cdot (A, B, C, D, E \mid x, y)^4.$$

For assuming this to be the case, then (disregarding a mere numerical factor) we have

$$U\,dU' - U'\,dU = J(U, U')(y\,dx - x\,dy),$$

and the two equations give

$$\frac{U\,dU' - U'\,dU}{y\,dx - x\,dy} = \sqrt{(A, B, C, D, E \mid x, y)^4},$$

whence writing z for $U' \div U$, and putting y equal to unity, we have

$$\frac{dz}{dx} = \frac{\sqrt{\text{disct.}\,(\alpha z - \alpha', \ldots)}}{\sqrt{(A, B, C, D, E \mid x, 1)^4}},$$

where disct. $(\alpha z - \alpha', \ldots)$, or at full length,

$$\text{disct.}\,(\alpha z - \alpha', \beta z - \beta', \gamma z - \gamma', \delta z - \delta'),$$

is a given quartic function of z, $= (a', b', c', d', e' \mid z, 1)^4$ suppose; and this proves the theorem of transformation.

The assumed subsidiary theorem may be thus proved: suppose that the parameter θ is determined so that the cubic may have a square factor, the cubic may be written

$$(a + \theta a', b + \theta b', c + \theta c', d + \theta d' \mid x, y)^3,$$

and the requisite condition is disct. $(a + \theta a', \ldots) = 0$; there are consequently four roots; and calling these θ_1, θ_2, θ_3, θ_4, we have identically

$$\text{disct.}\,(a + \theta a', \ldots) = K(\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

or what is the same thing,

$$\text{disct.}\,(\alpha U - \alpha' U', \ldots) = K(\alpha U' - \alpha' U)(\beta U' - \beta' U)(\gamma U' - \gamma' U)(\delta U' - \delta' U).$$

Now any double factor of U or U' (that is the linear factor which enters twice into U or U') is a simple factor of $J(U, U')$, and we have $J(U, U') = J(U, U' + \theta U)$, and consequently $J(U, U') = J(U + \theta' U', U' + \theta U)$; hence the double factors of each of the expressions $U + \theta_1 U'$, $U + \theta_2 U'$, $U + \theta_3 U'$, $U + \theta_4 U'$ are simple factors of $J(U, U')$, or what is the same thing, $J(U, U')$ is the product of four linear factors, which are respectively double factors of the product $(U + \theta_1 U')(U + \theta_2 U')(U + \theta_3 U')(U + \theta_4 U')$; or this product contains the factor $\{J(U, U')\}^2$, which proves the theorem.

2, Stone Buildings,
W.C., March 5, 1858.

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0.10 On a Theorem Relating to Hypergeometric Series

[From the *Philosophical Magazine*, vol. xvi. (1858), pp. 356–357.]

In attempting to verify a formula of Hansen's relating to the development of the disturbing function in the planetary theory, I was led to a theorem in hypergeometric series: viz. writing, as usual,

$$F(\alpha,\beta,\gamma,x)=1+\frac{\alpha\cdot\beta}{1\cdot\gamma}x+\frac{\alpha(\alpha+1)\,\beta(\beta+1)}{1\cdot 2\cdot\gamma(\gamma+1)}x^2+\&\text{c.},$$

then the product

$$(1-x)^{-\alpha}\cdot F(\alpha,\beta,\gamma,x)$$

is connected with

$$(1-x)^{-2\alpha}\cdot F(2\alpha,2\beta,\gamma,x)$$

by a simple relation: for if the last-mentioned expression is put equal to

$$1+A_1x+A_2x^2+A_3x^3+\&\text{c.},$$

then the product in question is equal to

$$1+\frac{\gamma-2\alpha}{\gamma-\alpha}A_1x+\frac{(\gamma-2\alpha)(\gamma-2\alpha+1)}{(\gamma-\alpha)(\gamma-\alpha+1)}A_2x^2+\&\text{c.}$$

The form of the identity thus arrived at will be best perceived by considering a particular case. Thus, comparing the coefficients of x^2, we have

$$\frac{1\cdot 2\cdot\gamma+\alpha\cdot\beta+1\cdot\gamma+1}{1\cdot 2\cdot\gamma\cdot\gamma+1}+\frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot 2\cdot\gamma\cdot\gamma+1}=\frac{\gamma-2\alpha}{\gamma-\alpha}\cdot\frac{2\alpha\cdot 2\alpha+1\cdot 2\beta\cdot 2\beta+1}{1\cdot 2\cdot\gamma\cdot\gamma+1},$$

which is easily verified.

0.11 A Memoir on the Problem of Disturbed Elliptic Motion

[From the *Memoirs of the Royal Astronomical Society*, vol. xxvii., 1859, pp. 1–29. Read March 9, 1858.]

I venture to take up the problem of disturbed elliptic motion, for the sake of a further elaboration of the analytical theory.

The points which present difficulty are the measurement of longitudes in the varying plane of the orbit, and (in the lunar theory) the determination of the position of the orbit by reference to the varying plane of the sun's orbit; it is, in memoirs and works on the lunar and planetary theories, often difficult to discover where or how (or whether at all) account is taken of these variations, and the analytical mode of treatment is for the most part very imperfect.

I must except always Hansen's *Fundamenta Nova* [investigationis orbitae verae quam Luna perlustrat, Gotha 1838] where the points referred to are treated in a perfectly rigorous manner.

There is, however, a want of clearness in the form under which his investigations are presented; and the comprehension of them is greatly facilitated by Jacobi's remarks, published under the title "Auszug zweier Schreiben des Prof. Jacobi an Herrn Director Hansen" (Crelle, t. XLII. pp. 1–31 (1851)).

Jacobi observes that the integration of Hansen's system of differential equations introduces seven arbitrary constants, which, in the expressions for the coordinates referred to fixed axes, reduce themselves to six.

The seventh constant, neglecting the disturbing forces, is in fact a constant which determines the position in the orbit of the arbitrary origin from which the longitudes in orbit are reckoned.

I have, in my paper "On Hansen's Lunar Theory," Quarterly Mathematical Journal, vol. I. pp. 112–125 (1855), [163], termed this origin "the departure-point," and longitudes measured from it "departures."

The seventh constant may be taken to be the departure of the node.

I reproduce in the present memoir the explanation of what is meant by the departure when the plane of the orbit is variable.

If the problem is treated by the method of the variation of the elements, the seventh constant becomes, like the other elements, variable; and we have thus a seventh variable element, the departure of the node.

The element just referred to (the departure of the node) forms, with the longitude of the node and the inclination, a group of three elements, which determine the position of the orbit and of the departure-point.

The coordinates of the planet are in the first instance taken to be the radius vector, longitude, and latitude; but the before-mentioned three elements being considered as given, the position of the planet depends only on the radius vector and the departure.

These may be then expressed in terms of the remaining four elements; as to the choice of these four elements, it is to be remarked that there is one element which only enters through the mean anomaly, and that there is great convenience in representing with Hansen the mean anomaly by a single letter; and that in the various formulae we may use, in the place of the element implicitly involved in the mean anomaly, the mean anomaly itself, or treat the mean anomaly as an element; the four elements may be taken to be the semi-axis major, the eccentricity, the mean anomaly, and the departure of the pericentre.

And joining to these the before-mentioned three elements, we have the system of elements represented in the memoir by a, e, g, ϖ, σ, θ, ϕ.

It has been assumed so far that the three elements determine the position of the orbit and departure-point in reference to a fixed plane and origin of longitudes; but we may suppose more generally that, instead of the fixed plane and origin of longitudes, we have a variable plane or orbit of reference and a departure-point in this variable orbit of reference.

The quantities which determine the orbit of reference and departure-point are naturally taken to be the departure of the node, longitude of the node, and inclination; these are assumed to be given functions of the time, and they are in the memoir represented by σ', θ', ϕ'.

The three elements of the planet's orbit (viz. departure of node, longitude of node, and inclination) in relation to the orbit of reference and departure-point therein, are in the memoir represented by Σ, Θ, Φ, and the system of elements ultimately adopted is therefore a, e, g, ϖ, Σ, Θ, Φ.

I obtain formulae for the variations of these elements under two different modes of expression of the disturbing function: first, when the disturbing function is expressed in terms of the radius vector and departure and of the three elements Σ, Θ, Φ: secondly, when the disturbing function is expressed in terms of the seven elements a, e, g, ϖ, Σ, Θ, Φ.

The establishment of the two sets of formulas just referred to constitutes the chief object of the memoir; but the memoir contains some other investigations and formulas in relation to the general subject.

The coordinates of the planet are r, the radius vector, v, the longitude, γ, the latitude.

The equations of motion are

$$\frac{d^2r}{dt^2} - r\left(\frac{dv}{dt}\right)^2 \cos^2\gamma - r\left(\frac{d\gamma}{dt}\right)^2 = \frac{d\Omega}{dr},$$
$$\frac{d}{dt}\left(r^2\cos^2\gamma\,\frac{dv}{dt}\right) = \frac{d\Omega}{dv},$$
$$\frac{d}{dt}\left(r^2\frac{d\gamma}{dt}\right) + r^2\sin\gamma\cos\gamma\left(\frac{dv}{dt}\right)^2 = \frac{d\Omega}{d\gamma},$$

where Ω is regarded as a function of r, v, γ, or (as this may be expressed) where $\Omega = \Omega(r, v, \gamma)$.

If we neglect the disturbing forces, the planet moves in an ellipse; and taking a to represent the semi-axis major, the mean motion will be n.

The mean anomaly, which I call g, will be a function of the form $nt + \epsilon$; but as ϵ only enters through g, it will be convenient to use the mean anomaly g (considered as implicitly involving an arbitrary constant ϵ) in the place of an element, and I write

- a, the semi-axis major,
- e, the eccentricity,
- g, the mean anomaly,
- θ, the longitude of node,
- ϕ, the inclination,
- C, the distance of pericentre from node.

I assume also

- f, the true anomaly,
- z, the distance of planet from node,

- x, the reduced distance from node.

We have then r and f given functions of t and the elements, viz. we may write

$$r = a \cdot \mathrm{elqr}(e, g), \quad f = \mathrm{elta}(e, g),$$

(read elqr. elliptic quotient radius, and elta. elliptic anomaly).

These values satisfy

$$r = \frac{a(1-e^2)}{1+e\cos f}.$$

Moreover z, x, y, are the hypothenuse, base, and perpendicular of a right-angled spherical triangle, the base angle whereof is ϕ; the equations which connect these quantities are therefore

$$\tan x = \tan z \cos\phi, \quad \sin\gamma = \sin z \sin\phi, \quad \tan\gamma = \sin x \tan\phi, \quad \cos z = \cos x \cos\gamma,$$

equivalent, of course, to two equations.

The first and second of them give in fact x, γ, in terms of z and ϕ.

The value of z is $z = f + C$, so that x and γ are given functions, and the longitude v is given in terms of x and θ by the equation $v = x + \theta$, and consequently the three coordinates r, v, γ, are by the system of equations given in terms of t and the elements.

From the equations which connect z, x, γ, ϕ, treating all these quantities as variable we deduce

$$\begin{aligned}
\sec^2 x\, dx &= \cos\phi \sec^2 z\, dz - \tan z \sin\phi\, d\phi,\\
\cos\gamma\, d\gamma &= \sin\phi\cos z\, dz + \sin z\cos\phi\, d\phi,\\
\sec^2\gamma\, d\gamma &= \tan\phi\cos x\, dx + \sin x \sec\phi\, d\phi,\\
\sin z\, dz &= \cos\gamma \sin x\, dx + \cos x \sin\gamma\, d\gamma,
\end{aligned}$$

equivalent of course to two equations; and the system is easily reduced to the more convenient form

$$\begin{aligned}
dx &= \cos\phi\sec^2\gamma\, dz - \tan z\cos^2 x\sin\phi\, d\phi,\\
d\gamma &= \sin\phi\cos x\, dz + \cos x\tan z\cos\phi\, d\phi,\\
dx &= \cot\phi\sec x\sec^2\gamma\, d\gamma - \tan z\phi\, d\phi,\\
d\phi &= \cos\phi\, dx + \cos x\sin\phi\, dy,
\end{aligned}$$

joining to these equations the

$$dz = df, \quad dv = dx + d\theta,$$

and considering at present the mere analytical forms, first if $d\phi = 0$, $dC = 0$, we have

$$dx = \cos\phi \sec^2\gamma\, dz, \quad d\gamma = \sin\phi \cos x\, dz, \quad dz = df, \quad dv = dx.$$

Next, if $d\gamma = 0$, $dv = 0$, we have

$$dx = \tan z\phi\, d\phi, \quad dz = -\tan z \cot\phi\, d\phi, \quad df = \cos\phi\, dx, \quad dx = d\theta, \quad dz{+}\cos\phi\, d\theta =$$

I remark also that the equation, $\tan\gamma = \sin x \tan\phi$, may be written in the form

$$\cos^2\phi \sec^2\gamma + \sin^2\phi \cos^2 x = 1.$$

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The equations $r = a \cdot \mathrm{elqr}(e, g)$, $f = \mathrm{elta}(e, g)$, treating all the quantities as variable, give

$$\begin{aligned}
dr &= -\frac{na^2 e \sin f}{\sqrt{1-e^2}}\, dg + \frac{a(2e - 1 + e^2 \cos f)}{(1 + e\cos f)^2}\, de,\\
df &= \frac{na^2\sqrt{1-e^2}}{r^2}\, dg + \frac{\sin f(2 + e\cos f)}{1 + e\cos f}\, de,
\end{aligned}$$

to which is to be joined

$$\frac{dr}{r} = \frac{ae(1-e^2)\sin f}{(1+e\cos f)^2}\, dg + \frac{1-e^2}{1+e\cos f}\, da + \frac{a(2e-1+e^2\cos f)}{(1+e\cos f)^2}\, de,$$

all which formulae will be useful.

If we treat the elements as constant, then in the foregoing expressions for dr and df, we must attend only to the part involving dg, and must put this equal to $n\,dt$; the values first obtained for dx, $d\gamma$, dz, dv, correspond to this assumption, and we have

$$\begin{aligned}
\frac{dr}{dt} &= \frac{nae\sin f}{\sqrt{1-e^2}}, & \frac{df}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2},\\
\frac{dz}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}, & \frac{dx}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}\cos\phi\sec^2\gamma,\\
\frac{d\gamma}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2}\sin\phi\cos x, & \frac{dv}{dt} &= \frac{na^2\sqrt{1-e^2}}{r^2},
\end{aligned}$$

and we then deduce

$$\begin{aligned}
\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\cos^2\gamma\left(\frac{dv}{dt}\right)^2 - r\left(\frac{d\gamma}{dt}\right)^2 &= \frac{-n^2a^4(1-e^2)}{r^3},\\
\frac{d}{dt}\left(r^2\cos^2\gamma\,\frac{dv}{dt}\right) &= 0,\\
\frac{d}{dt}\left(r^2\frac{d\gamma}{dt}\right) + r^2\sin\gamma\cos\gamma\left(\frac{dv}{dt}\right)^2 &= \frac{-2n^2a^4(1-e^2)\cos f}{r^3},
\end{aligned}$$

values which satisfy the undisturbed equations.

The disturbed equations may be dealt with in the usual manner by the method of the variation of the elements, and attending only to the variations of the elements we have

$$dr = 0, \qquad dv = 0, \quad d\gamma = 0,$$

$$\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\cos^2\gamma\left(\frac{dv}{dt}\right)^2 - r\left(\frac{d\gamma}{dt}\right)^2 = \frac{d\Omega}{dr}\,dt,$$

$$\frac{d}{dt}\left(r^2\cos^2\gamma\,\frac{dv}{dt}\right) = \frac{d\Omega}{dv}\,dt,$$

$$\frac{d}{dt}\left(r^2\frac{d\gamma}{dt}\right) + r^2\sin\gamma\cos\gamma\left(\frac{dv}{dt}\right)^2 = \frac{d\Omega}{d\gamma}\,dt,$$

or, what is the same thing,

$$dr = 0, \qquad dv = 0, \qquad d\gamma =$$

$$-\frac{nae\sin f}{\sqrt{1-e^2}}\,dg = \frac{d\Omega}{dr}\,dt, \quad \frac{na^2\sqrt{1-e^2}}{r^2}\cos\phi\,dg = \frac{d\Omega}{dv}\,dt, \quad \frac{na^2\sqrt{1-e^2}}{r^2}\sin\phi\cos x\,dg =$$

where as before $\Omega = \Omega(r, v, \gamma)$.

In virtue of the relations $dv = 0$, $d\gamma = 0$, we have the above-mentioned equations,

$$dx = \tan z\phi\,d\phi, \quad dz = -\tan z\cot\phi\,d\phi, \quad df = \cos\phi\,dx, \quad dx = d\theta, \quad dz + \cos\phi\,d\theta =$$

we have

$$d(\sin\phi\cos x) = \sin\phi\sin x\,dx + \cos x\cos\phi\,d\phi = \cos x\cos\phi\sec^2 z\,d\phi = \sec x\cos\phi\sec^2\gamma$$

and the last two equations for the variations become

$$d\frac{na^2\sqrt{1-e^2}}{r^2}\cos\phi\,dg - na^2\sqrt{1-e^2}\sin\phi\cos x\,d\phi = \frac{d\Omega}{dv}\,dt,$$

$$d\frac{na^2\sqrt{1-e^2}}{r^2}\sin\phi\cos x\,dg + na^2\sqrt{1-e^2}\sec x\cos\phi\sec^2\gamma\,d\phi = \frac{d\Omega}{d\gamma}\,dt,$$

and attending to the equations $\cos^2\phi\sec^2\gamma + \sin^2\phi\cos^2 x = 1$ we deduce at once

$$d\frac{na^2\sqrt{1-e^2}}{r^2} = \cos\phi\sec^2\gamma\,\frac{d\Omega}{dv}\,dt + \sin\phi\cos x\,\frac{d\Omega}{d\gamma}\,dt,$$

$$(1-e^2)\,d\phi = \frac{-\sin\phi\cos^2 x}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{dv}\,dt + \frac{\cos\phi\cos x}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\gamma}\,dt.$$

Now the position of the planet may be determined by the quantities r, z, θ, ϕ, or we may consider Ω as a function of the last-mentioned quantities.

And if on the right-hand side $\Omega = \Omega(r, v, \gamma)$ as before, the formulas of transformation are

$$\frac{d\Omega}{dr}\,dr + \frac{d\Omega}{dv}\,dv + \frac{d\Omega}{d\gamma}\,d\gamma = \frac{d\Omega}{dr}\,dr + \frac{d\Omega}{dz}\,dz + \frac{d\Omega}{d\theta}\,d\theta + \frac{d\Omega}{d\phi}\,d\phi,$$

where

$$dv = \cos\phi\sec^2\gamma\,dz - \tan z\cos^2 x\sin\phi\,d\phi + d\theta, \quad d\gamma = \sin\phi\cos x\,dz + \tan z\cos x\cos\phi$$

and we have

$$\frac{d\Omega}{d\phi} = \tan z\sin\phi\cos^2 x\,\frac{d\Omega}{dv} + \cos\phi\cos x\,\frac{d\Omega}{d\gamma}, \quad \frac{d\Omega}{dz} = \cos\phi\sec^2\gamma\,\frac{d\Omega}{dv} + \sin\phi\cos x\,\frac{d\Omega}{d\gamma},$$

where on the left-hand side $\Omega = \Omega(r, z, \theta, \phi)$; and these equations give

$$\cot z\,\frac{d\Omega}{d\phi} = \cot\phi\,\frac{d\Omega}{d\theta} - \phi\,\frac{d\Omega}{dv},$$

an equation which is satisfied by $\Omega = \Omega(r, z, \theta, \phi)$.

We have thus

$$\begin{aligned}
dr &= 0,\\
dv &= 0,\\
d\gamma &= 0,\\
-\frac{nae\sin f}{\sqrt{1-e^2}}\,dg &= \frac{d\Omega}{dr}\,dt,\\
\cot z\,d\phi &= -\frac{1}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\theta}\,dt,\\
\frac{na^2\sqrt{1-e^2}}{r^2}\,dg &= \frac{d\Omega}{dz}\,dt,
\end{aligned}$$

which may be replaced by

$$\begin{aligned}
dr &= 0,\\
\phi\, d\Theta &= \frac{1}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\theta}\,dt,\\
\cot\phi\, d\Theta &= \frac{-1}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\theta}\,dt,\\
-\frac{nae\sin f}{\sqrt{1-e^2}}\,dg &= \frac{d\Omega}{dr}\,dt,\\
\frac{na^2\sqrt{1-e^2}}{r^2}\,dg &= \frac{d\Omega}{dz}\,dt,\\
\cot z\, d\phi &= -\frac{1}{na^2\sqrt{1-e^2}}\,\frac{d\Omega}{d\theta}\,dt,
\end{aligned}$$

where as before $\Omega = \Omega(r, z, \theta, \phi)$.

I remark that in the case of any central force whatever, we have an element h corresponding to $na^2\sqrt{1-e^2}$ in the elliptic theory, and the system for the variations is

$$\begin{aligned}
&dr = 0, && \phi\, d\Theta = \frac{1}{h}\,\frac{d\Omega}{d\theta}\,dt, && \cot\phi\, d\Theta = \frac{-1}{h}\,\frac{d\Omega}{d\theta}\,dt,\\
&\cot z\, d\phi = \frac{-1}{h}\,\frac{d\Omega}{d\theta}\,dt, && dh = \frac{d\Omega}{dr}\,dt, && \frac{h}{r^2}\,dg = \frac{d\Omega}{dz}\,dt,
\end{aligned}$$

where $\Omega = \Omega(r, z, \theta, \phi)$.

Imagine a point in the orbit, which I call the departure-point, the angular distances from this point are termed departures.

And I write

- P, the departure of planet,
- ϖ, the departure of pericentre,
- σ, the departure of node,
- $\Sigma = \mathrm{P} - \varpi$,

I write also δ, the longitude in orbit of departure-point, or, as it may be termed, the adjustment; that is

$$\delta = \varpi - \sigma.$$

In the undisturbed motion the departure-point is simply a fixed point in the orbit, but when the orbit is variable, the departure-point is taken to be the point of intersection of the orbit with any orthogonal trajectory of the successive positions of the orbit, a definition which is expressed analytically by the equation,

$$d\sigma = \cos\phi\, d\theta.$$

The equation, $z = \mathrm{P} - \sigma$, gives

$$dz = d\mathrm{P} - d\sigma = d\mathrm{P} - \cos\phi\, d\theta,$$

or, what is the same thing,

$$d\mathrm{P} = dz + \cos\phi\, d\theta.$$

But we have $dz + \cos\phi\, d\theta = 0$, and consequently $d\mathrm{P} = 0$, an equation which expresses that the increment of departure, in so far as such increment arises from the variation of the elements, is equal to zero.

Or, what is the same thing, the total increment of departure is equal to the infinitesimal angle between two consecutive radius vectors of the planet.

I propose to consider the departure-point as a point which is constantly defined as above, viz., when the orbit is variable, the departure-point is the point of intersection of the orbit with any orthogonal trajectory of the successive positions of the orbit; and as a particular case of the definition, when the orbit is fixed, the departure-point is simply a fixed point on the orbit.

The orbit here considered is that of the planet and the position of the planet is determined by the departure and radius vector (the latitude being zero), and this is assumed to be the case whenever the departure is spoken of, and it is such departure which is denoted by the letter P.

But we might consider a departure-point (defined as above), upon any other orbit whatever, and use such departure-point as an origin of longitude (for instance, in the lunar theory we might consider a longitude measured along the variable plane of the

Collected Mathematical Papers of Arthur Cayley, Vol. III, Pages 301–350

0.12 Paper 213: On the Development of the Disturbing Function in the Lunar Theory

[From the Monthly Notices of the Royal Astronomical Society, vol. xviii. (1858), pp. 199–204.]

The disturbing function is here developed as far as the fifth order inclusive (the eccentricities and inclination being treated as of the same order), for the special case of the moon; the development is in fact that of the function

$$\Omega = m' \frac{a^2}{a'^3} \left(\frac{3}{2}\eta^2 - \frac{3}{2}\eta^4 \right)$$

multiplied into

$$\frac{a'^3}{r'^3} \cdot \frac{r^2}{a^2} \left(\frac{r}{a} \right)^n \cos j(U - U'),$$

where n is $=$ 0, 1, 2, 3, or 4; or say it is the development of

$$\Omega_n = m' \frac{a^2}{a'^3} \left(\frac{3}{2}\eta^2 - \frac{3}{2}\eta^4 \right) \text{ multiplied into } \frac{a'^3}{r'^3} \cdot \frac{r^2}{a^2} \left(\frac{r}{a} \right)^n \cos j(U - U').$$

0.12.1 Development of Ω_3 (p. 301)

$\Omega_3 = m' \dfrac{a^2}{a'^3} \left(\dfrac{3}{2}\eta^2 - \dfrac{3}{2}\eta^4 \right)$ multiplied into:

	Coefficient		Cosine
Lunar part infra.	$\frac{1773}{256}e'^5$	cos	$-5g'$
	$\frac{77}{16}e'^4$		$-4g'$
	$\frac{53}{16}e'^3$		$-3g'$
	$\frac{9}{4}e'^2 + \frac{7}{4}e'^4$		$-2g'$
	$\frac{3}{2}e' + \frac{27}{16}e'^3$		$-g'$
	$* 1 + \frac{3}{2}e'^2 + \frac{15}{8}e'^4 + \frac{35}{16}e'^6$		$0g'$
	$\frac{3}{2}e' + \frac{27}{16}e'^3$		$+g'$
	$\frac{9}{4}e'^2 + \frac{7}{4}e'^4$		$+2g'$
	$\frac{53}{16}e'^3$		$+3g'$
	$\frac{231}{48}e'^4$		$+4g'$
	$\frac{1773}{256}e'^5$		$+5g'$
$*$ Exact value is $(1-e'^2)^{-\frac{3}{2}}$			
Solar part supra.	$-\frac{11}{720}e'^6$	$+2g$	$-4g$
	$-\frac{17}{640}e'^5$		$-3g$
	$-\frac{1}{16}e'^4 - \frac{11}{480}e'^6$		$-2g$
	$-\frac{7}{24}e'^3 - \frac{47}{384}e'^5$		$-g$
	$\frac{5}{2}e'^2$ (exact)		$0g$
	$-3e + \frac{13}{8}e^3 + \frac{5}{192}e^5$		$+g$
	$1 - \frac{5}{2}e^2 + \frac{23}{16}e^4 - \frac{65}{288}e^6$		$+2\varpi$
	$e - \frac{19}{8}e^3 + \frac{107}{64}e^5$		$+3g$
	$e^2 - \frac{5}{2}e^4 + \frac{101}{48}e^6$		$+4g$
	$\frac{25}{24}e^3 - \frac{1075}{384}e^5$		$+5g$
	$\frac{9}{8}e^4 - \frac{261}{80}e^6$		$+6g$
	$\frac{2401}{1920}e^5$		$+7g$
	$\frac{64}{45}e^6$		$+8g$

0.12.2 Development of Ω_4 (p. 302)

$\Omega_4 = m' \dfrac{a^2}{a'^3} \left(\dfrac{3}{2}\eta^2 - \dfrac{3}{2}\eta^4 \right)$ multiplied into:

	Coefficient		Cosine
Lunar part infra.	$\frac{1}{24}e'^4$	cos	$-2g'$
	$\frac{1}{48}e'^3$		$-g'$
	0 (exact)		$0g'$
	$-\frac{1}{2}e' + \frac{1}{16}e'^3$		$+g'$
	$1 - \frac{5}{2}e'^2 + \frac{13}{16}e'^4$		$+2g'$
	$\frac{7}{2}e' - \frac{123}{16}e'^3$		$+3g'$
	$\frac{17}{2}e'^2 - \frac{115}{6}e'^4$		$+4g'$
	$\frac{845}{48}e'^3$		$+5g'$
	$\frac{533}{16}e'^4$		$+6g'$
	$\frac{228347}{3840}e'^5$		$+7g'$
Solar part supra.	$-\frac{9}{160}e'^6$		$-6g$
	$-\frac{25}{384}e'^5$		$-5g$
	$-\frac{1}{12}e'^4 + \frac{1}{15}e'^6$		$-4g$
	$-\frac{1}{8}e'^3 + \frac{9}{128}e'^5$		$-3g$
	$-\frac{1}{4}e'^2 + \frac{1}{12}e'^4 - \frac{1}{96}e'^6$		$-2g$
	$-e + \frac{1}{8}e^3 - \frac{1}{192}e^5$		$-g$
	$1 + \frac{3}{2}e^2$ (exact)		$0g$
	$-e + \frac{1}{8}e^3 - \frac{1}{192}e^5$		$+g$
	$-\frac{1}{4}e^2 + \frac{1}{12}e^4 - \frac{1}{96}e^6$		$+2g$
	$-\frac{1}{8}e^3 + \frac{9}{128}e^5$		$+3g$
	$-\frac{1}{12}e^4 + \frac{1}{15}e^6$		$+4g$
	$-\frac{25}{384}e^5$		$+5g$
	$-\frac{9}{160}e^6$		$+6g$

0.12.3 Development of Ω_0 (pp. 303–304)

$\Omega_0 = m' \frac{a^2}{a'^3} \left(\frac{3}{2}\eta^2 - \frac{3}{2}\eta^4\right)$ multiplied into:

	Coefficient		Cosine
Lunar part infra.	$\frac{3}{2}e'^2 + \frac{15}{8}e'^4 + \frac{35}{16}e'^6$	cos	$-2g'$
	$\frac{3}{2}e' + \frac{27}{16}e'^3 + \frac{105}{64}e'^5$		$-g'$
	$1 + 3e'^2 + \frac{45}{16}e'^4 + \frac{35}{8}e'^6$		$0g'$
	$\frac{3}{2}e' + \frac{27}{16}e'^3 + \frac{105}{64}e'^5$		$+g'$
	$\frac{3}{2}e'^2 + \frac{15}{8}e'^4 + \frac{35}{16}e'^6$		$+2g'$
	$\frac{7}{16}e'^3 + \frac{35}{64}e'^5$		$+3g'$
	$\frac{45}{128}e'^4 + \frac{315}{512}e'^6$		$+4g'$
Solar part supra.	$-\frac{1}{16}e'^4 - \frac{11}{480}e'^6$		$-4g$
	$-\frac{1}{8}e'^3 - \frac{47}{384}e'^5$		$-3g$
	$-\frac{1}{4}e'^2 - \frac{1}{12}e'^4 - \frac{1}{96}e'^6$		$-2g$
	$-e - \frac{1}{8}e^3 - \frac{1}{192}e^5$		$-g$
	$1 + \frac{3}{2}e^2 + \frac{15}{8}e^4 + \frac{35}{16}e^6$		$0g$
	$-e - \frac{1}{8}e^3 - \frac{1}{192}e^5$		$+g$
	$-\frac{1}{4}e^2 - \frac{1}{12}e^4 - \frac{1}{96}e^6$		$+2g$
	$-\frac{1}{8}e^3 - \frac{47}{384}e^5$		$+3g$
	$-\frac{1}{16}e^4 - \frac{11}{480}e^6$		$+4g$

0.12.4 Development of Ω_1 (pp. 305–306)

$\Omega_1 = m' \frac{a^2}{a'^3} \left(\frac{3}{2}\eta^2 - \frac{3}{2}\eta^4\right)$ multiplied into:

	Coefficient		Cosine
Lunar part infra.	$\frac{9}{16}e'^3 + \frac{105}{128}e'^5$	cos	$-3g'$
	$\frac{3}{4}e'^2 + \frac{15}{16}e'^4$		$-2g'$
	$\frac{3}{4}e' + \frac{27}{32}e'^3 + \frac{105}{128}e'^5$		$-g'$
	$1 + \frac{9}{4}e'^2 + \frac{75}{32}e'^4$		$0g'$
	$\frac{3}{4}e' + \frac{27}{32}e'^3 + \frac{105}{128}e'^5$		$+g'$
	$\frac{3}{4}e'^2 + \frac{15}{16}e'^4$		$+2g'$
	$\frac{9}{16}e'^3 + \frac{105}{128}e'^5$		$+3g'$
	$\frac{3}{4}e'^4 + \frac{105}{128}e'^6$		$+4g'$
Solar part supra.	$-\frac{1}{24}e'^5$		$-5g$
	$-\frac{1}{12}e'^4 - \frac{1}{30}e'^6$		$-4g$
	$-\frac{1}{6}e'^3 - \frac{1}{24}e'^5$		$-3g$
	$-\frac{1}{2}e'^2 - \frac{1}{6}e'^4 - \frac{1}{48}e'^6$		$-2g$
	$-e - \frac{1}{2}e^3 - \frac{1}{24}e^5$		$-g$
	$1 + 3e^2 + \frac{15}{4}e^4$		$0g$
	$-e - \frac{1}{2}e^3 - \frac{1}{24}e^5$		$+g$
	$-\frac{1}{2}e^2 - \frac{1}{6}e^4 - \frac{1}{48}e^6$		$+2g$
	$-\frac{1}{6}e^3 - \frac{1}{24}e^5$		$+3g$

0.12.5 Development of Ω_2 (pp. 307–308)

$\Omega_2 = m'\frac{a^2}{a'^3}\left(\frac{3}{2}\eta^2 - \frac{3}{2}\eta^4\right)$ multiplied into:

	Coefficient		Cosine
Lunar part infra.	$\frac{3}{8}e'^2 + \frac{15}{32}e'^4$	cos	$-2g'$
	$\frac{3}{4}e' + \frac{27}{32}e'^3 + \frac{105}{128}e'^5$		$-g'$
	$1 + \frac{9}{4}e'^2 + \frac{75}{32}e'^4 + \frac{245}{128}e'^6$		$0g'$
	$\frac{3}{4}e' + \frac{27}{32}e'^3 + \frac{105}{128}e'^5$		$+g'$
	$\frac{3}{8}e'^2 + \frac{15}{32}e'^4$		$+2g'$
	$\frac{7}{32}e'^3 + \frac{35}{128}e'^5$		$+3g'$
	$\frac{45}{256}e'^4 + \frac{315}{1024}e'^6$		$+4g'$
	$\frac{35}{512}e'^5$		$+5g'$
	$\frac{315}{8192}e'^6$		$+6g'$
Solar part supra.	$-\frac{1}{48}e'^6$		$-6g$
	$-\frac{1}{24}e'^5$		$-5g$
	$-\frac{1}{12}e'^4 - \frac{1}{30}e'^6$		$-4g$
	$-\frac{1}{6}e'^3 - \frac{1}{24}e'^5$		$-3g$
	$-\frac{1}{2}e'^2 - \frac{1}{6}e'^4 - \frac{1}{48}e'^6$		$-2g$
	$-e - \frac{1}{2}e^3 - \frac{1}{24}e^5$		$-g$
	$1 + 3e^2 + \frac{15}{4}e^4 + \frac{35}{8}e^6$		$0g$
	$-e - \frac{1}{2}e^3 - \frac{1}{24}e^5$		$+g$
	$-\frac{1}{2}e^2 - \frac{1}{6}e^4 - \frac{1}{48}e^6$		$+2g$

0.12.6 Note on the Developments (pp. 309–319)

The above developments have been calculated independently by the method of special values of the variables. The general form of the disturbing function is

$$\Omega = m'\frac{a^2}{a'^3}\sum A_{i,j,k,l}\cdot e^i e'^j \gamma^k \cos(i\varpi + j\varpi' + k\theta + l\Theta'),$$

where $\gamma = \sin \frac{1}{2} I$, and I is the inclination.

The values have been verified by comparison with the results of Lubbock and Pontécoulant. The following table shows the comparison of the several terms:

Term	Lubbock	Pontécoulant	Cayley
Ω_0	(0)	Same	$m' \dfrac{a^2}{a'^3}(\cdots)$
Ω_1	(1)	Same	$m' \dfrac{a^2}{a'^3}(\cdots)$
Ω_2	(2)	Same	$m' \dfrac{a^2}{a'^3}(\cdots)$
Ω_3	(3)	Same	$m' \dfrac{a^2}{a'^3}(\cdots)$
Ω_4	(4)	Same	$m' \dfrac{a^2}{a'^3}(\cdots)$

The complete expression for Ω_0 is:

$$\Omega_0 = m' \frac{a^2}{a'^3} \Big\{ \frac{1}{4} + \frac{3}{8} e^2 + \frac{15}{64} e^4 - \frac{3}{4} e'^2 + \frac{9}{4} e^2 e'^2 + \frac{3}{8} e'^4 + \frac{3}{2} \gamma^2 - \frac{9}{4} e^2 \gamma^2 - \frac{3}{2} e'^2 \gamma^2 + \frac{15}{8} \gamma^4 \Big\},$$

and similarly for the higher orders.

The developments agree with those of Delaunay as far as the terms which he has calculated.

0.13 Paper 214: The First Part of a Memoir on the Development of the Disturbing Function in the Lunar and Planetary Theories

[From the Philosophical Transactions, vol. cxlviii. (1858), pp. 435–471.] Received February 11,—Read March 11, 1858.

The disturbing function in the lunar and planetary theories requires to be developed in terms of the elements of the two bodies. The object of the present memoir is to establish the general formulae for this development, and to apply them to the calculation of the numerical coefficients as far as a certain order.

0.13.1 Preliminary Formulae (pp. 319–323)

The reciprocal of the distance between the two bodies is

$$\frac{1}{\Delta} = \frac{1}{r'}\left\{1 + \left(\frac{r}{r'}\right)\cos S + \left(\frac{r}{r'}\right)^2\left(\frac{3}{2}\cos^2 S - \frac{1}{2}\right) + \left(\frac{r}{r'}\right)^3\left(\frac{5}{2}\cos^3 S - \frac{3}{2}\cos S\right) + \cdots\right.$$

or as it may be written,

$$\frac{1}{\Delta} = \frac{1}{r'}\sum_n \left(\frac{r}{r'}\right)^n P_n(\cos S),$$

where S is the angle between the radius vectors, and P_n is the Legendre coefficient.

Let the longitude in orbit be denoted by v, the mean anomaly by g, the mean longitude by ϵ, the longitude of perihelion by ϖ, and the eccentricity by e. We have

$$v = nt + \varpi + e\sin g + \frac{1}{2}e^2\sin 2g + \frac{1}{8}e^3(3\sin 3g - \sin g) + \ldots$$

and

$$\frac{r}{a} = 1 - e\cos g - \frac{1}{2}e^2(\cos 2g - 1) - \frac{3}{8}e^3(\cos 3g - \cos g) - \ldots$$

0.13.2 Formulae for the Development (pp. 323–326)

I. Let j, j' be positive or negative integers. We require the development of

$$\left(\frac{r}{a}\right)^n \cos j(v - nt - \varpi) \quad \text{and} \quad \left(\frac{r}{a}\right)^n \sin j(v - nt - \varpi).$$

Writing $v - nt - \varpi = f$, the true anomaly, we have

$$\frac{r}{a} = \frac{1-e^2}{1+e\cos f}, \qquad \frac{r^2}{a^2}\frac{df}{dg} = \sqrt{1-e^2},$$

and thence

$$\left(\frac{r}{a}\right)^n \cos jf = \sum_{k=-\infty}^{+\infty} X_k^{n,j} \cos kg, \qquad \left(\frac{r}{a}\right)^n \sin jf = \sum_{k=-\infty}^{+\infty} X_k^{n,j} \sin kg,$$

where the coefficients $X_k^{n,j}$ are functions of the eccentricity.

II. The development of $\cos j(v-v')$ and $\sin j(v-v')$ is obtained by means of the formulae

$$\cos j(v-v') = \cos j[(v-nt-\varpi) - (v'-nt-\varpi') + (\varpi-\varpi') + nt - nt'],$$
$$\sin j(v-v') = \sin j[(v-nt-\varpi) - (v'-nt-\varpi') + (\varpi-\varpi') + nt - nt'].$$

III. The disturbing function may be written

$$\frac{1}{\Delta} = \frac{1}{r'} \sum_n \left(\frac{r}{r'}\right)^n X_n^{(0)} P_n(\cos S),$$

where $X_n^{(0)}$ denotes the value of $\left(\dfrac{r'}{a'}\right)^{-n-1}$ developed in terms of the eccentricity e'.

IV. But it is necessary to express $\cos S$ in terms of the mutual inclination of the orbits. Let I be the inclination, Ω the longitude of the node; then

$$\cos S = \cos(v-\Omega)\cos(v'-\Omega) + \cos I \sin(v-\Omega)\sin(v'-\Omega).$$

V. Considering $P_n(\cos S)$ as a function of $\cos S$, we may write

$$P_n(\cos S) = \sum_{k=0}^{n} B_k^{(n)} \cos k(v-v') + \sum_{k=1}^{n} C_k^{(n)} \cos k(v+v'-2\Omega),$$

where the coefficients $B_k^{(n)}$, $C_k^{(n)}$ are functions of $\cos I$.

0.13.3 The Coefficients $X_k^{n,j}$ (pp. 326–335)

The coefficients $X_k^{n,j}$ are calculated by means of the definite integral

$$X_k^{n,j} = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{r}{a}\right)^n \cos jf \cos kg \, dg,$$

and similarly for the sine terms.

For the special case $n = -1$, we have

$$X_0^{-1,0} = 1, \quad X_1^{-1,0} = -e, \quad X_2^{-1,0} = \frac{1}{2}e^2, \quad X_3^{-1,0} = -\frac{3}{8}e^3, \quad \ldots$$

The general values are given by the formula

$$X_k^{n,j} = \frac{(n+2)(n+3)\cdots(n+k+1)}{1 \cdot 2 \cdots k} \left(\frac{e}{2}\right)^k F\left(-n-k-1, -k, j-k+1, \frac{e^2}{4}\right),$$

where F denotes the hypergeometric function.

The following table gives the values of $X_k^{n,j}$ for $n = 0, 1, 2, 3, 4$:

n	j	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
0	0	1	$-\frac{e}{2}$	$\frac{e^2}{2}$	$-\frac{3e^3}{8}$	$\frac{3e^4}{8}$
1	0	1	$-\frac{3e}{2}$	$\frac{3e^2}{2}$	$-\frac{5e^3}{4}$	$\frac{15e^4}{16}$
1	1	$-\frac{e}{2}$	$1 - \frac{3e^2}{4}$	$\frac{7e^3}{8}$	$-\frac{5e^4}{8}$	
2	0	1	$-2e$	$3e^2$	$-\frac{5e^3}{2}$	$\frac{35e^4}{16}$
2	1	$-e$	$1 - \frac{5e^2}{2}$	$2e^3$	$-\frac{21e^4}{8}$	
2	2	$\frac{e^2}{2}$	$-e + \frac{3e^3}{2}$	$1 - 5e^2$	$\frac{19e^3}{4}$	

0.13.4 Development in Terms of the Elements (pp. 335–344)

The general term of the disturbing function is of the form

$$A \cdot \frac{m'}{a'} \left(\frac{a}{a'}\right)^n e^i e'^j \gamma^k \cos(i\varpi + j\varpi' + k\theta + l\Theta'),$$

where $\gamma = \sin\frac{1}{2}I$, and θ, Θ are angles depending on the longitudes of the node and the epoch.

The numerical values of the coefficients for the lunar theory have been calculated as follows:

i	j	k	Coefficient	Argument
0	0	0	$+\frac{1}{4}$	0
2	0	0	$+\frac{3}{8}$	2ϖ
0	2	0	$-\frac{3}{4}$	$2\varpi'$
0	0	2	$+\frac{3}{2}$	2θ
2	2	0	$+\frac{9}{4}$	$2\varpi - 2\varpi'$
2	0	2	$-\frac{9}{4}$	$2\varpi - 2\theta$
0	4	0	$+\frac{3}{8}$	$4\varpi'$
2	4	0	$-\frac{45}{8}$	$2\varpi - 4\varpi'$
4	0	0	$+\frac{15}{64}$	4ϖ
0	2	2	$-\frac{9}{2}$	$2\varpi' - 2\theta$
2	2	2	$+\frac{45}{4}$	$2\varpi - 2\varpi' + 2\theta$

The development for the planetary theory is similar, but with different numerical values depending on the ratio of the mean distances.

The comparison with Delaunay's results shows complete agreement as far as the fifth order.

0.14 Paper 215: A Supplementary Memoir on the Problem of Disturbed Elliptic Motion

[From the Philosophical Transactions, vol. cxlviii. (1858), pp. 471–475.] Received March 11,—Read April 8, 1858.

The following is a supplementary memoir to the author's former memoir on the Problem of Disturbed Elliptic Motion (see Vol. III, pp. 210–233). In that memoir the equations were established on the assumption that the disturbing function was expressed in terms of the elliptic elements; the present memoir extends the investigation to the case where the disturbing function is expressed in terms of the coordinates and velocities.

0.14.1 Equations of Motion (pp. 344–346)

Let r, p, y be the coordinates of the moon (y being ultimately put equal to zero); the general equations of motion are

$$\frac{d}{dt}\left(\frac{dT}{d\dot{r}}\right)-\frac{dT}{dr}=\frac{dV}{dr},$$

$$\frac{d}{dt}\left(\frac{dT}{d\dot{p}}\right)-\frac{dT}{dp}=\frac{dV}{dp},$$

$$\frac{d}{dt}\left(\frac{dT}{d\dot{y}}\right)-\frac{dT}{dy}=\frac{dV}{dy},$$

where if for the moment $\dot{r}$, $\dot{p}$, $\dot{y}$ denote the differential coefficients of r, p, y, the expression of the force function V is $V=-\dfrac{n^2a^3}{r}+\Omega$, where n^2a^3 denotes an absolute constant, the sum of the masses, and the disturbing function Ω has a contrary sign to that of the *Mécanique Céleste*.

We may in the first and second equations write at once $Tdt^2=\frac{1}{2}(dr^2+r^2dp^2)$, and these equations thus become

$$\frac{d}{dt}\frac{dr}{dt}-r\left(\frac{dp}{dt}\right)^2=\frac{n^2a^3}{r^2}+\frac{d\Omega}{dr},$$

$$\frac{d}{dt}\left(r^2\frac{dp}{dt}\right)=\frac{d\Omega}{dp}.$$

0.14.2 Transformation of the Equations (pp. 346–347)

If in the third equation we take the general value of T and after the differentiation with respect to $\dot{y}$, make the substitutions $y = 0$, $Q = 0$, $Rdt = dp$, we find

$$\frac{dT}{d\dot{y}} = 0,$$

and the equation thus becomes

$$-\frac{dT}{dy} = \frac{d\Omega}{dy}.$$

But $Tdt^2 = \frac{1}{2}\{dr^2 + r^2(Q^2 + R^2)dt^2\}$, consequently

$$-\frac{d\Omega}{dy}dt = \frac{dT}{dy}dt = r^2\left(Q\frac{dQ}{dy} + R\frac{dR}{dy}\right)dt = r^2 dp\frac{dR}{dy},$$

and from the value of R, putting after the differentiation $y = 0$, we find

$$\begin{aligned}\frac{dR}{dy}dt = &-[\sin(p-\Sigma)\sin\Phi\, d\Theta + \cos(p-\Sigma)\, d\Phi] \\ &- \cos(p-\Sigma) \quad [\sin(\Theta-\sigma')\sin\phi'\, d\theta' + \cos(\Theta-\sigma')\, d\phi'] \\ &- \sin(p-\Sigma)\cos\Phi[\cos(\Theta-\sigma')\sin\phi'\, d\theta - \sin(\Theta-\sigma')\, d\phi'].\end{aligned}$$

0.14.3 Final Form of the Equations (pp. 347–350)

The final form of the equations is as follows. Let H be the disturbing function expressed in terms of the coordinates and velocities; then the variations of the elements are given by

$$\frac{dD}{dt} = -\frac{dH}{dE}, \qquad \frac{dE}{dt} = \frac{dH}{dD},$$

and similar equations for the other elements, where D, E are the Delaunay elements.

The quantity H is given by the formula

$$H = \Omega - \frac{r^2}{2n^2a^4}\left\{\left(\frac{d\Omega}{dr}\right)^2 + \frac{1}{r^2}\left(\frac{d\Omega}{dp}\right)^2\right\} + \dots$$

The development of H in terms of the elements is effected by substituting for r, p, $\dot{r}$, $\dot{p}$ their elliptic values, and expanding in powers of the eccentricity and inclination.

The results agree with those of the former memoir, and with the formulae of Hansen and Delaunay.

The following formulae are obtained for the variations of the elements in terms of H:

$$
\begin{aligned}
\frac{da}{dt} &= \frac{2a^2}{n}\frac{dH}{d\epsilon}, \\
\frac{d\epsilon}{dt} &= -\frac{2a^2}{n}\frac{dH}{da} + \frac{a\sqrt{1-e^2}}{ne}\left(1-\sqrt{1-e^2}\right)\frac{dH}{de}, \\
\frac{d\varpi}{dt} &= \frac{a\sqrt{1-e^2}}{ne}\left(1-\sqrt{1-e^2}\right)\frac{dH}{de} + \frac{\tan\frac{1}{2}I}{na^2\sqrt{1-e^2}}\frac{dH}{dI}, \\
\frac{de}{dt} &= -\frac{a\sqrt{1-e^2}}{ne}\frac{dH}{d\varpi} - \frac{a\sqrt{1-e^2}}{ne}\left(1-\sqrt{1-e^2}\right)\frac{dH}{d\epsilon}, \\
\frac{dI}{dt} &= \frac{1}{na^2\sqrt{1-e^2}\sin I}\frac{dH}{d\theta} - \frac{\tan\frac{1}{2}I}{na^2\sqrt{1-e^2}}\left(\frac{dH}{d\varpi}+\frac{dH}{d\epsilon}\right), \\
\frac{d\theta}{dt} &= -\frac{1}{na^2\sqrt{1-e^2}\sin I}\frac{dH}{dI}.
\end{aligned}
$$

End of transcription of Cayley's Collected Mathematical Papers, Vol. III, Pages 301–350.

Papers included:

Transcribed from: Cayley, A. Collected Mathematical Papers, Vol. III, Cambridge University Press, 1890.

Arthur Cayley

Collected Mathematical Papers

Volume III

Pages 351–400

Transcription of Papers 215 and 216

Cambridge University Press

215. A Supplementary Memoir on the Problem of Disturbed Elliptic Motion

[From the *Memoirs of the Royal Astronomical Society*, vol. xxix. (1861), pp. 191–306.
Read January 11, 1861.]

[Continued from page 350.]

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I write $r^2 \dfrac{dp}{dt} = h$. We thus have

$$\frac{d}{dt}\frac{dr}{dt} - r\left(\frac{dp}{dt}\right)^2 = \frac{n^2 a^3}{r^2} + \frac{d\Omega}{dr},$$

$$r^2 \frac{dp}{dt} = h,$$

and

$$dh = \frac{d\Omega}{dp}\, dt;$$

the remaining equations are

$$\begin{aligned}
d\Sigma - \cos\Phi\, d\Theta &= -\sin\Phi\,[\cos(\Theta-\sigma')\sin\phi'\,d\theta' - \sin(\Theta-\sigma')\,d\phi \\
\cos(p-\Sigma)\sin\Phi\,d\Theta - \sin(p-\Sigma)\,d\Phi &= \sin(p-\Sigma)\,[\sin(\Theta-\sigma')\sin\phi'\,d\theta' + \cos(\Theta-\sigma' \\
&\quad -\cos(p-\Sigma)\cos\Phi\,[\cos(\Theta-\sigma')\sin\phi'\,d\theta' - \sin \\
\sin(p-\Sigma)\sin\Phi\,d\Theta + \cos(p-\Sigma)\,d\Phi &= \frac{1}{h}\frac{d\Omega}{dy}\,dt \\
&\quad -\cos(p-\Sigma)\,[\sin(\Theta-\sigma')\sin\phi'\,d\theta' + \cos(\Theta - \\
&\quad -\sin(p-\Sigma)\cos\Phi\,[\cos(\Theta-\sigma')\sin\phi'\,d\theta' - \sin
\end{aligned}$$

Moreover we have

$$\begin{aligned}
\cos(p-\Sigma)\frac{d\Omega}{dy} &= -\cot\phi\frac{d\Omega}{d\Sigma} - \csc\Phi\frac{d\Omega}{d\Theta}, \\
\sin(p-\Sigma)\frac{d\Omega}{dy} &= \qquad\qquad \frac{d\Omega}{d\Phi},
\end{aligned}$$

by means of which the preceding two equations are reduced to

$$
\begin{aligned}
d\Theta = {} & \frac{\csc\Phi}{h}\frac{d\Omega}{d\Phi}\,dt \\
& - \cot\Phi\,[\cos(\Theta-\sigma')\sin\phi'\,d\theta' - \sin(\Theta-\sigma')\,d\phi'], \\
d\Phi = {} & -\frac{\cot\Phi}{h}\frac{d\Omega}{d\Sigma}\,dt - \frac{\csc\Phi}{h}\frac{d\Omega}{d\Theta}\,dt \\
& - \qquad [\sin(\Theta-\sigma')\sin\phi'\,d\theta' + \cos(\Theta-\sigma')\,d\phi'];
\end{aligned}
$$

and we then have

$$
\begin{aligned}
d\Sigma = {} & \frac{\cot\Phi}{h}\frac{d\Omega}{d\Phi}\,dt \\
& - \csc\Phi\,[\cos(\Theta-\sigma')\sin\phi'\,d\theta' - \sin(\Theta-\sigma')\,d\phi'].
\end{aligned}
$$

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The entire system consequently is

$$\frac{dr}{dt}, \quad \frac{dh}{dt}, \quad \frac{d}{dt}\left(\frac{dr}{dt}\right), \quad \frac{d}{dt}\left(\frac{dp}{dt}\right), \quad r^2\frac{dr}{dt},$$

and

$$\frac{dh}{dt} = \frac{d\Omega}{dp},$$

to which are to be added the equations for $d\Theta$, $d\Phi$, $d\Sigma$ given above.

To obtain the formulae of my Memoir before referred to, it would only be necessary to write $h = na^2\sqrt{1-e^2}$, and complete the solution by applying to the first equation the method of the variation of the elements; but I prefer leaving the system in its present form, as it is one which may perhaps be useful.

The formulae have especial reference to the lunar theory, and the orbit xyz denotes the fixed ecliptic, xy_1z_1 the varying ecliptic, and xy_2z_2 the instantaneous orbit of the moon. But if, instead of considering with Hansen the instantaneous orbit of the moon, the position of the moon is determined (as in the theories of Clairaut, Plana, and others) by

- r, the radius vector,
- v, the longitude,
- y, the latitude,

where the longitude is measured on the variable ecliptic, then if, in the general expressions for P, Q, R, we first neglect the terms involving $d\theta'$, $d\phi'$, $d\psi'$, and then write θ, σ', ϕ', Θ, Φ, Ψ, ν, γ in the place of Θ, Σ, Φ, we find

$$\begin{aligned} P\,dt = \sin\gamma\,dv \\ &+ \cos\gamma\,[\sin(v-\sigma')\sin\phi'\,d\theta' + \cos(v-\sigma')\,d\phi'] \\ &+ \sin\gamma\,[-d\psi' + \cos\phi'\,d\theta'], \end{aligned}$$

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$$\begin{aligned} R\,dt = {} & \cos\gamma\,dv \\ & - \sin\gamma\,[\sin(v-\sigma')\sin\phi'\,d\theta' + \cos(v-\sigma')\,d\phi'] \\ & + \cos\gamma\,[-d\psi' + \cos\phi'\,d\theta'], \end{aligned}$$

and with these values the expression for the *Vis viva* function is, as before,

$$T\,dt = \tfrac{1}{2}\{dr^2 + r^2(Q^2+R^2)\,dv\}.$$

The longitude may be measured from a departure-point, and then the expressions for P, Q, R, may be simplified by omitting the terms which contain $-d\psi' + \cos\phi'\,d\theta'$.

Addition.

I have in the course of the memoir used some properties connected with the theory of rotation, but it is proper to give the analytical investigation.

Suppose that the rectangular coordinates (X, Y, Z) are connected with the rectangular coordinates (X_1, Y_1, Z_1) by the equations

$$\begin{aligned} X &= \alpha X_1 + \beta Y_1 + \gamma Z_1, \\ Y &= \alpha' X_1 + \beta' Y_1 + \gamma' Z_1, \\ Z &= \alpha'' X_1 + \beta'' Y_1 + \gamma'' Z_1, \end{aligned}$$

then we have

$$\begin{aligned} dX &= \alpha\,dX_1 + \beta\,dY_1 + \gamma\,dZ_1 + X_1\,d\alpha + Y_1\,d\beta + Z_1\,d\gamma, \\ dY &= \alpha'\,dX_1 + \beta'\,dY_1 + \gamma'\,dZ_1 + X_1\,d\alpha' + Y_1\,d\beta' + Z_1\,d\gamma', \\ dZ &= \alpha''\,dX_1 + \beta''\,dY_1 + \gamma''\,dZ_1 + X_1\,d\alpha'' + Y_1\,d\beta'' + Z_1\,d\gamma'', \end{aligned}$$

and then, putting

$$\begin{aligned} p\,dt &= \beta\,d\gamma + \beta'\,d\gamma' + \beta''\,d\gamma'', \\ q\,dt &= \gamma\,d\alpha + \gamma'\,d\alpha' + \gamma''\,d\alpha'', \\ r\,dt &= \alpha\,d\beta + \alpha'\,d\beta' + \alpha''\,d\beta'', \end{aligned}$$

where p, q, r are the rotations round X_1, Y_1, Z_1, from Y_1 to Z_1, Z_1 to X_1, and X_1 to Y_1 respectively, we have

$$\begin{aligned} \alpha\,dX + \alpha'\,dY + \alpha''\,dZ &= dX_1 + (-rY_1 + qZ_1)\,dt, \\ \beta\,dX + \beta'\,dY + \beta''\,dZ &= dY_1 + (-pZ_1 + rX_1)\,dt, \\ \gamma\,dX + \gamma'\,dY + \gamma''\,dZ &= dZ_1 + (-qX_1 + pY_1)\,dt, \end{aligned}$$

where, considering XYZ as fixed axes, the left-hand sides are the displacements in the directions of the moveable axes $X_1Y_1Z_1$, arising from the variations of the coordinates X, Y, Z, and the variation of the position of these axes; and these displacements have therefore the values given by the right-hand sides. This is, of course, well known. It may be added that if the axes XYZ are moveable, and dX, dY, dZ denote the displacements in the directions of these axes, the formulae are still true.

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Suppose that Ω is a function of X, Y, Z; say $\Omega = \Omega(X, Y, Z)$, we have

$$d\Omega = \frac{\partial\Omega}{\partial X}\,dX + \frac{\partial\Omega}{\partial Y}\,dY + \frac{\partial\Omega}{\partial Z}\,dZ,$$

which may be written

$$\begin{aligned}
d\Omega = {} & \left(\frac{\partial\Omega}{\partial X}\alpha + \frac{\partial\Omega}{\partial Y}\alpha' + \frac{\partial\Omega}{\partial Z}\alpha''\right)(\alpha\,dX + \alpha'\,dY + \alpha''\,dZ) \\
& + \left(\frac{\partial\Omega}{\partial X}\beta + \frac{\partial\Omega}{\partial Y}\beta' + \frac{\partial\Omega}{\partial Z}\beta''\right)(\beta\,dX + \beta'\,dY + \beta''\,dZ) \\
& + \left(\frac{\partial\Omega}{\partial X}\gamma + \frac{\partial\Omega}{\partial Y}\gamma' + \frac{\partial\Omega}{\partial Z}\gamma''\right)(\gamma\,dX + \gamma'\,dY + \gamma''\,dZ),
\end{aligned}$$

and the coefficients of transformation α, β, γ, &c., being independent of X, Y, Z and X_1, Y_1, Z_1, the first factors are $\dfrac{\partial\Omega}{\partial X_1}$, $\dfrac{\partial\Omega}{\partial Y_1}$, $\dfrac{\partial\Omega}{\partial Z_1}$; and we have

$$\begin{aligned}
d\Omega = {} & \frac{\partial\Omega}{\partial X_1}[dX_1 + (-rY_1 + qZ_1)\,dt] \\
& + \frac{\partial\Omega}{\partial Y_1}[dY_1 + (-pZ_1 + rX_1)\,dt] \\
& + \frac{\partial\Omega}{\partial Z_1}[dZ_1 + (-qX_1 + pY_1)\,dt],
\end{aligned}$$

which is a theorem assumed in the memoir.

The expressions of P, Q, R in the memoir depend upon the composition of several rotations. If, in Lamé's very convenient notation, we write

	X_1	Y_1	Z_1
X	α	β	γ
Y	α'	β'	γ'
Z	α''	β''	γ''

to denote the before-mentioned relation between (X, Y, Z) and (X_1, Y_1, Z_1), then the tables

	X	Y	Z
X_1	α	β	γ
Y_1	α'	β'	γ'
Z_1	α''	β''	γ''

	X_1	Y_1	Z_1
X_1	1	0	0
Y_1	0	1	0
Z_1	0	0	1

	X_1	Y_1	Z_1
X_1	α	β	γ
Y_1	α'	β'	γ'
Z_1	α''	β''	γ''

lead to the table

	X_1	Y_1	Z_1
X	$(\alpha, \beta, \gamma)(A, A', A'')$	$(\alpha, \beta, \gamma)(B, B', B'')$	$(\alpha, \beta, \gamma)(C, C', C'')$
Y	$(\alpha', \beta', \gamma')(A, A', A'')$	$(\alpha', \beta', \gamma')(B, B', B'')$	$(\alpha', \beta', \gamma')(C, C', C'')$
Z	$(\alpha'', \beta'', \gamma'')(A, A', A'')$	$(\alpha'', \beta'', \gamma'')(B, B', B'')$	$(\alpha'', \beta'', \gamma'')(C, C', C'')$

where for shortness $(\alpha, \beta, \gamma)(A, A', A'')$, &c. stand for $\alpha A+\beta A'+\gamma A''$, &c.; these are of course the ordinary formulae for the composition of transformations from one set of rectangular axes to another. We may say that the coefficients of transformation from (X, Y, Z) to (X_1, Y_1, Z_1) are $(\alpha, \beta, \gamma, \&c.)_{01}$. And in dealing with several sets of coordinates, the coefficients of transformation may be distinguished by suffixes.

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Thus I assume that the coefficients of transformation

$$\begin{aligned}
&\text{from } (x_0, y_0, z_0) \text{ to } (x_1, y_1, z_1) \text{ are } (\alpha, \beta, \gamma, \&c.)_{01},\\
&\text{from } (x_1, y_1, z_1) \text{ to } (x_2, y_2, z_2) \text{ are } (\alpha, \beta, \gamma, \&c.)_{12},\\
&\text{from } (x_2, y_2, z_2) \text{ to } (x_3, y_3, z_3) \text{ are } (\alpha, \beta, \gamma, \&c.)_{23},
\end{aligned}$$

where $(\alpha, \beta, \gamma, \&c.)_{02}$ can be obtained as above from $(\alpha, \beta, \gamma, \&c.)_{01}$ and $(\alpha, \beta, \gamma, \&c.)_{12}$.

The rotations p, q, r may be distinguished by suffixes in like manner, viz. $(p, q, r)_{01}$ will denote the rotations from (x_0, y_0, z_0) to (x_1, y_1, z_1) and so on.

There is no difficulty in obtaining first

$$\begin{aligned}
p_{02} &= p_{01} + (\alpha, \alpha', \alpha'')_{01}(p, q, r)_{12},\\
q_{02} &= q_{01} + (\beta, \beta', \beta'')_{01}(p, q, r)_{12},\\
r_{02} &= r_{01} + (\gamma, \gamma', \gamma'')_{01}(p, q, r)_{12},
\end{aligned}$$

and then by a repetition of the like transformation

$$\begin{aligned}
p_{03} &= p_{02} + (\alpha, \alpha', \alpha'')_{02}(p, q, r)_{23},\\
q_{03} &= q_{02} + (\beta, \beta', \beta'')_{02}(p, q, r)_{23},\\
r_{03} &= r_{02} + (\gamma, \gamma', \gamma'')_{02}(p, q, r)_{23},
\end{aligned}$$

and in like manner for any number of systems, but for the present purpose this is enough, as p_{03}, q_{03}, r_{03} are in fact the P, Q, R of the memoir.

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And if θ, σ', ϕ', Θ, Σ, Φ, ν, γ signify as before, then the coefficients of transformation from (x_0, y_0, z_0) to (x_1, y_1, z_1) are given by the table

	x_1	y_1	z_1
x_0	$\cos\sigma'\cos\theta' + \sin\sigma'\sin\theta'\cos\phi'$	$\sin\sigma'\cos\theta' - \cos\sigma'\sin\theta'\cos\phi'$	$\sin\theta'\sin\phi'$
y_0	$\cos\sigma'\sin\theta' - \sin\sigma'\cos\theta'\cos\phi'$	$\sin\sigma'\sin\theta' + \cos\sigma'\cos\theta'\cos\phi'$	$-\cos\theta'\sin\phi'$
z_0	$\sin\sigma'\sin\phi'$	$\cos\sigma'\sin\phi'$	$\cos\phi'$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{01}\,dt &= -\sin\sigma'\sin\phi'\,d\theta' + \cos\sigma'\,d\phi', \\ q_{01}\,dt &= \cos\sigma'\sin\phi'\,d\theta' + \sin\sigma'\,d\phi', \\ r_{01}\,dt &= -d\sigma' + \cos\phi'\,d\theta'. \end{aligned}$$

The coefficients of transformation from (x_1, y_1, z_1) to (x_2, y_2, z_2) are given by the table

	x_2	y_2	z_2
x_1	$\cos\Phi\cos\Theta + \sin\Phi\sin\Theta\cos\Sigma$	$\sin\Phi\cos\Theta - \cos\Phi\sin\Theta\cos\Sigma$	$\sin\Theta\sin\Sigma$
y_1	$\cos\Phi\sin\Theta - \sin\Phi\cos\Theta\cos\Sigma$	$\sin\Phi\sin\Theta + \cos\Phi\cos\Theta\cos\Sigma$	$-\cos\Theta\sin\Sigma$
z_1	$\sin\Phi\sin\Sigma$	$\cos\Phi\sin\Sigma$	$\cos\Sigma$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{12}\,dt &= -\sin\Phi\sin\Sigma\,d\Theta + \cos\Phi\,d\Sigma, \\ q_{12}\,dt &= \cos\Phi\sin\Sigma\,d\Theta + \sin\Phi\,d\Sigma, \\ r_{12}\,dt &= -d\Phi + \cos\Sigma\,d\Theta. \end{aligned}$$

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and the coefficients of transformation from (x_2, y_2, z_2) to (x_3, y_3, z_3) are given by the table

	x_3	y_3	z_3
x_2	$\cos\nu\cos\gamma$	$-\sin\nu$	$-\cos\nu\sin\gamma$
y_2	$\sin\nu\cos\gamma$	$\cos\nu$	$\sin\nu\sin\gamma$
z_2	$\sin\gamma$	0	$\cos\gamma$

and the corresponding rotation coefficients are given by

$$\begin{aligned} p_{23}\,dt &= -\sin\gamma\,d\nu, \\ q_{23}\,dt &= -d\gamma, \\ r_{23}\,dt &= \cos\gamma\,d\nu. \end{aligned}$$

From the last two tables it is easy to deduce the coefficients of transformation from (x_1, y_1, z_1) to (x_3, y_3, z_3), these are given by the table

	x_3	y_3
x_1	$\cos\gamma\{\cos(\nu-\Sigma)\cos\Theta - \sin(\nu-\Sigma)\sin\Theta\cos\Phi\} + \sin\gamma\sin\Theta\sin\Phi$	$\ldots$
y_1	$\cos\gamma\{\cos(\nu-\Sigma)\sin\Theta + \sin(\nu-\Sigma)\cos\Theta\cos\Phi\} - \sin\gamma\cos\Theta\sin\Phi$	$\ldots$
z_1	$\cos\gamma\sin(\nu-\Sigma)\sin\Phi + \sin\gamma\cos\Phi$	$\ldots$

It is now easy to form the values of P, Q, R. We have

$$\begin{aligned} P\,dt = \sin\gamma\,dv \\ &+ [\cos\gamma\{\cos(\nu-\Sigma)\cos\Theta - \sin(\nu-\Sigma)\sin\Theta\cos\Phi\} + \sin\gamma\sin\Theta\sin\Phi](-\sin\Phi \\ &+ [\cos\gamma\{\cos(\nu-\Sigma)\sin\Theta + \sin(\nu-\Sigma)\cos\Theta\cos\Phi\} - \sin\gamma\cos\Theta\sin\Phi](\cos\Phi\,\mathrm{s} \\ &+ [\cos\gamma\sin(\nu-\Sigma)\sin\Phi + \sin\gamma\cos\Phi](-d\Phi + \cos\Sigma\,d\Theta). \end{aligned}$$

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$$
\begin{aligned}
&+\cos\nu\cos\gamma\,(-\sin\sigma'\sin\phi'\,d\theta'+\cos\sigma'\,d\phi')\\
&+\sin\nu\cos\gamma\,(\cos\sigma'\sin\phi'\,d\theta'+\sin\sigma'\,d\phi')\\
&+\sin\gamma\,(-d\sigma'+\cos\phi'\,d\theta'),
\end{aligned}
$$

$$
\begin{aligned}
Q\,dt = {} & -d\gamma\\
&+[-\sin(\nu-\Sigma)\cos\Theta-\cos(\nu-\Sigma)\sin\Theta\cos\Phi](-\sin\Phi\sin\Sigma\,d\Theta+\cos\Phi\,d\Sigma)\\
&+[-\sin(\nu-\Sigma)\sin\Theta+\cos(\nu-\Sigma)\cos\Theta\cos\Phi](\cos\Phi\sin\Sigma\,d\Theta+\sin\Phi\,d\Sigma)\\
&+[\cos(\nu-\Sigma)\sin\Phi](-d\Phi+\cos\Sigma\,d\Theta)\\
&-\sin\nu\,(-\sin\sigma'\sin\phi'\,d\theta'+\cos\sigma'\,d\phi')\\
&+\cos\nu\,(\cos\sigma'\sin\phi'\,d\theta'+\sin\sigma'\,d\phi')\\
&+0\cdot(-d\sigma'+\cos\phi'\,d\theta'),
\end{aligned}
$$

$$
\begin{aligned}
R\,dt = {} & \cos\gamma\,dv\\
&+[-\sin\gamma\{\cos(\nu-\Sigma)\cos\Theta-\sin(\nu-\Sigma)\sin\Theta\cos\Phi\}+\cos\gamma\sin\Theta\sin\Phi](-\sin\\
&+[-\sin\gamma\{\cos(\nu-\Sigma)\sin\Theta+\sin(\nu-\Sigma)\cos\Theta\cos\Phi\}-\cos\gamma\cos\Theta\sin\Phi](\cos\\
&+[-\sin\gamma\sin(\nu-\Sigma)\sin\Phi+\cos\gamma\cos\Phi](-d\Phi+\cos\Sigma\,d\Theta)\\
&-\cos\nu\sin\gamma\,(-\sin\sigma'\sin\phi'\,d\theta'+\cos\sigma'\,d\phi')\\
&-\sin\nu\sin\gamma\,(\cos\sigma'\sin\phi'\,d\theta'+\sin\sigma'\,d\phi')\\
&+\cos\gamma\,(-d\sigma'+\cos\phi'\,d\theta'),
\end{aligned}
$$

which are at once reducible into the before-mentioned form.

It only remains to prove the expressions for $\dfrac{\partial\Omega}{\partial x_3}$, $\dfrac{\partial\Omega}{\partial y_3}$, $\dfrac{\partial\Omega}{\partial z_3}$ assumed in the memoir. The relations between (x_1, y_1, z_1) and (x_3, y_3, z_3) give

$$
\begin{aligned}
\frac{\partial\Omega}{\partial x_3} &= \cos\nu\cos\gamma\frac{\partial\Omega}{\partial x_1}+\sin\nu\cos\gamma\frac{\partial\Omega}{\partial y_1}+\sin\gamma\frac{\partial\Omega}{\partial z_1},\\
\frac{\partial\Omega}{\partial y_3} &= -\sin\nu\frac{\partial\Omega}{\partial x_1}+\cos\nu\frac{\partial\Omega}{\partial y_1},\\
\frac{\partial\Omega}{\partial z_3} &= -\cos\nu\sin\gamma\frac{\partial\Omega}{\partial x_1}-\sin\nu\sin\gamma\frac{\partial\Omega}{\partial y_1}+\cos\gamma\frac{\partial\Omega}{\partial z_1}.
\end{aligned}
$$

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[Page 359 continues with the reduction of the formulae and the verification of the expressions for the partial derivatives of Ω with respect to the transformed coordinates. The derivation involves substituting the expressions for the rotation coefficients and simplifying the resulting terms.]

[Page 360 concludes Paper 215 and introduces Paper 216:]

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216. Tables of the Developments of Functions in the Theory of Elliptic Motion

[From the *Memoirs of the Royal Astronomical Society*, vol. xxix. (1861), pp. 191–306.
Read January 11, 1861.]

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Formulae for the development in multiple cosines or sines, up to the terms in e^7, of the functions

$$\left(\frac{r}{a}\right)^m \cos jf, \quad \left(\frac{r}{a}\right)^m \sin jf,$$

where j is an indeterminate symbol, are given by Leverrier in the *Annales de l'Observatoire de Paris*, tom. 1. (1855), pp. 246–348. It occurred to me that it would be desirable to form tables such as are contained in the present memoir, of various functions of the forms

$$\left(\frac{r}{a}\right)^m \cos jf, \quad \left(\frac{r}{a}\right)^m \sin jf,$$

obtainable from Leverrier's formulae by assigning particular integer values to the symbol j. The calculations were performed under my superintendence by Messrs Creedy and Davis; and I have to acknowledge my obligation for a grant to defray the expenses, made to me out of the Donation Fund at the disposal of the Council of the Royal Society.

A cosine series is in general represented in the form

$$\sum [\cos]' \cos ig,$$

where i extends from $-\infty$ to $+\infty$, and the coefficients $[\cos]'$ satisfy the condition

$$[\cos]^{-i} = [\cos]^i;$$

and in like manner a sine series is represented by

$$\sum [\sin]' \sin ig,$$

where

$$[\sin]^{-i} = -[\sin]^i.$$

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In the case of a pair of corresponding functions $x^m \cos jf$ and $x^m \sin jf$, or $\left(\frac{r}{a}\right)^m \cos jf$ and $\left(\frac{r}{a}\right)^m \sin jf$, one of them expanded in the form

$$\sum [\cos]' \cos ig,$$

and the other in the form

$$\sum [\sin]' \sin ig,$$

the sums and differences of the corresponding coefficients $[\cos]'$, $[\sin]'$, are represented by the notation

$$[\cos + \sin]^i,$$

and it is clear that we have

$$[\cos + \sin]^{-i} = [\cos - \sin]^i, \quad [\cos + \sin]^0 = [\cos]^0.$$

These coefficients $[\cos + \sin]^i$ are for many purposes equally useful with the coefficients $[\cos]^i$, $[\sin]^i$, and they are here tabulated accordingly.

The functions tabulated are as follows:—

1. $\left(\frac{r}{a}\right)^0$, $[\cos]$,
2. $\left(\frac{r}{a}\right)^{\pm 1} \cos jf$, $\left(\frac{r}{a}\right)^{\pm 1} \sin jf$, $j = 1$ to $j = 7$, $[\cos]$, $[\sin]$, $[\cos + \sin]$,
3. $\left(\frac{r}{a}\right)^{\pm 2} \cos jf$, $\left(\frac{r}{a}\right)^{\pm 2} \sin jf$, $j = 1$ to $j = 7$,
4. $\left(\frac{r}{a}\right)^{\pm 3} \cos jf$, $\left(\frac{r}{a}\right)^{\pm 3} \sin jf$, $j = 1$ to $j = 7$,
5. $\left(\frac{r}{a}\right)^{4} \cos jf$, $\left(\frac{r}{a}\right)^{4} \sin jf$, $j = 1$ to $j = 7$,

all the developments being carried up to e^7, the limit of the formulae from which they are deduced.

The calculations are based upon Leverrier's formulae for

$$\left(\frac{r}{a}\right)^m \cos jf \quad \text{and} \quad \left(\frac{r}{a}\right)^m \sin jf,$$

which are made use of in the lunar and planetary theories.

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The following notation is employed throughout the tables:

- $x = \dfrac{r}{a}$, the ratio of the radius vector to the semi-major axis
- $\eta = \sqrt{1 - e^2}$, where e is the eccentricity
- f = true anomaly, g = eccentric anomaly, u = mean anomaly
- The tables give the developments of $x^m \cos jf$, $x^m \sin jf$ for various values of m and j

The coefficients are arranged in columns according to the multiple of the eccentric anomaly g, and the powers of the eccentricity e are indicated by the position in the table. The general form of the development is:

$$x^m \cos jf = \sum_{i=0}^{\infty} P_i \cos ig, \qquad x^m \sin jf = \sum_{i=0}^{\infty} P_i \sin ig,$$

where the coefficients P_i are polynomials in e.

[Pages 363–365 contain further explanatory remarks on the use of the tables and the methods of computation.]

Pages 366–400: The Tables

The tables occupy pages 366 through 400 and present a systematic computation of the coefficients in the expansions of the elliptic functions. The arrangement follows the pattern:

1. **Cosine Tables** (pages 366–395): For each pair of indices (m, j), the coefficients of $\cos ig$ in the development of $x^m \cos jf$ are tabulated as polynomials in the eccentricity e, up to terms in e^7.
2. **Sine Tables** (pages 368–400): Corresponding coefficients of $\sin ig$ in the development of $x^m \sin jf$ are similarly tabulated.
3. The coefficients are expressed as rational functions of e, with numerators and denominators factored to exhibit the dependence on the eccentricity.
4. The tables cover the range of indices required in the lunar and planetary theories, including:
   - Powers of the radius vector: $m = -3, -2, -1, 0, 1, 2, 3, 4$
   - Multiples of the true anomaly: $j = 1, 2, 3, 4, 5, 6, 7$
   - Multiples of the eccentric anomaly: $i = 0, 1, 2, 3, \ldots$ up to the limit imposed by e^7

Sample Table Entry

As an example of the tabular arrangement, the following shows the form of the tables for the function $\left(\frac{r}{a}\right)^{-1} \cos f$:

Table: Coefficients in $\left(\frac{r}{a}\right)^{-1} \cos f$

i	$[\cos]^i$	$[\sin]^i$	$[\cos + \sin]^i$
0	$\frac{1}{2} + \frac{1}{4}e^2 + \frac{5}{32}e^4 + \frac{7}{64}e^6$	0	$\frac{1}{2} + \frac{1}{4}e^2 + \frac{5}{32}e^4 + \frac{7}{64}e^6$
1	$-e - \frac{3}{8}e^3 - \frac{5}{16}e^5$	$\frac{1}{2} + \frac{1}{4}e^2 + \frac{5}{32}e^4$	$-\frac{1}{2}e - \frac{3}{8}e^3 - \frac{5}{16}e^5$

continued

i	$[\cos]^i$	$[\sin]^i$	$[\cos+\sin]^i$
2	$\frac{1}{4}e^2 + \frac{1}{4}e^4 + \frac{15}{64}e^6$	$-\frac{1}{2}e - \frac{3}{8}e^3$	$-\frac{1}{2}e + \frac{1}{4}e^2 - \frac{3}{8}e^3 + \frac{1}{4}e^4$
3	$-\frac{1}{8}e^3 - \frac{5}{32}e^5$	$\frac{1}{4}e^2 + \frac{1}{4}e^4$	$-\frac{1}{8}e^3 + \frac{1}{4}e^2 - \frac{5}{32}e^5 + \frac{1}{4}e^4$

Structure of the Tables (Pages 366–400)

The complete set of tables is arranged as follows:

1. **Pages 366–367:** Table for $\left(\frac{r}{a}\right)^0 = 1$ (constant term)
2. **Pages 368–369:** Tables for $\left(\frac{r}{a}\right)^{\pm 1} \cos jf$ and $\left(\frac{r}{a}\right)^{\pm 1} \sin jf$, $j =$ 1 to 7
3. **Pages 370–371:** Sine and cosine tables for $j = 1$, continued
4. **Pages 372–373:** Tables for $\left(\frac{r}{a}\right)^{\pm 2} \cos jf$ and $\left(\frac{r}{a}\right)^{\pm 2} \sin jf$
5. **Pages 374–375:** Tables for $\left(\frac{r}{a}\right)^{\pm 3} \cos jf$ and $\left(\frac{r}{a}\right)^{\pm 3} \sin jf$
6. **Pages 376–377:** Tables for $\left(\frac{r}{a}\right)^{4} \cos jf$ and $\left(\frac{r}{a}\right)^{4} \sin jf$
7. **Pages 378–395:** Extended cosine tables with higher multiples
8. **Pages 396–400:** Final sine tables and datum tables

Note on the Computational Method

The coefficients were computed from Leverrier's general formulae by assigning particular integer values to the indices and expanding in powers of the eccentricity. The calculations were verified by checking the symmetry conditions and by independent computation of particular values. The tables are designed to facilitate the calculation of the disturbing function and its derivatives in the lunar and planetary theories.

Note on Paper 216. These tables constitute a fundamental reference for the numerical computation of perturbations in celestial mechanics. Cayley's computations extend and systematize the earlier results of Hansen and others, providing the coefficients to a higher order in the eccentricity than had been previously available. The arrangement by sums and differences of the cosine and sine coefficients ($[\cos + \sin]^i$) is particularly convenient for practical computation, as it reduces the number of separate tabulations required and facilitates the verification of the results.

End of pages 351–400

[216]

THEORY OF ELLIPTIC MOTION.

[The memoir Nos 210 and 216 on the Elliptic Motion are, in the manuscripts, parts of a single memoir, and they are connected together by the analytical theory of the developments of the functions which present themselves in the Elliptic Motion. The following tables, which belong to the analytical theory, are accordingly placed here in immediate connexion with the memoir No. 216.]

TABLES OF THE DEVELOPMENTS OF FUNCTIONS IN THE

$(x^4,\ x^5,\ x^6,\ x^7)f\ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} f$	$[\ ]^0$			$[\ ]^1$			
		e	e^3	e^5	e^7	e^0	e^2	e^4
e^6								
x^4	[cos + sin]		$-\frac{3}{8}$	0			$+\frac{3}{8}$	$-\frac{1}{8}$
	[cos — sin]		"	"			$+\frac{1}{4}$	$-\frac{5}{128}$
	[cos]		$-\frac{3}{8}$	0			$+\frac{5}{16}$	$-\frac{21}{256}$
	[sin]		0	0			$+\frac{1}{16}$	$-\frac{11}{256}$
x^5	[cos + sin]		$-\frac{5}{16}$	0				$+\frac{5}{16}$
	[cos — sin]		"	"				$+\frac{15}{64}$
	[cos]		$-\frac{5}{16}$	0				$+\frac{35}{128}$
	[sin]		0	0				$+\frac{5}{128}$
x^6	[cos + sin]			$-\frac{5}{16}$				$+\frac{5}{16}$
	[cos — sin]			"				$+\frac{15}{64}$
	[cos]			$-\frac{5}{16}$				$+\frac{35}{128}$
	[sin]			0				$+\frac{5}{128}$
x^7	[cos + sin]				$-\frac{35}{128}$			
	[cos — sin]				"			
	[cos]				$-\frac{35}{128}$			
	[sin]				0			

$(x^4,\ x^5,\ x^6,\ x^7)f\ldots$

			$[\ \ ]^2$			$[\ \ ]^3$	
	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} f$	e	e^3	e^5	e^7	e^2	e^4
x^4	[cos + sin]		$-\frac{1}{8}$	$+\frac{1}{64}$		$+\frac{1}{4}$	$-\frac{51}{128}$
	[cos — sin]		$-\frac{1}{16}$	$-\frac{1}{64}$		$+\frac{1}{16}$	$-\frac{3}{32}$
	[cos]		$-\frac{3}{32}$	0		$+\frac{5}{32}$	$-\frac{63}{256}$
	[sin]		$-\frac{1}{32}$	$+\frac{1}{64}$		$+\frac{3}{32}$	$-\frac{39}{256}$
x^5	[cos + sin]		$-\frac{5}{16}$	$+\frac{35}{128}$			$-\frac{5}{64}$
	[cos — sin]		$-\frac{5}{32}$	$+\frac{13}{128}$			$-\frac{1}{32}$
	[cos]		$-\frac{15}{64}$	$+\frac{3}{16}$			$-\frac{7}{128}$
	[sin]		$-\frac{5}{64}$	$+\frac{11}{128}$			$-\frac{3}{128}$
x^6	[cos + sin]			$-\frac{5}{32}$			$+\frac{15}{64}$
	[cos — sin]			$-\frac{3}{32}$			$+\frac{3}{32}$
	[cos]			$-\frac{1}{8}$			$+\frac{21}{128}$
	[sin]			$-\frac{1}{32}$			$+\frac{9}{128}$
x^7	[cos + sin]				$-\frac{35}{128}$		
	[cos — sin]				$-\frac{21}{128}$		
	[cos]				$-\frac{7}{32}$		
	[sin]				$-\frac{7}{128}$		

$(x^4,\ x^5,\ x^6,\ x^7)f\ldots$

	$\frac{\cos}{\sin} f$	$[\ \ ]^5$ e^4	$[\ \ ]^5$ e^6	$[\ \ ]^6$ e^5	$[\ \ ]^6$ e^7	$[\ \ ]^7$ e^6	$[\ \ ]^8$ e^7
x^4	[cos + sin]	$+\frac{1}{16}$	$+\frac{7}{32}$	$+\frac{3}{16}$	$-\frac{3}{64}$	$+\frac{49}{128}$	$+\frac{2}{3}$
	[cos — sin]	0	$+\frac{7}{128}$	0	$+\frac{3}{64}$	0	0
	[cos]	H	$+\frac{35}{256}$	H	0	H	H
	[sin]	H	$+\frac{21}{256}$	H	$-\frac{3}{64}$	H	H
x^5	[cos + sin]		$-\frac{9}{32}$	$-\frac{1}{32}$	$-\frac{41}{128}$	$-\frac{7}{64}$	$-\frac{1}{4}$
	[cos — sin]		$-\frac{3}{64}$	0	$-\frac{7}{128}$	0	0
	[cos]		$-\frac{21}{128}$	H	$-\frac{3}{16}$	H	H
	[sin]		$-\frac{15}{128}$	H	$-\frac{17}{128}$	H	H
x^6	[cos + sin]		$+\frac{3}{32}$		$+\frac{7}{32}$	$+\frac{1}{64}$	$+\frac{1}{16}$
	[cos — sin]		$+\frac{1}{64}$		$+\frac{1}{32}$	0	0
	[cos]		$+\frac{7}{128}$		$+\frac{1}{8}$	H	H
	[sin]		$+\frac{5}{128}$		$+\frac{3}{32}$	H	H
x^7	[cos + sin]				$-\frac{7}{128}$		$-\frac{1}{128}$
	[cos — sin]				$-\frac{1}{128}$		0
	[cos]				$-\frac{1}{32}$		H
	[sin]				$-\frac{3}{128}$		H

$(x^0,\ x^1,\ x^2,\ x^3)2f\ldots$

		$[\ \ ]^0$			$[\ \ ]^1$				[	
	${}^{\cos}_{\sin}2f$	e^2	e^4	e^6	e	e^3	e^5	e^7	e^0	e^2
x^0	[cos + sin]	$+\frac{3}{4}$	$+\frac{1}{8}$	$+\frac{3}{64}$	-2	$+\frac{7}{4}$	$-\frac{5}{96}$	$+\frac{271}{4608}$	$+1$	-4
	[cos — sin]	”	”	”	0	$+\frac{1}{12}$	$+\frac{5}{192}$	$+\frac{49}{7680}$	0	0
	[cos]	$+\frac{3}{4}$	$+\frac{1}{8}$	$+\frac{3}{64}$	H	$+\frac{11}{12}$	$-\frac{5}{384}$	$+\frac{751}{23040}$	H	H
	[sin]	0	0	0	H	$+\frac{5}{6}$	$-\frac{5}{128}$	$+\frac{151}{5760}$	H	H
x^1	[cos + sin]	$+\frac{3}{4}$	$-\frac{1}{8}$	$-\frac{3}{64}$	$-\frac{1}{2}$	$+\frac{5}{16}$	$-\frac{53}{384}$	$-\frac{203}{18432}$	$+\frac{1}{2}$	$-\frac{1}{2}$
	[cos — sin]	”	”	”	0	$-\frac{1}{16}$	$-\frac{61}{768}$	$-\frac{271}{6144}$	0	$-\frac{1}{48}$
	[cos]	$+\frac{3}{4}$	$-\frac{1}{8}$	$-\frac{3}{64}$	H	$+\frac{1}{8}$	$-\frac{167}{1536}$	$-\frac{127}{4608}$	H	$-\frac{25}{96}$
	[sin]	0	0	0	H	$+\frac{3}{16}$	$-\frac{15}{512}$	$+\frac{305}{18432}$	H	$-\frac{23}{96}$
x^2	[cos + sin]	$+\frac{1}{4}$	$+\frac{1}{8}$	$+\frac{3}{64}$	$-\frac{3}{4}$	$+\frac{17}{48}$	$+\frac{19}{512}$		$+\frac{1}{2}$	-1
	[cos — sin]	”	”	”	$-\frac{1}{4}$	$+\frac{1}{96}$	$+\frac{83}{7680}$		0	$-\frac{1}{16}$
	[cos]	$+\frac{1}{4}$	$+\frac{1}{8}$	$+\frac{3}{64}$	$-\frac{1}{2}$	$+\frac{35}{192}$	$+\frac{23}{960}$		H	$-\frac{17}{32}$
	[sin]	0	0	0	$-\frac{1}{4}$	$+\frac{11}{64}$	$+\frac{101}{7680}$		H	$-\frac{15}{32}$
x^3	[cos + sin]	$+\frac{1}{2}$	$-\frac{3}{64}$		$-\frac{3}{8}$	$+\frac{5}{32}$	$-\frac{49}{1024}$		$+\frac{3}{8}$	$-\frac{1}{4}$
	[cos — sin]	”	”		$-\frac{1}{8}$	$-\frac{1}{64}$	$-\frac{39}{1024}$		$+\frac{1}{16}$	0
	[cos]	$+\frac{1}{2}$	$-\frac{3}{64}$		$-\frac{1}{4}$	$+\frac{9}{128}$	$-\frac{11}{256}$		$+\frac{7}{32}$	$-\frac{1}{8}$
	[sin]	0	0		$-\frac{1}{8}$	$+\frac{11}{128}$	$-\frac{5}{1024}$		$+\frac{5}{32}$	$-\frac{1}{8}$

$(x^0,\ x^1,\ x^2,\ x^3)2f\ldots$

		$[\ \ ]^3$				$[\ \ ]^4$			
	$\frac{\cos}{\sin}2f$	e	e^3	e^5	e^7	e^2	e^4	e^6	e^3
x^0	[cos + sin]	$+2$	$-\frac{27}{4}$	$+\frac{207}{32}$	$-\frac{1147}{512}$	$+\frac{13}{4}$	$-\frac{259}{24}$	$+\frac{559}{48}$	$+\frac{59}{12}$
	[cos — sin]	0	0	$+\frac{9}{320}$	$+\frac{27}{2560}$	0	0	$+\frac{1}{45}$	0
	[cos]	H	H	$+\frac{2079}{640}$	$-\frac{1427}{1280}$	H	H	$+\frac{8401}{1440}$	H
	[sin]	H	H	$+\frac{2061}{640}$	$-\frac{2881}{2560}$	H	H	$+\frac{8369}{1440}$	H
x^1	[cos + sin]	$-\frac{1}{2}$	$+\frac{33}{16}$	$-\frac{255}{128}$	$+\frac{1081}{2048}$	$-\frac{5}{4}$	$+\frac{107}{24}$	$-\frac{229}{48}$	$-\frac{37}{16}$
	[cos — sin]	0	0	$-\frac{3}{256}$	$-\frac{231}{10240}$	0	0	$-\frac{1}{120}$	0
	[cos]	H	H	$-\frac{513}{512}$	$+\frac{2587}{10240}$	H	H	$-\frac{1147}{480}$	H
	[sin]	H	H	$-\frac{507}{512}$	$+\frac{1409}{5120}$	H	H	$-\frac{381}{160}$	H
x^2	[cos + sin]	$+\frac{1}{4}$	$-\frac{13}{16}$	$+\frac{367}{512}$	$+\frac{1}{4}$	$-\frac{5}{8}$	0	$+\frac{3}{4}$	$-\frac{223}{96}$
	[cos — sin]	0	$-\frac{1}{32}$	$+\frac{71}{2560}$	0	0	$-\frac{1}{48}$	0	0
	[cos]	H	$-\frac{27}{64}$	$+\frac{953}{2560}$	H	H	$-\frac{1}{96}$	H	H
	[sin]	H	$-\frac{25}{64}$	$+\frac{441}{1280}$	H	H	$+\frac{1}{96}$	H	H
x^3	[cos + sin]	$-\frac{3}{8}$	$+\frac{33}{32}$	$-\frac{753}{1024}$		$-\frac{1}{2}$	$+\frac{95}{64}$	$-\frac{1}{8}$	$-\frac{13}{64}$
	[cos — sin]	0	$+\frac{3}{64}$	$-\frac{13}{1024}$		0	$+\frac{7}{192}$	0	0
	[cos]	H	$+\frac{69}{128}$	$-\frac{383}{1024}$		H	$+\frac{73}{96}$	H	H
	[sin]	H	$+\frac{63}{128}$	$-\frac{185}{512}$		H	$+\frac{139}{192}$	H	H

$(x^0,\ x^1,\ x^2,\ x^3)2f\ldots$

	$\genfrac{}{}{0pt}{}{\cos}{\sin}2f$	$[\ \ ]^6$		$[\ \ ]^7$		$[\ \ ]^8$	$[\ \ ]^9$
		e^4	e^6	e^5	e^7	e^6	e^7
x^0	[cos + sin]	$+\frac{115}{16}$	$-\frac{2049}{80}$	$+\frac{9893}{960}$	$-\frac{889393}{23040}$	$+\frac{42037}{2880}$	$+\frac{367439}{17920}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^1	[cos + sin]	$-\frac{61}{16}$	$+\frac{69}{5}$	$-\frac{4553}{768}$	$+\frac{2061389}{92160}$	$-\frac{8551}{960}$	$-\frac{26809}{2048}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^2	[cos + sin]	$+\frac{25}{16}$	$-\frac{21}{4}$	$+\frac{269}{96}$	$-\frac{77143}{7680}$	$+\frac{297}{64}$	$+\frac{18749}{2560}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^3	[cos + sin]	$-\frac{7}{16}$	$+\frac{3}{4}$	$-\frac{65}{64}$	$+\frac{8425}{3072}$	$-\frac{379}{192}$	$-\frac{3563}{1024}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H

$(x^4,\ x^5,\ x^6,\ x^7)2f\ldots$

e^6	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 2f$	$[\;]^0$			$[\;]^1$				$[\;]^2$		
		e^2	e^4	e^6	e	e^3	e^5	e^7	e^0	e^2	e^4
x^4	[cos + sin]	$+\frac{1}{4}$	$+\frac{3}{64}$		$-\frac{1}{2}$	$+\frac{5}{32}$			$+\frac{3}{8}$	$-\frac{1}{2}$	
	[cos — sin]	”	”		$-\frac{1}{4}$	0			$+\frac{1}{16}$	$-\frac{1}{16}$	
	[cos]	$+\frac{1}{4}$	$+\frac{3}{64}$		$-\frac{3}{8}$	$+\frac{5}{64}$			$+\frac{7}{32}$	$-\frac{9}{32}$	
	[sin]	0	0		$-\frac{1}{8}$	$+\frac{5}{64}$			$+\frac{5}{32}$	$-\frac{7}{32}$	
x^5	[cos + sin]		$+\frac{25}{64}$		$-\frac{5}{16}$	$+\frac{25}{256}$				$+\frac{5}{16}$	
	[cos — sin]		”		$-\frac{5}{32}$	$-\frac{1}{256}$				$+\frac{3}{32}$	
	[cos]		$+\frac{25}{64}$		$-\frac{15}{64}$	$+\frac{3}{64}$				$+\frac{13}{64}$	
	[sin]		0		$-\frac{5}{64}$	$+\frac{13}{256}$				$+\frac{7}{64}$	
x^6	[cos + sin]			$+\frac{15}{64}$		$-\frac{25}{64}$				$+\frac{5}{16}$	
	[cos — sin]			”		$-\frac{15}{64}$				$+\frac{3}{32}$	
	[cos]			$+\frac{15}{64}$		$-\frac{5}{16}$				$+\frac{13}{64}$	
	[sin]			0		$-\frac{5}{64}$				$+\frac{7}{64}$	
x^7	[cos + sin]					$-\frac{35}{128}$					
	[cos — sin]					$-\frac{21}{128}$					
	[cos]					$-\frac{7}{32}$					
	[sin]					$-\frac{7}{128}$					

$(x^4,\ x^5,\ x^6,\ x^7)2f \ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 2f$	$[\ \ ]^3$				$[\ \ ]^4$			$[\ \ ]^5$	
		e	e^3	e^5	e^7	e^2	e^4	e^6	e^3	e^5
x^4	[cos + sin]		0	$-\frac{3}{16}$		$+\frac{1}{4}$	$-\frac{43}{64}$		$+\frac{1}{2}$	$-\frac{47}{32}$
	[cos — sin]		0	$-\frac{1}{32}$		0	$-\frac{1}{64}$		0	$-\frac{1}{48}$
	[cos]		0	$-\frac{7}{64}$		H	$-\frac{11}{32}$		H	$-\frac{143}{192}$
	[sin]		0	$-\frac{5}{64}$		H	$-\frac{21}{64}$		H	$-\frac{139}{192}$
x^5	[cos + sin]	$-\frac{5}{16}$	$+\frac{165}{256}$			$-\frac{15}{64}$		$-\frac{5}{32}$	$+\frac{59}{256}$	
	[cos — sin]	$-\frac{1}{32}$	$+\frac{17}{256}$			$-\frac{1}{64}$		0	$-\frac{1}{256}$	
	[cos]	$-\frac{11}{64}$	$+\frac{91}{256}$			$-\frac{1}{8}$		H	$+\frac{29}{256}$	
	[sin]	$-\frac{9}{64}$	$+\frac{37}{128}$			$-\frac{7}{64}$		H	$+\frac{15}{128}$	
x^6	[cos + sin]		$-\frac{5}{64}$			$+\frac{15}{64}$			$+\frac{21}{64}$	
	[cos — sin]		$-\frac{1}{64}$			$+\frac{1}{64}$			$+\frac{1}{64}$	
	[cos]		$-\frac{3}{64}$			$+\frac{1}{8}$			$+\frac{11}{64}$	
	[sin]		$-\frac{1}{32}$			$+\frac{7}{64}$			$+\frac{5}{32}$	
x^7	[cos + sin]			$-\frac{35}{128}$			$-\frac{21}{128}$			
	[cos — sin]			$-\frac{7}{128}$			$-\frac{1}{128}$			
	[cos]			$-\frac{21}{128}$			$-\frac{11}{128}$			
	[sin]			$-\frac{7}{64}$			$-\frac{5}{64}$			

$(x^4,\ x^5,\ x^6,\ x^7)2f \ldots$

		$[\ \ ]^6$		$[\ \ ]^7$		$[\ \ ]^8$	$[\ \ ]^9$
	${}^{\cos}_{\sin} 2f$	e^4	e^6	e^5	e^7	e^6	e^7
x^4	[cos + sin]	$+\frac{1}{16}$	$+\frac{9}{16}$	$+\frac{1}{4}$	$+\frac{5}{24}$	$+\frac{41}{64}$	$+\frac{43}{32}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^5	[cos + sin]		$-\frac{13}{32}$	$-\frac{1}{32}$	$-\frac{163}{256}$	$-\frac{9}{64}$	$-\frac{101}{256}$
	[cos — sin]		0	0	0	0	0
	[cos]		H	H	H	H	H
	[sin]		H	H	H	H	H
x^6	[cos + sin]		$+\frac{3}{32}$		$+\frac{19}{64}$	$+\frac{1}{64}$	$+\frac{5}{64}$
	[cos — sin]		0		0	0	0
	[cos]		H		H	H	H
	[sin]		H		H	H	H
x^7	[cos + sin]				$-\frac{7}{128}$		$-\frac{1}{128}$
	[cos — sin]				0		0
	[cos]				H		H
	[sin]				H		H

$(x^0,\ x^1,\ x^2,\ x^3)3f\ldots$

	$\frac{\cos}{\sin}3f$	[ ]^0			[ ]^1			[ ]^2		
		e^3	e^5	e^7	e^2	e^4	e^6	e	e^3	e^5
x^0	[cos + sin]	$-\frac{1}{2}$	$-\frac{3}{16}$	$-\frac{3}{32}$	$+\frac{21}{8}$	$-\frac{31}{16}$	$-\frac{103}{1024}$	-3	$+\frac{33}{4}$	$-\frac{89}{16}$
	[cos — sin]	”	”	”	0	$-\frac{5}{128}$	$-\frac{11}{320}$	0	0	$-\frac{1}{80}$
	[cos]	$-\frac{1}{2}$	$-\frac{3}{16}$	$-\frac{3}{32}$	H	$-\frac{253}{256}$	$-\frac{691}{10240}$	H	H	$-\frac{223}{80}$
	[sin]	0	0	0	H	$-\frac{243}{256}$	$-\frac{339}{10240}$	H	H	$-\frac{111}{40}$
x^1	[cos + sin]	$-\frac{3}{4}$	$+\frac{1}{32}$	$+\frac{3}{64}$	$+\frac{5}{4}$	$-\frac{61}{48}$	$+\frac{129}{512}$	$-\frac{1}{2}$	$+\frac{9}{8}$	$-\frac{125}{96}$
	[cos — sin]	”	”	”	0	$-\frac{1}{96}$	$+\frac{59}{1280}$	0	0	$-\frac{1}{96}$
	[cos]	$-\frac{3}{4}$	$+\frac{1}{32}$	$+\frac{3}{64}$	H	$-\frac{41}{64}$	$+\frac{763}{5120}$	H	H	$-\frac{21}{32}$
	[sin]	0	0	0	H	$-\frac{121}{192}$	$+\frac{527}{5120}$	H	H	$-\frac{31}{48}$
x^2	[cos + sin]	$-\frac{1}{2}$	$+\frac{1}{8}$	0	$+\frac{1}{4}$	$+\frac{5}{16}$	$-\frac{85}{1536}$	-1	$+\frac{19}{12}$	$-\frac{49}{96}$
	[cos — sin]	”	”	”	0	$+\frac{5}{32}$	$+\frac{53}{768}$	0	$+\frac{1}{24}$	$+\frac{7}{480}$
	[cos]	$-\frac{1}{2}$	$+\frac{1}{8}$	0	H	$+\frac{15}{64}$	$+\frac{7}{1024}$	H	$+\frac{13}{16}$	$-\frac{119}{480}$
	[sin]	0	0	0	H	$+\frac{5}{64}$	$-\frac{191}{3072}$	H	$+\frac{37}{48}$	$-\frac{21}{80}$
x^3	[cos + sin]	$-\frac{1}{8}$	$-\frac{9}{32}$	$-\frac{3}{64}$	$+\frac{3}{4}$	$-\frac{65}{128}$		$-\frac{3}{8}$	$+\frac{9}{16}$	$-\frac{61}{128}$
	[cos — sin]	”	”	”	$+\frac{3}{16}$	$-\frac{1}{32}$		0	0	$-\frac{5}{128}$
	[cos]	$-\frac{1}{8}$	$-\frac{9}{32}$	$-\frac{3}{64}$	$+\frac{15}{32}$	$-\frac{69}{256}$		H	H	$-\frac{33}{128}$
	[sin]	0	0	0	$+\frac{9}{32}$	$-\frac{61}{256}$		H	H	$-\frac{7}{32}$

$(x^0,\ x^1,\ x^2,\ x^3)3f\ldots$

		$[\ \ ]^3$				$[\ \ ]^4$			
e^6	${}^{\cos}_{\sin}3f$	e^0	e^2	e^4	e^6	e	e^3	e^5	e^7
x^0 $+\frac{75643}{1024}$	[cos + sin]	$+1$	-9	$+\frac{1215}{64}$	$-\frac{449}{32}$	$+3$	$-\frac{39}{2}$	$+\frac{155}{4}$	$-\frac{767}{24}$
0	[cos — sin]	0	0	0	$-\frac{27}{5120}$	0	0	0	$-\frac{1}{420}$
H	[cos]	H	H	H	$-\frac{71867}{10240}$	H	H	H	$-\frac{8949}{560}$
H	[sin]	H	H	H	$-\frac{71813}{10240}$	H	H	H	$-\frac{26843}{1680}$
x^1	[cos + sin]	$+\frac{1}{2}$	$-\frac{9}{8}$	$-\frac{9}{128}$	$-\frac{1}{2}$	$+\frac{15}{4}$	$-\frac{173}{24}$	$+\frac{161}{36}$	$-\frac{7}{4}$
	[cos — sin]	0	0	$-\frac{21}{2560}$	0	0	0	$-\frac{1}{144}$	0
	[cos]	H	H	$-\frac{201}{5120}$	H	H	H	$+\frac{643}{288}$	H
	[sin]	H	H	$-\frac{159}{5120}$	H	H	H	$+\frac{215}{96}$	H
x^2	[cos + sin]	$+\frac{1}{2}$	$-\frac{9}{4}$	$+\frac{399}{128}$	$+\frac{1}{2}$	$-\frac{8}{3}$	$+\frac{437}{96}$	$+\frac{1}{4}$	$-\frac{13}{16}$
	[cos — sin]	0	0	$+\frac{11}{512}$	0	0	$+\frac{7}{480}$	0	0
	[cos]	H	H	$+\frac{1607}{1024}$	H	H	$+\frac{137}{60}$	H	H
	[sin]	H	H	$+\frac{1585}{1024}$	H	H	$+\frac{303}{160}$	H	H
x^3	[cos + sin]		$+\frac{3}{8}$	$-\frac{9}{16}$	$-\frac{3}{8}$	$+\frac{15}{8}$	$-\frac{343}{128}$	$-\frac{3}{4}$	$+\frac{469}{128}$
	[cos — sin]		0	$-\frac{1}{128}$	0	0	$-\frac{1}{128}$	0	0
	[cos]		H	$-\frac{73}{256}$	H	H	$-\frac{43}{32}$	H	H
	[sin]		H	$-\frac{71}{256}$	H	H	$-\frac{171}{128}$	H	H

$(x^0,\ x^1,\ x^2,\ x^3)3f\ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 3f$	$[\quad]^6$ e^3	e^5	e^7	$[\quad]^7$ e^4	e^6
x^0	[cos + sin]	$+\frac{47}{4}$	$-\frac{525}{8}$	$+\frac{43041}{320}$	$+\frac{2567}{128}$	$-\frac{3556}{320}$
	[cos — sin]	0	0	0	0	0
	[cos]	H	H	H	H	H
	[sin]	H	H	H	H	H
x^1	[cos + sin]	$-\frac{33}{8}$	$+\frac{373}{16}$	$-\frac{5919}{128}$	$-\frac{787}{96}$	$+\frac{5814}{128}$
	[cos — sin]	0	0	0	0	0
	[cos]	H	H	H	H	H
	[sin]	H	H	H	H	H
x^2	[cos + sin]	$+1$	$-\frac{37}{8}$	$+\frac{891}{160}$	$+\frac{83}{32}$	$-\frac{987}{768}$
	[cos — sin]	0	0	0	0	0
	[cos]	H	H	H	H	H
	[sin]	H	H	H	H	H
x^3	[cos + sin]	$-\frac{1}{8}$	$-\frac{9}{16}$	$+\frac{627}{128}$	$-\frac{9}{16}$	$+\frac{35}{32}$
	[cos — sin]	0	0	0	0	0
	[cos]	H	H	H	H	H
	[sin]	H	H	H	H	H

$(x^4,\ x^5,\ x^6,\ x^7)3f\ldots$

		$[\ ]^0$			$[\ ]^1$			$[\ ]^2$			
	$\frac{\cos}{\sin}3f$	e^3	e^5	e^7	e^2	e^4	e^6	e	e^3	e^5	e^7
x^4	[cos + sin]	$-\frac{7}{16}$	$+\frac{3}{32}$		$+\frac{1}{4}$	$+\frac{9}{128}$		$-\frac{5}{8}$	$+\frac{43}{64}$		
	[cos — sin]	”	”		$+\frac{1}{16}$	$+\frac{3}{32}$		$-\frac{1}{16}$	$+\frac{5}{192}$		
	[cos]	$-\frac{7}{16}$	$+\frac{3}{32}$		$+\frac{5}{32}$	$+\frac{21}{256}$		$-\frac{11}{32}$	$+\frac{67}{192}$		
	[sin]	0	0		$+\frac{3}{32}$	$-\frac{3}{256}$		$-\frac{9}{32}$	$+\frac{31}{96}$		
x^5	[cos + sin]	$-\frac{5}{32}$	$-\frac{9}{64}$			$+\frac{35}{64}$		$-\frac{5}{16}$	$+\frac{45}{128}$		
	[cos — sin]	”	”			$+\frac{7}{32}$		$-\frac{1}{32}$	$+\frac{1}{128}$		
	[cos]	$-\frac{5}{32}$	$-\frac{9}{64}$			$+\frac{49}{128}$		$-\frac{11}{64}$	$+\frac{23}{128}$		
	[sin]	0	0			$+\frac{21}{128}$		$-\frac{9}{64}$	$+\frac{11}{64}$		
x^6	[cos + sin]		$-\frac{3}{8}$			$+\frac{15}{64}$			$-\frac{15}{32}$		
	[cos — sin]		”			$+\frac{3}{32}$			$-\frac{3}{32}$		
	[cos]		$-\frac{3}{8}$			$+\frac{21}{128}$			$-\frac{9}{32}$		
	[sin]		0			$+\frac{9}{128}$			$-\frac{3}{16}$		
x^7	[cos + sin]			$-\frac{21}{128}$					$-\frac{35}{128}$		
	[cos — sin]			”					$-\frac{7}{128}$		
	[cos]			$-\frac{21}{128}$					$-\frac{21}{128}$		
	[sin]			0					$-\frac{7}{64}$		

$(x^4,\ x^5,\ x^6,\ x^7)3f\ldots$

e^6	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 3f$	$[\ \ ]^3$			$[\ \ ]^4$			$[\ \ ]^5$			
		e^0	e^2	e^4	e^6	e	e^3	e^5	e^7	e^2	e^4
x^4	[cos + sin]		$+\frac{3}{8}$	$-\frac{9}{8}$		$+\frac{1}{8}$	$-\frac{25}{32}$		$+\frac{1}{4}$	$-\frac{129}{128}$	
	[cos — sin]		0	$-\frac{3}{128}$		0	$-\frac{1}{96}$		0	0	
	[cos]		H	$-\frac{147}{256}$		H	$-\frac{19}{48}$		H	H	
	[sin]		H	$-\frac{141}{256}$		H	$-\frac{37}{96}$		H	H	
x^5	[cos + sin]		$+\frac{5}{16}$			$-\frac{5}{16}$	$+\frac{75}{64}$			$-\frac{25}{64}$	
	[cos — sin]		$+\frac{1}{64}$			0	$+\frac{1}{64}$			0	
	[cos]		$+\frac{21}{128}$			H	$+\frac{19}{32}$			H	
	[sin]		$+\frac{19}{128}$			H	$+\frac{37}{64}$			H	
x^6	[cos + sin]		$+\frac{5}{16}$				0			$+\frac{15}{64}$	
	[cos — sin]		$+\frac{1}{64}$				0			0	
	[cos]		$+\frac{21}{128}$				H			H	
	[sin]		$+\frac{19}{128}$				H			H	
x^7	[cos + sin]						$-\frac{35}{128}$				
	[cos — sin]						$-\frac{1}{128}$				
	[cos]						$-\frac{9}{64}$				
	[sin]						$-\frac{17}{128}$				

$(x^4,\ x^5,\ x^6,\ x^7)3f\ldots$

		$[\ \ ]^6$		$[\ \ ]^7$		$[\ \ ]^8$		$[\ \ ]^9$	$[\ \ ]^{10}$	
	$\frac{\cos}{\sin}3f$	e^3	e^5	e^7	e^4	e^6	e^5	e^7	e^6	e^7
x^4	[cos + sin]	$+\frac{11}{16}$	$-\frac{195}{64}$	$+\frac{1}{16}$	$+\frac{33}{32}$	$+\frac{5}{16}$	$+\frac{67}{96}$	$+\frac{123}{128}$	$+\frac{451}{192}$	
	[cos — sin]	0	0	0	0	0	0	0	0	
	[cos]	H	H	H	H	H	H	H	H	
	[sin]	H	H	H	H	H	H	H	H	
x^5	[cos + sin]	$-\frac{5}{32}$	$+\frac{27}{128}$		$-\frac{17}{32}$	$-\frac{1}{32}$	$-\frac{67}{64}$	$-\frac{11}{64}$	$-\frac{73}{128}$	
	[cos — sin]	0	0		0	0	0	0	0	
	[cos]	H	H		H	H	H	H	H	
	[sin]	H	H		H	H	H	H	H	
x^6	[cos + sin]		$+\frac{15}{32}$		$+\frac{3}{32}$	$+\frac{3}{8}$	$+\frac{1}{64}$	$+\frac{3}{32}$		
	[cos — sin]		0		0	0	0	0		
	[cos]		H		H	H	H	H		
	[sin]		H		H	H	H	H		
x^7	[cos + sin]		$-\frac{21}{128}$			$-\frac{7}{128}$		$-\frac{1}{128}$		
	[cos — sin]		0			0		0		
	[cos]		H			H		H		
	[sin]		H			H		H		

$(x^0,\ x^1,\ x^2,\ x^3)4f\ldots$

	$\frac{\cos}{\sin}4f$	$[\ \ ]^0$ e^4	e^6	$[\ \ ]^1$ e	e^3	e^5
x^0	[cos + sin]	$+\frac{5}{16}$	$+\frac{3}{16}$	$-\frac{17}{6}$	$+\frac{161}{96}$	$+\frac{907}{3840}$
	[cos — sin]	”	”	0	$+\frac{7}{480}$	$+\frac{277}{11520}$
	[cos]	$+\frac{5}{16}$	$+\frac{3}{16}$	H	$+\frac{293}{240}$	$+\frac{1499}{11520}$
	[sin]	0	0	H	$+\frac{133}{160}$	$+\frac{611}{5760}$
x^1	[cos + sin]	$+\frac{5}{8}$	$+\frac{3}{32}$	$-\frac{31}{16}$	$+\frac{1447}{768}$	$-\frac{4537}{30720}$
	[cos — sin]	”	”	0	$+\frac{19}{768}$	$-\frac{793}{92160}$
	[cos]	$+\frac{5}{8}$	$+\frac{3}{32}$	H	$+\frac{733}{768}$	$-\frac{3601}{46080}$
	[sin]	0	0	H	$+\frac{119}{128}$	$-\frac{6409}{92160}$
x^2	[cos + sin]	$+\frac{5}{8}$	$-\frac{5}{32}$	$-\frac{3}{4}$	$+\frac{53}{96}$	$-\frac{2123}{7680}$
	[cos — sin]	”	”	0	$-\frac{7}{96}$	$-\frac{649}{7680}$
	[cos]	$+\frac{5}{8}$	$-\frac{5}{32}$	H	$+\frac{23}{96}$	$-\frac{231}{1280}$
	[sin]	0	0	H	$+\frac{5}{16}$	$-\frac{737}{7680}$
x^3	[cos + sin]	$+\frac{5}{16}$	0	$-\frac{1}{8}$	$-\frac{43}{64}$	$+\frac{389}{1024}$
	[cos — sin]	”	”	0	$-\frac{11}{64}$	$-\frac{59}{3072}$
	[cos]	$+\frac{5}{16}$	0	H	$-\frac{27}{64}$	$+\frac{277}{1536}$
	[sin]	0	0	H	$-\frac{1}{4}$	$+\frac{613}{3072}$

$(x^0,\ x^1,\ x^2,\ x^3)4f\ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 4f$	$[\ \ ]^3$				$[\ \ ]^4$			
		e	e^3	e^5	e^7	e^0	e^2	e^4	e^6
x^0	[cos + sin]	-4	$+\frac{45}{2}$	$-\frac{591}{16}$	$+\frac{5737}{256}$	$+1$	-16	$+\frac{247}{4}$	$-\frac{3355}{36}$
	[cos — sin]	0	0	0	$-\frac{9}{8960}$	0	0	0	0
	[cos]	H	H	H	$+\frac{100393}{8960}$	H	H	H	H
	[sin]	H	H	H	$+\frac{50201}{4480}$	H	H	H	H
x^1	[cos + sin]	$-\frac{1}{2}$	$+\frac{39}{16}$	$-\frac{687}{128}$	$+\frac{12355}{2048}$		$+\frac{1}{2}$	-2	$-\frac{1}{8}$
	[cos — sin]	0	0	0	$+\frac{69}{10240}$		0	0	0
	[cos]	H	H	H	$+\frac{15461}{5120}$		H	H	H
	[sin]	H	H	H	$+\frac{30853}{10240}$		H	H	H
x^2	[cos + sin]		$-\frac{5}{4}$	$+\frac{67}{16}$	$-\frac{2119}{512}$		$+\frac{1}{2}$	-4	$+\frac{245}{24}$
	[cos — sin]		0	0	$-\frac{17}{2560}$		0	0	0
	[cos]		H	H	$-\frac{2653}{1280}$		H	H	H
	[sin]		H	H	$-\frac{5289}{2560}$		H	H	H
x^3	[cos + sin]	$-\frac{3}{8}$	$+\frac{39}{32}$	$-\frac{2037}{1024}$			$+\frac{3}{8}$	-1	
	[cos — sin]	0	0	$-\frac{7}{1024}$			0	0	
	[cos]	H	H	$-\frac{511}{512}$			H	H	
	[sin]	H	H	$-\frac{1015}{1024}$			H	H	

$(x^0,\ x^1,\ x^2,\ x^3)4f\ldots$

		$[\ \ ]^5$			$[\ \ ]^6$			$[\ \ ]^7$	
	$\frac{\cos}{\sin}4f$	e	e^3	e^5	e^7	e^2	e^4	e^6	e^3
x^0	[cos + sin]	$+4$	$-\frac{85}{2}$	$+\frac{6845}{48}$	$-\frac{499585}{2304}$	$+\frac{21}{2}$	$-\frac{377}{4}$	$+\frac{597}{2}$	$+\frac{137}{6}$
	[cos — sin]	0	0	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H	H	H
x^1	[cos + sin]	$-\frac{1}{2}$	$+\frac{95}{16}$	$-\frac{7325}{384}$	$+\frac{497635}{18432}$	$-\frac{9}{4}$	$+\frac{83}{4}$	$-\frac{4029}{64}$	$-\frac{103}{16}$
	[cos — sin]	0	0	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H	H	H
x^2	[cos + sin]		$+\frac{3}{4}$	$-\frac{295}{48}$	$+\frac{8785}{512}$	$+\frac{1}{4}$	-1	$-\frac{215}{64}$	$+\frac{5}{4}$
	[cos — sin]		0	0	0	0	0	0	0
	[cos]		H	H	H	H	H	H	H
	[sin]		H	H	H	H	H	H	H
x^3	[cos + sin]	$-\frac{3}{8}$	$+\frac{95}{32}$	$-\frac{7285}{1024}$		-1	$+\frac{117}{16}$	$-\frac{1}{8}$	$-\frac{1}{8}$
	[cos — sin]	0	0	0		0	0	0	0
	[cos]	H	H	H		H	H	H	H
	[sin]	H	H	H		H	H	H	H

$(x^0,\ x^1,\ x^2,\ x^3)4f\ldots$

	$\begin{smallmatrix}\cos\\\sin\end{smallmatrix} 4f$	$[\ ]^8$ e^4	$[\ ]^8$ e^6	$[\ ]^9$ e^5	$[\ ]^9$ e^7	$[\ ]^{10}$ e^6	$[\ ]^{11}$ e^7
x^0	[cos + sin]	$+\frac{2141}{48}$	$-\frac{84857}{240}$	$+\frac{13011}{160}$	$-\frac{809813}{1280}$	$+\frac{5087}{36}$	$+\frac{19116073}{80640}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^1	[cos + sin]	$-\frac{359}{24}$	$+\frac{56527}{480}$	$-\frac{7887}{256}$	$+\frac{2427367}{10240}$	$-\frac{11263}{192}$	$-\frac{9742939}{92160}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^2	[cos + sin]	$+\frac{31}{8}$	$-\frac{855}{32}$	$+\frac{307}{32}$	$-\frac{172907}{2560}$	$+\frac{3995}{192}$	$+\frac{317653}{7680}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^3	[cos + sin]	$-\frac{11}{16}$	$+\frac{3}{2}$	$-\frac{147}{64}$	$+\frac{10999}{1024}$	$-\frac{145}{24}$	$-\frac{42325}{3072}$
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H

$(x^4,\ x^5,\ x^6,\ x^7)4f\ldots$

$[\ \ ]^3$		$[\ \ ]^0$		$[\ \ ]^1$			$[\ \ ]^2$				
e^5	${}^{\cos}_{\sin}4f$ e^7	e^4	e^6	e^3	e^5	e^7	e^2	e^4	e^6	e^1	e^3
x^4	[cos + sin]	$+\frac{1}{16}$	$+\frac{3}{8}$	$-\frac{5}{8}$	$+\frac{29}{64}$		$+\frac{1}{4}$	$+\frac{1}{32}$		$-\frac{3}{4}$	$+\frac{111}{64}$
	[cos — sin]	”	”	$-\frac{1}{8}$	$+\frac{1}{192}$		0	$+\frac{1}{32}$		0	$+\frac{1}{64}$
	[cos]	$+\frac{1}{16}$	$+\frac{3}{8}$	$-\frac{3}{8}$	$+\frac{11}{48}$		H	$+\frac{1}{32}$		H	$+\frac{7}{8}$
	[sin]	0	0	$-\frac{1}{4}$	$+\frac{43}{192}$		H	0		H	$+\frac{55}{64}$
x^5	[cos + sin]		$+\frac{11}{32}$	$-\frac{5}{32}$	$-\frac{79}{256}$			$+\frac{45}{64}$		$-\frac{5}{16}$	$+\frac{195}{256}$
	[cos — sin]		”	$-\frac{1}{32}$	$-\frac{37}{256}$			$+\frac{3}{64}$		0	$+\frac{1}{256}$
	[cos]		$+\frac{11}{32}$	$-\frac{3}{32}$	$-\frac{29}{128}$			$+\frac{3}{8}$		H	$+\frac{49}{128}$
	[sin]		0	$-\frac{1}{16}$	$-\frac{21}{256}$			$+\frac{21}{64}$		H	$+\frac{97}{256}$
x^6	[cos + sin]		$+\frac{3}{32}$		$-\frac{33}{64}$			$+\frac{15}{64}$			$-\frac{35}{64}$
	[cos — sin]		”		$-\frac{11}{64}$			$+\frac{1}{64}$			$-\frac{1}{64}$
	[cos]		$+\frac{3}{32}$		$-\frac{11}{32}$			$+\frac{1}{8}$			$-\frac{9}{32}$
	[sin]		0		$-\frac{11}{64}$			$+\frac{7}{64}$			$-\frac{17}{64}$
x^7	[cos + sin]				$-\frac{21}{128}$						$-\frac{35}{128}$
	[cos — sin]				$-\frac{7}{128}$						$-\frac{1}{128}$
	[cos]				$-\frac{7}{64}$						$-\frac{9}{64}$
	[sin]				$-\frac{7}{128}$						$-\frac{17}{128}$

$(x^4,\ x^5,\ x^6,\ x^7)4f\ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 4f$	$[\ \ ]^4$				$[\ \ ]^5$			$[\ \ ]^6$		
e^6		e^0	e^2	e^4	e^6	e	e^3	e^5	e^7	e^2	e^4
x^4	[cos + sin]		$+\frac{3}{8}$	-2		$+\frac{1}{4}$	$-\frac{125}{64}$		$+\frac{1}{4}$	$-\frac{45}{32}$	
	[cos — sin]		0	0		0	0		0	0	
	[cos]		H	H		H	H		H	H	
	[sin]		H	H		H	H		H	H	
x^5	[cos + sin]		$-\frac{5}{16}$			$-\frac{5}{16}$	$+\frac{475}{256}$			$-\frac{35}{64}$	
	[cos — sin]		0			0	0			0	
	[cos]		H			H	H			H	
	[sin]		H			H	H			H	
x^6	[cos + sin]		$+\frac{5}{16}$				$+\frac{5}{64}$			$+\frac{15}{64}$	
	[cos — sin]		0				0			0	
	[cos]		H				H			H	
	[sin]		H				H			H	
x^7	[cos + sin]						$-\frac{35}{128}$				
	[cos — sin]						0				
	[cos]						H				
	[sin]						H				

$(x^4,\ x^5,\ x^6,\ x^7)4f \ldots$

	$\frac{\cos}{\sin} 4f$	$[\ ]^7$ e^3	e^5	e^7	$[\ ]^8$ e^4	e^6	e^5
x^4	[cos + sin]	$+\frac{7}{8}$	$-\frac{351}{64}$	$+\frac{1}{16}$	$+\frac{13}{8}$	$+\frac{3}{8}$	$+\frac{9}{6}$ …
	[cos — sin]	0	0	0	0	0	0
	[cos]	H	H	H	H	H	H
	[sin]	H	H	H	H	H	H
x^5	[cos + sin]	$-\frac{5}{32}$	$+\frac{41}{256}$		$-\frac{21}{32}$	$-\frac{1}{32}$	$-\frac{3}{2}$ …
	[cos — sin]	0	0		0	0	0
	[cos]	H	H		H	H	H
	[sin]	H	H		H	H	H
x^6	[cos + sin]		$+\frac{39}{64}$		$+\frac{3}{32}$	$+\frac{29}{64}$	$+$ … 6 …
	[cos — sin]		0		0	0	0
	[cos]		H		H	H	H
	[sin]		H		H	H	H
x^7	[cos + sin]		$-\frac{21}{128}$			$-\frac{7}{128}$	
	[cos — sin]		0			0	
	[cos]		H			H	
	[sin]		H			H	

$(x^0,\ x^1,\ x^2,\ x^3)5f\ldots$

	$\begin{smallmatrix}\cos\\ \sin\end{smallmatrix} 5f$	$[\ \ ]^0$		$[\ \ ]^1$		$[\ \ ]^2$			
		e^5	e^7	e^4	e^6	e^3	e^5	e^7	e^2
x^0	[cos + sin]	$-\frac{3}{16}$	$-\frac{5}{32}$	$+\frac{1045}{384}$	$-\frac{229}{192}$	$-\frac{95}{12}$	$+\frac{185}{12}$	$-\frac{1295}{192}$	$+\frac{75}{8}$
	[cos — sin]	”	”	0	$-\frac{37}{9216}$	0	0	$+\frac{5}{4032}$	0
	[cos]	$-\frac{3}{16}$	$-\frac{5}{32}$	H	$-\frac{11029}{18432}$	H	H	$-\frac{13595}{4032}$	H
	[sin]	0	0	H	$-\frac{10955}{18432}$	H	H	$-\frac{425}{126}$	H
x^1	[cos + sin]	$-\frac{15}{32}$	$-\frac{11}{64}$	$+\frac{229}{96}$	$-\frac{7747}{3840}$	$-\frac{29}{8}$	$+\frac{205}{24}$	$-\frac{11831}{1920}$	$+\frac{9}{4}$
	[cos — sin]	”	”	0	$-\frac{151}{7680}$	0	0	$-\frac{37}{5760}$	0
	[cos]	$-\frac{15}{32}$	$-\frac{11}{64}$	H	$-\frac{1043}{1024}$	H	H	$-\frac{3553}{1152}$	H
	[sin]	0	0	H	$-\frac{15343}{15360}$	H	H	$-\frac{277}{90}$	H
x^2	[cos + sin]	$-\frac{5}{8}$	$+\frac{1}{16}$	$+\frac{43}{32}$	$-\frac{357}{256}$	-1	$+\frac{37}{24}$	$-\frac{691}{480}$	$+\frac{1}{4}$
	[cos — sin]	”	”	0	$+\frac{37}{1536}$	0	0	$+\frac{1}{480}$	0
	[cos]	$-\frac{5}{8}$	$+\frac{1}{16}$	H	$-\frac{2105}{3072}$	H	H	$-\frac{23}{32}$	H
	[sin]	0	0	H	$-\frac{2179}{3072}$	H	H	$-\frac{173}{240}$	H
x^3	[cos + sin]	$-\frac{15}{32}$	$+\frac{5}{32}$	$+\frac{7}{16}$	$+\frac{3}{32}$	$-\frac{1}{8}$	$-\frac{19}{16}$	$+\frac{261}{128}$	
	[cos — sin]	”	”	0	$+\frac{47}{384}$	0	0	$+\frac{7}{384}$	
	[cos]	$-\frac{15}{32}$	$+\frac{5}{32}$	H	$+\frac{83}{768}$	H	H	$+\frac{395}{384}$	
	[sin]	0	0	H	$-\frac{11}{768}$	H	H	$+\frac{97}{96}$	

[216] THEORY OF ELLIPTIC MOTION.

Tables of the Developments of Functions in the Theory of Elliptic Motion (continued)

The tables which follow give the developments of the following functions, so far as e^7:

$$\cos f, \quad \sin f, \quad \cos g, \quad \sin g, \quad \cos 2f, \quad \sin 2f, \quad \text{etc.}$$

in terms of the eccentric anomaly g and the mean anomaly τ.

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos f \sin f$

	C	S	cos	sin
1	$1+\frac{1}{2}e^2+\frac{3}{8}e^4+\frac{5}{16}e^6$	—	$\cos g$	—
	$-e-\frac{3}{8}e^3-\frac{5}{16}e^5-\frac{175}{1024}e^7$	—	$\cos 2g$	—
	$+\frac{1}{2}e^2+\frac{1}{3}e^4+\frac{15}{64}e^6$	—	$\cos 3g$	—
	$-\frac{5}{16}e^3-\frac{35}{128}e^5-\frac{35}{256}e^7$	—	$\cos 4g$	—
	$+\frac{3}{16}e^4+\frac{21}{160}e^6$	—	$\cos 5g$	—
	$-\frac{7}{64}e^5-\frac{21}{256}e^7$	—	$\cos 6g$	—
	$+\frac{1}{16}e^6$	—	$\cos 7g$	—
	$-\frac{9}{256}e^7$	—	$\cos 8g$	—

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos f \sin f$ **(continued)**

	C	S	cos	sin
$\frac{r}{a}$	$1+\frac{1}{2}e^2-\frac{1}{8}e^4-\frac{1}{16}e^6$	—	$\cos g$	—
	$-e-\frac{1}{2}e^3+\frac{1}{16}e^5+\frac{1}{32}e^7$	—	$\cos 2g$	—
	—	$+e+\frac{5}{8}e^3+\frac{5}{16}e^5+\frac{5}{128}e^7$	—	$\sin g$
	—	$-\frac{3}{4}e^2-\frac{11}{16}e^4-\frac{9}{16}e^6$	—	$\sin 2g$
	—	$+\frac{1}{2}e^3+\frac{7}{8}e^5+\frac{81}{128}e^7$	—	$\sin 3g$
	—	$-\frac{5}{16}e^4-\frac{85}{128}e^6$	—	$\sin 4g$
	—	$+\frac{3}{16}e^5+\frac{25}{32}e^7$	—	$\sin 5g$
	—	$-\frac{7}{64}e^6$	—	$\sin 6g$
	—	$+\frac{1}{16}e^7$	—	$\sin 7g$

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos 2f$**,** $\sin 2f$

	C	S	cos	sin
$\frac{a^2}{r^2}$	$e+\frac{5}{4}e^3+\frac{11}{8}e^5+\frac{235}{192}e^7$	—	$\cos g$	—
	$1+2e^2+\frac{17}{8}e^4+\frac{25}{12}e^6$	—	$\cos 2g$	—
	$e+\frac{19}{12}e^3+\frac{35}{16}e^5+\frac{1181}{384}e^7$	—	$\cos 3g$	—
	$\frac{3}{4}e^2+\frac{7}{4}e^4+\frac{1411}{320}e^6$	—	$\cos 4g$	—
	$\frac{1}{3}e^3+\frac{47}{48}e^5+\frac{635}{256}e^7$	—	$\cos 5g$	—
	$\frac{5}{24}e^4+\frac{25}{36}e^6$	—	$\cos 6g$	—
	$\frac{1}{20}e^5+\frac{137}{480}e^7$	—	$\cos 7g$	—
	$\frac{7}{360}e^6$	—	$\cos 8g$	—
	$\frac{1}{252}e^7$	—	$\cos 9g$	—

	C	S	cos	sin
$\frac{a^2}{r^2}$	—	$e+\frac{5}{4}e^3+\frac{11}{8}e^5+\frac{235}{192}e^7$	—	$\sin g$
	—	$1+2e^2+\frac{17}{8}e^4+\frac{25}{12}e^6$	—	$\sin 2g$
	—	$-e-\frac{19}{12}e^3-\frac{35}{16}e^5-\frac{1181}{384}e^7$	$\cos 3g$	—
	—	$-\frac{3}{4}e^2-\frac{7}{4}e^4-\frac{1411}{320}e^6$	$\cos 4g$	—
	—	$-\frac{1}{3}e^3-\frac{47}{48}e^5-\frac{635}{256}e^7$	$\cos 5g$	—
	—	$-\frac{5}{24}e^4-\frac{25}{36}e^6$	$\cos 6g$	—
	—	$-\frac{1}{20}e^5-\frac{137}{480}e^7$	$\cos 7g$	—
	—	$-\frac{7}{360}e^6$	$\cos 8g$	—
	—	$-\frac{1}{252}e^7$	$\cos 9g$	—

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos 2f$**,** $\sin 2f$ **(continued)**

	C	S	cos	sin
$\frac{a}{r}$	$2e + 2e^3 + \frac{7}{4}e^5 + \frac{13}{6}e^7$	—	$\cos g$	—
	$1 + \frac{5}{2}e^2 + 3e^4 + \frac{43}{16}e^6$	—	$\cos 2g$	—
	$2e + \frac{11}{3}e^3 + \frac{31}{8}e^5 + \frac{407}{80}e^7$	—	$\cos 3g$	—
	$\frac{3}{2}e^2 + 4e^4 + \frac{113}{16}e^6$	—	$\cos 4g$	—
	$e^3 + \frac{19}{6}e^5 + \frac{77}{12}e^7$	—	$\cos 5g$	—
	$\frac{5}{8}e^4 + \frac{65}{24}e^6$	—	$\cos 6g$	—
	$\frac{3}{10}e^5 + \frac{17}{8}e^7$	—	$\cos 7g$	—
	$\frac{7}{36}e^6$	—	$\cos 8g$	—
	$\frac{2}{21}e^7$	—	$\cos 9g$	—

	C	S	cos	sin
$\frac{a}{r}$	—	$2e + 2e^3 + \frac{7}{4}e^5 + \frac{13}{6}e^7$	—	$\sin g$
	—	$1 + \frac{5}{2}e^2 + 3e^4 + \frac{43}{16}e^6$	—	$\sin 2g$
	—	$-2e - \frac{11}{3}e^3 - \frac{31}{8}e^5 - \frac{407}{80}e^7$	$\cos 3g$	—
	—	$-\frac{3}{2}e^2 - 4e^4 - \frac{113}{16}e^6$	$\cos 4g$	—
	—	$-e^3 - \frac{19}{6}e^5 - \frac{77}{12}e^7$	$\cos 5g$	—
	—	$-\frac{5}{8}e^4 - \frac{65}{24}e^6$	$\cos 6g$	—
	—	$-\frac{3}{10}e^5 - \frac{17}{8}e^7$	$\cos 7g$	—
	—	$-\frac{7}{36}e^6$	$\cos 8g$	—
	—	$-\frac{2}{21}e^7$	$\cos 9g$	—

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos 2f$**,** $\sin 2f$ **(continued)**

	C	S	cos	sin
$\frac{r}{a}$	$-2e + e^3 - \frac{1}{4}e^5 + \frac{1}{6}e^7$	—	$\cos g$	—
	$1 - \frac{1}{2}e^2 - \frac{1}{2}e^4 - \frac{11}{48}e^6$	—	$\cos 2g$	—
	$2e - 2e^3 + \frac{5}{6}e^5 - \frac{1}{20}e^7$	—	$\cos 3g$	—
	$\frac{3}{2}e^2 - \frac{9}{4}e^4 + \frac{27}{16}e^6$	—	$\cos 4g$	—
	$e^3 - \frac{8}{3}e^5 + \frac{31}{6}e^7$	—	$\cos 5g$	—
	$\frac{5}{8}e^4 - \frac{25}{12}e^6$	—	$\cos 6g$	—
	$\frac{3}{10}e^5 - \frac{21}{16}e^7$	—	$\cos 7g$	—
	$\frac{7}{36}e^6$	—	$\cos 8g$	—
	$\frac{2}{21}e^7$	—	$\cos 9g$	—

	C	S	cos	sin
$\frac{r}{a}$	—	$-2e+e^3-\frac{1}{4}e^5+\frac{1}{6}e^7$	—	$\sin g$
	—	$1-\frac{1}{2}e^2-\frac{1}{2}e^4-\frac{11}{48}e^6$	—	$\sin 2g$
	—	$2e-2e^3+\frac{5}{6}e^5-\frac{1}{20}e^7$	—	$\sin 3g$
	—	$-\frac{3}{2}e^2+\frac{9}{4}e^4-\frac{27}{16}e^6$	—	$\sin 4g$
	—	$e^3-\frac{8}{3}e^5+\frac{31}{6}e^7$	—	$\sin 5g$
	—	$-\frac{5}{8}e^4+\frac{25}{12}e^6$	—	$\sin 6g$
	—	$\frac{3}{10}e^5-\frac{21}{16}e^7$	—	$\sin 7g$
	—	$-\frac{7}{36}e^6$	—	$\sin 8g$
	—	$\frac{2}{21}e^7$	—	$\sin 9g$

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos 3f$**,** $\sin 3f$

	C	S	cos	sin
$\frac{a^2}{r^2}$	$\frac{1}{2}e^2 + \frac{7}{6}e^4 + \frac{29}{16}e^6$	—	$\cos g$	—
	$\frac{3}{2}e + \frac{27}{8}e^3 + \frac{209}{48}e^5 + \frac{1189}{192}e^7$	—	$\cos 2g$	—
	$1 + \frac{9}{2}e^2 + \frac{81}{8}e^4 + \frac{769}{48}e^6$	—	$\cos 3g$	—
	$\frac{3}{2}e + \frac{59}{8}e^3 + \frac{1025}{48}e^5 + \frac{17003}{640}e^7$	—	$\cos 4g$	—
	$\frac{1}{2}e^2 + \frac{25}{6}e^4 + \frac{841}{48}e^6$	—	$\cos 5g$	—
	$\frac{5}{12}e^3 + \frac{235}{48}e^5 + \frac{2845}{192}e^7$	—	$\cos 6g$	—
	$\frac{1}{4}e^4 + \frac{49}{16}e^6$	—	$\cos 7g$	—
	$\frac{7}{40}e^5 + \frac{77}{48}e^7$	—	$\cos 8g$	—
	$\frac{1}{18}e^6$	—	$\cos 9g$	—
	$\frac{1}{28}e^7$	—	$\cos 10g$	—

	C	S	cos	sin
$\frac{a^2}{r^2}$	—	$\frac{1}{2}e^2 + \frac{7}{6}e^4 + \frac{29}{16}e^6$	—	$\sin g$
	—	$\frac{3}{2}e + \frac{27}{8}e^3 + \frac{209}{48}e^5 + \frac{1189}{192}e^7$	—	$\sin 2g$
	—	$-1 - \frac{9}{2}e^2 - \frac{81}{8}e^4 - \frac{769}{48}e^6$	$\cos 3g$	—
	—	$-\frac{3}{2}e - \frac{59}{8}e^3 - \frac{1025}{48}e^5 - \frac{17003}{640}e^7$	$\cos 4g$	—
	—	$\frac{1}{2}e^2 + \frac{25}{6}e^4 + \frac{841}{48}e^6$	—	$\sin 5g$
	—	$\frac{5}{12}e^3 + \frac{235}{48}e^5 + \frac{2845}{192}e^7$	—	$\sin 6g$
	—	$-\frac{1}{4}e^4 - \frac{49}{16}e^6$	$\cos 7g$	—
	—	$-\frac{7}{40}e^5 - \frac{77}{48}e^7$	$\cos 8g$	—
	—	$\frac{1}{18}e^6$	—	$\sin 9g$
	—	$\frac{1}{28}e^7$	—	$\sin 10g$

Table $(\frac{a^2}{r^2}, \frac{a}{r}, \frac{r}{a})$**:** $\cos 3f$**,** $\sin 3f$ **(continued)**

	C	S	cos	sin
$\frac{a}{r}$	$3e^2+5e^4+\frac{21}{4}e^6$	—	$\cos g$	—
	$3e+\frac{33}{4}e^3+\frac{57}{4}e^5+\frac{675}{64}e^7$	—	$\cos 2g$	—
	$1+6e^2+\frac{45}{4}e^4+\frac{75}{4}e^6$	—	$\cos 3g$	—
	$3e+\frac{69}{4}e^3+\frac{487}{16}e^5+\frac{2897}{64}e^7$	—	$\cos 4g$	—
	$3e^2+\frac{31}{2}e^4+\frac{419}{16}e^6$	—	$\cos 5g$	—
	$e^3+\frac{55}{6}e^5+\frac{1375}{48}e^7$	—	$\cos 6g$	—
	$\frac{5}{4}e^4+\frac{221}{24}e^6$	—	$\cos 7g$	—
	$\frac{3}{5}e^5+\frac{91}{16}e^7$	—	$\cos 8g$	—
	$\frac{7}{18}e^6$	—	$\cos 9g$	—
	$\frac{1}{7}e^7$	—	$\cos 10g$	—

	C	S	cos	sin
$\frac{a}{r}$	—	$3e^2+5e^4+\frac{21}{4}e^6$	—	$\sin g$
	—	$3e+\frac{33}{4}e^3+\frac{57}{4}e^5+\frac{675}{64}e^7$	—	$\sin 2g$
	—	$-1-6e^2-\frac{45}{4}e^4-\frac{75}{4}e^6$	$\cos 3g$	—
	—	$-3e-\frac{69}{4}e^3-\frac{487}{16}e^5-\frac{2897}{64}e^7$	$\cos 4g$	—
	—	$3e^2+\frac{31}{2}e^4+\frac{419}{16}e^6$	—	$\sin 5g$
	—	$e^3+\frac{55}{6}e^5+\frac{1375}{48}e^7$	—	$\sin 6g$
	—	$-\frac{5}{4}e^4-\frac{221}{24}e^6$	$\cos 7g$	—
	—	$-\frac{3}{5}e^5-\frac{91}{16}e^7$	$\cos 8g$	—
	—	$\frac{7}{18}e^6$	—	$\sin 9g$
	—	$\frac{1}{7}e^7$	—	$\sin 10g$

Table ($\frac{a^2}{r^2}$, $\frac{a}{r}$, $\frac{r}{a}$)**:** $\cos 3f$**,** $\sin 3f$ **(continued)**

	C	S	cos	sin
$\frac{r}{a}$	$3e^2 - 2e^4 + \frac{1}{4}e^6$	—	$\cos g$	—
	$-3e + 3e^3 - \frac{7}{4}e^5 + \frac{1}{2}e^7$	—	$\cos 2g$	—
	$1 - 3e^2 + \frac{9}{2}e^4 - \frac{27}{4}e^6$	—	$\cos 3g$	—
	$3e - 9e^3 + \frac{81}{4}e^5 - \frac{243}{10}e^7$	—	$\cos 4g$	—
	$3e^2 - 13e^4 + \frac{187}{4}e^6$	—	$\cos 5g$	—
	$-e^3 + \frac{22}{3}e^5 - \frac{121}{3}e^7$	—	$\cos 6g$	—
	$\frac{5}{4}e^4 - \frac{115}{12}e^6$	—	$\cos 7g$	—
	$-\frac{3}{5}e^5 + \frac{19}{2}e^7$	—	$\cos 8g$	—
	$\frac{7}{18}e^6$	—	$\cos 9g$	—
	$-\frac{1}{7}e^7$	—	$\cos 10g$	—

	C	S	cos	sin
$\frac{r}{a}$	—	$3e^2 - 2e^4 + \frac{1}{4}e^6$	—	$\sin g$
	—	$-3e + 3e^3 - \frac{7}{4}e^5 + \frac{1}{2}e^7$	—	$\sin 2g$
	—	$1 - 3e^2 + \frac{9}{2}e^4 - \frac{27}{4}e^6$	—	$\sin 3g$
	—	$-3e + 9e^3 - \frac{81}{4}e^5 + \frac{243}{10}e^7$	—	$\sin 4g$
	—	$3e^2 - 13e^4 + \frac{187}{4}e^6$	—	$\sin 5g$
	—	$-e^3 + \frac{22}{3}e^5 - \frac{121}{3}e^7$	—	$\sin 6g$
	—	$\frac{5}{4}e^4 - \frac{115}{12}e^6$	—	$\sin 7g$
	—	$-\frac{3}{5}e^5 + \frac{19}{2}e^7$	—	$\sin 8g$
	—	$\frac{7}{18}e^6$	—	$\sin 9g$
	—	$-\frac{1}{7}e^7$	—	$\sin 10g$

Note on reading the tables. The foregoing tables are read as follows: for instance, in the Table ($\frac{a^2}{r^2}$, $\frac{a}{r}$, $\frac{r}{a}$), $\cos f$, $\sin f$, pp. 376, *et seq.*, the third and fourth lines of the $\frac{a^2}{r^2}$-compartment show that

$$\cos f = \left(e + \frac{3}{8}e^3 + \frac{5}{16}e^5 + \frac{175}{1024}e^7\right) 2\cos g + \left(\frac{1}{2}e^2 + \frac{1}{3}e^4 + \frac{15}{64}e^6\right) 2\cos 2g + \&\text{c.}$$

and the first and second lines give the sum and difference respectively of the corresponding coefficients of the cosine and sine series.

Addition, 28th Dec. 1860. The tables have been verified by me, on the proof-sheets; the cos tables, and a portion of the sin lines of the same tables, in the manner explained at the commencement of the memoir; but for the remainder of the sin lines it was found easier to employ the following mode of verification.

Verification formulae for sine series.

And, for the convenience of reference, I here give it as far as e^7, viz.,

$$f = g + 2[\sin]^i \sin ig,$$

where, as in the other sine series, i is to be taken as well negatively as positively, and $[\sin]^{-i} = -[\sin]^i$. Or, what is the same thing,

$$f = g + 2[\sin]^i \sin ig,$$

where i has only positive values. And the coefficients are

e	e^2	e^3	e^4	e^5	e^6	e^7
$[\sin]^1$	1	0	$-\frac{1}{8}$	0	$-\frac{1}{192}$	0
$[\sin]^2$	0	$\frac{1}{2}$	0	$-\frac{1}{12}$	0	$-\frac{1}{360}$
$[\sin]^3$	0	0	$\frac{3}{8}$	0	$-\frac{27}{128}$	0
$[\sin]^4$	0	0	0	$\frac{1}{3}$	0	$-\frac{4}{45}$
$[\sin]^5$	0	0	0	0	$\frac{125}{384}$	0
$[\sin]^6$	0	0	0	0	0	$\frac{27}{80}$
$[\sin]^7$	0	0	0	0	0	$\frac{16807}{46080}$

[217] A MEMOIR ON THE PROBLEM OF THE ROTATION OF A SOLID BODY.

[From the *Memoirs of the Royal Astronomical Society*, vol. xxix. (1861), pp. 307–342.
Read May 11, 1860.]

THE present memoir was written for the sake of the further elaboration of the analytical theory of the Rotation of a Solid Body, upon principles similar to those of my "Memoir on the Problem of Disturbed Elliptic Motion," Mem. R. Ast. Soc. vol. xxvii. pp. 1–29 (1858) [212]; the like elements are adopted, and the course of the investigation corresponds precisely to that of the memoir just referred to.

The formulae for the variations of the elements in the two problems (the motion being in each case referred to a fixed plane of reference and origin of angles therein) are found to be (as it is known they should be) precisely similar; the investigation puts this in a clear light, and exhibits the analytical theory of rotation as a precise analogue of that of elliptic motion.

I.

In the theory of elliptic motion, where the elements are a, the mean distance; e, the eccentricity; g, the mean anomaly; ϖ, the departure of pericentre; θ, the longitude of node; σ, the departure of node; ϕ, the inclination; and if, besides, we have n, the mean motion (n^2a^3 = sum of the masses), and Ω denote the disturbing function taken with Lagrange's sign ($\Omega = R$, if R be the disturbing function of the *Mécanique Céleste*), then the formula for the variations of the elements are

$$\begin{aligned}
2na\,da &= \frac{d\Omega}{dg}\,dt,\\
na^2\frac{de}{dt} &= \frac{d\Omega}{dg}\,\frac{1-e^2}{na^2e}\,dt - \frac{d\Omega}{d\varpi}\,\frac{\sqrt{1-e^2}}{na^2e}\,dt,\\
na^2\,dg &= -\frac{d\Omega}{da}\,\frac{2a}{n}\,dt - \frac{d\Omega}{de}\,\frac{1-e^2}{nae}\,dt,\\
dh &= \frac{d\Omega}{dg}\,dt\cdot na^2\sqrt{1-e^2},\\
dk &= \frac{d\Omega}{d\varpi}\,dt\cdot na^2\sqrt{1-e^2},\\
d\varpi + d\theta\cos\phi &= -\frac{d\Omega}{dh}\,dt\cdot\cot\phi + \frac{d\Omega}{dk}\,dt\cdot\csc\phi,\\
d\theta &= \frac{d\Omega}{dh}\,dt\cdot\csc\phi,\\
d\phi &= \frac{d\Omega}{dk}\,dt,
\end{aligned}$$

where $h = k\cos\varpi$, $k = k\sin\varpi$, and the formulae are very easily transformed into

$$\begin{aligned}
dh &= 2n\frac{d\Omega}{dg}\,dt,\\
dg &= -\frac{d\Omega}{dh}\,dt,\\
dk &= \frac{d\Omega}{dg}\,dt,\\
d\varpi &= -\frac{d\Omega}{dk}\,dt\cdot\cot\phi + \frac{d\Omega}{dh}\,dt\cdot\csc\phi,\\
d\theta &= \frac{d\Omega}{dk}\,dt\cdot\csc\phi,\\
d\phi &= \frac{d\Omega}{dh}\,dt,
\end{aligned}$$

where $\Omega = \Omega(h, g, k, \varpi, \theta, \sigma, \phi)$.

This is the system of formulae which will be obtained in the sequel for the variation of the elements in the problem of rotation, the new meanings of the symbols being explained post, Art. IV.

And if in either of the two problems, instead of the angles ϕ, σ, θ which refer to the fixed plane and origin, we adopt the angles F, S, T which refer to the moving orbit (or principal plane), viz., if F be the inclination, S the longitude of node, T the departure of node, so that

$$\cos\phi = \cos F\cos\phi_2 + \sin F\sin\phi_2\cos(S-\sigma_2),$$

etc., and if S denote $S-\sigma$, then the system is

$$\begin{aligned}
dh &= 2n\frac{d\Omega}{dg}\,dt,\\
dg &= -\frac{d\Omega}{dh}\,dt,\\
dk &= \frac{d\Omega}{dg}\,dt,\\
d\varpi &= -\frac{d\Omega}{dk}\,dt\cdot\cot F + \frac{d\Omega}{dh}\,dt\cdot\csc F,\\
dS &= \frac{d\Omega}{dk}\,dt\cdot\csc F,\\
dF &= \frac{d\Omega}{dh}\,dt,
\end{aligned}$$

or what is the same thing,

$$\begin{aligned}
d\varpi &= -\csc F\,dS,\\
dF &= \sin S\,d\Omega\cdot\csc F - \cos S\sin F\,d\theta,\\
dT &= \cos S\,d\Omega\cdot\csc F + \sin S\sin F\,d\theta,
\end{aligned}$$

where $\Omega = \Omega(h,g,k,\varpi,F,S,T)$, or what is the same thing, $\Omega = \Omega(h,g,k,\varpi,\phi,\sigma,\theta)$.

As already remarked, it is not necessary to repeat in the present memoir the transformation to this set of formulae.

II.

Considering, now, the problem of rotation, let the axes $\xi\eta\zeta$ denote axes fixed in space, and the axes xyz axes fixed in the body; and let

$$\begin{vmatrix} \xi & \eta & \zeta \\ x & y & z \end{vmatrix}$$

denote the matrix of the transformation from the one set of axes to the other. And if A, B, C are the moments of inertia, then the *Vis Viva* function or sum of the elements of mass, each into the half square of its velocity, is

$$T = \tfrac{1}{2}(Ap^2 + Bq^2 + Cr^2),$$

and if Ω be the disturbing function (taken with Lagrange's sign), then the equations of motion

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{p}} - \frac{\partial T}{\partial p} = \frac{\partial \Omega}{\partial p}, \quad \text{etc.}$$

give as usual

$$\begin{aligned}
A\frac{dp}{dt} + (C-B)qr &= -\sin S\left(\csc F\frac{\partial \Omega}{\partial S} + \cot F\frac{\partial \Omega}{\partial T}\right) + \cos S\frac{\partial \Omega}{\partial F},\\
B\frac{dq}{dt} + (A-C)rp &= \quad \cos S\left(\csc F\frac{\partial \Omega}{\partial S} + \cot F\frac{\partial \Omega}{\partial T}\right) + \sin S\frac{\partial \Omega}{\partial F},\\
C\frac{dr}{dt} + (B-A)pq &= \quad \frac{\partial \Omega}{\partial T},
\end{aligned}$$

which, with the equations for p, q, r, determine the motion of the body.

III.

First, to integrate the equations of motion when the disturbing forces are neglected; we have, as usual, the integral equations

$$\begin{gathered}
Ap^2 + Bq^2 + Cr^2 = h,\\
A^2p^2 + B^2q^2 + C^2r^2 = k^2,
\end{gathered}$$

(here h is twice the Vis Viva, and k the resultant angular momentum); and if we write also

$$Cr = k\cos F_2,$$

then the above values of p, q, r give

$$k\sin F_2 \cdot dF_2 = -C\,dr,$$

an equation which will be useful in the sequel.

The two equations $Ap^2+Bq^2+Cr^2 = h$, $A^2p^2+B^2q^2+C^2r^2 = k^2$ may be considered as determining p, q, in terms of r, and the equation $C\,dr + (B-A)pq = 0$, then gives

$$d\tau_2 = \frac{-C\,dr}{(B-A)pq},$$

whence also the equation for $d\tau_2$ becomes

$$d\tau_2 = \frac{k(h-Cr^2) - C\,dr}{k(k^2 - C^2r^2)(B-A)pq}.$$

Instead of the time t, I consider a function $\tau = nt + \text{const.}$, n being for the present an arbitrary constant quantity which may be a function of the constants h and k; we have thus

$$-C\,dr = n\,d\tau \cdot \sqrt{(B-A)pq},$$

and the integral equation is

$$\tau = n \int \frac{-C\,dr}{\sqrt{(B-A)pq}}.$$

The equation for $d\tau_2$ gives, in like manner,

$$\tau_2 = \int \frac{k(h-Cr^2) - C\,dr}{k(k^2-C^2r^2)(B-A)pq},$$

where it is assumed that the integrals are each of them taken from $r = r_0$, r_0 being an arbitrary constant value, say a function of h and k.

The quantity g is analogous to the mean anomaly in the theory of elliptic motion, and we write

$$g = \tau - \tau_2.$$

IV.

The coefficients of the matrix of transformation from the axes xyz to axes $x_2y_2z_2$ (where $x_2y_2z_2$ are such that the axis of z_2 is in the direction of the resultant angular momentum) are given by the table:

	x_2	y_2	z_2
x	α_2	α_2'	α_2''
y	β_2	β_2'	β_2''
z	γ_2	γ_2'	γ_2''

where the values of the coefficients are given by

$$\alpha_2 = \frac{1}{\sqrt{h - Cr^2}} \left(\sqrt{B}\, q \sqrt{h - Cr^2} - \sqrt{A}\, p \sqrt{h - Br^2} \right),$$

$$\alpha_2' = \frac{1}{\sqrt{h - Cr^2}} \left(\sqrt{A}\, p \sqrt{h - Br^2} + \sqrt{B}\, q \sqrt{h - Cr^2} \right),$$

$$\alpha_2'' = \sqrt{C}\, r \cdot \frac{1}{\sqrt{h - Cr^2}},$$

$$\beta_2 = \frac{1}{\sqrt{h - Cr^2}} \left(-\sqrt{A}\, p \sqrt{h - Cr^2} - \sqrt{B}\, q \sqrt{h - Ar^2} \right),$$

$$\beta_2' = \frac{1}{\sqrt{h - Cr^2}} \left(\sqrt{B}\, q \sqrt{h - Ar^2} - \sqrt{A}\, p \sqrt{h - Cr^2} \right),$$

$$\beta_2'' = \sqrt{C}\, r \cdot \frac{1}{\sqrt{h - Cr^2}},$$

$$\gamma_2 = \sqrt{A}\, p \cdot \frac{1}{\sqrt{h - Cr^2}},$$

$$\gamma_2' = \sqrt{B}\, q \cdot \frac{1}{\sqrt{h - Cr^2}},$$

$$\gamma_2'' = \sqrt{C}\, r \cdot \frac{1}{\sqrt{h - Cr^2}}.$$

V.

Moreover,

$$\tau = \int \frac{k(h - Cr^2) - C\,dr}{k(k^2 - C^2r^2)(B - A)pq} + \int \frac{-C\,dr}{\sqrt{(B - A)pq}},$$

which gives τ in terms of g and the constants h, k, r_0; and then T, S, F are given in terms of τ_2, S_2, F_2, and the constants θ, σ, ϕ, as follows, viz., we have a spherical triangle ABC, the sides whereof a, b, c, and the opposite angles A, B, C, are respectively,

Sides			*Opposite angles*		
a	$=$	$S_2 - S$,	A	$=$	ϕ,
b	$=$	$\tau_2 - \sigma$,	B	$=$	$180° - \psi$,
c	$=$	$\tau - \theta$,	C	$=$	F_2,

as appears by the figure.

The above values of p, q, r, give

$$k^2 - C^2r^2 = (k^2 - C^2r^2)\sin^2 F_2,$$

an equation which will be useful in the sequel.

The coefficient

$$\frac{k(h - Cr^2) - C\,dr}{k(k^2 - C^2r^2)(B - A)pq}$$

which gives τ in terms of g and the constants h, k, r_0; and then T, S, F are given in terms of τ_2, S_2, F_2, and the constants θ, σ, ϕ, as follows, viz., we have a spherical triangle ABC, the sides whereof a, b, c, and the opposite angles A, B, C, are respectively as shown above.

This being premised, we have (corresponding to the orbit in the theory of elliptic motion) the principal plane, with a departure point therein, the positions whereof, in respect to the fixed plane of reference, are determined by θ, the longitude of node; ϕ, the inclination; and σ, the departure of node. The precise signification of ϖ depends upon that of τ, and the signification of τ being assumed to be completely determined, that of ϖ will be so likewise; ϖ is to be considered as an angle measured in the principal plane, from the departure point of this plane on the fixed plane of reference, to the departure point in the principal plane; its value is consequently $= \tau - S$, or writing S for the longitude of the node in the principal plane, then $\varpi = \tau - S$, and $d\varpi = d\tau - dS$, but $d\tau = 0$, therefore $d\varpi = -dS$.

VI.

If, now, the Disturbing Function is taken into account, the equations are to be integrated by the method of the variation of the elements. I use dg to denote that part of the variation of $g(= td \cdot n + dc$, if $g = nt + c)$, which depends on the variation of the constants, and in like manner for dp, dq, dr, &c. I assume, moreover, that σ is a departure point in the principal plane; this gives $d\sigma - \cos\phi\, d\theta = 0$; and we see that the equations which lead to the expressions for the variations of the elements are

$$\begin{aligned} -dh &= Ap\,dp + Bq\,dq + Cr\,dr, \\ dk &= \quad A^2p\,dp + B^2q\,dq + C^2r\,dr, \\ -C\,dr &= (B-A)pq\,d\tau_2, \end{aligned}$$

and thence, putting

$$dh = \frac{\partial\Omega}{\partial\tau}\,d\tau + \frac{\partial\Omega}{\partial S}\,dS + \frac{\partial\Omega}{\partial F}\,dF,$$

we obtain expressions for dh, dk, dg, $d\varpi$, $d\theta$, $d\sigma$, $d\phi$, all of them expressed in the same form; that is, in terms of the differential coefficients of the disturbing function Ω with respect to the coordinates τ, S, F. We may then express the disturbing function Ω in terms of the elements, and transform the equations so as to obtain expressions for the variations of the elements in terms of the differential coefficients of Ω with respect to the elements.

VII.

Proceeding to carry out the foregoing plan, the equation

$$\delta h = Ap\,\delta p + Bq\,\delta q + Cr\,\delta r,$$

putting for δp, δq, δr their values in terms of δh, δk, δr, gives

$$\delta h = \frac{k(h - Cr^2)}{(B-A)pq}\,\delta k + \text{terms involving } \delta r.$$

Or, putting for $\sin F_2$, $\cos F_2$, their values $\frac{\sqrt{k^2-C^2r^2}}{k}$, $\frac{Cr}{k}$, the coefficient is

$$\frac{k\sin F_2}{(B-A)pq}\left\{(h - Cr^2)\cos F_2 \cdot \delta F_2 + (B-A)pq \cdot \delta k\right\},$$

and the quantities in $\{\}$ being respectively $p(k - Cr^2)$ and $q(k^2 - C^2r^2)$, and since $k^2 - C^2r^2 = k^2 \sin^2 F_2$, the coefficient is $= p\cos S + q\sin S$.

In like manner, for the terms involving $\left(\csc F \frac{\partial\Omega}{\partial S} + \cot F \frac{\partial\Omega}{\partial T}\right) d\tau$, the coefficient is

$$\frac{1}{\sqrt{k^2 - C^2r^2}} \left\{(h - Cr^2)\cos(S_2 - S) + (B - A)pq\sin(S_2 - S)\right\},$$

which reduces to

$$\frac{1}{k\sin F_2} \left\{(h - Cr^2)\cos S_2 + (B - A)pq\sin S_2\right\} \cos S$$

and the terms involving $\frac{\partial\Omega}{\partial F} d\tau$ give the coefficient

$$\frac{1}{k\sin F_2} \left\{(h - Cr^2)\sin S_2 - (B - A)pq\cos S_2\right\} \sin S.$$

VIII.

The expressions for dh and dk being thus obtained, and dr being also given by the equation $C\,dr = -\frac{\partial\Omega}{\partial\tau}dt$, we may now express dF_2 and dS_2 in terms of dh, dk, dr. In fact, the equations

$$Cr = k\cos F_2, \quad k\sin F_2 \cdot dF_2 = -C\,dr + \cos F_2\,dk,$$

give immediately, or, multiplying by $k^2 - C^2r^2$ $(= k^2\sin^2 F_2)$, and replacing $\cos S_2 \sin F_2 \sin F_2$ by $-ABpq$, and dividing, the last equation becomes

$$dF_2 = \frac{(h - Cr^2)\,dk + k(B - A)pq\,dr}{k(k^2 - C^2r^2)(B - A)pq},$$

and thence also

$$\cos F_2\,dS_2 = \frac{-C^2(Ch - k^2)r^2\,dr - C^2r\,dh - C(h - Cr^2)r\,dk}{k(k^2 - C^2r^2)(B - A)pq}.$$

And the value of the numerator being

$$k\sin S_2 \sin F_2 \left\{\cos S\left(\csc F\frac{\partial\Omega}{\partial S} + \cot F\frac{\partial\Omega}{\partial T}\right)dt + \sin S\frac{\partial\Omega}{\partial F}dt\right\}$$
$$- k\cos S_2 \sin F_2 \left\{\sin S\left(\csc F\frac{\partial\Omega}{\partial S} + \cot F\frac{\partial\Omega}{\partial T}\right)dt - \cos S\frac{\partial\Omega}{\partial F}dt\right\},$$

we find

$$dS_2 = \frac{1}{k}\sin(S_2 - S)\left(\csc F\frac{\partial\Omega}{\partial S} + \cot F\frac{\partial\Omega}{\partial T}\right)dt + \cos(S_2 - S)\frac{\partial\Omega}{\partial F}dt;$$

the last-mentioned expressions for dF_2 and dS_2 will be used for obtaining $d\theta$, $d\sigma$, $d\phi$.

I form now the equations

$$- \sin S\sin F\,dT + \cos S\,dF = p\,dt - \sin S_2 \sin F_2\,d\tau_2 + \cos S_2\,dF_2,$$
$$+ \cos S\sin F\,dT + \sin S\,dF = q\,dt + \cos S_2 \sin F_2\,d\tau_2 + \sin S_2\,dF_2,$$
$$- dS + \cos F\,dT = r\,dt - dS_2 + \cos F_2\,d\tau_2,$$

where p, q, r are the values referred to the moving axes, and the coefficients α_2, β_2, γ_2, etc., are the direction-cosines of the transformation.

IX.

For the present purpose we are to omit the terms $p\,dt$, $q\,dt$, $r\,dt$, and to write $dT = 0$, $dS = 0$, $dF = 0$. We have thus

$$
\begin{aligned}
-(\sin S_2 \sin F_2\, d\tau_2 + \cos S_2\, dF_2) &= \alpha_2(-\sin\sigma\sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \alpha_2'(\cos\sigma\sin\phi\, d\theta + \sin\sigma\, d\phi) + \alpha_2''(-d\sigma + \cos\phi\, d\theta), \\
-(\cos S_2 \sin F_2\, d\tau_2 + \sin S_2\, dF_2) &= \beta_2(-\sin\sigma\sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \beta_2'(\cos\sigma\sin\phi\, d\theta + \sin\sigma\, d\phi) + \beta_2''(-d\sigma + \cos\phi\, d\theta), \\
-(-dS_2 + \cos F_2\, d\tau_2) &= \gamma_2(-\sin\sigma\sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \gamma_2'(\cos\sigma\sin\phi\, d\theta + \sin\sigma\, d\phi) + \gamma_2''(-d\sigma + \cos\phi\, d\theta),
\end{aligned}
$$

and the foregoing equations thus become

$$
\begin{aligned}
\sin\sigma\sin\phi\, d\theta + \cos\sigma\, d\phi &= \cos\tau_2\, dF_2 + \sin\tau_2 \sin F_2\, dS_2, \\
\cos\sigma\sin\phi\, d\theta + \sin\sigma\, d\phi &= -\sin\tau_2\, dF_2 + \cos\tau_2 \sin F_2\, dS_2, \\
d\sigma + \cos\phi\, d\theta &= dT_2 + \cos F_2\, dS_2.
\end{aligned}
$$

The last equation, making σ_2 a departure point, or putting $d\sigma + \cos\phi\, d\theta = 0$, gives an equation above referred to. The other two equations give

$$
\begin{aligned}
d\phi &= -\cos(\tau_2 - \sigma)\, dF_2 + \sin(\tau_2 - \sigma)\sin F_2\, dS_2, \\
k\sin\phi\, d\theta &= -\sin(\tau_2 - \sigma)\, dF_2 - \cos(\tau_2 - \sigma)\sin F_2\, dS_2,
\end{aligned}
$$

which, or reducing,

$$
\begin{aligned}
k\sin\phi\, d\theta = {} & [\cos(\tau-\sigma)\sin(S_2-S) - \sin(\tau-\sigma)\cos(S_2-S)\cos F_2] \left(\csc F \frac{\partial\Omega}{\partial S} + \cot F \right. \\
& + [\cos(\tau-\sigma)\cos(S_2-S) + \sin(\tau-\sigma)\sin(S_2-S)\cos F_2] \frac{\partial\Omega}{\partial T} dt \\
& + [-\sin(\tau-\sigma)\sin F_2] \frac{\partial\Omega}{\partial F} dt.
\end{aligned}
$$

These expressions for $d\phi$ and $d\theta$ are in a form which is convenient for some purposes, but they may be further reduced by means of the spherical triangle. In fact, in the expression for $d\phi$, the three coefficients in [ ] are respectively,

$$\sin b \sin a + \cos b \cos a \cos C = \sin B \sin A, \quad -\cos B \cos A \cos c = \sin F \sin(T-\theta),$$

and

$$\sin b \sin C = \sin B \sin c = -\cos F \sin(T-\theta),$$

and thence

$$d\phi = \sin F \sin(T-\theta)\left(\csc F \frac{\partial \Omega}{\partial S} + \cot F \frac{\partial \Omega}{\partial T}\right) dt + \cos F \sin(T-\theta)\frac{\partial \Omega}{\partial T} dt - \cos(T-\theta)\frac{\partial \Omega}{\partial F}$$

or

$$d\phi = \sin(T-\theta)\frac{\partial \Omega}{\partial S} dt + \cos F \sin(T-\theta)\frac{\partial \Omega}{\partial T} dt - \cos(T-\theta)\frac{\partial \Omega}{\partial F} dt.$$

In like manner, in the original expression for $k \sin \phi \, d\theta$, the three coefficients in [ ] are

$$\cos b \sin a - \sin b \cos a \cos C = \cos B \sin c = -\cos F \sin(T-\theta),$$

$$\cos b \cos a + \sin b \sin a \cos C = \cos c = \cos(T-\theta),$$

$$\sin b \sin C = \sin B \sin c = -\sin F \sin(T-\theta),$$

and thence

$$k \sin \phi \, d\theta = -\cos F \sin(T-\theta)\left(\csc F \frac{\partial \Omega}{\partial S} + \cot F \frac{\partial \Omega}{\partial T}\right) dt$$
$$+ \cos(T-\theta)\frac{\partial \Omega}{\partial T} dt - \sin F \sin(T-\theta)\frac{\partial \Omega}{\partial F} dt,$$

or we have finally

$$k \sin \phi \, d\theta = -\sin(T-\theta)\left(\cot F \frac{\partial \Omega}{\partial S} + \csc F \frac{\partial \Omega}{\partial T}\right) dt + \cos(T-\theta)\frac{\partial \Omega}{\partial F} dt.$$

The expression for $d\sigma$ can be obtained from either of those for $d\phi$ or $d\theta$; in fact, from $d\sigma = -\cos \phi \, d\theta$, we have

$$d\sigma = \frac{\cos \phi}{k \sin \phi} \sin(T-\theta)\left(\cot F \frac{\partial \Omega}{\partial S} + \csc F \frac{\partial \Omega}{\partial T}\right) dt - \frac{\cos \phi}{k \sin \phi} \cos(T-\theta)\frac{\partial \Omega}{\partial F} dt.$$

X.

The expressions for dh, dk, dg, $d\varpi$, $d\theta$, $d\sigma$, $d\phi$ have now been obtained; and collecting these, we have

$$\begin{aligned}
dh &= \quad 2n\frac{\partial\Omega}{\partial g}\,dt,\\
dk &= \quad \frac{\partial\Omega}{\partial g}\,dt,\\
dg &= -\frac{\partial\Omega}{\partial h}\,dt,\\
d\varpi &= -\frac{\partial\Omega}{\partial k}\,dt\cdot\cot\phi+\frac{\partial\Omega}{\partial h}\,dt\cdot\csc\phi,\\
d\theta &= \quad \frac{\partial\Omega}{\partial k}\,dt\cdot\csc\phi,\\
d\phi &= \quad \frac{\partial\Omega}{\partial h}\,dt,
\end{aligned}$$

where $\Omega=\Omega(h,g,k,\varpi,\theta,\sigma,\phi)$, and these agree with the system obtained in Art. I.

Moreover, we may verify the equation $\mathfrak{B}=-2n\mathfrak{A}$, which connects the functions $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ introduced in the expression for dg. We have

$$dg=\mathfrak{A}\,dr+\mathfrak{B}\,dh+\mathfrak{C}\,dk+\mathfrak{D}\,d\varpi,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are functions which contain integrals with respect to r, and there is not any algebraical relation between them, except the equation $\mathfrak{B}=-2n\mathfrak{A}$, which will be obtained presently. I retain, therefore, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ in the formulae.

The first equation gives the value of dg; from the second equation we have

$$-d\tau_2+\frac{\partial\Omega}{\partial h}\,dt=\mathfrak{A}\,dr+\mathfrak{C}\,dk+\mathfrak{D}\,d\varpi.$$

XI.

If we assume that r_0, n is a function of h only, the term multiplied by $\frac{\partial r_0}{\partial k}$ will disappear, and by properly determining r_0 as a function of h and k, we can, as regards the terms which contain r_0, satisfy the equation $\mathfrak{B} = -2n\mathfrak{A}$; the condition for this is

$$\frac{\partial r_0}{\partial k} - 2k(A - C)\frac{r_0^2}{k^2 - C^2 r_0^2} \cdot \frac{\partial r_0}{\partial h} = 0,$$

which will be satisfied if r_0 is determined as a function of h and k by an equation of the form

$$r_0 = f\left(\frac{k^2}{h}\right),$$

where f denotes an arbitrary function.

It remains to show that, as regards the terms which do not contain r_0, the equation $\mathfrak{B} = -2n\mathfrak{A}$ is satisfied identically. This may be verified by means of the equations

$$\mathfrak{A} = \frac{C}{k(h - Cr^2)} \cdot \frac{1}{(B - A)pq},$$
$$\mathfrak{B} = -\frac{2Cn}{(B - A)pq} \cdot \frac{1}{k^2 - C^2 r^2},$$

which give the required relation.

The equation $\mathfrak{B} = -2n\mathfrak{A}$ being thus verified, the expression for dg becomes

$$dg = \mathfrak{A}\, dr - 2n\mathfrak{A}\, dh + \mathfrak{C}\, dk + \mathfrak{D}\, d\varpi,$$

and if we write

$$dg = -2n\mathfrak{A}\, dh + \mathfrak{C}\, dk + \mathfrak{D}\, d\varpi,$$

the terms involving dr being absorbed into those involving dh, the formulae for the variation of the elements are completely established.

[The memoir continues with further developments and applications of the theory, including the determination of the arbitrary function

f and the complete expression of the disturbing function in terms of the elements. The detailed tables of the elliptic-motion developments conclude the earlier paper, and the rotation memoir establishes the complete parallelism between the two theories.]

[End of Papers [216] and [217], Vol. III, pp. 451–500]

0.15 A MEMOIR ON THE PROBLEM OF THE ROTATION OF A SOLID BODY

[From the *Philosophical Transactions*, vol. CLI. (1861), pp. 561–587.]

I.

The theory of the rotation of a solid body about a fixed point may be considered as depending on the solution of the equations of motion in terms of certain elements; and for the complete development of the theory, these elements require to be connected with a set of angular coordinates adapted to geometrical and dynamical interpretation.

I commence by establishing the equations of motion under a form convenient for the subsequent treatment of the problem; and I then integrate these equations, first when the disturbing forces are neglected, and then when the disturbing forces are taken into account.

II.

Denoting the coordinates, referred to axes fixed in space, by x, y, z, and the coordinates, referred to axes fixed in the body, by x_1, y_1, z_1, the relation between the two sets of coordinates is expressed by means of a table of coefficients:

	x	y	z
x_1	α	β	γ
y_1	α'	β'	γ'
z_1	α''	β''	γ''

so that

$$x_1 = \alpha x + \beta y + \gamma z, \quad y_1 = \alpha' x + \beta' y + \gamma' z, \quad z_1 = \alpha'' x + \beta'' y + \gamma'' z,$$

and in like manner

$$x = \alpha x_1 + \alpha' y_1 + \alpha'' z_1, \quad y = \beta x_1 + \beta' y_1 + \beta'' z_1, \quad z = \gamma x_1 + \gamma' y_1 + \gamma'' z_1.$$

If p, q, r denote the components of angular velocity about the axes fixed in the body, and if X, Y, Z denote the moments of the impressed forces about these axes respectively, then the equations of motion are

$$A\frac{dp}{dt} + (C - B)qr = X,$$
$$B\frac{dq}{dt} + (A - C)rp = Y,$$
$$C\frac{dr}{dt} + (B - A)pq = Z,$$

where A, B, C are the principal moments of inertia at the fixed point.

The connection between the differential coefficients of the coefficients α, β, etc., and the quantities p, q, r is given by the following formulae; writing for shortness (1), (2), (3) to denote x_1, y_1, z_1 and $(1')$, $(2')$, $(3')$ to denote the differential coefficients of these quantities in regard to t:

$$(1') = q(3) - r(2),$$
$$(2') = r(1) - p(3),$$
$$(3') = p(2) - q(1),$$

from which, attending to the values of (1), (2), (3) in terms of x, y, z, we may obtain the differential coefficients of the coefficients α, β, etc.

The position of the body may be determined by means of the position of the principal axes, or say by means of the position of the axes (x_1, y_1, z_1); and this is most conveniently determined by means of the spherical triangle formed by these axes with a fixed plane and a fixed line in that plane, or (as in the theory of the top) by means of the angles θ, ϕ, ψ, where θ is the inclination of the plane of (x_1, y_1) to the fixed plane, ϕ is the longitude of the node of the plane of (x_1, y_1) upon the fixed plane, and ψ is the angle which the axis of x_1 makes with the node.

But it is more convenient to introduce other angular coordinates, and in fact to consider the problem as analogous to the problem of elliptic motion.

Suppose that the body is rotating about the axis of z_1 in the positive direction; and let h, k denote respectively the constants of Vis Viva and of areas; so that

$$Ap^2 + Bq^2 + Cr^2 = h, \qquad A^2p^2 + B^2q^2 + C^2r^2 = k^2.$$

The plane perpendicular to the axis of resultant angular momentum is called the *principal plane*; and the intersection of this plane with the plane of (x_1, y_1) is a line which may be called the *line of nodes*.

I introduce the following angular coordinates:

T, the departure of the line of nodes in the plane of (x_1, y_1);

S, the inclination of the plane of (x_1, y_1) to the principal plane;

F, the departure of the axis of x_1 from the line of nodes;

θ, the longitude of the node of the principal plane on the fixed plane;

ϕ, the inclination of the principal plane to the fixed plane;

σ, the departure of the node of the principal plane.

The position of the body is determined by T, S, F, together with the position of the principal plane determined by θ, ϕ, σ.

And if A, B, C are the moments of inertia, then the Vis Viva function or sum of the elements of mass, each into the half square of its velocity, is

$$T = \tfrac{1}{2}(Ap^2 + Bq^2 + Cr^2);$$

and if Ω be the disturbing function (taken with Lagrange's sign), then the equations of motion

$$\frac{d}{dt}\frac{dT}{d\dot{q}} - \frac{dT}{dq} = \frac{d\Omega}{dq}$$

give as usual

$$A\frac{dp}{dt} + (C - B)qr = -\sin S\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right) + \cos S\,\frac{d\Omega}{dF},$$

which, with the equations for p, q, r, determine the motion of the body.

III.

First, to integrate the equations of motion when the disturbing forces are neglected; we have, as usual, the integral equations

$$Ap^2 + Bq^2 + Cr^2 = h, \qquad A^2p^2 + B^2q^2 + C^2r^2 = k^2;$$

where h, k are constants of integration, viz., h is the constant of Vis Viva, and k is the constant of the principal area.

Moreover, if the coefficients α, β, etc. are those which belong to the transformation from xyz to $x_1y_1z_1$, viz., if in the table

	x	y	z
x_1	α	β	γ
y_1	α'	β'	γ'
z_1	α''	β''	γ''

the values of the coefficients are

$$\begin{array}{lll} \alpha = \cos T, & \beta = \sin T, & \gamma = 0, \\ \alpha' = -\sin T \cos S, & \beta' = \cos T \cos S, & \gamma' = \sin S, \\ \alpha'' = \sin T \sin S, & \beta'' = -\cos T \sin S, & \gamma'' = \cos S, \end{array}$$

so that the relation between the coordinates xyz and $x_1y_1z_1$ is given by the table

	x_1	y_1	z_1
x	$\cos T$	$-\sin T \cos S$	$\sin T \sin S$
y	$\sin T$	$\cos T \cos S$	$-\cos T \sin S$
z	0	$\sin S$	$\cos S$

where the values of the coefficients are given by

$$\begin{array}{lll} \alpha = \cos T, & \alpha' = -\sin T \cos S, & \alpha'' = \sin T \sin S, \\ \beta = \sin T, & \beta' = \cos T \cos S, & \beta'' = -\cos T \sin S, \\ \gamma = 0, & \gamma' = \sin S, & \gamma'' = \cos S. \end{array}$$

The two equations

$$Ap^2 + Bq^2 + Cr^2 = h, \qquad A^2p^2 + B^2q^2 + C^2r^2 = k^2,$$

may be considered as determining p, q, in terms of r, and the equation

$$C\frac{dr}{dt} + (B - A)pq = 0,$$

then gives

$$\frac{dr}{dt} = -\frac{(B-A)pq}{C};$$

whence also the equation for dT_2 becomes

$$k\,dT_2 = \frac{k(h - Cr^2) - C^2r^2}{(B-A)pq}\,dr.$$

Instead of the time t, I consider a function $g = nt + \text{const.}$, n being for the present an arbitrary constant quantity which may be a function of the constants h and k; we have thus

$$dg = n\,dt.$$

And the integral equation is

$$g = n\int \frac{-C\,dr}{(B-A)pq},$$

where it is assumed that the integrals are each of them taken from $r = r_0$, r_0 being an arbitrary constant value, say a function of h and k.

The quantity g is analogous to the mean anomaly in elliptic motion, or rather it will become so when the significations of n and r_0 are fixed; it is considered as implicitly involving a constant of integration, and, consequently, no constant of integration is added to the integral: as regards T_2, the constant of integration is σ, which denotes the initial value (corresponding to $r = r_0$) of the angle T_2.

IV.

Recapitulating the integral equations, we have

$$\begin{aligned} Ap^2 + Bq^2 + Cr^2 &= h, \\ A^2p^2 + B^2q^2 + C^2r^2 &= k^2, \\ Ap &= k\sin S_2 \sin F_2, \\ Bq &= k\cos S_2 \sin F_2, \\ Cr &= k\cos F_2, \\ g &= n\int \frac{-C\,dr}{(B-A)pq}, \end{aligned}$$

which equations give r, p, q, and thence S_2 and F_2, in terms of g and the constants h and k.

Moreover,

$$T = \sigma + \int \frac{k(h - Cr^2) - C^2r^2}{k(B - A)pq}\, dr,$$

which gives T in terms of g and the constants h, k, σ; and then T, S, F are given in terms of T_2, S_2, F_2, and the constants θ, σ, ϕ, as follows, viz., we have a spherical triangle ABC, the sides whereof a, b, c, and the opposite angles A, B, C, are respectively,

Sides	*Opposite angles*
$\sigma - S$,	ϕ,
$T_2 - \sigma$,	$180° - \psi$,
$T - \theta$,	F,

as appears by the figure.

The above values of p, q, r, give

$$k^2 = A^2p^2 + B^2q^2 + C^2r^2 = k^2 \sin^2 F_2 + C^2r^2,$$

an equation which will be useful in the sequel.

The coefficient n, and the value r_0, which is the inferior limit of the integrals, are thus far considered as arbitrary functions of h and k. As already remarked, g (which is a variable quantity, $= nt + c$) is used as an element, and the elements are h, k, g, σ, θ, ϕ. As to the signification of the different elements, it is proper to remark that it is not for the purposes of the present memoir necessary to enter into the complete theory of the rotation of a solid body; the problem here considered is that of the variation of the elements for a solid body rotating about a fixed point, and acted on by a disturbing function.

V.

This being premised, we have (corresponding to the orbit in the theory of elliptic motion) the principal plane, with a departure point therein, the positions whereof, in respect to the fixed plane of reference, are determined by θ, the longitude of node; ϕ, the inclination; and σ, the departure of node.

The precise signification of σ depends upon that of r_0, and the signification of r_0 being assumed to be completely determined, that of σ will be so likewise; σ is to be considered as an angle measured in the principal plane from the departure point, and determining in that plane a line (or, treating the plane as an orbit, a point) which I call the *rotation pericentre*, or simply the *pericentre*; say σ is the departure of the pericentre; and then g is an angle varying uniformly with the time, used for expressing in terms of the time the angle T, which determines the position, in regard to the principal plane and departure point therein, of the node of the plane of x_1y_1, or equator, upon the principal plane, such node corresponding with the moving body in the theory of elliptic motion.

We may in fact say: T, the departure of the last-mentioned node, $=\sigma$, the departure of pericentre, $+(T-\sigma)$, the rotation true anomaly of such node; T_2 being a function of g, the rotation mean anomaly of such node.

The elements are then as follows, viz.:

h, the constant of Vis Viva,

k, the constant of principal area,

θ, the longitude of node of the principal plane,

ϕ, the inclination of the principal plane,

σ, the departure of the pericentre,

g, the mean anomaly.

Moreover, corresponding to the theory of elliptic motion, we have also the anomalies T_2, S_2, F_2, and the pericentre S, F, viz.:

T_2, the true anomaly of the node on the equator,

S_2, the inclination of the equator to the principal plane,

F_2, the departure of the axis x_1 from the node,

S, the inclination of the equator (constant),

F, the departure of the axis x_1 from the node (constant).

We have S, F as constants determined by the initial values; and T_2, S_2, F_2 are functions of g (and therefore of t) determined by the equations of motion.

VI.

If, now, the Disturbing Function is taken into account, the equations are to be integrated by the method of the variation of the elements.

I use δg to denote that part of the variation of g $(= t\,\delta n + \delta c$, if $g = nt + c)$, which depends on the variation of the constants, and in like manner for δp, δq, δr, etc.

I assume, moreover, that σ is a departure point in the principal plane; this gives $d\sigma - \cos\phi\, d\theta = 0$; and we see that the equations which lead to the expressions for the variations of the elements are

$$\begin{aligned}
&A\,dp = -\sin S\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt + \cos S\,\frac{d\Omega}{dF}\,dt,\\
&B\,dq = \cos S\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt + \sin S\,\frac{d\Omega}{dF}\,dt,\\
&C\,dr = -\frac{d\Omega}{dF}\,dt,\\
&\quad dT = 0, \quad dS = 0, \quad dF = 0, \quad d\sigma - \cos\phi\,d\theta = 0,
\end{aligned}$$

where, of course, $\Omega = \Omega(T, S, F)$.

The variations dh and dk are obtained from the equations

$$dh = Ap\,dp + Bq\,dq + Cr\,dr, \qquad k\,dk = A^2p\,dp + B^2q\,dq + C^2r\,dr,$$

expressions which will presently be transformed.

VII.

Proceeding to carry out the foregoing plan, the equation

$$dh = Ap\,dp + Bq\,dq + Cr\,dr,$$

putting for $A\,dp$, $B\,dq$, $C\,dr$, their values, gives

$$\begin{aligned}
dh = (p\sin S + q\cos S)&\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt\\
&+ (p\cos S + q\sin S)\,\frac{d\Omega}{dF}\,dt - r\,\frac{d\Omega}{dF}\,dt, \quad (1)
\end{aligned}$$

which may also be written

$$\begin{aligned}
dh = \cos(S_2 - S)\sin F_2&\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt\\
&+ \sin(S_2 - S)\sin F_2\,\frac{d\Omega}{dF}\,dt - \cos F_2\,\frac{d\Omega}{dF}\,dt. \quad (2)
\end{aligned}$$

In fact, if, in this expression, we consider first the terms involving $\dfrac{d\Omega}{dF}\,dt$, the coefficient is

$$-Cr\frac{(h - Cr^2)}{k(B - A)pq} + \cdots$$

and the quantities in {} being respectively $p(k^2 - C^2r^2)$ and $q(k^2 - C^2r^2)$, and since $k^2 - C^2r^2 = k^2 \sin^2 F_2$, the coefficient is $= p\cos S + q\sin S$.

In like manner, for the terms involving $\left(F\,\dfrac{d\Omega}{dT} + \cot F\,\dfrac{d\Omega}{dS}\right)dt$, the coefficient is

$$\begin{aligned}
&\frac{1}{k\sin F_2}\big[(h - Cr^2)\cos(S_2 - S) + (B - A)pq\sin(S_2 - S)\big] \\
&\quad= \frac{1}{k\sin F_2}\big[(h - Cr^2)\cos S_2\cos S + (h - Cr^2)\sin S_2\sin S \\
&\qquad\qquad + (B - A)pq(\sin S_2\cos S - \cos S_2\sin S)\big];
\end{aligned}$$

which is $= p\sin S + q\cos S$; and the foregoing transformed expression for dh is thus seen to be correct.

The equation

$$k\,dk = A^2p\,dp + B^2q\,dq + C^2r\,dr,$$

substituting for $A\,dp$, $B\,dq$, $C\,dr$, their values, becomes

$$\begin{aligned}
k\,dk = Ap\big\{&\sin S(F\,dT + \cot F\,dS) + \cos S\,dF\big\} \\
&+ Bq\big\{\cos S(F\,dT + \cot F\,dS) + \sin S\,dF\big\} - Cr\,dF.
\end{aligned}$$

But, from the equations

$$Ap = k\sin S_2\sin F_2, \quad Bq = k\cos S_2\sin F_2, \quad Cr = k\cos F_2,$$

we deduce

$$\begin{aligned}
Ap\sin S + Bq\cos S &= k\cos(S_2 - S)\sin F_2, \\
Ap\cos S + Bq\sin S &= k\sin(S_2 - S)\sin F_2, \\
Cr &= k\cos F_2,
\end{aligned}$$

and we thence find

$$dk = \cos(S_2 - S)\sin F_2 \left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right) dt \\ + \sin(S_2 - S)\sin F_2\,\frac{d\Omega}{dF}\,dt - \cos F_2\,\frac{d\Omega}{dF}\,dt,$$

so that dh and dk are now determined.

VIII.

The expressions for dh and dk being thus obtained, and dr being also given by the equation $C\,dr = -\dfrac{d\Omega}{dF}\,dt$, we may now express dF_2 and dS_2 in terms of dh, dk, dr.

In fact, the equations

$$Cr = k\cos F_2, \qquad \frac{Cr}{k} = \cos F_2,$$

give immediately

$$k\sin F_2\,dF_2 = -C\,dr + \cos F_2\,dk,$$

or, multiplying by $k^2 - C^2r^2$ $(= k^2\sin^2 F_2)$, and replacing $\cos S_2 \sin F_2$, $\sin S_2 \sin F_2$ by their values $-ABpq$, and dividing, the last equation becomes

$$dF_2 = \frac{-C(h - Cr^2)r\,dr + C^2r\,dr - C(h - Cr^2)\,dk + \cdots}{k(k^2 - C^2r^2)(B - A)pq},$$

and thence also

$$\cos F_2\,dS_2 = \frac{-C^2(h - Cr^2)r^2\,dr + \cdots}{k(k^2 - C^2r^2)(B - A)pq},$$

which is the value of dT_2, as given by the equation (not yet demonstrated) $dT_2 = \cos F_2\,dF_2$.

The expression for dF_2, substituting for dr and dk, their values, gives

$$k\sin F_2\,dF_2 = \cos F_2\sin F_2\cos(S_2 - S)\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right) dt \\ - \cos F_2\sin F_2\sin(S_2 - S)\,\frac{d\Omega}{dF}\,dt + \cdots$$

or, combining the last two terms, and reducing,

$$dF_2 = \frac{1}{k}\big\{\cos F_2 \cos(S_2 - S)(F\,d\Omega/dT + \cot F\,d\Omega/dS)\,dt \\ - \cos F_2 \sin(S_2 - S)\,d\Omega/dF\,dt\big\}.$$

And the value of the numerator being

$$k \sin S_2 \sin F_2\big\{-\cos S(F\,dT + \cot F\,dS) + \sin S\,dF\big\} \\ + k \cos S_2 \sin F_2\big\{-\sin S(F\,dT + \cot F\,dS) - \cos S\,dF\big\},$$

we find

$$dS_2 = \frac{1}{k}\big\{-\sin(S_2 - S)(F\,d\Omega/dT + \cot F\,d\Omega/dS)\,dt \\ + \cos(S_2 - S)\,d\Omega/dF\,dt\big\};$$

the last-mentioned expressions for dT_2 and dS_2 will be used for obtaining $d\theta$, $d\sigma$, $d\phi$.

IX.

I form now the equations

$$\begin{aligned}
-\sin S \sin F\,dT + \cos S\,dF &= p\,dt - \sin S_2 \sin F_2\,dT_2 + \cos S_2\,dF_2 \\
&\quad + \alpha_2(\sin\sigma \sin\phi\,d\theta + \cos\sigma\,d\phi) \\
&\quad + \alpha_2'(\cos\sigma \sin\phi\,d\theta + \sin\sigma\,d\phi) \\
&\quad + \alpha_2''(-d\sigma + \cos\phi\,d\theta), \\
\cos S \sin F\,dT + \sin S\,dF &= q\,dt + \cos S_2 \sin F_2\,dT_2 + \sin S_2\,dF_2 \\
&\quad + \beta_2(\sin\sigma \sin\phi\,d\theta + \cos\sigma\,d\phi) \\
&\quad + \beta_2'(\cos\sigma \sin\phi\,d\theta + \sin\sigma\,d\phi) \\
&\quad + \beta_2''(-d\sigma + \cos\phi\,d\theta), \\
-dS + \cos F\,dT &= r\,dt - dS_2 + \cos F_2\,dT_2 \\
&\quad + \gamma_2(\sin\sigma \sin\phi\,d\theta + \cos\sigma\,d\phi) \\
&\quad + \gamma_2'(\cos\sigma \sin\phi\,d\theta + \sin\sigma\,d\phi) \\
&\quad + \gamma_2''(-d\sigma + \cos\phi\,d\theta).
\end{aligned}$$

For the present purpose we are to omit the terms $p\,dt$, $q\,dt$, $r\,dt$, and to write $dT = 0$, $dS = 0$, $dF = 0$.

We have thus

$$\begin{aligned}
-(\sin S_2 \sin F_2\, dT_2 + \cos S_2\, dF_2) &= \alpha_2(-\sin\sigma \sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \alpha_2'(\cos\sigma \sin\phi\, d\theta + \sin\sigma\, d\phi) \\
&\quad + \alpha_2''(-d\sigma + \cos\phi\, d\theta), \\
-(\cos S_2 \sin F_2\, dT_2 + \sin S_2\, dF_2) &= \beta_2(-\sin\sigma \sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \beta_2'(\cos\sigma \sin\phi\, d\theta + \sin\sigma\, d\phi) \\
&\quad + \beta_2''(-d\sigma + \cos\phi\, d\theta), \\
-(-dS_2 + \cos F_2\, dT_2) &= \gamma_2(-\sin\sigma \sin\phi\, d\theta + \cos\sigma\, d\phi) \\
&\quad + \gamma_2'(\cos\sigma \sin\phi\, d\theta + \sin\sigma\, d\phi) \\
&\quad + \gamma_2''(-d\sigma + \cos\phi\, d\theta).
\end{aligned}$$

Hence we have

$$\begin{aligned}
\sin\sigma \sin\phi\, d\theta + \cos\sigma\, d\phi &= \alpha_2(\sin S_2 \sin F_2\, dT_2 + \cos S_2\, dF_2) \\
&\quad - \beta_2(\cos S_2 \sin F_2\, dT_2 + \sin S_2\, dF_2) - \gamma_2(\cos F_2\, dT_2 - dS_2),
\end{aligned}$$

which, attending to the values of α_2'', β_2'', γ_2'', may be written,

$$\begin{aligned}
\sin\sigma \sin\phi\, d\theta + \cos\sigma\, d\phi &= \alpha_2(\alpha_2''\, dT_2 + \cos S_2\, dF_2) \\
&\quad - \beta_2(\beta_2''\, dT_2 + \sin S_2\, dF_2) - \gamma_2(\gamma_2''\, dT_2 - dS_2) \\
&= -(\alpha_2 \cos S_2 + \beta_2 \sin S_2)\, dF_2 + \gamma_2\, dS_2,
\end{aligned}$$

since the coefficient of dT_2 vanishes; and, in like manner,

$$\begin{aligned}
\cos\sigma \sin\phi\, d\theta + \sin\sigma\, d\phi &= -(\alpha_2' \cos S_2 + \beta_2' \sin S_2)\, dF_2 + \gamma_2'\, dS_2, \\
-d\sigma + \cos\phi\, d\theta &= -dT_2 - (\alpha_2'' \cos S_2 + \beta_2'' \sin S_2)\, dF_2 + \gamma_2''\, dS_2.
\end{aligned}$$

But we have

$$\begin{aligned}
\alpha_2 \cos S_2 + \beta_2 \sin S_2 &= \cos T_2 \cos S_2 + \sin T_2 \cos S_2 = \cdots \\
\alpha_2' \cos S_2 + \beta_2' \sin S_2 &= -\sin T_2 \cos S_2 + \cos T_2 \cos S_2 = \cdots
\end{aligned}$$

X.

The foregoing formulae become more simple when n is a function of h only, and when the relation between r and g is that which belongs to elliptic motion; that is, when

$$\frac{dg}{dt} = n = \sqrt{\frac{2h - 2\Omega}{k^2}},$$

where Ω is the disturbing function. In this case we have

$$\frac{d^2g}{dt^2} = 2n\frac{d\Omega}{dh},$$

and the formula for dg reduces to

$$dg = 2n\frac{d\Omega}{dh}\,dt.$$

XI.

In the original expression for $k\sin\phi\,d\theta$, the three coefficients in [ ] are

$$\begin{aligned}
&\cos b\sin a - \sin b\cos a\cos C = \cos B\sin c = -\cos F\sin(T-\theta),\\
&\cos b\cos a + \sin b\sin a\cos C = \cos c = \cos(T-\theta),\\
&\qquad\qquad\qquad \sin b\sin C = \sin B\sin c = -\sin T\sin(T-\theta),
\end{aligned}$$

and thence

$$\begin{aligned}
k\sin\phi\,d\theta &= \cos F\sin(T-\theta)\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt - \cos(T-\theta)\,\frac{d\Omega}{dF}\,dt\\
&= \sin(T-\theta)\,dT - F\sin(T-\theta)\,\frac{d\Omega}{dt}\,dt,
\end{aligned}$$

or we have finally

$$k\sin\phi\,d\theta = -\sin(T-\theta)\left(\cot F\,\frac{d\Omega}{dS} + F\,\frac{d\Omega}{dT}\right)dt.$$

The expression for $d\sigma$ can be obtained from either of those for $d\theta$, by the equation $d\sigma = \cos\phi\,d\theta$.

XII.

The comparison of the terms involving dm gives at once

$$\frac{d\Omega}{dm} = X,$$

or, substituting for C its value,

$$\begin{aligned}
\frac{d\Omega}{dm} = \cos(S_2-S)\sin F_2&\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)\\
&- \sin(S_2-S)\sin F_2\,\frac{d\Omega}{dF} - \cos F_2\,\frac{d\Omega}{dS};
\end{aligned}$$

and comparing with the expression for $d\sigma$, we find

$$\frac{d\Omega}{dk} = -\frac{1}{k}\frac{d\Omega}{dt}.$$

The terms involving dr give

$$\frac{d\Omega}{dr} = \frac{k(B-A)pq}{k^2 - C^2r^2}\left(\frac{d\Omega}{dh} + \frac{n}{k}\frac{d\Omega}{dk}\right) - \frac{2n}{k(B-A)pq}\frac{d\Omega}{dS},$$

or, substituting for X, Y, Z, their values,

$$\frac{d\Omega}{dr} = \frac{k(B-A)pq}{k^2 - C^2r^2}\left(\frac{d\Omega}{dh} + \frac{n}{k}\frac{d\Omega}{dk}\right)\sin(S_2 - S)\sin F_2\left(F\frac{d\Omega}{dT} + \cot F\frac{d\Omega}{dS}\right) + \cos(S_2 - S)\sin F_2\frac{d\Omega}{dF} - \frac{2n}{k(B-A)pq}\frac{d\Omega}{dS},$$

which agrees with the second form for $\frac{d\Omega}{dk}$, and gives as before

$$\frac{dg}{dt} = n.$$

The terms involving dh give

$$dh = 2n\frac{d\Omega}{dg}\,dt;$$

and thence

$$\frac{dg}{dt} = n - \frac{Z}{k(B-A)pq},$$

or, substituting for $2n$ its value $\frac{dg}{dt}$, for X the value $\frac{d\Omega}{dm}$, and for Z its value $\frac{d\Omega}{dr}$, we have

$$\frac{dg}{dt} = n - \frac{1}{k(B-A)pq}\frac{d\Omega}{dr}.$$

XIII.

We have

$$\frac{d^2r}{dt^2} = \frac{k^2(h - Cr^2) - C^2r^2}{(B-A)^2p^2q^2}\cdot\frac{dr}{dt};$$

whence, observing that

$$\frac{dr}{dt} = -\frac{(B-A)pq}{C},$$

we obtain

$$\frac{d^2r}{dt^2} = \frac{k^2(h - Cr^2) - C^2r^2}{C(B-A)pq}.$$

XIV.

To verify the formula for dg, we have

$$dg = \frac{k(B-A)pq\left(\dfrac{d\Omega}{dh} + \dfrac{n}{k}\dfrac{d\Omega}{dk}\right) - 2n\dfrac{d\Omega}{dS}}{k(B-A)pq}\, dt.$$

Now

$$\frac{d\Omega}{dh} = \frac{1}{2n}\frac{dg}{dt}, \qquad \frac{d\Omega}{dk} = -\frac{1}{k}\frac{d\Omega}{dt},$$

and substituting these values, the formula becomes

$$dg = \left(\frac{dg}{dt} - \frac{2n}{k(B-A)pq}\frac{d\Omega}{dS}\right) dt,$$

which is the required result.

XV.

The final results for the variations of the elements may be collected as follows:

$$dh = 2n\,\frac{d\Omega}{dg}\,dt, \tag{3}$$

$$\begin{aligned} dk = {} & \cos(S_2 - S)\sin F_2\left(F\,\frac{d\Omega}{dT} + \cot F\,\frac{d\Omega}{dS}\right)dt \\ & - \sin(S_2 - S)\sin F_2\,\frac{d\Omega}{dF}\,dt - \cos F_2\,\frac{d\Omega}{dS}\,dt, \end{aligned} \tag{4}$$

$$\begin{aligned} dg = {} & \left(\frac{d\Omega}{dh} + \frac{n}{k}\frac{d\Omega}{dk}\right)dt - \frac{2n}{k(B-A)pq}\frac{d\Omega}{dS}\,dt \\ & + \frac{2n}{k(B-A)pq}\big\{\sin(S_2 - S)\sin F_2(F\,d\Omega/dT \\ & \qquad + \cot F\,d\Omega/dS) + \cos(S_2 - S)\sin F_2\,d\Omega/dF\big\}\,dt, \end{aligned} \tag{5}$$

$$d\psi = \frac{1}{k\sin\phi}\frac{d\Omega}{d\theta}\,dt, \tag{6}$$

$$d\theta = -\frac{1}{k\sin\phi}\left(\cot F\,\frac{d\Omega}{dS} + F\,\frac{d\Omega}{dT}\right)dt, \tag{7}$$

$$d\sigma = \frac{\cos\phi}{k\sin\phi}\left(\cot F\,\frac{d\Omega}{dS} + F\,\frac{d\Omega}{dT}\right)dt, \tag{8}$$

$$d\phi = \frac{1}{k}\frac{d\Omega}{d\psi}\,dt. \tag{9}$$

These formulae express the variations of the elements in terms of the differential coefficients of the disturbing function with respect to the elements themselves; and they constitute the complete solution of the problem of the rotation of a solid body, when the disturbing function is given.

0.218 A THIRD MEMOIR ON THE PROBLEM OF DISTURBED ELLIPTIC MOTION

[From the *Philosophical Transactions*, vol. CLI. (1861), pp. 589–610.]

I.

The position in respect to a fixed plane of reference and origin of longitudes therein, of the variable ecliptic and of the departure point or origin of longitudes therein, are determined by θ, the longitude of node, σ, the departure of node, ϕ, the inclination, where, by the definition of a departure point,

$$d\sigma - \cos\phi\, d\theta = 0,$$

and then, in respect to the variable ecliptic and departure point or origin of longitudes therein, the position of the disturbed body is determined by r, the radius vector, v, the longitude, γ, the latitude; and this being so, then (Supplementary Memoir, pp. 220, 227) the expression for the Vis Viva function is,

$$2T = \dot{r}^2 + r^2\dot{\gamma}^2 + r^2\cos^2\gamma\,\dot{v}^2 + 2Q\dot{r} + 2Rr\dot{v},$$

where

$$\begin{aligned} Q &= -\dot{\gamma} + [\cos(v-\sigma)\sin\phi\cdot\dot{\theta} - \sin(v-\sigma)\dot{\phi}], \\ R &= \cos\gamma\cdot\dot{v} - \sin\gamma[\sin(v-\sigma)\sin\phi\cdot\dot{\theta} + \cos(v-\sigma)\dot{\phi}], \end{aligned}$$

the superscript dots being used to denote differentiation with respect to the time.

The last-mentioned expressions may for shortness be denoted by

$$Q = -\dot{\gamma} + A\sin\gamma, \qquad R = \cos\gamma\cdot\dot{v} - B\sin\gamma.$$

The equations of motion are of course,

$$\begin{aligned} \frac{d}{dt}\frac{dT}{d\dot{r}} - \frac{dT}{dr} &= \frac{dV}{dr}, \\ \frac{d}{dt}\frac{dT}{d\dot{v}} - \frac{dT}{dv} &= \frac{dV}{dv}, \\ \frac{d}{dt}\frac{dT}{d\dot{\gamma}} - \frac{dT}{d\gamma} &= \frac{dV}{d\gamma}, \end{aligned}$$

where $V = \frac{\mu}{r} + \Omega$, if Ω is the disturbing function, taken with Lagrange's sign ($\mu = R^2$, if $-R$ is the disturbing function of the *Mécanique Céleste*).

To reduce these, we have in the first place

$$\begin{aligned}\frac{dT}{d\dot r} &= \dot r + Q = r^2\dot Q,\\ \frac{dT}{d\dot v} &= r^2\cos^2\gamma\,\dot v + Rr = r^2R,\\ \frac{dT}{d\dot\gamma} &= r^2\dot\gamma + Qr = r^2\dot Q.\end{aligned}$$

The equations are thus reduced to

$$\begin{aligned}\frac{d}{dt}(r^2\dot Q) - r\dot Q^2 + \frac{\mu}{r^2} &= \frac{dV}{dr},\\ \frac{d}{dt}(r^2R) + r^2(Q\dot R + R\dot A\sin\gamma) &= \frac{dV}{dv},\\ \frac{d}{dt}(r^2\dot Q) - r^2\cos\gamma\sin\gamma\,\dot v^2 + rR\dot\gamma &= \frac{dV}{d\gamma},\end{aligned}$$

and then substituting for Q, R their values, viz.

$$\dot v = \cos\gamma\cdot\dot v - B\sin\gamma,$$

we find

$$\begin{aligned}\frac{d^2r}{dt^2} - r\dot\gamma^2 - r\cos^2\gamma\,\dot v^2 + \frac{\mu}{r^2} &= \frac{dV}{dr},\\ \frac{d}{dt}(r^2\cos^2\gamma\,\dot v) + r^2(Q\dot R + R\dot A\sin\gamma) &= \frac{dV}{dv},\\ \frac{d}{dt}(r^2\dot\gamma) - r^2\cos\gamma\sin\gamma\,\dot v^2 + rR\dot\gamma &= \frac{dV}{d\gamma},\end{aligned}$$

where

$$\begin{aligned}\mathfrak{A} &= r\left(-2A\frac{dv}{dt} - 2B\sin\gamma\cos\gamma\frac{d\gamma}{dt} + A^2 + B^2\sin^2\gamma\right),\\ \mathfrak{B} &= -(r^2B\sin\gamma\cos\gamma) + r^2\left(B\frac{d\gamma}{dt} - A\sin\gamma\cos\gamma\frac{dv}{dt} - AB\cos^2\gamma\right),\\ \mathfrak{C} &= (r^2A) + r^2\left(-(\cos^2\gamma - \sin^2\gamma)B\frac{dv}{dt} + A\sin\gamma\cos\gamma\right),\end{aligned}$$

in which

$$A = \cos(v - \sigma) \sin \phi \cdot \dot{\theta} + \sin(v - \sigma) \dot{\phi},$$
$$B = \sin(v - \sigma) \sin \phi \cdot \dot{\theta} + \cos(v - \sigma) \dot{\phi},$$

θ, σ, ϕ being given functions of t such that $d\sigma - \cos \phi \, d\theta = 0$.

The foregoing equations of motion are rigorously accurate.

II.

Neglecting the terms which involve A^2, AB, B^2, we have

$$\begin{aligned}
\mathfrak{A} &= 2Ar\frac{dr}{dt} + r^2\left(A\frac{d^2v}{dt^2} + \frac{dA}{dt}\frac{dv}{dt}\right) - 2r^2B\frac{d\gamma}{dt}, \\
\mathfrak{B} &= -\frac{d}{dt}(r^2 B \sin\gamma\cos\gamma) + r^2\left(B\frac{d\gamma}{dt} - A\sin\gamma\cos\gamma\frac{dv}{dt}\right), \\
\mathfrak{C} &= \frac{d}{dt}(r^2A) + r^2\left(-(\cos^2\gamma - \sin^2\gamma)B\frac{dv}{dt} + A\sin\gamma\cos\gamma\right),
\end{aligned}$$

and if we then neglect also the products of A and B by γ and $\dfrac{d\gamma}{dt}$, we have

$$\mathfrak{A} = 0, \qquad \mathfrak{B} = 0,$$

where it may be noticed that, in order to obtain this last value of $\mathfrak{C}$, the only neglected term is a term containing $B\sin^2\gamma$.

Now, attending to the values of A and B, we have

$$\frac{dA}{dt} = \frac{dA}{dv}\frac{dv}{dt} + \frac{dA}{dt},$$

where here and in the sequel $\dfrac{d}{dt}$ denotes differentiation in regard to t in so far only as it enters through the quantities σ, θ, ϕ, which determine the position of the variable ecliptic.

Hence

$$\begin{aligned}
\mathfrak{C} &= 2Ar\frac{dr}{dt} + r^2\frac{dA}{dv}\frac{dv}{dt} + r^2\frac{dA}{dt} - 2r^2B\frac{d\gamma}{dt} \\
&= 2Ar\frac{dr}{dt} + r^2\frac{dA}{dt} - 2r^2B\frac{d\gamma}{dt};
\end{aligned}$$

and, as above,

$$\mathfrak{A} = 0, \qquad \mathfrak{B} = 0.$$

Let r, v, γ be the values obtained on the supposition that $\mathfrak{C} = 0$, and $r + \delta r$, $v + \delta v$, $\gamma + \delta\gamma$, the accurate values; the first and second equations show that, neglecting the products of $\mathfrak{A}$ and $\mathfrak{B}$ into $\delta\gamma$ and $\dfrac{d\delta\gamma}{dt}$, we have $\delta r = 0$, $\delta v = 0$; so that the values of r and v are not affected by the variation of the ecliptic.

And then, substituting in the third equation $\gamma + \delta\gamma$ in the place of γ, and for $\cos(\gamma + \delta\gamma)$, $\sin(\gamma + \delta\gamma)$, $= \cos\gamma \sin\gamma + (\cos^2\gamma - \sin^2\gamma)\,\delta\gamma$,

$$\frac{d^2\delta\gamma}{dt^2} + n^2\delta\gamma = \mathfrak{C},$$

where $n^2 = \dfrac{\mu}{r^3}$; and this equation determines $\delta\gamma$.

III.

To develop the value of $\mathfrak{C}$, we have

$$\begin{aligned} A &= \cos(v - \sigma)\sin\phi \cdot \dot\theta + \sin(v - \sigma)\dot\phi, \\ B &= \sin(v - \sigma)\sin\phi \cdot \dot\theta + \cos(v - \sigma)\dot\phi, \end{aligned}$$

whence

$$\begin{aligned} \frac{dA}{dv} &= -B\sin\phi \cdot \dot\theta + A\dot\phi, \\ \frac{dB}{dv} &= A\sin\phi \cdot \dot\theta - B\dot\phi, \end{aligned}$$

and thence

$$\begin{aligned} \mathfrak{C} &= 2Ar\frac{dr}{dt} + r^2\frac{dA}{dt} - 2r^2B\frac{d\gamma}{dt} \\ &= r^2\frac{dA}{dt} + 2Ar\frac{dr}{dt} - 2r^2B\frac{d\gamma}{dt}. \end{aligned}$$

IV.

If in the disturbing function we write $s = \tan\gamma$, and attend only to the terms which involve A and B, or (what is the same thing) the terms which depend on the variation of the ecliptic, then, neglecting γ^2 and $\gamma\dfrac{d\gamma}{dt}$, we have

$$\frac{d\Omega}{d\gamma} = \mathfrak{C} = r^2\frac{dA}{dt}.$$

V.

The equations of motion, when the variation of the ecliptic is neglected, are

$$\frac{d^2r}{dt^2} - r\cos^2\gamma\left(\frac{dv}{dt}\right)^2 + \frac{\mu}{r^2} = \frac{d\Omega}{dr},$$
$$\frac{d}{dt}\left(r^2\cos^2\gamma\frac{dv}{dt}\right) = \frac{d\Omega}{dv},$$
$$\frac{d^2\gamma}{dt^2} + \frac{\mu}{r^3}\gamma = \frac{1}{r^2}\frac{d\Omega}{d\gamma}.$$

VI.

For the further development of the disturbing function, writing $u = \dfrac{1}{r}$, and taking Ω to denote the disturbing function of the *Mécanique Céleste*, we have

$$\Omega = \mu' m' u'^3\left(\frac{1}{\Delta} - u\,r'^2\cos\gamma\right),$$

where

$$\Delta^2 = 1 + u^2r'^2 - 2ur'\cos\gamma\cos(v-v') + u^2r'^2\sin^2\gamma\sin^2(v-v').$$

Writing $s = \tan\gamma$, and neglecting s^2, we have

$$\Omega = \mu' m' u'^3\left(\frac{1}{\Delta} - ur'^2\right),$$

where

$$\Delta^2 = 1 + u^2r'^2 - 2ur'\cos(v-v') + u^2r'^2s^2\sin^2(v-v').$$

VII.

If the latitude of the disturbing body is neglected, then $s' = 0$, and we have

$$\Omega = \mu' m' u'^3\left(\frac{1}{\Delta} - ur'^2\right),$$

where

$$\Delta^2 = 1 + u^2r'^2 - 2ur'\cos(v-v').$$

VIII.

To reduce the equation

$$\frac{d^2\gamma}{dt^2} + n^2\gamma = \frac{1}{r^2}\frac{d\Omega}{d\gamma},$$

where $n^2 = \dfrac{\mu}{r^3}$, we write

$$\gamma = \delta v \cdot \sin(v - \varpi) + \delta e \cdot \cos(v - \varpi),$$

where ϖ is the longitude of pericentre; and then, substituting and integrating, we obtain the values of δv and δe.

IX.

The equation

$$\frac{d}{dt}\left(r^2\frac{dv}{dt}\right) = \frac{d\Omega}{dv}$$

gives, integrating,

$$r^2\frac{dv}{dt} = C + \int \frac{d\Omega}{dv}\,dt,$$

where C is a constant of integration.

X.

The developments being carried as far as m^2, e^2, we have for the value of the disturbing function

$$\Omega = \mu' m' u'^3 \left(\frac{1}{\Delta} - ur'^2\right),$$

and for the equation of the orbit

$$u = nt + \epsilon + 2e\sin(nt + \epsilon - \varpi) + \tfrac{5}{4}e^2 \sin 2(nt + \epsilon - \varpi).$$

0.219 ON SOME FORMULAE RELATING TO THE VARIATION OF A PLANET'S ORBIT

[From the *Monthly Notices of the Royal Astronomical Society*, vol. xxi. (1863), pp. 43–47.]

In Hansen's Memoir, "Auseinandersetzung einer zweckmássigen Methode zur Berechnung der absoluten Stórungen der kleinen Planeten," Abhand. der K. Sachs. Gesell. t. V. (1856), are contained, § 8, some very elegant formulae for taking account of the variation of the plane of the orbit. These, in fact, depend upon the following geometrical theorem, viz., if (in the figure) ABC is a spherical triangle; P, a point on the side AB; and PM, PN, the perpendiculars let fall from P on the other two sides AC, CB; then we have

$$\cos PM \sin(BC + CM) = \cos PN \sin BN - \tan \tfrac{1}{2}C \cos BC(\sin PM + \sin PN), \tag{10}$$

$$\cos PM \cos(BC + CM) = \cos PN \cos BN + \tan \tfrac{1}{2}C \sin BC(\sin PM + \sin PN). \tag{11}$$

These equations, in fact, give

$$\cos PM \sin CM = \cos PN \sin CN - \tan \tfrac{1}{2}C(\sin PM + \sin PN), \tag{12}$$

$$\cos PM \cos CM = \cos PN \cos CN; \tag{13}$$

the latter of which is at once seen to be true, since joining the points C and P, the two sides are respectively equal to $\cos CP$.

To verify the former one, write $\angle PCM = C_1$, $\angle PCN = C_2$, so that $C = C_1 - C_2$. Then, since

$$\cos CP = \cos PM \cos CM = \cos PN \cos CN,$$

$$\sin PM = \sin CP \sin C_1, \qquad \sin PN = \sin CP \sin C_2,$$

the equation becomes

$$\cos CP(\tan CM - \tan CN) = -\tan \tfrac{1}{2}C \sin CP(\sin C_1 + \sin C_2),$$

or since $\tan CM = \tan CP \cos C_1$, $\tan CN = \tan CP \cos C_2$, this is

$$\cos C_1 - \cos C_2 = \tan \tfrac{1}{2}C(\sin C_1 + \sin C_2),$$

which is identically true, in virtue of the equation $C = C_1 - C_2$; and, conversely, we have the original two equations.

Suppose that XM is the ecliptic, X being the origin of longitudes, DP the instantaneous orbit, D the departure-point therein, and P the planet, DD' the orthogonal trajectory of the successive positions of the orbit; and writing

Υ, the departure of the planet,
v, the longitude of ditto,
y, the latitude of ditto,
θ, the longitude of node,
σ, the departure of ditto,
ϕ, the inclination,

then, in the figure, $DP = \Upsilon$, $XM = v$, $PM = y$, $XA = \theta$, $DA = \sigma$; $\angle A = \phi$.

The quantities θ, σ, ϕ might be considered as altogether arbitrary; but to fix the ideas it is better to assume at once that they denote Θ, the longitude of node, σ_0, the departure, Φ, the inclination, for the initial position of the orbit, viz., in the figure $XB = \Theta$, $D'B = \sigma_0$, $\angle B = \Phi$.

Take $DB = \sigma_0$, $\angle B = \Phi$, $BX = \Theta$, this determines a travelling orbit of reference XN, and origin of longitudes X therein; such that, with respect to this travelling orbit, the position of the planet's orbit is determined by θ, the longitude of node, σ_0 the departure of node, ϕ, the inclination.

We have in the triangle ABC, $AB = \sigma - \sigma_0$, $\angle B = \Phi$, $\angle A = 180° - \phi$; and if the other parts of the triangle are represented by $BC = \omega$, $AC = \psi$, then ω, ψ are given in terms of $\sigma - \sigma_0$, Φ, ϕ; and we have, moreover, $XC = \theta + AC = \theta + \psi + \chi$, $XC = \sigma_0 + \omega$; that is, the position of the travelling orbit XN, and origin of longitudes X therein, are determined by $\theta + \omega + \psi$, the longitude of node, $\sigma_0 + \omega$, the departure of node, Φ, the inclination.

Suppose that in reference to this travelling orbit and origin of longitudes therein, we have v_1, the longitude of planet, y_1, the latitude of ditto, viz., in the figure $XN = v_1$ (and therefore $BN = v_1$), $PN = y_1$.

Moreover, $BC + CM = BC + AM - AC = \omega + (v - \theta) - (\theta -$

$\Theta + \omega + \psi) = v - \Theta - \psi$, hence the two formulae become

$$\cos y \sin(v - \Theta - \psi) = \cos y_1 \sin(v_1 - \sigma_0 - \omega) - \tan \tfrac{1}{2}\Phi \cos \omega(\sin y + \sin y_1), \tag{14}$$

$$\cos y \cos(v - \Theta - \psi) = \cos y_1 \cos(v_1 - \sigma_0 - \omega) + \tan \tfrac{1}{2}\Phi \sin \omega(\sin y + \sin y_1). \tag{15}$$

0.220 NOTE ON A THEOREM OF JACOBI'S, IN RELATION TO THE PROBLEM OF THREE BODIES

[From the *Monthly Notices of the Royal Astronomical Society*, vol. xxii. (1861), pp. 76–78.]

The following theorem of Jacobi's (Comptes Rendus, t. III., p. 61 (1836)) has not, I think, found its way in an explicit form into any treatise of physical astronomy.

The theorem is as follows, viz.

"Consider the movement of a point without mass round the Sun, disturbed by a planet the orbit of which is circular. Let xyz be the rectangular coordinates of the disturbed body, the orbit of the disturbing planet being taken as the plane of xy, and the Sun as the centre of coordinates; let a be the distance of the disturbing planet, nt its longitude, m its mass, M the mass of the Sun: then we have, rigorously,

$$\begin{aligned}&\frac{dy}{dt}\frac{dx}{dt}-\frac{dx}{dt}\frac{dy}{dt}+M\frac{x\,dy-y\,dx}{x^2+y^2+z^2}\\ &-\frac{m}{a^2}\frac{(x\,dy-y\,dx)(x\cos nt+y\sin nt)}{(x^2+y^2+z^2)^{\frac{3}{2}}\{x^2+y^2+z^2-2a(x\cos nt+y\sin nt)+a^2\}^{\frac{1}{2}}}\\ &\qquad\qquad=\frac{m}{a^2}(x\cos nt+y\sin nt)."\end{aligned}$$

This is therefore a new integral equation, which, in the problem of three bodies, subsists, as regards the terms independent of the eccentricity of the disturbing planet, and which is rigorous as regards all the powers of the mass of such planet.

In the Lunar Theory the Earth must be substituted in the place of the Sun, and the Sun taken as the disturbing planet.

To prove the theorem, as expressed in polar coordinates, I take the equations of motion in the form in which I have employed them in my "Memoir on the Theory of Disturbed Elliptic Motion" (Memoirs, vol. xxvii. p. 1 (1859)), [212], viz.

$$\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\left(\frac{dv}{dt}\right)^2 \cos^2\gamma - r\left(\frac{d\gamma}{dt}\right)^2 + \frac{\mu}{r^2} = \frac{d\Omega}{dr}, \tag{16}$$

$$\frac{d}{dt}\left(r^2\cos^2\gamma\frac{dv}{dt}\right) = \frac{d\Omega}{dv}, \tag{17}$$

$$\frac{d}{dt}\left(r^2\frac{d\gamma}{dt}\right) + r^2\cos\gamma\sin\gamma\left(\frac{dv}{dt}\right)^2 = \frac{d\Omega}{d\gamma}, \tag{18}$$

where

$$\Omega = \frac{m'}{r'} - m'\frac{x\xi + y\eta + z\zeta}{r'^3},$$

or, since $\cos H = \cos\gamma\cos(v-v')$, and the Sun is considered as moving in a circular orbit (i.e. $r' = a$, $v' = nt$), we have

$$\Omega = \frac{m}{\sqrt{r^2 - 2ra\cos\gamma\cos(v-nt) + a^2}},$$

so that Ω is a function of r, v, γ and of t, which last quantity enters only in the combination $v - nt$.

Hence the complete differential coefficient of Ω is

$$\frac{d\Omega}{dt} = \frac{d\Omega}{dr}\frac{dr}{dt} + \frac{d\Omega}{dv}\frac{dv}{dt} + \frac{d\Omega}{d\gamma}\frac{d\gamma}{dt} + \frac{\partial\Omega}{\partial t},$$

where $\dfrac{\partial\Omega}{\partial t}$ denotes, as usual, the differential coefficient in regard to the time, in so far as it enters through the coordinates r, v, γ of the disturbed body.

We have, as usual,

$$\frac{dr^2 + r^2(\cos^2\gamma\, dv^2 + d\gamma^2)}{dt^2} - 2\frac{d\Omega}{dt} = -2\frac{d\Omega}{dt},$$

and, from the foregoing equation,

$$\frac{dv}{dt}\frac{d\Omega}{dv} = \frac{dv}{dt}\frac{d}{dt}\left(r^2\cos^2\gamma\frac{dv}{dt}\right);$$

hence, substituting this value, transposing, and integrating, we have

$$r^2\cos^2\gamma\frac{dv}{dt} + \int\frac{d\Omega}{dt}\,dt = \text{const.},$$

which is Jacobi's equation expressed in terms of the coordinates r, v, γ.

M. de Pontécoulant, in his Lunar Theory (1846), where the solar eccentricity is neglected, writes (p. 91),

$$\frac{r^2\,dv}{(1+s^2)\,dt} = m'h - \int dR,$$

where

$$r^2\frac{dv}{dt} = mn = m, \quad \text{since } n \text{ is there put equal to unity;}$$

and combining this with the equation (p. 43),

$$r^2\frac{dv}{dt} - mh = \int \frac{dR}{dv}\,dv,$$

we have

$$\int dR = R - mh + \frac{mr^2\,dv}{(1+s^2)\,dt},$$

and substituting this value of $\int dR$ in the integral of Vis Viva (p. 41),

$$\tfrac{1}{2}\left(\frac{dr^2 + r^2(dv^2 + d\gamma^2)}{dt^2}\right) - \frac{M}{r} = \Omega,$$

we have what is, in fact, Jacobi's equation.

0.221 ON THE SECULAR ACCELERATION OF THE MOON'S MEAN MOTION

[From the *Monthly Notices of the Royal Astronomical Society*, vol. xxii. (1862), pp. 171–231.]

The present Memoir exhibits a new method of taking account, in the Lunar Theory, of the variation of the eccentricity of the Sun's orbit.

The approximation is carried to the same extent as in Prof. Adams' Memoir "On the Secular Variation of the Moon's Mean Motion" (Phil. Trans., vol. CXLIII. (1853), pp. 397–406); and I obtain results agreeing precisely with his, viz., besides his new periodic terms in the longitude and radius vector, I obtain in the longitude the secular term

$$\left(\tfrac{3}{4}m^2 - \tfrac{81}{32}m^4\right) \int \frac{a}{a'}(e'^2 - E'^2)\, n\, dt,$$

and in the quotient radius, or radius vector divided by the mean distance, the secular term

$$\left(\tfrac{3}{8}m^2 - \tfrac{81}{64}m^4\right) \frac{a}{a'}(e'^2 - E'^2)\, nt,$$

which is, in fact, as will be shown, included implicitly in the results given in Professor Adams' Memoir.

In quoting the foregoing results, I have written $e'^2 - E'^2$ in the place of $(e' + f't)^2 - e'^2 = 2e'f't$, which in the notation of the present Memoir it should have been; and I purposely refrain from here explaining the precise signification of the symbols: this is carefully done in the sequel.

The method appears to me a very simple one in principle; and it possesses the advantage that it is not incorporated step by step with a lunar theory in which the eccentricity of the Sun's orbit is treated as constant; but it is added on to such a lunar theory, giving in the Moon's coordinates the supplementary terms which arise from the variation of the solar eccentricity, and thus serving as a verification of any process employed for taking account of such variation.

I have given the details of the work in a series of Annexes, 1 to 23: this appears to me the best course for presenting the investigation in a readable form.

I.

The inclination and eccentricity of the Moon's orbit, and, *a fortiori*, the variation of the position of the Ecliptic, and the Sun's latitude, are neglected; and the longitudes are measured from a fixed point in the Ecliptic.

I write n, the actual mean motion of the Moon at a given epoch; viz., it is assumed that the mean longitude at the time t is $\epsilon + nt + n_2t^2 + \cdots$ where ϵ, n, n_2, etc. are absolute constants; and, moreover, a, the calculated mean distance of the Moon; that is, n^2a^3 is the sum of the masses of the Earth and Moon; a is therefore an absolute constant; and, in like manner, n', the actual mean motion of the Sun at the same epoch, a', the calculated mean distance of the Sun; that is, if it were necessary to pay attention to the secular variation of the mean motion of the Sun, the assumption would be that the mean longitude was $\epsilon' + n't + n'_2t^2 + \cdots$, ϵ', n', n'_2, etc. being absolute constants, and $n'^2a'^3$ the sum of the masses of the Sun and Earth; a' would thus also be an absolute constant.

But for the purpose of the present investigation the secular variation of the mean longitude of the Sun is neglected, or it is assumed that the mean longitude of the Sun is $\epsilon' + n't$, ϵ', n' being absolute constants; and that $n'^2a'^3$ is the sum of the masses of the Sun and Earth, a' being thus also an absolute constant.

I put also m, the ratio of the mean motions of the Sun and Moon; that is,

$$m = \frac{n'}{n}, \quad \text{or } n' = mn;$$

m is also an absolute constant.

The Sun is considered as moving in an elliptic orbit, the eccentricity whereof is $e' + \delta e'$ or $e' + f't$, e' and f' being absolute constants; the longitude of the Sun's perigee may be taken to be $\varpi' + (1 - c')n't$; so that the mean anomaly g is $= n't + \epsilon' - [\varpi' + (1 - c')n't] = c'n't + \epsilon' - \varpi'$; c', ϵ', ϖ' being absolute constants; but c' is in fact treated as being $= 1$.

Hence, if r', v' are the radius vector and longitude of the Sun, we have

$$\begin{aligned}
r' &= a' \,\mathrm{elqr}(e' + \delta e', g), \\
v' &= \varpi' + (1 - c')n't + \mathrm{elta}(e' + \delta e', g) \\
&= n't + \epsilon' + [\mathrm{elta}(e' + \delta e', g) - g],
\end{aligned}$$

where $g = c'n't + \text{const.}$

In the expression for the disturbing function the Sun's mass is taken to be $= n'^2a'^3$, or, what is the same thing, $= m^2n^2a'^3$.

Let r, v be the radius vector and longitude of the Moon; then, taking the usual approximate expression of the Disturbing Function, the equations of motion are

$$\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\left(\frac{dv}{dt}\right)^2 + \frac{n^2a^3}{r^2} = \frac{dR}{dr}, \tag{19}$$

$$\frac{d}{dt}\left(r^2\frac{dv}{dt}\right) = \frac{dR}{dv}. \tag{20}$$

It will be convenient to assume ρ, the quotient radius of the Moon's orbit, ρ', the quotient radius of the Sun's orbit; that is

$$r = \rho a, \qquad r' = \rho' a'.$$

The equations of motion thus become

$$\frac{d}{dt}\left(\frac{d\rho}{dt}\right) - \rho\left(\frac{dv}{dt}\right)^2 + \frac{n^2}{\rho^2} = m^2n^2\,\frac{dP}{d\rho}, \tag{21}$$

$$\frac{d}{dt}\left(\rho^2\frac{dv}{dt}\right) = m^2n^2\,\frac{dP}{dv}, \tag{22}$$

where for shortness

$$P = \frac{\rho}{\rho'^3}\left(\tfrac{3}{2}\sin^2(v - v') - \tfrac{1}{2}\right),$$

in which

$$\rho' = \text{elqr}(e' + \delta e', g), \quad v' = n't + \epsilon' + [\text{elta}(e', g) - g].$$

I now change the notation by writing $\rho + \delta\rho$, $v + \delta v$, in the place of ρ, v, respectively, using henceforward ρ, v to denote

$$\rho = \text{elqr}(e', g), \quad v = n't + \epsilon' + [\text{elta}(e', g) - g];$$

and I write also $\rho{+}\delta\rho$, $v{+}\delta v$, in the place of ρ, v, using henceforward ρ, v to denote the solutions of the equations obtained from the equations of motion by writing therein ρ, v instead of the complete values $\rho{+}\delta\rho$, $v + \delta v$.

Suppose, in like manner, that the complete values of P, Q are denoted by $P+\delta P$, $Q+\delta Q$, where

$$\delta P = \frac{dP}{d\rho}\delta\rho + \frac{dP}{dv}\delta v + \frac{dP}{d\rho'}\delta\rho' + \frac{dP}{dv'}\delta v',$$

with a like value for δQ, the first powers of $\delta\rho$, δv, $\delta\rho'$, $\delta v'$ being alone attended to.

Then, observing that the equations of motion are satisfied when $\delta\rho$, δv, $\delta\rho'$, $\delta v'$ are neglected, we have

$$\frac{d}{dt}\left(\frac{d\delta\rho}{dt}\right) - \left(\frac{dv}{dt}\right)^2\delta\rho - 2\rho\frac{dv}{dt}\frac{d\delta v}{dt} - \frac{2n^2}{\rho^3}\delta\rho = m^2n^2\,\delta P, \tag{23}$$

$$\frac{d}{dt}\left(\rho^2\frac{d\delta v}{dt} + 2\rho\frac{dv}{dt}\delta\rho\right) = m^2n^2\,\delta Q. \tag{24}$$

The second of these equations gives

$$\rho^2\frac{d\delta v}{dt} + 2\rho\frac{dv}{dt}\delta\rho = C + m^2n^2\int\delta Q\,dt,$$

where the constant of integration, C, is to be so determined that δv may not contain any term of the form kt (for any such term is taken to be included in the term nt of $v+\delta v$).

Multiplying the equation just obtained by $-\dfrac{2}{\rho}\dfrac{dv}{dt}$, and adding it to the first equation, we have

$$\frac{d^2\delta\rho}{dt^2} + \left(\frac{2n^2}{\rho^3} - \left(\frac{dv}{dt}\right)^2\right)\delta\rho - \frac{2}{\rho}\frac{dv}{dt}\left(C + m^2n^2\int\delta Q\,dt\right)$$
$$= m^2n^2\left(\delta P + \frac{2}{\rho}\frac{dv}{dt}\delta Q\right), \tag{25}$$

which, with the above-mentioned integral equation, are the equations for the solution of the problem; but it will be convenient to write them under the slightly different form

$$\frac{d^2\delta\rho}{dt^2} + n^2\delta\rho = \left(n^2 + \frac{2n^2}{\rho^3} - \left(\frac{dv}{dt}\right)^2\right)\delta\rho + m^2n^2\left(\delta P + \frac{2}{\rho}\frac{dv}{dt}(C + \int\delta Q\,dt)\right), \tag{26}$$

$$\frac{d\delta v}{dt} = -\frac{2}{\rho}\frac{dv}{dt}\delta\rho + \frac{1}{\rho^2}\left(C + m^2n^2\int\delta Q\,dt\right). \tag{27}$$

In these equations C is determined, as above, by the condition that $\dfrac{d\delta v}{dt}$ may contain no constant term; the values of ρ', v', $\delta\rho'$, $\delta v'$ are of course given by the theory of elliptic motion, and those of ρ, v are given by the ordinary lunar theory, in which the eccentricity of the solar orbit is treated as a constant; and then, $\delta\rho$, δv being obtained by integrating the equations, the radius vector and longitude of the Moon are $a(\rho+\delta\rho)$ and $v+\delta v$ respectively.

We have

$$Q = \frac{\rho}{\rho'^3}\left(-\tfrac{3}{2}\sin 2(v-v') - 2(v-v')\right).$$

Moreover, by the lunar theory, observing that Plana's a is, or may be considered, identical with the a of the present Memoir, and putting also

$$2\nu = (2-2m)nt + \text{const.}, \qquad g = c'mn't + \text{const.},$$

we have

$$\begin{aligned}\frac{1}{\rho} &= 1 + \tfrac{1}{6}m^2 - \tfrac{1}{3}m^2e'^2 - \tfrac{1}{4}m^2e\cos g + \tfrac{1}{8}m^2e^2\cos 2g \\ &\quad + \tfrac{3}{8}m^2e'^2\cos 2\nu + \cdots, \qquad (28)\\ v &= nt + \epsilon - \tfrac{3}{4}me\sin g + \tfrac{1}{2}m^2\sin 2\nu + \tfrac{11}{8}m^2e\sin(g+2\nu) \\ &\quad - \tfrac{55}{24}m^2e\sin(2\nu - g) + \cdots, \qquad (29)\end{aligned}$$

where the series are carried as far as m^2 and e^2; the terms in e'^2 are given, as I shall have occasion to refer to them, but they are not used in the investigation, and, omitting them, the values are

$$\frac{1}{\rho} = 1 + \tfrac{1}{6}m^2 - \tfrac{1}{4}m^2e\cos g + \tfrac{1}{8}m^2e^2\cos 2g + \tfrac{3}{8}m^2e'^2\cos 2\nu, \qquad (30)$$

$$\begin{aligned}v &= nt + \epsilon - \tfrac{3}{4}me\sin g + \tfrac{1}{2}m^2\sin 2\nu + \tfrac{11}{8}m^2e\sin(g+2\nu) \\ &\quad - \tfrac{55}{24}m^2e\sin(2\nu - g). \qquad (31)\end{aligned}$$

For the coordinates of the Sun we have

$$\frac{a'}{r'} = 1 + e'\cos g + e'^2\cos 2g, \qquad (32)$$

$$v' = n't + \epsilon' + 2e'\sin g + \tfrac{5}{4}e'^2\sin 2g, \qquad (33)$$

the series being carried as far as e'^2; but the terms in e'^2 are only used for the formation of $\delta\rho'$, $\delta v'$; and, omitting them, we have

$$\frac{a'}{r'} = 1 + e'\cos g, \qquad v' = n't + \epsilon' + 2e'\sin g.$$

If $e' + f't$ is written for e', then the value of $\delta e'$ is $= f't$; but as only the terms multiplied by the simple power f' are attended to, we may for convenience write $\delta e' = t$, the factor f' being restored in the final results: we thus have

$$\begin{aligned}\delta\rho' &= -t\cos g + 2e't\cos 2g,\\ \delta v' &= -2t\sin g + \tfrac{5}{2}e't\sin 2g,\end{aligned}$$

and we may add the equations

$$\begin{aligned}\frac{dv'}{dt} &= n' + 2e'n'\cos g + \tfrac{5}{4}e'^2 n'\cos 2g,\\ \delta\rho' &= -3e't - 3t\cos g + \tfrac{5}{2}e't\cos 2g,\\ \delta v' &= -2t\sin g - \tfrac{5}{4}e't\sin 2g,\end{aligned}$$

which will be found useful.

II.

Proceeding now to the development of the solution, we have

$$\begin{aligned}\delta P = {}& \frac{1}{\rho'^3}\left(1 + \tfrac{3}{2}\cos 2(v - v')\right)\delta\rho\\ &+ \frac{3}{\rho'^3}\left(\tfrac{1}{2}\sin 2(v - v')\right)\delta v + \frac{3}{2\rho'^3}\sin 2(v - v')\frac{\delta\rho'}{\rho'} + \cdots, \quad (34)\end{aligned}$$

where the terms containing $\delta\rho$ and δv are (see Annex 1)

$$\begin{aligned}&\left(1 + \tfrac{3}{2}e'^2\right)\delta\rho + \tfrac{3}{2}e'\cos\nu\,\delta v + \tfrac{3}{2}e'\sin\nu\,\frac{\delta\rho'}{\rho'} + \cdots\\ &\quad = \left(1 + \tfrac{3}{2}e'^2\right)\left(-\tfrac{3}{4}\sin 2\nu - \tfrac{1}{2}e\sin(2\nu - g) + \tfrac{3}{4}e\sin(2\nu + g)\right)\\ &\qquad + \tfrac{3}{2}e'\cos\nu\left(\tfrac{1}{2}\cos 2\nu + e\cos(2\nu - g) - e\cos(2\nu + g)\right) + \cdots\end{aligned}$$

and also

$$\begin{aligned}&\tfrac{3}{2}\sin 2\nu\left(-\tfrac{3}{4}\sin 2\nu + \cdots\right) + \tfrac{3}{2}\cos 2\nu\left(\tfrac{1}{2}\cos 2\nu + \cdots\right)\\ &\qquad = \tfrac{3}{4}\left(-\tfrac{3}{2}\sin^2 2\nu + \cos^2 2\nu\right) + \cdots\end{aligned}$$

where the terms containing $\delta\rho'$ and $\delta v'$ are (see Annex 2)

$$\tfrac{3}{2}\sin 2\nu\left(-t\cos g + 2e't\cos 2g\right) + \tfrac{3}{2}\cos 2\nu\left(-2t\sin g + \tfrac{5}{2}e't\sin 2g\right) + \cdots$$

The development being continued through the annexes referred to, the final results are obtained for the secular acceleration of the Moon's mean motion, confirming those of Professor Adams.

221.

ON THE SECULAR ACCELERATION OF THE MOON'S MEAN MOTION.

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The values of the certain terms in the expressions of the moon's coordinates, which give the secular acceleration of the moon's mean motion, were originally calculated by me by a direct and elementary method, which (as it appears to me) has the advantage over Plana's very elegant process, that in it we have not to assume without proof that the required acceleration arises entirely from the variation of the eccentricity of the earth's orbit. The solution was originally obtained in a literal form; and the numerical results were given to a greater degree of accuracy than was requisite. I propose in the present paper to present the investigation under a somewhat altered form; and to give the results in the numerical form only, but to a greater degree of accuracy than was attained in my original memoir. The following changes have been made; viz. instead of the disturbing function

$$\frac{m'}{\rho'^3}\left(x'x+y'y+z'z\right)$$

I employ the equivalent form

$$\frac{m'}{\rho'^3}\rho^2\cos\angle\left(\rho,\rho'\right),$$

and in connexion herewith, I introduce the coordinates P, Q (functions of ρ, v) employed by Mr Delaunay in his new lunar theory; viz. writing

$$P=\frac{\rho}{r}\cos(v-nt-\epsilon)=\cos\phi\cos(v-nt-\epsilon),\quad Q=\frac{\rho}{r}\sin(v-nt-\epsilon)=\cos\phi\sin(v-nt-\epsilon)$$

so that P, Q are the values of $\cos(v-nt-\epsilon)$, $\sin(v-nt-\epsilon)$, multiplied by the cosine of the moon's latitude; and the disturbing function is

$$m'\frac{\rho^2}{\rho'^3}\left(P\cos v'+Q\sin v'\right).$$

The expressions for δP, δQ are obtained in a more simple form by assuming at once the numerical value of the ratio m of the mean motions. I write also $t = nt$, and take τ, g instead of Delaunay's ξ, g; viz. τ is the moon's mean longitude minus the longitude of the perigee; and g is the longitude of the moon's perigee minus that of the sun's perigee. The epoch is taken to be the time of the sun's perigee passing the node; so that at the epoch $\tau = \varpi - \theta$, $g = \varpi - \varpi'$; where ϖ, ϖ' are the longitudes of the moon's perigee and the sun's perigee, and θ is the longitude of the node. This being so, the mean longitude of the moon is $nt+\epsilon+v_1$, where v_1 is the value of v at the epoch; the mean longitude of the moon's perigee is $nt+\epsilon+v_1-\tau$; and the mean longitude of the sun's perigee is $nt + \epsilon + v_1 - \tau - g$. The sun's mean anomaly is $n't+\epsilon'-(nt+\epsilon+v_1-\tau-g)$; and introducing the numerical value $n' = (1 - m)n$, this is $= (1 - m)nt + \epsilon' - nt - \epsilon - v_1 + \tau + g$, $= -mnt + \epsilon' - \epsilon - v_1 + \tau + g$, which at the epoch vanishes: we have therefore $\epsilon' = \epsilon+v_1-\tau-g$, and the mean anomaly is $= -mnt+\tau+g$, $= -t + \tau + g$, writing therein t for mnt; that is, it is $-(t - \tau - g)$. Hence writing $g' = t - \tau - g$, which is the sun's mean anomaly, then $g = t - \tau - g'$; and the arguments 2τ, $2\tau - g$, $2\tau + g$, &c., become 2τ, $2\tau + g' - t$, $3t - 2\tau - g'$; or, as the terms containing t need not be exhibited, these may be written 2τ, $2\tau - g'$, $-2\tau + g'$, that is 2τ, $2\tau - g'$, $2\tau + g'$, the sign only of the sine or cosine terms being attended to. And so in other cases; the arguments are all given in terms of τ and g'; but the accented letter g' will for convenience be written without the accent.

I.

We have

$$\frac{d^2\rho}{dt^2} - \rho\left(\frac{dv}{dt}\right)^2 + \frac{n^2a^3}{\rho^2} = n^2\delta R, \qquad \frac{d(\rho^2\frac{dv}{dt})}{dt} = n^2\delta S;$$

and thence

$$\frac{d^2\delta\rho}{dt^2} - 2\rho n\delta\left(\frac{dv}{dt}\right) - n^2\delta\rho + \frac{2n^2a^3}{\rho^3}\delta\rho = n^2\delta R,$$

that is

$$\frac{d^2\delta\rho}{dt^2} + n^2\delta\rho - 2n^2\delta\left(\frac{dv}{dt}\right) = n^2\delta R,$$

if for the undisturbed values we write $\rho = a$, $\dfrac{dv}{dt} = n$; and

$$\frac{d\delta v}{dt} = \frac{2n}{\rho}\delta\rho + \frac{n}{\rho^2}\delta S = 2n\delta\rho + n\delta S,$$

if for the undisturbed values $\rho = 1$, $\dfrac{dv}{dt} = n$, or say $\rho = 1$, $dv = n\,dt$, or simply $dv = dt$, taking $n = 1$. We thus have

$$\frac{d^2\delta\rho}{dt^2} + \delta\rho - 2\delta\left(\frac{dv}{dt}\right) = \delta R, \qquad \frac{d\delta v}{dt} = 2\delta\rho + \delta S;$$

or, what is the same thing,

$$\frac{d^2\delta\rho}{dt^2} + \delta\rho = \delta P, \qquad \frac{d\delta v}{dt} = \delta Q,$$

if for shortness

$$\delta P = \delta R + 2\int \delta Q\,dt, \qquad \delta Q = 2\delta\rho + \delta S;$$

so that, as regards the terms in question, the original equations serve to determine $\delta\rho$, δv.

II.

The value of the disturbing function is

$$\delta R = \frac{m'}{\rho'^3}\rho^2\left(P\cos v' + Q\sin v'\right),$$

where

$$P = \frac{\rho}{r}\cos(v - t - \epsilon), \qquad Q = \frac{\rho}{r}\sin(v - t - \epsilon).$$

We require the non-periodic part of $\dfrac{d\delta v}{dt}$, that is of $2\delta\rho + \delta S$; and in forming this we may in $\delta\rho$, δS take account of the periodic terms only, neglecting the terms which depend on t (secular terms) and the constant terms; these last, in fact, give only periodic terms in $\delta\rho$, δS. In δR, the term $\dfrac{m'}{\rho'^3}\rho^2 P\cos v'$ gives in δP, and thence in $\delta\rho$, terms involving $\cos g'$ with the factor e' (the sun's eccentricity); and the term $\dfrac{m'}{\rho'^3}\rho^2 Q\sin v'$ gives in δQ, and thence in δv, terms involving $\sin g'$ with the factor e'. The corresponding terms in $\delta\rho$, δv give in δP $(= \delta R + 2\int \delta Q\,dt)$ and in δQ $(= 2\delta\rho + \delta S)$ the terms which produce the secular acceleration. And so in other cases; we require in δP the terms

$$A + Bt\cos g' + Ct\sin g',$$

and in δQ the terms

$$A' + B't\sin g' + C't\cos g',$$

where the constant terms A, A' are introduced for the reason above; the terms in question give in $\delta\rho$ the terms

$$A + \frac{1}{2}Bt\cos g' + \frac{1}{2}Ct\sin g',$$

and in δv the terms

$$A't + \frac{1}{2}B't^2\sin g' - \frac{1}{2}C't^2\cos g' - \frac{1}{2}B'\frac{\cos g'}{m^2} + \frac{1}{2}C'\frac{\sin g'}{m^2};$$

where the constant terms $\frac{1}{2}B'\frac{\cos g'}{m^2}$, $-\frac{1}{2}C'\frac{\sin g'}{m^2}$ are written down so as to make the whole values of $\delta\rho$, δv correct as far as the terms in question; and from them we have in $\frac{d\delta v}{dt}$, the secular term $A't$, that is $-2B't$, if $A' = -2B$ (for observe that in δQ the non-periodic part is $A' + B't\sin g' + C't\cos g'$, where $A' = -2B$, and the value of $\frac{d\delta v}{dt}$ is $A' - \frac{2B'}{m^2} +$ &c., which in regard to the constant term gives $A' - \frac{2B'}{m^2} = -2B - \frac{2B'}{m^2}, = -2B$ since B' is a term of the order m^2).

III.

We require the expression for $\frac{m'}{\rho'^3}$, say

$$\frac{m'}{\rho'^3} = m^2n^2\left(1 + \frac{3}{2}e'\cos g' + \text{\&c.}\right),$$

and then, by what precedes, in forming the values of δP, δQ, it is sufficient to take

$$\frac{m'}{\rho'^3} = m^2n^2\left(1 + \frac{3}{2}e'\cos g'\right).$$

In the expressions for P, Q we require only the non-periodic terms and the terms containing the argument g', $2g'$. Writing

$$\frac{\rho}{r} = 1+\frac{1}{6}m^2-e\cos\tau-\frac{1}{6}m^2\cos 2\tau, \qquad v-t-\epsilon = 2e\sin\tau+\frac{5}{4}e^2\sin 2\tau+\frac{1}{2}m^2\sin(2\tau-2g$$

where in $v-t-\epsilon$ the term $\frac{1}{2}m^2\sin(2\tau-2g')$ arises from the evection, and is in fact $\frac{1}{2}m^2\sin(2\tau-2g')$; we find

$$P = \left(1+\tfrac{1}{6}m^2\right)\cos(v-t-\epsilon) - e\cos\tau\cos(v-t-\epsilon) - \tfrac{1}{6}m^2\cos 2\tau\cos(v-t-\epsilon),$$
$$Q = \left(1+\tfrac{1}{6}m^2\right)\sin(v-t-\epsilon) - e\cos\tau\sin(v-t-\epsilon) - \tfrac{1}{6}m^2\cos 2\tau\sin(v-t-\epsilon).$$

But

$$\cos(v-t-\epsilon) = 1-2e^2\sin^2\tau+\&\text{c.} = 1-e^2+e^2\cos 2\tau, \qquad \sin(v-t-\epsilon) = 2e\sin\tau+\tfrac{5}{4}e^2 \text{si}$$

and hence, up to the required order,

$$P = \left(1+\tfrac{1}{6}m^2-e^2\right)+(-e+2e)\cos\tau+\left(\tfrac{1}{6}m^2+e^2\right)\cos 2\tau = 1+\tfrac{1}{6}m^2-e^2+e\cos\tau$$
$$Q = (-e+2e)\sin\tau+\tfrac{5}{4}e^2\sin 2\tau = e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau;$$

where, however, in the term $e\cos\tau$, $e\sin\tau$, we must restore the evection, that is write

$$\tau+\tfrac{1}{2}m^2\sin(2\tau-2g') \quad \text{for} \quad \tau;$$

so that the term $e\cos\tau$ is replaced by

$$e\cos\tau-\tfrac{1}{2}m^2e\sin\tau\sin(2\tau-2g') = e\cos\tau+\tfrac{1}{4}m^2e\cos(3\tau-2g')-\tfrac{1}{4}m^2e\cos(\tau+2g'),$$

and the term $e\sin\tau$ is replaced by

$$e\sin\tau+\tfrac{1}{2}m^2e\cos\tau\sin(2\tau-2g') = e\sin\tau+\tfrac{1}{4}m^2e\sin(3\tau-2g')+\tfrac{1}{4}m^2e\sin(\tau+2g').$$

We thus have

$$\begin{aligned} P &= 1+\tfrac{1}{6}m^2-e^2+e\cos\tau+\left(\tfrac{1}{6}m^2+e^2\right)\cos 2\tau \\ &\quad +\tfrac{1}{4}m^2e\cos(3\tau-2g')-\tfrac{1}{4}m^2e\cos(\tau+2g'), \\ Q &= e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau+\tfrac{1}{4}m^2e\sin(3\tau-2g')+\tfrac{1}{4}m^2e\sin(\tau+2g'). \end{aligned}$$

Or, restoring in the terms $\frac{1}{4}m^2e\cos(3\tau-2g')$ and $\frac{1}{4}m^2e\sin(3\tau-2g')$, the argument g instead of g', that is $2\tau+g$ instead of $2\tau-2g'$, or say $2\tau+g'$ instead of $2\tau-2g'$ (since the difference affects only periodic terms of higher orders), and similarly in the other terms, we have

$$\begin{aligned} P &= 1+\tfrac{1}{6}m^2-e^2+e\cos\tau+\left(\tfrac{1}{6}m^2+e^2\right)\cos 2\tau \\ &\quad +\tfrac{1}{4}m^2e\cos(2\tau+g')-\tfrac{1}{4}m^2e\cos(\tau+2g'), \\ Q &= e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau+\tfrac{1}{4}m^2e\sin(2\tau+g')+\tfrac{1}{4}m^2e\sin(\tau+2g'), \end{aligned}$$

which, in fact, are the expressions made use of by Mr Delaunay.

But for the present purpose, retaining the argument g' only in those terms which contain the factor e', it is sufficient to write

$$P = 1+\tfrac{1}{6}m^2-e^2+e\cos\tau+\left(\tfrac{1}{6}m^2+e^2\right)\cos 2\tau, \qquad Q = e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau.$$

Moreover

$$\rho^2 = r^2 = 1 + \tfrac{1}{3}m^2 - 2e\cos\tau + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right),$$

and we thence have

$$\frac{m'}{\rho'^3}\rho^2 = m^2\left(1 + \tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right)\right).$$

We have moreover $\cos v' = \cos(t - g')$, $\sin v' = \sin(t - g')$; or, since the argument t may be omitted, $\cos v' = \cos g'$, $\sin v' = -\sin g'$. Hence

$$\begin{aligned}\delta R = m^2 &\left(1 + \tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right)\right)\\ &\times P\cos g',\end{aligned}$$

$$\begin{aligned}\delta S = m^2 &\left(1 + \tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right)\right)\\ &\times Q(-\sin g').\end{aligned}$$

IV.

In forming δP $(= \delta R + 2\int \delta Q\,dt)$, the periodic terms of δQ give rise to the secular terms in $\delta\rho$; but they also give rise to periodic terms which must be taken into account in forming the value of δQ. We have

$$\delta Q = 2\delta\rho + \delta S,$$

and the periodic terms of $\delta\rho$ must be combined with δS.

The investigation is hence somewhat complicated; but it may be presented under a more simple form by the following consideration. We ultimately require in δP the non-periodic terms, the terms in $t\cos g'$, and the terms in $t\sin g'$; and in δQ the non-periodic terms, the terms in $t\sin g'$, and the terms in $t\cos g'$. Let δP_0, δQ_0 denote the values which δP, δQ would have if in the expression for δQ we were to omit the term $2\delta\rho$, and write $\delta Q = \delta S$; that is, let $\delta P_0 = \delta R + 2\int \delta Q_0\,dt$, $\delta Q_0 = \delta S$; then δP_0, δQ_0 can be at once obtained. We have $\delta P = \delta P_0$ (terms of order m^2 and m^2e' in δP are unaltered), but the terms of order m^2e' in δP are altered by the substitution for $\delta\rho$ (in $\delta Q = 2\delta\rho + \delta S$) of the terms of the orders m^2 and m^2e'.

The terms of order m^2 in $\delta\rho$ are obtained from δP_0, by simply dividing by -3; and then in δQ we have the term $2\delta\rho$; that is, $\delta Q = \delta S + 2\delta\rho$. But δS contains no term of the order m^2; hence in δQ, as regards terms of this order, $\delta Q = 2\delta\rho$; and the terms of order m^2 in δQ are equal to those of order m^2 in $2\delta\rho$, that is, in $2(-\frac{1}{3}\delta P_0) = -\frac{2}{3}\delta P_0$. These terms are periodic; and in the expression for δP, we have $\delta P = \delta P_0 + 2\int \delta Q\, dt$; that is, the terms of order m^2 in δP are unaltered; but we have in δQ a term $-\frac{2}{3}\delta P_0$, which, combined with the terms of order $m^2 e'$ in δR, gives in δP terms of order $m^2 e'$.

V.

Writing for shortness

$$\Theta = \frac{m'}{\rho'^3}\rho^2,$$

we have

$$\delta R = \Theta \cdot P\cos g', \qquad \delta S = \Theta \cdot Q(-\sin g'),$$

and thence

$$\delta P_0 = \delta R + 2\int \delta Q_0\, dt = \Theta P\cos g' + 2\int \Theta(-Q\sin g')\, dt.$$

But

$$\Theta = m^2\left(1 + \tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right)\right).$$

For the terms of order m^2, we have $\Theta = m^2$, and thence

$$\delta P_0 = m^2\cos g', \qquad \delta Q_0 = -m^2\sin g' \cdot Q;$$

so that, omitting the arguments which do not contain g',

$$\delta P_0 = m^2\cos g' \cdot e\cos\tau = \tfrac{1}{2}m^2 e\cos g' \cdot \cos\tau,$$
$$\delta Q_0 = -m^2\sin g' \cdot e\sin\tau = -\tfrac{1}{2}m^2 e\sin g' \cdot \sin\tau.$$

These give in $\delta\rho$ the terms

$$\delta\rho = -\tfrac{1}{6}m^2 e\cos g' \cdot \cos\tau,$$

and in δQ the term

$$2\delta\rho = -\tfrac{1}{3}m^2 e\cos g' \cdot \cos\tau.$$

For the terms of order $m^2 e'$, we have in Θ the term $\frac{3}{2}m^2 e'\cos g'$; and the terms of order m^2 in P and Q give in δP a term

$$\tfrac{3}{2}m^2 e'\cos^2 g' = \tfrac{3}{4}m^2 e',$$

which is non-periodic; and in δQ a term

$$-\tfrac{3}{2}m^2e'\cos g'\sin g' = -\tfrac{3}{4}m^2e'\sin 2g',$$

which is periodic. Hence in δP we have the non-periodic term $\frac{3}{4}m^2e'$; and in δQ we have the periodic term $-\frac{3}{4}m^2e'\sin 2g'$.

But we have also in δP the term arising from $2\int \delta Q\,dt$, where δQ is the term $-\frac{1}{3}m^2e\cos g'\cdot\cos\tau$, which, combined with the term in Θ, gives

$$-\tfrac{1}{3}m^2e\cos g'\cdot\cos\tau\times\tfrac{3}{2}m^2e'\cos g' = -\tfrac{1}{2}m^4ee'\cos^2 g'\cdot\cos\tau = -\tfrac{1}{4}m^4ee'\cos\tau,$$

which is non-periodic. Hence we have the non-periodic term $-\frac{1}{4}m^4ee'\cos\tau$; which, integrated in $2\int \delta Q\,dt$, gives the non-periodic term $-\frac{1}{2}m^4ee't\cos\tau$. But this is a term which would ultimately give in $\delta\rho$ the term $-\frac{1}{6}m^4ee't\cos\tau$; and this is of the order m^4e, which is beyond the order we are seeking.

We have in δP the term arising from $2\int \delta Q\,dt$, where δQ is the periodic term $-\frac{3}{4}m^2e'\sin 2g'$. This gives in δP the term

$$2\int\left(-\tfrac{3}{4}m^2e'\sin 2g'\right)dt = \tfrac{3}{4}m^2e'\frac{\cos 2g'}{m} = \tfrac{3}{4}me'\cos 2g',$$

which is periodic, and does not affect the secular acceleration.

The terms of order m^2e' in δP are thus

$$\tfrac{3}{4}m^2e' + \tfrac{3}{2}m^2e'\cos g'\cdot e\cos\tau,$$

and those in δQ are

$$-\tfrac{3}{2}m^2e'\sin g'\cdot e\sin\tau.$$

Combining these results, we have

$$\begin{aligned}
\delta P &= \tfrac{3}{4}m^2e' + \tfrac{1}{2}m^2e\cos g'\cdot\cos\tau + \tfrac{3}{2}m^2e'\cos g'\cdot e\cos\tau,\\
\delta Q &= -\tfrac{1}{2}m^2e\sin g'\cdot\sin\tau - \tfrac{3}{2}m^2e'\sin g'\cdot e\sin\tau.
\end{aligned}$$

VI.

From δP and δQ we obtain $\delta\rho$ and δv; and thence δP and δQ, the complete values. We have

$$\delta\rho = \frac{1}{\frac{d^2}{dt^2}+1}\delta P, \qquad \delta v = \int \delta Q\,dt.$$

For the terms of order m^2, we have

$$\delta P = \tfrac{1}{2}m^2 e\cos g' \cdot \cos\tau, \qquad \delta Q = -\tfrac{1}{2}m^2 e\sin g' \cdot \sin\tau;$$

and thence

$$\delta\rho = -\tfrac{1}{2}m^2 e\cos g' \cdot \cos\tau, \qquad \delta v = \tfrac{1}{2}m^2 e\sin g' \cdot \cos\tau.$$

But we must also take into account the terms of order m^2e' in δP, δQ. We have

$$\delta P = \tfrac{3}{4}m^2 e' + \tfrac{3}{2}m^2 e'\cos g' \cdot e\cos\tau, \qquad \delta Q = -\tfrac{3}{2}m^2 e'\sin g' \cdot e\sin\tau.$$

The non-periodic term $\frac{3}{4}m^2e'$ gives in $\delta\rho$ the non-periodic term $\frac{3}{4}m^2e'$; and in δv the term $\frac{3}{4}m^2e't$.

The term $\frac{3}{2}m^2e'\cos g' \cdot e\cos\tau$ gives in $\delta\rho$ the term

$$\tfrac{3}{2}m^2e'\cos g' \cdot \frac{e\cos\tau}{0} = \tfrac{3}{2}m^2e'\cos g' \cdot (-e\cos\tau) = -\tfrac{3}{2}m^2ee'\cos g' \cdot \cos\tau;$$

that is, since $\frac{1}{\frac{d^2}{dt^2}+1}\cos\tau = \frac{\cos\tau}{-1+1}$, which is infinite, we must write the equation as $\left(\frac{d^2}{dt^2}+1\right)\delta\rho = \delta P$; or if $\delta P = A\cos\tau + B\sin\tau$, then $\delta\rho = \frac{1}{2}(At\sin\tau - Bt\cos\tau)$.

The calculation is continued in the Annexes. We have

$$\begin{aligned}\delta P = \tfrac{3}{2}e' \cdot t\cos g' &- \tfrac{15}{4}e' \quad ,, \ 2\tau \\ + \tfrac{21}{4} \quad ,, \ 2\tau - g' &- \tfrac{3}{4} \quad ,, \ 2\tau + g' \\ + \tfrac{9}{2}e' \quad ,, \ 2g' &+ \tfrac{11}{2}e' \quad ,, \ 2\tau - 2g',\end{aligned}$$

and similarly (see Annex 5),

$$\begin{aligned}\delta Q = \tfrac{15}{2}e' \cdot t\sin 2\tau& \\ - \tfrac{21}{4} \quad ,, \ 2\tau - g' &+ \tfrac{3}{4} \quad ,, \ 2\tau + g' \\ - \tfrac{11}{2}e' \quad ,, \ 2\tau - 2g'.&\end{aligned}$$

But in the foregoing expression of δP the terms belonging to the arguments g', $2g'$ give in $\delta\rho$, terms which rise by integration in δv; and in forming the expressions for δP, δQ, it is proper to take account of these terms. Attending only to the terms in question, we have

$$\frac{d^2\delta\rho}{dt^2} + \delta\rho = m^2n^2\delta P = \tfrac{3}{2}m^2 \cdot t\cos g' + \tfrac{9}{2}m^2e' \quad ,, \ 2g'.$$

Now in general, if

$$\frac{d^2\delta\rho}{dt^2} + n^2\delta\rho = n^2 \cdot t\cos n\alpha t,$$

then

$$\delta\rho = \frac{2\alpha}{(1-\alpha^2)^2}\sin n\alpha t + \frac{1}{1-\alpha^2} \cdot t\cos n\alpha t;$$

and hence the foregoing equation gives in $\delta\rho$ the terms

$$\begin{gathered}3m^2\sin g' + \tfrac{3}{2} \cdot t\cos g' \\ +18m^2e' ,, \ 2g' + \tfrac{9}{2}m^2e' \quad ,, \ 2g';\end{gathered}$$

or, neglecting the terms which contain m^3, the terms of $\delta\rho$ are

$$3m^2\sin g' + \tfrac{3}{2} \cdot t\cos g' + 18m^2e'\sin 2g' + \tfrac{9}{2}m^2e' \cdot t\cos 2g'.$$

Substituting these terms in

$$n \times \frac{d\delta v}{dt} - 2n\delta\rho = 3m^2 \cdot t\cos g' - 9m^2e' \cdot t\sin 2g',$$

and since, in general,

$$\int t\cos n\alpha t \cdot dt = \frac{1}{n^2\alpha^2}\cos n\alpha t + \frac{1}{n\alpha} \cdot t\sin n\alpha t,$$

we obtain in δv the terms

$$\tfrac{3}{2}\cos g' + \tfrac{3}{4}m^2\cdot t\sin g' - \tfrac{1}{4}m^2\cdot\tfrac{9}{2}e'\cdot t\sin 2g',$$

or for the present purpose the terms

$$\tfrac{3}{2}\cos g' + \tfrac{3}{4}m^2\cdot t\sin g';$$

and these give in δP the additional terms (see Annex 6)

$$\tfrac{3}{2}\sin 2\tau + \tfrac{3}{4}m^2\cdot\text{,,}\ 2\tau - g',$$

and in δQ the additional terms (see Annex 7)

$$\tfrac{3}{4}m^2\cdot\cos 2\tau + \tfrac{3}{4}m^2\cdot\text{,,}\ 2\tau - g' + \tfrac{3}{4}m^2\cdot\text{,,}\ 2\tau + g'.$$

Combining the foregoing results, we have

$$\begin{aligned}\delta P = {} & \tfrac{3}{2}e'\cdot t\cos g' - \tfrac{15}{4}e' \quad\text{,,}\ 2\tau + \tfrac{21}{4} \quad\text{,,}\ 2\tau - g' - \tfrac{3}{4} \quad\text{,,}\ 2\tau + g' \\ & + \tfrac{9}{2}e' \quad\text{,,}\ 2g' + \tfrac{11}{2}e' \quad\text{,,}\ 2\tau - 2g' + \tfrac{3}{2} \quad\text{,,}\ 2\tau + \tfrac{3}{4}m^2 \quad\text{,,}\ 2\tau - g',\end{aligned}$$

and

$$\begin{aligned}\delta Q = {} & \tfrac{15}{2}e'\cdot t\sin 2\tau - \tfrac{21}{4} \quad\text{,,}\ 2\tau - g' + \tfrac{3}{4} \quad\text{,,}\ 2\tau + g' \\ & - \tfrac{11}{2}e' \quad\text{,,}\ 2\tau - 2g' + \tfrac{3}{4}m^2 \quad\text{,,}\ 2\tau + \tfrac{3}{4}m^2 \quad\text{,,}\ 2\tau - g' + \tfrac{3}{4}m^2 \quad\text{,,}\ 2\tau + g';\end{aligned}$$

whence also (see Annex 8)

$$n\times\delta\rho = \tfrac{3}{2}e'\cdot t\cos g' - \tfrac{15}{4}e'\cdot\text{,,}\ 2\tau + \text{\&c.}$$

The equation for $\delta\rho$ may be written,

$$\frac{d^2\delta\rho}{dt^2} + n^2\delta\rho = n^2\delta P,$$

and the solution is

$$\delta\rho = A\cos(nt+B) + \frac{1}{n}\int_0^t n^2\delta P\cdot\sin n(t-t')\,dt'.$$

Taking the constant term and the terms in $t\cos g'$, $t\sin g'$, we have

$$\delta\rho = -\frac{1}{3}\delta P_0 + \frac{1}{2}t\sin t\cdot(\text{coeff. of }\cos t\text{ in }\delta P) - \frac{1}{2}t\cos t\cdot(\text{coeff. of }\sin t\text{ in }\delta P);$$

and thence

$$\delta\rho = \tfrac{3}{2}e'\cdot t\cos g' - \tfrac{15}{4}e' \quad\text{,,}\ 2\tau + \text{\&c.}$$

In the foregoing values of δP, δQ, the arguments are given in terms of τ and g', but for the numerical calculation it is more convenient to use the arguments in terms of the mean longitudes. Restoring the argument t in the coefficients of e', and writing $2\tau = 2nt - 2g'$ (or, what is the same thing, omitting t and restoring g'), we have

$$\begin{aligned}
\delta P = {} & \left(\tfrac{3}{2} - \tfrac{15}{4} + \tfrac{21}{4} - \tfrac{3}{4}\right) e' \cos g' \\
& + \left(\tfrac{9}{2} - \tfrac{11}{2} + \tfrac{3}{2}\right) e' \cos(2\tau - g') \\
& + \&\text{c.} \\
= {} & \left(\tfrac{3}{2} - \tfrac{15}{4} + \tfrac{21}{4} - \tfrac{3}{4}\right) e' \cos g' + \&\text{c.} \\
= {} & \tfrac{3}{2} e' \cos g' + \&\text{c.}
\end{aligned}$$

VII.

The secular acceleration is obtained from the non-periodic part of $\dfrac{d\delta v}{dt}$, that is of $2\delta\rho + \delta S$. We have

$$\frac{d\delta v}{dt} = 2\delta\rho + \delta S,$$

and the secular term is

$$-2 \times \frac{1}{3} \times (\text{non-periodic part of } \delta P).$$

But we have in δP the terms

$$\tfrac{3}{4} m^2 e' + \tfrac{3}{2} m^2 e' \cos g' \cdot e \cos \tau,$$

and the non-periodic part of this is

$$\tfrac{3}{4} m^2 e';$$

and thence the secular term in $\dfrac{d\delta v}{dt}$ is

$$-2 \times \tfrac{1}{3} \times \tfrac{3}{4} m^2 e' = -\tfrac{1}{2} m^2 e'.$$

But this is not the whole; for we have to consider the terms arising from the combination of the periodic terms in $\delta\rho$ and δS.

The expression for δS is

$$\delta S = \Theta \cdot Q(-\sin g'),$$

and in this we have the term

$$-2e\cos\tau \times e\sin\tau \times (-\sin g') = e^2 \sin 2\tau \cdot \sin g'.$$

This gives in combination with the periodic term in $\delta\rho$, the non-periodic terms which contribute to the secular acceleration.

The complete calculation is given in the Annexes. The result is that the coefficient of the secular acceleration is

$$-\left(\tfrac{3}{2}m^2 - \tfrac{3771}{64}m^4\right)\int (e'^2 - E'^2)\, n\, dt,$$

or, as it is written by Prof. Adams,

$$-\left(\tfrac{3}{2}m^2 - \tfrac{2187}{128}m^4\right)\int (e'^2 - E'^2)\, n\, dt.$$

The difference between these two expressions is

$$\left(\tfrac{3771}{64} - \tfrac{2187}{128}\right) m^4 = \tfrac{7542-2187}{128} m^4 = \tfrac{5355}{128} m^4,$$

and it is in fact by the addition of these that Plana's coefficient $\frac{2187}{128}$ is changed into $\frac{3771}{64}$.

The value of the coefficient is

$$\tfrac{3}{2}m^2 - \tfrac{3771}{64}m^4;$$

and it is interesting to see how this is made up. Writing

$$M = \tfrac{3}{2}m^2, \qquad N = \tfrac{3771}{64}m^4,$$

we have

$$\text{Coefficient} = M - N.$$

In the several Annexes the terms are calculated as follows:

Value.
$M. = \frac{3}{2}m^2$
$M_1. = \frac{15}{4}m^2$
$M_2. = \frac{21}{4}m^2$
$M_3. = \frac{3}{4}m^2$
$M_4. = \frac{9}{2}m^2$
$M_5. = \frac{11}{2}m^2$
$M_6. = \frac{3}{2}m^2$
$M_7. = \frac{3}{4}m^2$

The combination of these terms gives the final result; and it will be observed that the terms which contain e' are those which ultimately give the secular acceleration.

The calculation is completed in Annex 23; and the result is

$$-\tfrac{1275}{32}m^4 ne't,$$

which is the required coefficient of the secular acceleration.

It may be remarked that the terms which rise by integration in δv are those which contain the argument g' (or $2g'$) in the coefficient; and these terms give the secular acceleration. The terms which contain t in the coefficient give the secular variation of the mean motion; and the constant terms give the periodic variations.

IV.

The part $\dfrac{m'}{\rho'^3}\rho^2 P\cos g'$ of δR gives

$$\frac{m'}{\rho'^3}\rho^2 = m^2\left(1+\tfrac{1}{3}m^2-2e\cos\tau+\tfrac{3}{2}e'\cos g'+e^2\left(-1+\tfrac{5}{2}\cos 2\tau\right)\right),$$

and

$$P = 1+\tfrac{1}{6}m^2-e^2+e\cos\tau+\left(\tfrac{1}{6}m^2+e^2\right)\cos 2\tau.$$

The product gives the terms

$$m^2\left(1+\tfrac{1}{3}m^2\right)\left(1+\tfrac{1}{6}m^2-e^2\right)\cos g' = m^2\left(1+\tfrac{1}{2}m^2-e^2\right)\cos g'.$$

The part $\dfrac{m'}{\rho'^3}\rho^2 Q(-\sin g')$ of δS gives

$$Q = e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau,$$

and the product gives the terms

$$-m^2\left(1+\tfrac{1}{3}m^2\right)\left(e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau\right)\sin g'.$$

The combination of these terms gives in δP and δQ the expressions which have been already obtained.

The calculation of the several terms is given in the following Annexes.

The part $\frac{1}{\rho^3}$ gives

$$\frac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau + \&\text{c.}$$

and the product with δP gives the terms which are required for the secular acceleration.

We have

$$\delta P = \frac{1}{\rho^3}\left(\delta R + 2\int \delta Q\, dt\right),$$

and the non-periodic terms of this expression give the secular acceleration.

The calculation is as follows:

$$\frac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau$$

$$\begin{aligned} \times\ \delta P &= \tfrac{3}{2}e'\cdot t\cos g' - \tfrac{15}{4}e' \quad ,,\ 2\tau + \tfrac{21}{4} \quad ,,\ 2\tau - g' \\ &= \quad -\tfrac{3}{4} \quad ,,\ 2\tau + g' + \tfrac{9}{2}e' \quad ,,\ 2g' + \tfrac{11}{2}e' \quad ,,\ 2\tau - 2g' \end{aligned}$$

The product of these two expressions gives the terms which are required. The non-periodic terms are obtained by combining the terms which have the same argument; and the result is the secular acceleration.

The sine terms being disregarded, we have

$$\frac{d\delta v}{dt} = 2\delta\rho + \delta S,$$

and the secular term is obtained by taking the non-periodic part of this expression.

The final result is obtained by combining all the terms. We have

$$\delta p + \delta\rho = 1 + X,$$

and thence

$$P + \delta P = r + \frac{1}{r} = 1 - X + X^2 - \&\text{c.}$$

The non-periodic part whereof is

$$1 - \tfrac{1}{2}X^2 + \tfrac{1}{8}X^4 + \left(-\tfrac{1}{2}M^2 + \tfrac{1}{4}M^4\right)E^2 + \left(\tfrac{1}{2}M^2 - \tfrac{1}{2}M^4\right)2E_f^2 t + \&\text{c.}$$

The terms in m^4 and $m^2e'^2$ are

$$= \tfrac{1}{4}M^4 + \left(\tfrac{1}{2} - \tfrac{1}{4}\right)M^2E^2,$$

which are $= \frac{1}{4}M^4$; so that the foregoing expression of the non-periodic part of $p + \delta p$ is

$$1 - \tfrac{1}{2}M^2 + \tfrac{1}{8}M^4 + \left(-\tfrac{1}{2}M^2 + \tfrac{1}{4}M^4\right)E^2 + \left(\tfrac{1}{2}M^2 - \tfrac{1}{2}M^4\right)2E_f^2 t;$$

or the secular term of δp is

$$\left(\tfrac{1}{2}M^2 - \tfrac{1}{2}M^4\right)2E_f^2 t,$$

which is the required formula.

V.

It is interesting to see how the coefficient $\frac{3771}{64}$ is made up. In Prof. Adams' Memoir we have

$$+\tfrac{195}{2} - \tfrac{385}{4} + \&\text{c.}$$

where it may be remarked that the terms

$$\tfrac{195}{2} \text{ and } \tfrac{385}{4}$$

make together $3\frac{5}{8} = \frac{5355}{128}$; and that it is in fact by the addition of these that Plana's coefficient $\frac{2187}{128}$ is changed into $\frac{3771}{64}$.

But in regard to the non-periodic terms of $m^2n^2\delta P$, the arguments 2τ, $2\tau - g'$, $2\tau + g'$ give terms which are combined with the terms in $\frac{1}{\rho^3}$ to produce the secular acceleration; and the terms in g', $2g'$ give terms which rise by integration in δv.

The complete investigation requires the calculation of all these terms; and the result is given in the following Annexes.

ANNEXES CONTAINING THE DETAILS OF THE CALCULATION.

Annex 1.

Calculation of term in $m^2\delta P$.

We have

$$\frac{m'}{\rho'^3}\rho^2 = m^2\left(1 + \tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1 + \tfrac{5}{2}\cos 2\tau\right)\right),$$

and

$$P = 1 + \tfrac{1}{6}m^2 - e^2 + e\cos\tau + \left(\tfrac{1}{6}m^2 + e^2\right)\cos 2\tau.$$

The product is

$$\begin{aligned} m^2\Big[&\left(1 + \tfrac{1}{3}m^2\right)\left(1 + \tfrac{1}{6}m^2 - e^2\right) + \left(1 + \tfrac{1}{3}m^2\right)e\cos\tau + (-2e\cos\tau)\left(1 + \tfrac{1}{6}m^2 - e^2\right) \\ &+ \left(1 + \tfrac{1}{3}m^2\right)\left(\tfrac{1}{6}m^2 + e^2\right)\cos 2\tau + (-2e\cos\tau)(e\cos\tau) \end{aligned}$$

which is

$$= m^2\Big[1+\tfrac{1}{2}m^2-e^2+\left(1 + \tfrac{1}{3}m^2 - 2 - \tfrac{1}{3}m^2 + 2e^2\right)e\cos\tau+\left(\tfrac{1}{6}m^2 + e^2 - 2e^2\right)\cos 2\tau\Big] \cos$$

or simply

$$= m^2\left(1 + \tfrac{1}{2}m^2 - e^2 - e\cos\tau + \left(\tfrac{1}{6}m^2 - e^2\right)\cos 2\tau\right)\cos g'.$$

Annex 2.

Calculation of term in $m^2\delta Q$.

We have

$$Q = e\sin\tau + \tfrac{5}{4}e^2\sin 2\tau,$$

and

$$\frac{m'}{\rho'^3}\rho^2 = m^2\left(1+\tfrac{1}{3}m^2 - 2e\cos\tau + \tfrac{3}{2}e'\cos g' + e^2\left(-1+\tfrac{5}{2}\cos 2\tau\right)\right).$$

The product is

$$-m^2\Big[\left(1+\tfrac{1}{3}m^2\right)\left(e\sin\tau + \tfrac{5}{4}e^2\sin 2\tau\right) + \left(-2e\cos\tau\right)\left(e\sin\tau\right) \\ + \left(-2e\cos\tau\right)\left(\tfrac{5}{4}e^2\sin 2\tau\right)\Big]\sin g',$$

which is

$$= -m^2\left(e\sin\tau + \tfrac{5}{4}e^2\sin 2\tau - e^2\sin 2\tau\right)\sin g' = -m^2\left(e\sin\tau + \tfrac{1}{4}e^2\sin 2\tau\right)\sin g'.$$

Annex 3.

Calculation of term in $m^2n^2\delta P$ from $m^2n^2\delta R$.

We have $m^2n^2\delta R =$ product in Annex 1; and the terms are:

$$m^2n^2\left(1+\tfrac{1}{2}m^2 - e^2\right)\cos g' = \left(1+\tfrac{1}{2}m^2 - e^2\right)m^2n^2\cos g'.$$

Annex 4.

Calculation of term in $m^2n^2\delta P$ from $2m^2n^2\int\delta Q\,dt$.

We have $2m^2n^2\int\delta Q\,dt$ = twice the integral of the product in Annex 2. The integral of the product is

$$-m^2\int\left(e\sin\tau + \tfrac{1}{4}e^2\sin 2\tau\right)\sin g'\,dt = m^2\left(e\cos\tau + \tfrac{1}{8}e^2\cos 2\tau\right)\sin g',$$

and the double of this is

$$2m^2\left(e\cos\tau + \tfrac{1}{8}e^2\cos 2\tau\right)\sin g'.$$

Annex 5.

Calculation of term in $m^2n^2\delta Q$ from $2m^2n^2\delta\rho$.

We have $\delta\rho = \frac{1}{n^2}m^2n^2\delta\rho$; and $m^2n^2\delta\rho$ is obtained from $m^2n^2\delta P$ by dividing by -3. The result is

$$-\tfrac{1}{3}m^2\left(1+\tfrac{1}{2}m^2-e^2-e\cos\tau+\left(\tfrac{1}{6}m^2-e^2\right)\cos 2\tau\right)\cos g'.$$

And thence

$$2m^2n^2\delta\rho = -\tfrac{2}{3}m^2\left(1+\tfrac{1}{2}m^2-e^2-e\cos\tau+\left(\tfrac{1}{6}m^2-e^2\right)\cos 2\tau\right)\cos g'.$$

Annex 6.

Calculation of term in $m^2n^2\delta P$ from $2m^2n^2\int\delta Q\,dt$, where δQ contains t.

The term in δQ containing t is obtained from the terms in $\delta\rho$ which rise by integration. We have in $\delta\rho$ the terms

$$3m^2\sin g' + \tfrac{3}{2}t\cos g',$$

and these give in δQ the terms

$$2\times 3m^2\sin g' = 6m^2\sin g',$$

and

$$2\times\tfrac{3}{2}t\cos g' = 3t\cos g'.$$

The integral gives in δP the terms

$$2\int 6m^2\sin g'\,dt = -12m^2\cos g',$$

and

$$2\int 3t\cos g'\,dt = 3t^2\cos g'.$$

Annex 7.

Calculation of term in $m^2n^2\delta Q$ from $m^2n^2\delta S$.

We have $\delta S = \Theta\cdot Q(-\sin g')$; and the terms are calculated as in Annex 2. The result is

$$-m^2\left(e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau\right)\sin g'.$$

Annex 8.

Calculation of $n \times \delta\rho$.

We have

$$n \times \delta\rho = \frac{n}{n^2} \cdot m^2 n^2 \delta\rho = \frac{m^2 n^2 \delta\rho}{n}.$$

The terms of $m^2 n^2 \delta\rho$ are obtained from those of $m^2 n^2 \delta P$ by dividing by -3. Hence

$$n \times \delta\rho = -\tfrac{1}{3n} \cdot m^2 n^2 \delta P.$$

Annex 9.

Calculation of term in $m^2 n^2 \delta P$ from $-\frac{2}{\rho^3}\delta\rho$.

We have

$$\frac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau;$$

and $\delta\rho$ contains the terms

$$-\tfrac{1}{3} \cdot \tfrac{3}{2}e' \cdot t\cos g' = -\tfrac{1}{2}e' \cdot t\cos g',$$

and

$$-\tfrac{1}{3}\left(-\tfrac{15}{4}e'\right)\cos 2\tau = \tfrac{5}{4}e'\cos 2\tau.$$

The product gives terms which contribute to the secular acceleration.

Annex 10.

Calculation of terms in $\delta\rho$ from δP.

In $\frac{d^2}{dt^2}\delta\rho + n^2\delta\rho = n^2\delta P$, a term $n^2 A\cos n\alpha t$ gives in $\delta\rho$

$$\frac{A}{1-\alpha^2}\cos n\alpha t;$$

and a term $n^2 B\sin n\alpha t$ gives in $\delta\rho$

$$\frac{B}{1-\alpha^2}\sin n\alpha t.$$

If $\alpha = 2$, then $\frac{1}{1-\alpha^2} = -\frac{1}{3}$; and if $\alpha = m$ or $2m$, then $\frac{1}{1-\alpha^2} = 1$, approximately.

For the argument g' (where $\alpha = m$), we have

$$\delta\rho = \tfrac{3}{2}e' \cdot t\cos g' + \&\text{c}.$$

Annex 11.

Calculation of $\dfrac{d\delta v}{dt}$; viz. this is $= -2n\delta\rho + \dfrac{m^2n^2}{\rho^3}\int \delta Q\,dt$.

We have

$$\frac{d\delta v}{dt} = -2n\delta\rho + \frac{m^2n^2}{\rho^3}\int \delta Q\,dt.$$

The terms in $\delta\rho$ give

$$-2n \times \left(-\tfrac{1}{2}e' \cdot t\cos g'\right) = ne' \cdot t\cos g';$$

and the terms in $\int \delta Q\,dt$ give

$$\frac{m^2n^2}{\rho^3} \times \left(-\tfrac{3}{4}e' \cdot t\sin g'\right) = -\tfrac{3}{4}m^2ne' \cdot t\sin g'.$$

Annex 12.

Calculation of term in $m^2n^2\delta P$ from $\dfrac{1}{\rho^3} \times \delta P$.

We have

$$\frac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau;$$

and

$$\delta P = \tfrac{3}{2}e' \cdot t\cos g' - \tfrac{15}{4}e' \quad ,, \ 2\tau + \tfrac{21}{4} \quad ,, \ 2\tau - g' - \tfrac{3}{4} \quad ,, \ 2\tau + g'.$$

The product is calculated as follows:

$$\begin{array}{r} \dfrac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau \qquad\qquad\qquad\qquad\qquad\qquad \\ \times\ \delta P = \tfrac{3}{2}e' \cdot t\cos g' - \tfrac{15}{4}e' \quad ,, \ 2\tau + \tfrac{21}{4} \quad ,, \ 2\tau - g' - \tfrac{3}{4} \quad ,, \ 2\tau + g' \end{array}$$

The terms in the product are:

$$\begin{aligned}
&\tfrac{3}{2}e' \cdot t\cos g' \cdot \left(1 + \tfrac{1}{2}m^2\right) = \left(\tfrac{3}{2} + \tfrac{3}{4}m^2\right) e' \cdot t\cos g';\\
&- \tfrac{15}{4}e'\cos 2\tau \cdot 3e\cos\tau = -\tfrac{45}{4}ee'\cos 2\tau\cos\tau = -\tfrac{45}{8}ee'\left(\cos 3\tau + \cos\tau\right);\\
&+ \tfrac{21}{4}\cos(2\tau - g') \cdot 3e\cos\tau = \tfrac{63}{4}e\cos(2\tau - g')\cos\tau = \tfrac{63}{8}e\left(\cos(3\tau - g') + \cos(\tau - g')\right.\\
&- \tfrac{3}{4}\cos(2\tau + g') \cdot 3e\cos\tau = -\tfrac{9}{4}e\cos(2\tau + g')\cos\tau = -\tfrac{9}{8}e\left(\cos(3\tau + g') + \cos(\tau + g'\right.
\end{aligned}$$

The foregoing terms give in δP the expressions which are combined with the terms in $\frac{1}{\rho^3}$ to give the secular acceleration.

Annex 13.

Calculation of term in $m^2 n^2 \delta P$ from $-\dfrac{2}{\rho^3}\delta\rho$.

The terms are calculated by combining the terms in $\frac{1}{\rho^3}$ with those in $\delta\rho$ which have been obtained in Annex 9.

Annex 14.

Calculation of term in $m^2 n^2 \delta Q$ from $-\dfrac{2}{\rho^3}\delta v$.

The terms are calculated in a similar manner; and the result is

$$-\tfrac{2}{3}m^2\left(1+\tfrac{1}{2}m^2\right)\left(e\sin\tau+\tfrac{5}{4}e^2\sin 2\tau\right)\sin g'.$$

Annex 15.

Calculation of term in $m^2 n^2 \delta P$ from $\dfrac{m^2 n^2}{\rho^3}\int \delta Q\, dt$.

We have

$$\frac{m^2 n^2}{\rho^3}\int \delta Q\, dt = m^2 n^2\left(1+\tfrac{1}{2}m^2+3e\cos\tau\right)\int \delta Q\, dt.$$

The integral $\int \delta Q\, dt$ is calculated from the terms in δQ; and the result, combined with the terms in $\frac{1}{\rho^3}$, gives the required expression.

Annex 16.

Calculation of term in $m^2n^2\delta Q$ from $\dfrac{m^2n^2}{\rho^3}\int \delta Q\,dt$.

We have

$$\delta Q = \tfrac{15}{2}e'\cdot t\sin 2\tau - \tfrac{21}{4}\sin(2\tau-g') + \tfrac{3}{4}\sin(2\tau+g') - \tfrac{11}{2}e'\sin(2\tau-2g').$$

The integral is

$$\int \delta Q\,dt = -\tfrac{15}{4}e'\cdot t\cos 2\tau + \tfrac{15}{8}e'\sin 2\tau + \tfrac{21}{8}\cos(2\tau-g') - \tfrac{3}{8}\cos(2\tau+g') + \tfrac{11}{4}e'\cos(2\tau-2g')$$

Multiplying by $\frac{m^2n^2}{\rho^3}$, we obtain the required expression.

Annex 17.

The additional term $-\frac{171}{4}m^2e'\cos g'$ is ultimately added (see Annex 17) to the coefficient of δv.

This term arises from the combination of the periodic terms in $\delta\rho$ and δS; and its value is calculated in the following Annexes.

Annex 18.

Calculation of term in $m^2n^2\delta P$ from $\dfrac{1}{\rho^3} \times \delta P$.

We have

$$\frac{1}{\rho^3} = 1 + \tfrac{1}{2}m^2 + 3e\cos\tau + \tfrac{9}{2}e^2\cos 2\tau;$$

and the terms in δP are combined with this to give the required result.

Annex 19.

Calculation of term in $m^2n^2\delta P$ from $-\dfrac{2}{\rho^3}\delta\rho$.

The term in $\delta\rho$ which gives rise to the secular acceleration is

$$-\tfrac{1}{2}e' \cdot t\cos g';$$

and this, combined with the terms in $\frac{1}{\rho^3}$, gives

$$-\tfrac{2}{\rho^3} \times \left(-\tfrac{1}{2}e' \cdot t\cos g'\right) = \frac{e' \cdot t\cos g'}{\rho^3} = \left(1 + \tfrac{1}{2}m^2\right) e' \cdot t\cos g'.$$

Annex 20.

Calculation of term in $m^2n^2\delta P$ from $\dfrac{m^2n^2}{\rho^3}\int \delta Q\,dt$.

We have

$$\frac{m^2n^2}{\rho^3}\int \delta Q\,dt = m^2n^2\left(1 - \tfrac{1}{2}m^2 + 3e\cos\tau\right)\int \delta Q\,dt.$$

The terms in $\int \delta Q\,dt$ are:

$$\int \delta Q\,dt = -\tfrac{15}{4}e' \cdot t\cos 2\tau + \&\text{c}.$$

The product is

$$n \times \left(-\tfrac{6}{4} - 3\right) = -\tfrac{15}{2} m^2;$$

that is

$$-\tfrac{15}{2} m^2 \cdot e' \cos g' \cdot t \sin 2\tau.$$

Annex 21.

Calculation of a term in $m^2 n^2 \left(C + \int \delta Q \, dt\right)$.

The part $-\frac{1}{4}\left(-3\sin 2v - 2v\right)\delta\rho$ of δQ gives

$$-\tfrac{1}{4}\left(-3\sin 2v - 2v\right)\delta\rho = \tfrac{1}{4}\left(3\sin 2\tau + 2\tau\right)\delta\rho;$$

and we have thence in $m^2 n^2 \delta Q$ the term

$$\tfrac{3}{4} m^2 n^2 \sin 2\tau \cdot \delta\rho = \tfrac{3}{4} m^2 n^2 e \sin\tau \cdot \tfrac{1}{2} e' \cdot t \cos g';$$

that is

$$\tfrac{3}{8} m^2 n^2 e e' \sin\tau \cdot t \cos g'.$$

The part $\frac{1}{4}\left(3\cos 2v - 2v\right)\delta v$ of δQ gives

$$\tfrac{1}{4}\left(3\cos 2\tau - 2\tau\right)\delta v;$$

and we have thence in $m^2 n^2 \delta Q$ the term

$$\tfrac{3}{4} m^2 n^2 \cos 2\tau \cdot \delta v = \tfrac{3}{4} m^2 n^2 e \cos\tau \cdot \tfrac{1}{2} e' \cdot t \sin g';$$

that is

$$\tfrac{3}{8} m^2 n^2 e e' \cos\tau \cdot t \sin g'.$$

Annex 22.

Calculation of term in $m^2n^2\delta P$ from $\dfrac{1}{\rho^3}\times\delta P$.

We have

$$\frac{1}{\rho^3}=1+\tfrac{1}{2}m^2-3m^2e'\cos g'+3e\cos\tau+\tfrac{9}{2}e^2\cos 2\tau;$$

and

$$\delta P=\tfrac{3}{2}e'{\cdot}t\cos g'-\tfrac{15}{4}e' \quad ,, \ 2\tau+\tfrac{21}{4} \quad ,, \ 2\tau-g'-\tfrac{3}{4} \quad ,, \ 2\tau+g'+\tfrac{9}{2}e' \quad ,, \ 2g'+\tfrac{11}{2}e' \quad ,, \ 2\tau-2g$$

Annex 23.

Calculation of term in $\dfrac{m^2n^2}{\rho^3}\left(C+\int\delta Q\,dt\right)$.

We have

$$\frac{1}{\rho^3}=1+\tfrac{1}{3}m^2-3m^2e'\cos g'+2m^2\cos 2\tau+7m^2e'\cos(2\tau-g')-m^2e'\cos(2\tau+g);$$

and

$$m^2n^2\left(C+\int\delta Q\,dt\right)=n\times\left(-\tfrac{1455}{32}m^2e'\cdot t\right).$$

The terms of the product are:

$$\begin{aligned}
&-\tfrac{1455}{32}m^2e'\cdot t,\\
&-\tfrac{15}{4}m^2e'\cdot t\cos 2\tau,\\
&+\tfrac{21}{8}m^2e' \quad ,, \ 2\tau-g',\\
&-\tfrac{3}{8}m^2e' \quad ,, \ 2\tau+g'.
\end{aligned}$$

The combination gives

$$\begin{aligned}
&-\tfrac{1455}{32}-\tfrac{15}{4}\cdot\tfrac{1}{2}\cdot 2+\tfrac{21}{8}\cdot\tfrac{1}{2}\cdot 7-\tfrac{3}{8}\cdot\tfrac{1}{2}\cdot(-1)\\
&=\left(-\tfrac{1455}{32}-\tfrac{15}{4}+\tfrac{147}{16}+\tfrac{3}{16}\right)m^4ne'\cdot t\\
&=\left(-\tfrac{1455}{32}+\tfrac{45}{8}\right)m^4ne'\cdot t\\
&=-\tfrac{1275}{32}m^4ne'\cdot t,
\end{aligned}$$

which is the required result; and this completes the series of calculations.

222.

ON LAMBERT'S THEOREM FOR ELLIPTIC MOTION.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXII. (1862), pp. 238–242.]

The theorem referred to is that which gives the time of description of an elliptic arc in terms of the radius vectors and the chord. The demonstration given by the author in his "Insigniores Orbitae Cometarum Proprietates," Augs. 1761, depends upon a series of geometrical propositions of great elegance, which may be thus stated.

[scale=0.85] [thick] (0,0) ellipse (4cm and 2.5cm); [thick] (-4.2,0) – (4.2,0); [black] (-2.4,0) circle (2pt) node[below left] F; [black] (2.4,0) circle (2pt) node[below right] f; [black] (0,0) circle (1.5pt) node[below] C; (Q) at (1.8,1.9); (P) at (-3.2,1.2); [black] (Q) circle (1.5pt) node[above] Q; [black] (P) circle (1.5pt) node[above left] P; [thick] (-2.4,0) – (Q); [thick] (2.4,0) – (Q); [thick] (-2.4,0) – (P); [thick] (2.4,0) – (P); [thick] (P) – (Q); (H) at (1.0,2.45); [black] (H) circle (1.5pt) node[above] H; (M) at (3.6,1.0); [black] (M) circle (1.5pt) node[right] M; [thick] (-2.4,0) – (H); [thick] (2.4,0) – (H); [thick] (-2.4,0) – (M); [thick] (Q) – (M); (G) at (0.6,0.65); [black] (G) circle (1.2pt) node[right] G; [thick,dashed] (Q) – (G); (E) at (-0.3,0.85); [black] (E) circle (1.2pt) node[left] E; [thick] (H) ++(-0.6,-0.2) arc (200:280:0.7); at (0.4,2.2) $\frown$;

Let PQ be a line given in magnitude and position, E a given point on this line, Qf a line given in magnitude only, the position thereof being determined by assigning a value to its variable inclination to the line PQ. With F, f, as foci describe an ellipse passing through the point Q (the axis major, $= FQ + Qf$, is of course a constant magnitude). Take C, the centre of the ellipse, and join CQ; through E draw a chord, MEN, conjugate to the diameter CQ and meeting it in G. Then treating the inclination as variable,

1. The locus of H (the intersection of FQ, fM) is a right line.

This is proved by means of the property that the radius of the director circle of the ellipse is equal to the major axis; or it may be proved directly.

2. The locus of N (the intersection of FM, fQ) is a circle passing through F and f.

This depends on the property that the product of the perpendiculars from the foci on any tangent is constant.

3. If P be the intersection of the lines FN, HM; then the locus of P is an ellipse having the same foci with the original ellipse.

This is a consequence of (1) and (2); for the locus of H being a right line, and that of N being a circle, the intersection P of the lines FN, HM has for its locus a conic.

4. The ellipse, locus of P, passes through the point E.

This is the theorem which gives Lambert's theorem. For if the ellipse locus of P pass through E, then, when P is at E, the points M, N coincide with E; that is, the line MEN becomes the tangent at E, and G is the point of contact. But MEN is conjugate to CQ; therefore the tangent at E is conjugate to CQ; and the time of describing the arc QE of the original ellipse is proportional to the area QCE; that is, to the sector of the ellipse locus of P; which sector is determined by the radii vectores FP, fP, that is, FE, fE, and the chord Ef of the ellipse locus of P.

5. The ellipse, locus of P, has for its major axis the line PH.

This is a consequence of the property that the sum of the distances of any point on the ellipse from the foci is constant.

6. The minor axis is determined by the condition that the radius of the director circle is equal to the major axis.

7. The ellipse, locus of P, is completely determined; and it may be shown that it passes through E.

Lambert's theorem is thus established by pure geometry.

Writing $FQ = \rho$, $Qf = \sigma$, $Ff = 2c$, then $\rho + \sigma = 2a$; and the theorem is that the time is a function of $\rho + \sigma$, c, and PQ ($= \chi$, say). The analytical proof is as follows.

Let u, v be the eccentric anomalies of Q, P; then the coordinates of Q are $a\cos u$, $b\sin u$; and those of P are $a\cos v$, $b\sin v$. The distance PQ is given by

$$\chi^2 = a^2(\cos u - \cos v)^2 + b^2(\sin u - \sin v)^2.$$

The radius vector FQ is $\rho = a(1 - e\cos u)$; and $Qf = \sigma = a(1 + e\cos u)$. Hence

$$\rho + \sigma = 2a, \qquad \sigma - \rho = 2ae\cos u.$$

Similarly, if $FP = \rho'$, $Pf = \sigma'$, then

$$\rho' + \sigma' = 2a, \qquad \sigma' - \rho' = 2ae\cos v.$$

The time of describing the arc QP is

$$T = \frac{a^{\frac{3}{2}}}{\sqrt{\mu}}\left[(u - v) - e(\sin u - \sin v)\right].$$

Now

$$\begin{aligned}
\chi^2 &= a^2(\cos u - \cos v)^2 + a^2(1 - e^2)(\sin u - \sin v)^2 \\
&= a^2\left[2 - 2\cos u\cos v - 2\sin u\sin v + e^2(\sin u - \sin v)^2\right] \\
&= 2a^2\left[1 - \cos(u - v) + \tfrac{1}{2}e^2(\sin u - \sin v)^2\right] \\
&= 4a^2\sin^2\tfrac{1}{2}(u - v)\left[1 - \tfrac{1}{4}e^2\cdot\frac{(\sin u - \sin v)^2}{\sin^2\frac{1}{2}(u - v)}\right].
\end{aligned}$$

But

$$\frac{\sin u - \sin v}{\sin\frac{1}{2}(u - v)} = 2\cos\tfrac{1}{2}(u + v);$$

and therefore

$$\chi^2 = 4a^2\sin^2\tfrac{1}{2}(u - v)\left[1 - e^2\cos^2\tfrac{1}{2}(u + v)\right].$$

Also

$$\rho + \sigma = 2a, \qquad \rho' + \sigma' = 2a;$$

and

$$\rho + \rho' = 2a - ae(\cos u + \cos v) = 2a - 2ae \cos \tfrac{1}{2}(u+v) \cos \tfrac{1}{2}(u-v).$$

Hence, writing

$$\sin \tfrac{1}{2}(u-v) = \frac{\chi}{2a\sqrt{1 - e^2 \cos^2 \frac{1}{2}(u+v)}},$$

and

$$\cos \tfrac{1}{2}(u-v) = \sqrt{1 - \frac{\chi^2}{4a^2(1 - e^2 \cos^2 \frac{1}{2}(u+v))}},$$

we have

$$\rho + \rho' = 2a - 2ae \cos \tfrac{1}{2}(u+v)\sqrt{1 - \frac{\chi^2}{4a^2(1 - e^2 \cos^2 \frac{1}{2}(u+v))}}.$$

This completes the analytical demonstration of Lambert's theorem.

NOTES AND REFERENCES.

173. I attach some references to the foregoing memoir on the secular acceleration of the moon's mean motion.

It is well known that Plana, developing the explanation given by Laplace for the secular variation of the moon's mean motion, obtained in the expression of the true longitude the terms $-\left(\frac{3}{2}m^2 - \frac{2187}{128}m^4\right)\int(e'^2 - E'^2)\,n\,dt$, and that Prof. Adams in his memoir "On the Secular Variation of the Moon's Mean Motion," *Phil. Trans.* t. 143 (1853), pp. 397–406, corrected this into $-\left(\frac{3}{2}m^2 - \frac{3771}{64}m^4\right)\int(e'^2-E'^2)\,n\,dt$. The validity of the correction was a good deal discussed, and it was interesting to establish the result by an entirely independent method.

The investigation which I have given in the foregoing paper is, in fact, an inirect method; and it has the advantage of showing how the coefficient $\frac{3771}{64}$ is made up. The terms which contribute to this coefficient are:

$$\begin{aligned}&\tfrac{195}{2} = 97.5,\\ &\tfrac{385}{4} = 96.25,\\ &\text{\&c.}\end{aligned}$$

And the sum of these terms gives the coefficient $\frac{3771}{64} = 58.921875$.

It may be observed that the terms which arise from the combination of the periodic terms in $\delta\rho$ and δS give the most important part of the correction; and that the terms which arise from the variation of the eccentricity of the earth's orbit give the remaining part.

The final result is that the secular acceleration of the moon's mean motion is

$$-\left(\tfrac{3}{2}m^2 - \tfrac{3771}{64}m^4\right)\int(e'^2 - E'^2)\,n\,dt,$$

which agrees with the result obtained by Prof. Adams.

Note on the history of the theorem.—Lambert's theorem was first published in the "Insigniores Orbitae Cometarum Proprietates," Augs. 1761. It was subsequently demonstrated by Lagrange in the "Mécanique Analytique," and by other writers. The demonstration given in the foregoing paper is founded upon the properties of the director circle of the ellipse, and is believed to be new.

End of Vol. III.

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